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1 Periodic Table, Periodic Properties and Variations of Properties

SYLLABUS

Periodic properties and variations of properties — Physical and Chemical

(i) Periodic properties and their variations in groups and periods.

Definitions of following periodic properties and trends in these properties in groups and periods should be studied;

- *atomic size*
- *metallic character*
- *non-metallic character*
- *ionisation potential*
- *electron affinity*
- *electronegativity*

(ii) Periodicity on the basis of atomic number for elements.

Relation between atomic number for light elements (proton number) and atomic mass for light elements; the modern periodic table up to period 3 (students to be exposed to the complete modern periodic table but no questions will be asked on elements beyond period 3 – Argon); periodicity and other related properties to be described in terms of shells (not orbitals); special reference to the alkali metals and halogen groups.

Note : *According to the recommendation of International Union of Pure and Applied Chemistry (IUPAC), the groups are numbered from 1 to 18 replacing the older notation of groups IA ..... VIIA, VIII, IB ... . VIIB and 0. However, for the examination both notations will be accepted.*

Old notation	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
New notation	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

1.1 INTRODUCTION

Elements are pure substances made up of one type of atoms. They are the basic units of all types of matter. In order to study elements in an organised manner, they need to be classified.

You have studied in class IX contribution of Dobereiner, Newland and Mendeleev to the classification of elements.

Dobereiner, grouped the elements in three (*triads*).

Newland, observed that when elements are arranged in increasing order of their atomic mass, every *eighth element beginning from any element resembles the first* element in its physical and chemical properties.

Dmitri Mendeleev, a Russian chemist gave the first periodic table of elements base on his law which states that *"the properties of elements are the periodic functions of their atomic masses"*. This arrangement enabled Mendeleev to place 63 elements known that time in vertical columns (***groups***), and in horizontal rows (***periods***). He predicted the existence of those elements which were yet to be discovered. *But this method could not explain the positions of certain elements, the rare earth metals and the isotopes.* These defects were removed when **Henry Moseley** put forward the **modern periodic table**. He stated that **"the physical and chemical properties of elements are the periodic functions of their atomic number".**

Later on Niel Bohr gave the extended form of the table known as *long form of the modern periodic table* given on page 2.

A tabular arrangement of elements in groups (vertical columns) and periods (horizontal rows) highlighting the regular trends in properties of elements is called a PERIODIC TABLE.

1.2 SALIENT FEATURES OF THE MODERN PERIODIC TABLE (SEE PAGE 2)

GROUPS

The modern periodic table has **eighteen vertical columns**. They are known as groups, arranged from left to right in the order.

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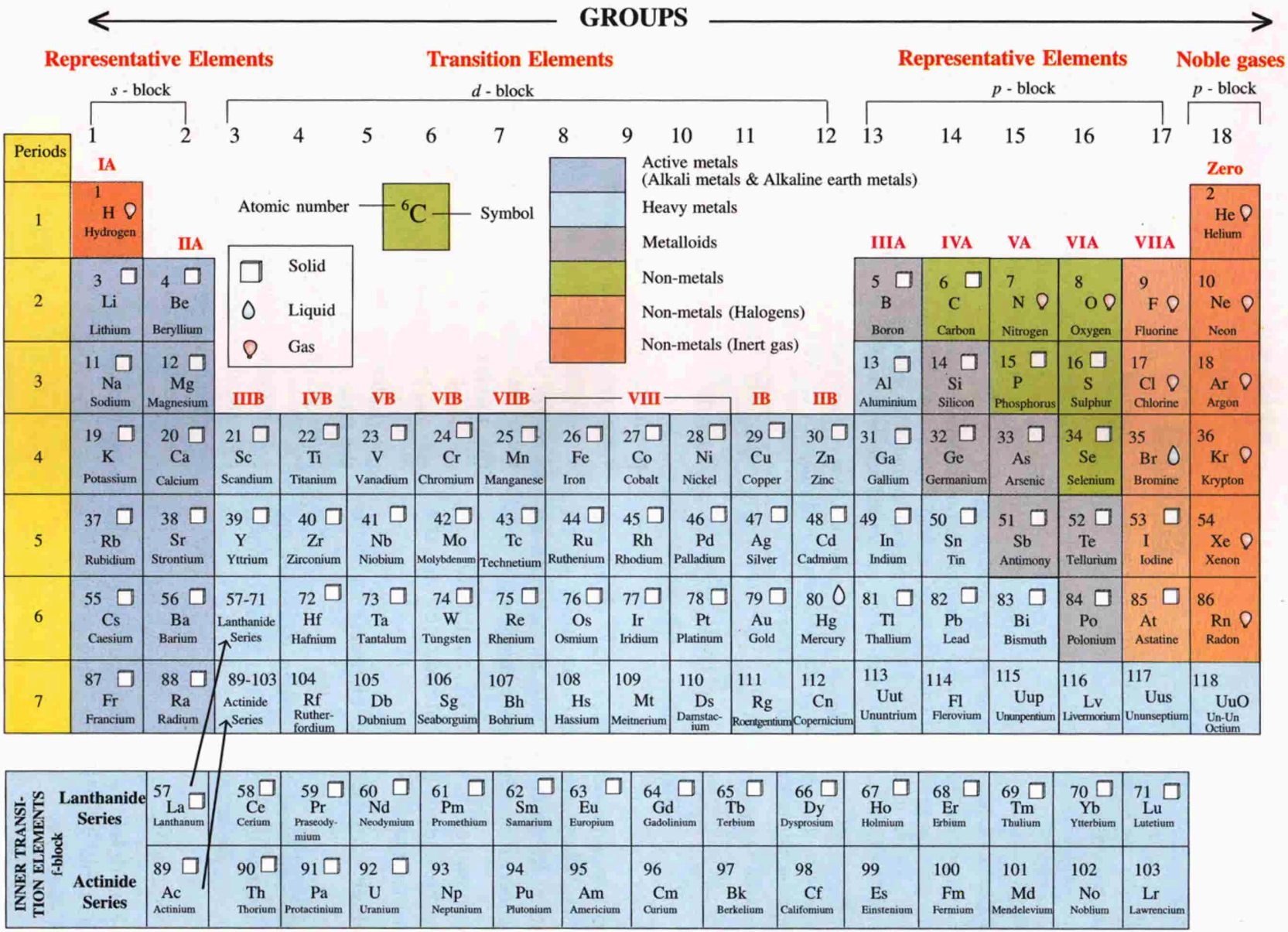
TABLE 1.1 : THE LONG FORM OF PERIODIC TABLE OF ELEMENTS



Elements with atomic number 110 and above have been reported but not yet fully authenticated and named.
As of 2005, the table contains 116 chemical elements whose discoveries have been confirmed;
94 are found naturally on earth and the rest are synthetic elements.

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Group 1 : These elements are known as **Alkali metals** as they form strong alkalis with water.

Group 2 : Alkaline earth metals – They form weaker alkalis as compared to group1 elements.

Groups 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12 are known as the **transition elements.** They have their ***two*** *outermost shells incomplete.*

Group 13 : Boron family – Boron is the first member of the group.

Group 14 : Carbon family.

Group 15 : Nitrogen family.

Group 16 : Oxygen family also known as chalcogens meaning ore forming.

Group 17 : These elements form salts and so known as **Halogen** (meaning – salt former).

Group 18 : (Zero group) – Elements of this group are called the ***noble gases*** *or* ***inert gases***. These elements have their outermost orbit complete. Due to stable electronic configuration they hardly react with other elements.

Note : The elements of groups 1, 2, 13, 14, 15, 16 and 17 are known as the **main group elements** or **representative elements** or **normal elements.**

The outermost shell of all the elements of these groups are incomplete.

PERIODS

There are **seven horizontal rows** in the modern periodic table. They are known as **periods** (see table given below).

The number of shells present in an atom determines its period. For example :

Elements of period one have one shell, elements of period two have two shells, and that of period three have three shells and so on.

Period	Type of period	Number of elements	Atomic no. of elements	No. of shell (s)	Elements in Group								
					1	2	3 - 12	13	14	15	16	17	18
1	Shortest period	2	1 – 2	1	H 1								He 2
2	Short period	8	3 – 10	2	Li 3	Be 4		B 5	C 6	N 7	O 8	F 9	Ne 10
3	Short period	8	11 –18	3	Na 11	Mg 12		Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
4	Long period	18	19 – 36	4	K 19								Kr 36
5	Long period	18	37 – 54	5	Rb 37								Xe 54
6	Longest period	32	55 – 86	6	Cs 55		La Hg 57 ↔ 80						Rn 86
7	Longest period	32	87 – 118	7	Fr 87		Ac Cn 89 ↔ 112						Uuo 118

Note : Lanthanides **Group 3 of the sixth period** and actinides **Group 3 of the seventh period** have similar properties because they belong to the same Group 3. They are shown at the bottom of the periodic table because they are large in number, and if shown in the main body of the table will distort its shape.

The **third period** elements, Na, Mg, Al, Si, P, S and Cl, summarise the properties of their respective groups and are called **typical elements.**

A period is determined by the number of shells and a group is determined by the number of electrons present in the outermost shell. For example : Sodium has atomic number 11 and its electronic configuration is 2, 8, 1. It has three orbits (shells) and has one electron in the outermost orbit hence it is placed in third period and group 1. Similarly, calcium, atomic number 20, electronic configuration 2, 8, 8, 2 is placed in fourth period and group 2.

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1.3 PERIODICITY

The properties that reappear at regular intervals, or in which there is gradual variation (*i.e.* increase or decrease) at regular intervals, are called *'periodic properties'* and the phenomenon is known as the *periodicity of elements.*

Cause of periodicity

The cause of periodicity is the recurrence of similar electronic configuration i.e. having same number of electrons in the outermost orbit.

In a particular group, **electrons in the outermost orbit** remain the same *i.e.* **electronic configuration is similar.** Since **chemical properties of elements depend upon the number of electrons in their outermost shell, thus elements of the same group have similar properties.**

For example in group 17, *i.e.,* halogens, all elements have seven electrons (see electronic configuration given in 1.4) in their respective outermost shells, therefore, they show similar properties, such as:

(i) they are coloured non-metals.

(ii) they form negative ions carrying a single charge. For example chloride ions (Cl^-).

(iii) they are very reactive and are, therefore, found in combined state.

(iv) they are only slightly soluble in water, but they dissolve much better in organic solvents like carbon disulphide, chloroform, alcohol, etc.

(v) their melting and boiling points increase regularly moving down the group.

(vi) they are good oxidising agents.

It is thus concluded that periodicity is due to the same number of electron(s) in the outermost orbit of different elements.

1.4 SHELLS (ORBITS) AND VALENCY

Orbits : Electrons revolve around the nucleus in certain definite circular paths called **orbits** or **shells.**

(1) Number of shells :

(a) Down a group, i.e., from top to bottom.

The number of shells increases successively, *i.e.,* one by one, such that the number of shells that an element has, *equals the number of the period* to which that element belongs.

For example, in halogens (Group 17).

Element with atomic numbers	No. of shells	Electronic configuration K L M N O P	Period to which the element belongs
F (9)	2	2, 7	Second
Cl (17)	3	2, 8, 7	Third
Br (35)	4	2, 8, 18, 7	Fourth
I (53)	5	2, 8, 18, 18, 7	Fifth
At (85)	6	2, 8, 18, 32, 18, 7	Sixth

(b) Across a period, i.e., from left to right.

On moving from left to right in a given period, the number of shells remains the same. *For example*, in the 2nd period, the number of shells remains two, *i.e.*, equal to the number of the period. Similarly in the third period the number of shells remains three and so on.

(2) Valency

Valency denotes the combining capacity of the atom of an element. It is equal to the number of electrons an atom can donate or accept or share.

On moving down a given group, the number of electrons in the outermost shell, i.e., valence electron, remains the same. Therefore, **valency,** in a group, also remains the same.

In a given period, the number of electrons in the valence (outermost) shell increases from left to right. But the valency increases only upto Group 14, where it becomes 4, and then it decreases, *i.e.*, in Group 17 it becomes 1.

Note 1 : Valency depends on the number of electrons in the outermost shell (*i.e.* valence shell). If the number of electrons present in the outermost shell are 1, 2, 3 or 4, then their valency

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is 1, 2, 3 or 4 respectively. If the number of electrons present in the outermost shell are 5, 6 or 7, then their valency is 8 – 5 = 3, 8 – 6 = 2 and 8 – 7 = 1 respectively.

Note 2 : Valency is the combining capacity so it is always positive.

Groups →	IA 1	IIA 2	IIIA 13	IVA 14	VA 15	VIA 16	VIIA 17	Zero 18
Elements of the 2nd period	Li	Be	B	C	N	O	F	Ne
Atomic No.	3	4	5	6	7	8	9	10
Electronic configuration	K L 2, 1	K L 2, 2	K L 2, 3	K L 2, 4	K L 2, 5	K L 2, 6	K L 2, 7	K L 2, 8
No. of shells	2	2	2	2	2	2	2	2
Valency	1	2	3	4	3	2	1	0
Formula of Hydride	LiH	BeH_2	BH_3	CH_4	NH_3	H_2O	HF	-

In the periodic table

- elements are arranged in order of increasing atomic number (proton number).
- the vertical columns of elements with similar properties are called groups.
- the horizontal rows are called periods.

Intext Questions

1. (a) State modern periodic law. Name the scientist who stated the law.
   (b) What is a periodic table ? How many groups and periods does modern periodic table have?
2. Elements of group 1 and elements of group 17 both have valency 1. Explain.
3. What are horizontal rows and vertical columns in a periodic table known as ?
4. Periodicity is observed due to the similar ..... (number of valence electrons/atomic number/electronic configuration).
5. How does the electronic configuration in atoms change
   (i) in a period from left to right ?
   (ii) in a group top to bottom ?
6. Correct the statements.
   (i) Elements in the same period have the same valency.
   (ii) Valency depends upon the number of shells in an atom.
   (iii) Copper and zinc are representative elements.
   (iv) Transition elements are placed at extreme right of the periodic table.
7. Name two elements in each case :
   (i) Alkali metals (ii) Alkaline earth metals
   (iii) Halogens (iv) Inert gas
   (v) Transition element (vi) Lanthanides
   (vii) Actinides
8. What do you understand by ?
   (i) Periodicity (ii) Typical elements
   (iii) Orbits
9. Name two elements that you would expect to show chemical reactions similar to calcium. What is the basis of your choice ?
10. Name the (i) metals (ii) metalloids and (iii) non-metals in the first twenty elements.
11. Fluorine, chlorine and Bromine are put in one group on the basis of their similar properties.
    (i) What are those similar properties ?
    (ii) What is the common name of this group or family ?
12. What is the main characteristic of the last element in each period of the Periodic Table ? What is the general name of such elements ?
13. According to atomic structure, what determines which element will be the first and which will be the last in a period?
14. How does the number of :
    (i) valence electrons and
    (ii) valency vary on moving from left to right in the second period of the periodic table ?
15. Name the type of elements, which have their
    (i) outermost shell complete
    (ii) outermost shell incomplete
    (iii) two outermost shell incomplete
    (iv) one electron short of octet
    (v) two electrons in the outermost orbit.
16. An element has 2 electrons in its N shell.
    (i) What is its atomic number ?
    (ii) State its position in periodic table
    (iii) Is it metal or non-metal ?
    (iv) State the name assigned to this group.
17. Answer the following in respect of element $^{32}_{16}S$.
    (i) Give its electronic configuration.
    (ii) To which group and period does it belong ?
    (iii) What is its valency ?
    (iv) Is it metal or non-metal ?
    (v) Is it a reducing agent or an oxidising agent ?
    (vi) Give its formula with chlorine.

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1.5 PERIODIC PROPERTIES

The properties of elements which are directly or indirectly related to their electronic configurations and show a regular gradation as we move across a period, from left to right or down the group from top to bottom, are called **Periodic Properties**.

Important periodic properties are :

(i) ***atomic size*** (atomic radius),

(ii) ***metallic character,***

(iii) ***non-metallic character,***

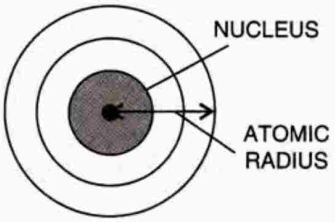
(iv) ***ionisation potential*** (ionisation energy),

(v) ***electron affinity,***

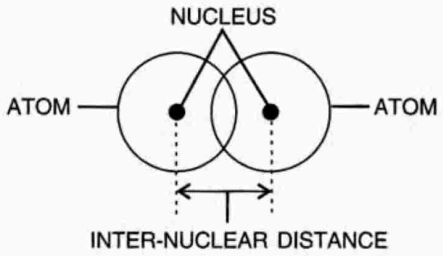
(vi) ***electronegativity.***

1.5.1 Atomic size (atomic radius)

It is the distance between the centre of the nucleus of an atom and its outermost shell.



Atomic radius can also be defined as half the inter-nuclear distance between the combined atoms in a molecule.



Unit : Angstrom : $1A = 10^{-10}$ m
Picometre : $1\ pm = 10^{-12}$ m.

Atomic size depends upon :

(i) number of shells and (ii) nuclear charge.

(i) *Number of shells*

An increase in the number of shells increases the size of an atom because the distance between the outermost shell and the nucleus increases.

(ii) *Nuclear charge*

It is the positive charge present in the nucleus of an atom, which is equal to the number of protons in the nucleus, *i.e.*, the atomic number.

An increase in nuclear charge decreases the size of the atom because the electrons are then attracted towards the nucleus with a greater force, thereby bringing the outermost shell closer to the nucleus.

Trends in atomic size (atomic radius)

(a) Down a group :

In a group, the size of an atom increases as one proceeds from top to bottom. This is due to the successive addition of shells (which overweighs the increased nuclear charge) as one moves from one period to the next in a group. For example :

Elements in Group 1

In group 1 the size of hydrogen is smallest.

Elements	Atomic No.	Atomic radius	Electronic configuration K L M N O P	Size
Hydrogen	1	37 pm	1	
Lithium	3	152 pm	2, 1	
Sodium	11	186 pm	2, 8, 1	
Potassium	19	231 pm	2, 8, 8, 1	
Rubidium	37	244 pm	2, 8, 18, 8, 1	
Caesium	55	262 pm	2, 8, 18, 18, 8, 1	

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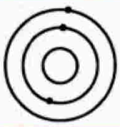
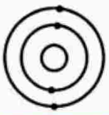
In group 17 the size of fluorine is smallest.

F	<	Cl	<	Br	<	I	<	At
64 pm		99 pm		114 pm		133 pm		140 pm

(b) Across a period :

In a period, the size of an atom decreases from left to right. This is because the nuclear charge, *i.e.*, the atomic number increases from left to right in the same period, thereby bringing the outermost shell closer to the nucleus. Therefore, considering the second period given above, it has been found that lithium (Li) has the largest atomic size while fluorine (F) has the smallest.

Trends in atomic size across a period.

Elements	Lithium	Beryllium	Boron	Carbon	Nitrogen	Oxygen	Fluorine
Atomic No.	3	4	5	6	7	8	9
Atomic Radius	152 pm	112 pm	88 pm	77 pm	70 pm	66 pm	64 pm
Electronic configuration	K L 2, 1	K L 2, 2	K L 2, 3	K L 2, 4	K L 2, 5	K L 2, 6	K L 2, 7
Size							

In third period, sodium is largest in size

Na	>	Mg	>	Al	>	Si	>	P	>	S	>	Cl
186 pm		160 pm		143 pm		117 pm		110 pm		104 pm		99 pm

Note : As an exception the size of the atoms of inert gases are bigger. This is because the outer shell of inert gases is complete They have the maximum number of electrons in their outer most orbit thus the electronic repulsions are maximum. Hence the size of the atom of an inert gas is bigger.

Note : **1. Cation is always smaller than the parent atom,** from which it is formed.

Reason :

Cation is formed by the loss of electron(s), hence proton(s) are more than electron(s) in a cation. So electrons are strongly attracted by the nucleus and are pulled inward. Hence the size decreases.

$$\underset{(2,8,1)}{Na - e^-} \rightarrow \underset{(2,8)}{Na^+}$$

2. Anion is larger than the parent atom.

Reason : Anion is formed by the gain of electron(s). Thus, the number of electron(s) are more than proton(s). The effective positive charge in the nucleus is less, so less inward pull is experienced. Hence the size expands.

$$\underset{(2,8,7)}{Cl} + e^- \rightarrow \underset{(2,8,8)}{Cl^-}$$

3. Size of isoelectronic ions *i.e.* the ions having the same number of electrons depends upon the nuclear charge (no. of protons). Greater is the nuclear charge smaller is the size .

Example :

Isoelectronic ions	Mg^{2+}	Na^+	F^-	O^{2-}
No. of electrons	10	10	10	10
No. of protons	12	11	9	8
Size in A°	0·65	0·95	1·26	1·42

1.5.2 Metallic character

Those elements, which have a tendency to lose their valence electrons (electrons of the outermost orbit) *and form a positive ion, are considered metals.*

$$\underset{(2,8,1)}{Na - e^-} \rightarrow \underset{(2,8)}{Na^+}$$

$$\underset{(2,8,2)}{Mg - 2e^-} \rightarrow \underset{(2,8)}{Mg^{2+}}$$

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Note : Hydrogen is an element (non metal) which does not have a neutron, it has only one electron and one proton. On losing this electron it forms its cation which has only one proton, therefore its cation can also be called a **proton**. $H - e^- \rightarrow H^+$

The metallic character of elements depends on:

(i) atomic size and (ii) nuclear charge.

(i) ***Atomic size*** **:** The greater the **atomic size**, the farther the outermost orbit, and thus, lesser the nuclear pull exerted on it. As a result, electron(s) can be removed more easily from the valence shell, thus making the elements more metallic.

(ii) ***Nuclear charge*** **:** The greater the **nuclear charge**, the greater is the force exerted by the nucleus on the electron(s) of the outermost orbit. This makes it difficult to remove the electron(s) from the outermost orbit. Thus metallic nature decreases.

Elements which lose electron(s) to complete their octet are **reducing agents**. Metals are good reducing agents. Greater the tendency to lose electron(s) stronger is the reducing agent.

Trends in metallic character

Down a group

On moving ***down a group***, the atomic size increases and the nuclear charge also increases. The effect of an increased atomic size is greater as compared to the increased nuclear charge. Therefore, ***metallic nature increases as one moves down a group,*** *i.e.,* they can lose electrons easily.

Li
Na
K
Rb
Cs
↓ Metallic nature increases

Example : **In group 1,** lithium is the least metallic element.

Note : Francium is radioactive element so its properties are not known.

Table 1.2

Group	*GROUP 1*	*GROUP 2*	*GROUP 13*	*GROUP 14*	*GROUP 15*	*GROUP 16*	*GROUP 17*	*GROUP 18*
Element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Electronic configuration	K, L, M 2, 8, 1	K, L, M 2, 8, 2	K, L, M 2, 8, 3	K, L, M 2, 8, 4	K, L, M 2, 8, 5	K, L, M 2, 8, 6	K, L, M 2, 8, 7	K, L, M 2, 8, 8
Metallic property	Metal	Metal	Metal	Metalloid	Non-metal	Non-metal	Non-metal	Noble gas
Valency	1	2	3	4	3	2	1	0

Across a period

On moving ***across a period***, nuclear pull increases due to the increase in atomic number, and thus the atomic size decreases. Hence, elements cannot lose electrons easily.

Therefore, ***the metallic nature decreases across a period, moving from left to right.***

Example : **In the 2nd period,** lithium is the most metallic.

Li Be — Metals; B — Metalloid; C N O — Non-metals

In the 3rd period, *sodium is the most metallic element.*

1.5.3 Non-metallic character

Those elements, which have a tendency to gain electrons, in order to attain octet in their outermost orbit, are considered as ***non-metals.***

Non-metals usually have 5, 6 or 7 electrons in their outermost orbits. They can gain 3, 2 or 1 electron respectively, in order to attain octet or to complete their valence shells (outermost orbits) and form an anion.

For example, chlorine (Cl) and oxygen (O) gain 1 and 2 electron(s) respectively to form anions.

$$\underset{(2,\,8,\,7)}{Cl} + e^- \rightarrow \underset{(2,\,8,\,8)}{Cl^-}$$

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$$O + 2e^- \rightarrow O^{2-}$$
$$(2, 6) \qquad (2, 8)$$

Non-metallic character also depends on :

(i) atomic size and

(ii) nuclear charge.

(i) ***Atomic size*** : The smaller the **atomic size,** the greater is the nuclear pull and the tendency to gain electrons. Therefore, the element is more non-metallic by nature.

(ii) ***Nuclear charge :*** The greater the nuclear charge, the greater is the tendency to gain electron(s), hence more non-metallic is the element.

Trends in non-metallic character

Down a group

The atomic size increases, due to the addition of new shells over successive periods. Though the nuclear charge increases, due to an increase in atomic number, the effect of an increasing atomic size is greater. Therefore, ***non-metallic nature decreases down the group.***

Example : **Group 14**

C	Non metal
Si	Metalloid
Ge	Metalloid
Sn	Metal
Pb	Metal

Across the period (left to right)

On moving across a period, the tendency to gain electron(s) increases due to an increase in nuclear pull and a decrease in the atomic size. Therefore, *non-metallic character increases across a period, i.e., from left to right.*

For example, in the 3rd period,

$$Na < Mg < Al < Si < P < S < Cl$$

Metals — Metalloid — Non-metals →

Non-metallic character increases, or metallic character decreases.

Note : Non-metals are oxidising agents.

The nature of oxides also show periodicity.

Across a period : The oxides of elements in a particular *period show decreasing basic nature* and finally become acidic. *For example*, the oxides of the elements of the third period.

Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
Strongly basic	Basic	Amphoteric	Feebly acidic	Acidic	More acidic	Most acidic

Down a group : The basic nature of oxides of metals increases.

Chemical reactivity

The reactivity of elements depend upon their tendency to lose or gain electrons to complete their outermost orbit. Greater the tendency to lose electron(s), greater is the reactivity in the case of metals. Similarly greater the tendency to gain electron(s), greater is the reactivity of non-metals.

Trends in chemical reactivity

Across a period :

On moving from left to right in a period, the chemical reactivity of elements first decreases and then increases.

The group 1 element can lose electrons easily as compared to group 2. Similarly, group 2 elements can also lose electrons easily but only in comparison to group 13 and not at all in comparison to group 1. As the tendency to lose electrons decreases, reactivity also decreases and thus silicon is the least reactive element in third period. As we move from phosphorus to chlorine, the tendency to gain electrons increases, hence reactivity again increases., *e.g.*,

Third period :

Na (Most reactive metal) — Mg — Al → (reactivity decreases) — Si (least reactive) — P — S ← (reactivity decreases) — Cl (Most reactive Non-metal)

Down a group

The tendency of loosing electrons increases down the group. Since chemical reactivity in metals depends upon the tendency to lose electrons, thus reactivity increases on going down the group.

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The most reactive metal is at the bottom of group 1.

The chemical reactivity of non-metals decreases on going down the group as it depends upon the tendency to gain electrons, which decreases down the group. The most reactive non-metal is at the top of group 17 *i.e.*, Fluorine.

Gradation in physical properties

The melting and boiling points of metals **decrease** on going down the group, *e.g.*,

Metals	m.p.	b.p.
Li	180·5°C	1347°C
Na	94·5°C	883°C
K	63·5°C	774°C

↓ decreases

The melting and boiling points of non-metals **increase** on going down the group.

Non-Metals	m.p.	b.p.	Physical state
Fluorine	–219·6°C	–187°C	gas
Chlorine	–101°C	–34·6°C	gas
Bromine	–7·2°C	+58·8°C	liquid
Iodine	+113·6°C	+183°C	solid

↓ increases

Note : **Across a period**, left to right, melting point and boiling point usually increases upto group 14 (IV A) and then decreases.

Element of 3rd period	Na	Mg	Al	Si	P	S
M.P. (°C)	98	650	660	1410	44·2	115·2

Density of elements across a period increases gradually to maximum and then slight decrease may be noticed.

Element	Na	Mg	Al	Si	P	S
Density g/cc	1·0	1·7	2·7	2·3	1·8	2·1

Down a group density of elements increases gradually.

Element	Li	Na	K	Rb	Cs
Density g/cc	0·54	0·97	0·86	1·53	1·87

Intext Questions

1. What do you understand by atomic size ? State its unit.
2. Give the trends in atomic size on moving :
   (i) down the group,
   (ii) across the period left to right.
3. Arrange the elements of second and third period in increasing order of their atomic size.
4. Why is the size of (i) neon greater than fluorine? (ii) sodium is greater than magnesium ?
5. Which is greater in size ?
   (i) an atom or a cation (ii) an atom or an anion
   (iii) Fe^{2+} or Fe^{3+}
6. Arrange :
   (i) Be, Li, C, B, N, O, F (in increasing metallic character).
   (ii) Si, Na, Al, Mg, Cl, P, S (in decreasing non-metallic character).
7. State the trend in chemical reactivity :
   (i) across the period left to right,
   (ii) down the group.
8. A metal M forms an oxide having the formula M_2O_3. It belongs to third period. Write the atomic number and valency of the metal.
9. An element X belong to 4^{th} period and 17^{th} group, state
   (i) no. of valence electrons in it.
   (ii) name of the element.
   (iii) name the family to which it belong.
   (iv) write the formula of the compound formed when X reacts with $^{27}_{13}Y$.
10. The given table shows elements with same number of electrons in its valence shell.

Elements	A	B	C
m.p.	63·0	180·0	97·0

State :
   (i) Whether these elements belong to same group or period.
   (ii) Arrange them in order of increasing metallic character.
11. Which one of the following has the largest atomic radius ?
   (i) Sodium (ii) Potassium
   (iii) Magnesium (iv) Aluminium
12. Which one has the largest size ?
   (i) Br (ii) I (iii) I^- (ii) Cl

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13. The metals of group 2 from top to bottom are Be, Mg, Ca, Sr and Ba
    (i) Which one of these elements will form ions most readily and why ?
    (ii) State the common feature in their electronic configuration.
14. Write the number of protons, neutrons and electronic configuration of $^{39}_{19}K$, $^{31}_{15}P$. Also state their position in periodic table.
15. The electronic configuration of an element T is 2, 8, 8, 1.
    (i) What is the group number of T ?
    (ii) What is the period number of T ?
    (iii) How many valence electrons are there in an atom of T ?
    (iv) What is the valency of T ?
    (v) Is it a metal or a non-metal ?
16. Complete the following sentences choosing the correct word or words from those given in brackets at the end of each sentence :
    (i) The properties of the elements are a periodic function of their _____ (atomic number, mass number, relative atomic mass).
    (ii) Moving across a _____ of the Periodic Table the elements show increasing _____ character (group, period, metallic, non-metallic).
    (iii) The elements at the bottom of a group would be expected to show _____ metallic character than the element at the top (less, more).
    (iv) The similarities in the properties of a group of elements are because they have the same _____ (electronic configuration, number of outer electrons, atomic numbers).
17. Give reasons for the following :
    (i) The size of a Cl^- ion is greater than the size of a Cl atom.
    (ii) Argon atom is bigger than chlorine atom.
    (iii) Ionisation potential of the element increases across a period.
    (iv) Alkali metals are good reducing agents.
18. Name the element which has :
    (i) two shells, both of which are completely filled with electrons ?
    (ii) the electronic configuration 2, 8, 3 ?
    (iii) a total of three shells with five electrons in its valence shell ?
    (iv) a total of four shells with two electrons in its valence shell ?
    (v) twice as many electrons in its second shell as in its first shell ?
19. (i) State the number of elements in Period 1, Period 2, and Period 3 of the periodic table. Name them.
    (ii) What is the common feature of the electronic configuration of the elements at the end of Period 2 and Period 3 ?
    (iii) If an element is in Group 17, it is likely to be _____ [metallic/non-metallic] in character, while with one electron in its outermost energy level (shell), then it is likely to be _____ [metallic/non-metallic].

1.5.4 Ionisation potential or Ionisation energy (I.E.) or ionisation enthalpy

We know that the electrons in an atom are attracted by the positively charged nucleus. So, if we want to remove an electron from an atom, some energy has to be supplied to overcome the strong attraction of the nucleus. And after the electron gets removed from the atom, the atom then acquires one unit positive charge and becomes a positive ion (or cation).

The energy required to remove an electron from a neutral isolated gaseous atom and convert it into a positively charged gaseous ion is called ionisation potential (I.P.) or ***ionisation energy (I.E.)*** or **first ionisation energy** (IE_1).

$M(g) + I.E. \rightarrow M^+(g) + e^-$

M can be any element.

Unit : *I.E. is measured in electron volts per atom (eV/atom) and its S.I. unit is kilojoule per mole (kJ mol^{-1}).*

Ionisation energy depends on :

(i) atomic size (ii) nuclear charge

(i) ***Atomic size :*** The greater the atomic size, the lesser the force of attraction. Since the electrons of the outermost shell lie further away from the nucleus, thus make their removal easier, *i.e.*, the ionisation energy required is less.

(ii) ***Nuclear charge :*** The greater the nuclear charge, greater is the attraction for the electrons of the outermost shell. Therefore, the electrons

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in the outermost shell are more firmly held because of which greater energy is required to remove the electron(s).

Trends in ionisation energy

Across a period

The ionisation energy tends to increase as one moves from left to right across a period (with exceptions), because the atomic size decreases due to an increase in the nuclear charge, and thus, more energy is required to remove the electron(s).

The elements of the 2nd period	Li	Be	B	C	N	O	F	Ne
Ionisation energy in kJ mol^{-1}	520	899*	801	1088	1402*	1314	1681	2080

The elements of the 3rd period	Na	Mg	Al	Si	P	S	Cl	Ar
I.E. in kJ mol^{-1}	496	737*	577	786	1011*	999	1256	1520

Down a group

There is an increase in atomic number (nuclear charge) and atomic size down the group due to the addition of extra shells. This increase in the atomic size overcomes the effect of an increase in the nuclear charge.

Therefore, *ionisation energy decreases with an increase in the atomic size, i.e., it decreases as one moves down a group.*

The elements of Group 1	Ionisation energy in kJ mol^{-1}
H	1312
Li	520
Na	498
K	419
Rb	403
Cs	375

The elements of Group 17	Ionisation energy in kJ mol^{-1}
F	1681
Cl	1256
Br	1143
I	1008

* Exceptions

Note : • Helium will have the highest ionisation energy 2372·0 kJ mol^{-1} while caesium (Cs) will have the lowest ionisation energy of 375·0 KJ mol^{-1} (I.E. of Fr is not determined correctly as it is radioactive).

• Metals usually have low I.E. whereas non-metals have high I.E.

1.5.5 Electron Affinity (E.A.) or Electron Gain Enthalpy

We have just studied that to remove an electron from an atom, energy is required. Similarly, when an extra electron is added to an atom, energy is released.

The amount of energy released while converting a neutral gaseous isolated atom into a negatively charged gaseous ion (anion) by the addition of electron is called Electron Affinity (E.A.)

$$X(g) + e^- \rightarrow X^-(g) + E.A.$$

X is any element taken in its gaseous state.

Unit : *electron volts per atom (eV/atom) or kJ mol^{-1}. Electron affinity is represented by negative sign [–] e.g.,*

$$Cl(g) + e^- \rightarrow Cl^-(g) + 349 \text{ KJ/mol.}$$

Therefore, electron affinity of chlorine is – 349 KJ/mol.

Electron affinity depends on :

(i) atomic size and (ii) nuclear charge.

(i) Atomic size : *The smaller the atomic size, the greater the electron affinity,* because the effective attractive force between the nucleus and the valence electrons is greater in smaller atoms, and so the electrons are held firmly.

(ii) Nuclear charge : *The greater the nuclear charge, greater is the electron affinity,* because increase in nuclear charge increases the effective attractive force on the valence electrons.

Variation (Trends) in electron affinity

In a period, i.e., from left to right in a horizontal row of the periodic table, the atomic size decreases and the nuclear charge increases, so the

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electron affinity increases. Thus electron affinity (E.A.) is the highest for group 17 elements (halogens) and the least for group 1 (alkali metals).

Electron Affinity Values in KJ mol^{-1}

	IA (1)	IIA (2)	IIIA (13)	IVA (14)	VA (15)	VIA (16)	VIIA (17)	Zero (18)
I	H −72·8							He*
II	Li −59·8	Be*	B −88·0	C −122	N*	O −140·9	F −327·9	Ne*
III	Na −53·1	Mg*	Al −50·0	Si −119	P −74	S −200	Cl −349	Ar*

Down a group

Moving from the top to the bottom in a group, the atomic size increases more than the nuclear charge, thereby causing a net decrease in E.A.

Group I	
Li	−59·8
Na	−53·1
K	−48·4
Rb	−47·0

Note : In group 17 fluorine has lower E.A. than chlorine and in group 16 oxygen has lower E.A. than sulphur. This is because the size of Fluorine and oxygen atom is very small. As a result, there are strong inter electronic repulsions and thus incoming electron does not feel much attraction.

1.5.6 Electronegativity (E.N.)

The tendency of an atom in a molecule to attract the shared pair of electrons towards itself is called its ***electronegativity.***

Electronegativity is a *dimensionless property,* since it is only a tendency. It only indicates the net result of the tendencies of different elements to attract the bond-forming electron pair.

Electronegativity is measured on several scales. The most widely used scale of electronegativity was devised by ***Linus Pauling*** (1932) in which the highest value of electronegativity [for fluorine] is taken as 4·0 and the lowest one [for caesium] as 0·7.

* *Inert gases have zero electron affinity due to their stable electronic configuration.*

Groups 2 and 15 do not show negative values. They are exceptions.

ELECTRONEGATIVITY VALUES

DECREASING ↓	1	2	13	14	15	16	17
	H 2·1						
	Li 1·0	Be 1·5	B 2·0	C 2·5	N 3·0	O 3·5	F 4
	Na 0·9	Mg 1·2	Al 1·5	Si 1·8	P 2·1	S 2·5	Cl 3·0
	K 0·8	Ca 1·0	Ga 1·6	Ge 1·8	As 2·0	Se 2·4	Br 2·8
	Rb 0·8	Sr 1·0	In 1·7	Sn 1·8	Sb 1·9	Te 2·1	I 2·5
	Cs 0·7	Ba 0·9	Tl 1·8	Pb 1·9	Bi 1·9	Po 2·0	At 2·2
	INCREASING →						

Electronegativity values too depend on :

(i) size of atom and

(ii) nuclear charge.

(i) **Atomic size :** *The greater the size of the atom, the lesser the electronegativity,* since the electrons *being farther away from the nucleus, experience a lesser force of attraction.*

(ii) **Nuclear charge :** *The greater the nuclear charge, the greater the electronegativity,* because increase in nuclear charge causes electron attraction with a greater force.

Trends in electronegativity

Across a period

Since the nuclear charge increases due to an increase in atomic number, ***electronegativity increases from left to right in a period.***

For example, in the second period, electronegativity increases from lithium to fluorine.

Down a group

There is an increase in atomic number down a group, *i.e.*, nuclear charge increases, but due to the addition of extra shells, the atomic size increases. The effect of an increase in the atomic size overcomes the effect of an increase in the nuclear charge, hence ***electronegativity decreases down a group.***

For example, in the first group, it significantly decreases from lithium downwards to francium.

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Note : 1. Generally, metals show lower electronegativity as compared to non-metals. Thus metals are electropositive and non-metals are electronegative.

2. The elements of second period differs in properties from their respective groups. This is due to small size of the atom and their high electronegativity.

Diagonal relationship : The elements of the **second period** show resemblance in properties with the elements of the next group of the **third period**, due to very less electronegativity difference. This leads to a **diagonal relationship**, *viz.* Li & Mg, Be & Al, B & Si. These elements are called **bridge elements.**

Group →	1	2	13	14
Period 2	Li	Be	B	C
Period 3	Na	Mg	Al	Si

Noble gases have complete octet so they do not have tendency to attract electrons.

Note : Greater the value of electron affinity easier it is to gain electron(s) and more non metallic or more electronegative or more oxidising is the element.

Intext Questions

1. (a) Define the term 'ionisation potential'. **(2010)**
   (b) Represent it in the form of an equation. In which unit it is measured?
2. Ionisation Potential values depends on (a) atomic size (b) nuclear pull. Explain.
3. State the trends in ionisation energy :
   (a) across the period, (b) down the group.
4. Name the elements with highest and lowest ionisation energies in first three periods.
5. Arrange the elements of second and third period in increasing order of ionisation energy.
6. (a) Define the term 'electron affinity'. State its unit. **(2010)**
   (b) Arrange the elements of second period in increasing order of their electron affinity. Name the elements which do not follow the trend in this period.
7. Electron affinity values generally ..... across the period left to right and ..... down the group top to bottom.
8. (a) Define the term 'Electronegativity'. State its unit.
   (b) Among the elements given below, the element with least Electronegativity is :
   (i) Lithium (ii) Boron (iii) Carbon (iv) Fluorine
9. Explain the following :
   (a) Group 17 elements are strong non-metals, while group I elements are strong metals.
   (b) Metallic character of elements decreases from left to right in a period while it increases in moving down a group.
   (c) Halogens have a high electron affinity.
   (d) The reducing power of element increases down in the group while decreases in a period.
   (e) Size of atoms progressively becomes smaller when we move from sodium (Na) to chlorine (Cl) in the third period of the Periodic Table.
10. Name the periodic property which relates to the :
    (a) amount of energy required to remove an electron from an isolated gaseous atom,
    (b) character of element which loses one or more electrons when supplied with energy,
    (c) tendency of an atom to attract the shared pair of electron.
11. This question refers to the elements of the Periodic Table with atomic numbers from 3 to 18. Some of the elements are shown by letters, but the letters are not the usual symbols of the elements.

3	4	5	6	7	8	9	10
A	B	C	D	E	F	G	H
11	12	13	14	15	16	17	18
I	J	K	L	M	N	O	P

Which of these :
(a) are most electronegative element.
(b) is a halogen ?
(c) is an alkali metal ?
(d) is an element with valency 4 ?
(e) have least Ionisation Energy.
(f) have least atomic size in period 3.

1.6 RELATION BETWEEN ATOMIC NUMBER AND MASS NUMBER OF LIGHT ELEMENTS

Atomic number (Z)

The atomic number of an element is equal to the number of protons in the nucleus. ***Atomic number is the unique property of an element, because no two elements have the same atomic number.***

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The significance of atomic number (Z)

1. Atomic number = Number of protons.
   = Number of electrons.
2. It distinguishes an element from other elements, because no two elements have the same atomic number.
3. It gives the electronic configuration of an element, *e.g.*, an element with atomic number 13 will have electronic configuration 2, 8, 3.
4. It helps us in finding the position of an element in the periodic table.

   For example, the element with atomic number 17 will have electronic configuration 2, 8, 7. This element will be placed in the 3rd period of Group 17 *i.e.*, VIIA, because it has : (i) three energy shells and (ii) seven electrons in its outermost shell.

Table 1.4 : Arrangement of electrons, protons and neutrons of elements from atomic number 1 to 20

Elements	*Symbol*	*Atomic No. (Z) No. of protons (or electrons)*	*Mass No. (A) or Protons + neutrons*	*No. of neutrons (A – Z)*
Hydrogen	H	1	1	—
Helium	He	2	4	2
Lithium	Li	3	7	4
Beryllium	Be	4	9	5
Boron	B	5	11	6
Carbon	C	6	12	6
Nitrogen	N	7	14	7
Oxygen	O	8	16	8
Fluorine	F	9	19	10
Neon	Ne	10	20	10
Sodium	Na	11	23	12
Magnesium	Mg	12	24	12
Aluminium	Al	13	27	14
Silicon	Si	14	28	14
Phosphorus	P	15	31	16
Sulphur	S	16	32	16
Chlorine	Cl	17	35, 37	18, 20
Argon	Ar	18	40	22
Potassium	K	19	39	20
Calcium	Ca	20	40	20

Mass number (A)

The mass number of an element is the sum of the number of protons and neutrons in the nucleus of the atom of that element.

Mass number (A) = No. of protons (p)
+ No. of neutrons (n)

A close look at the electronic configurations of lighter elements reveals that those elements which have an ***even*** *number of protons, for example, atomic numbers like* $^{4}_{2}He$, $^{12}_{6}C$, etc., *have their mass numbers twice the atomic numbers except* $^{9}_{4}Be$ and $^{40}_{18}Ar$.

Elements which have an **odd** *number of protons like* $^{7}_{3}Li$, $^{11}_{5}B$, $^{19}_{9}F$, $^{23}_{11}Na$, *etc., have their mass numbers twice the atomic numbers* +1 (A = 2Z + 1) *except* $^{14}_{7}N$ and $^{1}_{1}H$.

Note : (i) Elements with *n/p* (neutron/proton) ratio around 1 are stable, *e.g.*, light metals like sodium, potassium, calcium, etc.

(ii) Elements with *n/p* ratio 1·5 and above are radioactive, *i.e.*, they emit radiations. They are unstable elements, *e.g.*, heavy metals like uranium.

1.7 COMPARISON OF ALKALI METALS AND HALOGENS

	Alkali metals [Group 1]	Halogens [Group 17]
Elements	Li, Na, K, Rb, Cs, Fr	F, Cl, Br, I, At
Occurrence	*Combined state.* Due to their reactive nature.	Combined state as *salts.* Due to their reactive nature.
Physical state	Shining white solid metals. Soft, and can be cut with a knife. Lithium is the hardest. They are shiny when freshly cut but soon becomes dull as they react with air.	Non-metals; diatomic in the gaseous state. Flourine (very reactive poisonous yellow gas) chlorine (poisonous yellow green gas) Bromine (poisonous red brown volatile liquid) Iodine (dark grey crystalline solid).

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Valence electrons	Possess *one* valence electron and therefore show similar properties.	Possess *seven* valence electrons each and therefore show similar properties.
Conduction	*Good conductors* of electricity.	*Non-conductors* of electricity.
Nature	Highly reactive, *electropositive* metals. Metallic character, *increases* from lithium (Li) to francium (Fr).	Highly reactive, *electronegative* non-metals. Non-metallic character, decreases from fluorine (F) to iodine (I).
Melting point and boiling point	Decreases down the group.	Increases down the group.
Atomic size	They have the largest atomic size in their period (except inert gases). The atomic size further increases down the group.	They have the smallest atomic size in their period. Atomic size increases down the group.
Ionisation energy	They have lowest I.E. in their period. It decreases down the group.	They have high I.E. (lower than noble gases) in their period.
Electron affinity	They have low E.A. values which further decrease down the group.	They have high E.A. values. They too decrease down the group.
Electronegativity	They have the lowest E.N. in their period. E.N. decreases down the group.	They have high E.N. highest in their period.
Reactivity	They are reactive metals. Reactivity further *increases* down the group.	They are reactive non-metals. The reactivity *decreases* down the group.
Reaction with water and acids	They react vigorously with water and acids liberating hydrogen. Reactivity further increases down the group.	Generally they do not react with dil. acids and water.
Reducing/ oxidising agents	Strong *reducing agents* as they lose electrons to complete their octet.	Strong *oxidising agents* as they accept electrons to complete their octet.
Compound formation	Form *electrovalent* compounds with non-metals. Example : NaCl, KBr.	Form *electrovalent compounds* with metals. *e.g.*, KCl, $CaCl_2$. Form ***covalent compounds*** with hydrogen and other non-metals, *e.g.*, HBr, HCl, HI,CCl_4.

Period No.	Type of period	No. of elements	Atomic number	1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIII	10	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 0
1	Shortest	2	1 and 2	H_1 1																	He_2 2
2.	Short	8	3 to 10	Li_3 2,1	Be_4 2,2											B_5 2,3	C_6 2,4	N_7 2,5	O_8 2,6	F_9 2,7	Ne_{10} 2,8
3.	Short	8	11 to 18	Na_{11} 2,8,1	Mg_{12} 2,8,2											Al_{13} 2,8,3	Si_{14} 2,8,4	P_{15} 2,8,5	S_{16} 2,8,6	Cl_{17} 2,8,7	Ar_{18} 2,8,8
4.	Long	18	19 to 36	K_{19} 2,8,8,1	Ca_{20} 2,8,8,2			TRANSITION ELEMENTS												Br_{35}	Kr_{36}
5.	Long	18	37 to 54	Rb_{37}																I_{53}	Xe_{54}
6.	Longest	32	55 to 86	Cs_{55}		La_{57}														At_{85}	Rn_{86}
7.	Incomplete		87 to	Fr_{87}		AC_{89}														Uus_{117}	Uuo_{118}
						Lanthanide series	Ce_{58}					INNER TRANSITION ELEMENTS									Lu_{71}
						Actinide series	Th_{90}														Lr_{103}

Group 1 are Alkali metals (except hydrogen)
Group 2 Alkaline earth metals
Group 3 to 12 Transition elements
Group 17 Halogens
Group 18 Inert gases
Group IIIB 6th period Lanthanides
Group IIIB 7th period Actinides
] Inner Transition Elements

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Properties	*Across the period (Left to right)*	*Down the group (Top to bottom)*
No. of valence electrons	Increases	Remains same
Atomic size (radius)	Decreases	Increases
Ionisation energy	Increases	Decreases
Metallic character	Decreases	Increases
Non-metallic character	Increases	Decreases
Electron affinity	Increases	Decreases
Electronegativity	Increases	Decreases
Basic nature of oxides	Decreases	Increases
Melting point	Increases from group I to group IV and then decreases	Group I and II decreases Group III and IV decreases Group V to VII increases
Boiling point	Increases from group I to group IV and then decreases	Group I and II decreases Group III and IV decreases Group V to VII increases
Oxidising nature	increases	Decreases
Reducing nature	decreases	Increases

EXERCISE

1. An element Barium has atomic number 56. Look up its position in the Periodic Table and answer the following questions.
- (a) Is it a metal or a non-metal ?
- (b) Is it more or less reactive than calcium ?
- (c) What is its valency ?
- (d) What will be the formula of its phosphate ?
- (e) Is it larger or smaller than caesium (Cs) in size ?

2. Choose the most appropriate answer from
[SO_2, SiO_2, Al_2O_3, CO, MgO, Na_2O]
- (a) A covalent oxide of a metalloid.
- (b) An oxide which when dissolved in water form acid.
- (c) A basic oxide.
- (d) An amphoteric oxide.

3. In group I of the Periodic Table, three elements X, Y and Z have ionic radii 1.33 Å, 0.95 Å and 0.60 Å respectively. Giving a reason, arrange them in the order of increasing atomic numbers in the group.

4. Arrange the following as per **instructions** given in the brackets.
- (a) Mg, Cl, Na, S, Si (increasing order of atomic size)
- (b) K, Na, Cl, S, Si (increasing non-metallic character)
- (c) Na, K, Cl, S, Si (increasing ionisation potential)
- (d) Cl, F, Br, I (increasing electron affinity)
- (e) Cs, Na, Li, K, Rb (decreasing electronegativity)

5. Name
- (a) An alkali metal in period 3 and halogen in period 2.
- (b) The noble gas with 3 shells.
- (c) The non-metals present in period 2 and metals in period 3.
- (d) The element of period 3 with valency 4.
- (e) The element in period 3 which does not form oxide.
- (f) The element of lower nuclear charge out of Be and Mg.
- (g) Which has higher E.A., Fluorine or Neon.
- (h) Which has maximum metallic character Na, Li or K.

6. Chlorine in the Periodic Table is surrounded by the elements with atomic number 9, 16, 18 and 35.
- (a) Which of these have Physical and Chemical properties resembling chlorine.
- (b) Which is more electronegative than chlorine.

7. First ionisation enthalpy of two elements X and Y are 500 kJ mol^{-1} and 375 kJ mol^{-1} respectively. Comment about their relative position in a group as well as in a period.

8. Explain why are the following statements not correct :
- (a) All groups contain metals and non metals.
- (b) Atoms of elements in the same group have the same number of electron(s).
- (c) Non-metallic character decreases across a period with increase in atomic number.

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(d) Reactivity increases with atomic number in a group as well as in a period.

9. Arrange the following in order of increasing radii :

(a) Cl^-, Cl (b) Mg^{2+}, Mg, Mg^+ (c) N, O, P

10. Which element from the following has the highest ionisation energy ?

(a) P, Na, Cl (b) F, O, Ne (c) Ne, He, Ar

Explain your choice.

11. The electronegativities (according to Pauling) of the elements in Period 3 of the Periodic Table are as follows with the elements arranged in alphabetical order:

Al	Cl	Mg	Na	P	S	Si
1·5	3·0	1·2	0·9	2·1	2·5	1·8

Arrange the elements in the order in which they occur in the Periodic Table from left to right.

(The group 1 element first, followed by the group 2 element and so on, up to group 7).

12. Choose the word or phrase from the brackets which correctly completes each of the following statements :-

(a) The element below sodium in the same group would be expected to have a ..... (lower/higher) electro-negativity than sodium and the element above chlorine would be expected to have a ..... (lower/higher) ionization potential than chlorine.

(b) On moving from left to right in a given period, the number of shells ..... (remains the same/ increases decreases).

(iii) On moving down a group, the number of valence electrons ..... (remains the same/increases/ decreases).

(iv) Metals are good ..... (oxidising agent/ reducing agent) because they are electron ..... (acceptors/donors). **(2016)**

13. Parts (a) to (e) refer to changes in the properties of elements on moving from left to right across a period of the Periodic Table. For each property, choose the correct answer.

(a) The non-metallic character of the elements :

(i) decreases, (ii) increases,

(iii) remains the same, (iv) depends on the period

(b) The electronegativity :

(i) depends on the number of valence electrons,

(ii) remains the same,

(iii) decreases,

(iv) increases.

(c) The ionization potential :

(i) goes up and down (ii) decreases

(iii) increases (iv) remains the same

(d) The atomic size :

(i) decreases,

(ii) increases,

(iii) remains the same,

(iv) sometimes increases and sometimes decreases.

(e) The electron affinity of the elements in groups 1 to 7 :

(i) goes up and then down.

(ii) decreases and then increases,

(iii) increases,

(iv) decreases.

14. The elements of one short period of the Periodic Table are given below in order from left to right :

Li Be B C O F Ne

(a) To which period do these elements belong ?

(b) One element of this period is missing. Which is the missing element and where should it be placed ?

(c) Which one of the elements in this period shows the property of catenation ?

(d) Place the three elements fluorine, beryllium and nitrogen in the order of increasing electronegativity.

(e) Which one of the above elements belongs to the halogen series ?

2007

A group of elements in the Periodic Table are given below (boron is the first member of the group and Thallium is the last).

Boron, Aluminium, Gallium, Indium, Thallium

Answer the following questions in relation to the above group of elements :

(a) Which element has the most metallic character ?

(b) Which element would be expected to have the highest electronegativity ?

(c) If the electronic configuration of aluminium is 2, 8, 3, how many electrons are there in the outer shell of thallium ?

(d) The atomic number of boron is 5. Write the chemical formula of the compound formed when boron reacts with chlorine.

(e) Will the elements in the group to the right of this boron group be more metallic or less metallic in character ? Justify your answer.

2008

Select the correct answer from the choices A, B, C, D which are given. Write down only the letter corresponding to the correct answer.

With reference to the variation of properties in the Periodic Table, which of the following is generally true?

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(a) Atomic size increases from left to right across a period.

(b) Ionization potential increases from left to right across a period.

(c) Electron affinity increases going down a group.

(d) Electro-negativity increases going down a group.

2009

(a) Among period 2 elements A, B, C and D, the one which has high electron affinity is

A. Lithium B. Carbon

C. Fluorine D. Neon

(b) Group No.'s

IA	IIA	IIIA	IVA	VA	VIA	VIIA	O
1	2	13	14	15	16	17	18
Li		D			O	J	Ne
A	Mg	E	Si		H	K	
B	C		F	G			L

Select from the table

(i) Which is most electronegative.

(ii) How many valence electrons are present is G.

(iii) Write the formula of the compound between B and H.

(iv) In the compound between F and J what type of bond will be formed.

(v) Draw the electron dot structure for the compound formed between C and K.

2010

(a) The number of electrons in the valence shell of a halogen is ..... A – 1, B – 3, C – 5, D – 7.

(b) Electronegativity across the period ..... [increases/ decreases].

(c) Non-metallic character down the group ..... [increase/decreases].

(d) Atomic number of an element is 16. State

(i) to which period it belongs

(ii) the number of valence electron in the element

(iii) is the element metal or non-metal.

2011

(a) Give reasons – The oxidising power of elements increases from left to right along a period.

(b) Select the correct answer –

(i) Across a period, the ionization potential ..... [increase, decreases, remains same]

(ii) Down the group, electron affinity ..... [increases, decreases, remains same].

(c) Choose the correct answer from the choice given :

(i) In the periodic table alkali metals are placed in the group – A : 1, B : 11, C : 17, D : 18.

(ii) Which of the following properties do not match with elements of the halogen family –

A. They have seven electrons in their valence shell.

B. They are highly reactive chemically.

C. They are metallic in nature.

D. They are diatomic in their molecular form.

(d) State the group and period, of the element having three shells with three electron in valence shell.

2012

(a) Choose the correct answer from the option : An element in period 3 whose electrons affinity is zero.

A. Neon B. Sulphur

C. Sodium D. Agron

(b) Give reason :

(i) Ionisation potential of the element increases across a period.

(ii) Alkali metals are good reducing agents.

(c) There are three elements E, F, G with atomic numbers 19, 8 and 17 respectively –

Classifiy the above elements as metals and non-metals.

(d) Name : A metal present in period 3, group I of the periodic table.

2013

(a) Among the period 2 elements, the element which has high electron affinity is

A. Lithium B. Carbon

C. Chlorine D. Fluorine

(b)

Group No.	1-1A	2-IIA	13-IIIA	14-IVA	15-VA	16-VIA	17-VIIA	18-O
2nd period	Li		D			O	I	Ne
3rd period	A	Mg	E	Si		H	M	
4th period	R	T	I		Q	U		Y

In the above tabe H does not represent hydrogen. Some elements are given in their own symbol and position in the periodic table while others are shown with a letter. Answer the following questions.

(i) Identify the most electronegative element.

(ii) Identify the most reactive element of group I.

(iii) Identify the element from period 3 with least atomic size.

(iv) How many valence electrons are present in Q.

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(v) Which element from group 2 would have the least ionization energy.

(vi) Identify the noble gas of the fourth period.

(vii) In the compound between A and H what type of bond is formed and give its molecular formula.

(c) Identify : The element which has the highest ionization potential.

2014

(a) Choose the correct answer from the choice given :

(i) Ionisation potential increases over a period from left to right because the :

A. Atomic radius and nuclear charge increases

B. Atomic radius and nuclear charge decreases

C. Atomic radius increases and nuclear charge decreases

D. Atomic radius decreases and nuclear charge increases

(ii) An element A belonging to period 3 and group II will have,

A. 3 shells and 2 valence electrons

B. 2 shells and 3 valence electrons

C. 3 shells and 3 valence electrons

D. 2 shells and 2 valence electrons

(b) Atomic number of an element Z is 16. Answer the following :

(i) State the period and group to which Z belongs.

(ii) Is Z a metal or a non-metal.

(c) State the formula of the compound between Z and hydrogen. What kind of compound is this formula M_2O when dissolved in water forms the corresponding hydroxide which is a good conductor of element in the above context answer the following :

(i) What kind of combination exists between M and O.

(ii) State the number of electrons in the outermost shell of M.

(iii) Name the group to which M belongs.

(d) Give one word or phrase for : The amount of energy *released* when an atom in the gaseous state accepts an electron to from an anion.

(e) Match the option A and B with the statements (i) and (ii) :

A. metal	(i) The metal that forms two types of ion
B. iron	(ii) An element with electronic configuration 2, 8, 8, 3

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2 Chemical Bonding

SYLLABUS

Electrovalent, covalent and co-ordinate bonding, structures of various compounds – orbit structure and electron dot structure.

Definition of Electrovalent Bond.

Structure of Electrovalent compounds NaCl, $MgCl_2$, CaO;

Characteristic properties of electrovalent compounds – state of existence, melting and boiling points, conductivity (heat and electricity), ionisation in solution, dissociation in solution and in molten state to be linked with electrolysis.

Covalent Bond – definition and examples, structure of Covalent molecules on the basis of duplet and octet of electrons (example : hydrogen, chlorine, nitrogen, water, ammonia, carbon tetrachloride, methane.)

Characteristic properties of Covalent compounds – state of existence, melting and boiling points, conductivity (heat and electricity), ionisation in solution.

Comparison of Electrovalent and Covalent compounds.

Definition of Coordinate Bond : The lone pair effect of the oxygen atom of the water molecule and the nitrogen atom of the ammonia molecule to explain the formation of H_3O^+ and OH^- ions in water and NH_4^+ ion. The meaning of lone pair; the formation of hydronium ion and ammonium ion must be explained with the help of electron dot diagrams.

2.1 INTRODUCTION

Everything in this world wants stability so is the case with atoms. For atoms, stability means having the electron arrangement of an inert gas, *i.e.*, octet in its outermost shell. Helium has two electrons (DUPLET) while all other inert gases, *i.e.*, Neon, Argon, Krypton, Xenon and Radon have eight electrons (OCTET) in their outermost shell, as given in Table 2.1 below.

Table 2.1 Electronic configurations of the inert gases.

Inert gas	*Atomic No.*	*Electronic configuration*						*No. of valence electrons*
		K	L	M	N	O	P	
He	2	2						2
Ne	10	2	8					8
Ar	18	2	8	8				8
Kr	36	2	8	18	8			8
Xe	54	2	8	18	18	8		8
Rn	86	2	8	18	32	18	8	8

It is found that the elements with their complete outermost shell do not react or are least reactive. We, therefore, conclude that the atoms having 8 electrons (or 2 electrons, Helium configuration) in their outermost shells are very stable and unreactive. Therefore, *to attain stability, atoms tend to combine chemically by redistribution of electrons in the outermost shell or valence electrons so that each is left with a* ***stable electronic configuration*** **(duplet** or **octet).**

Cause of chemical combination is the tendency of elements to acquire the nearest noble gas configuration in their outermost orbit and become stable.

During redistribution of electrons, a bond (force) of attraction develops between atoms, which binds them together to form molecules. This bond is known as the **chemical bond.**

A ***chemical bond*** *may be defined as the force of attraction between any two atoms, in a molecule, to maintain stability.*

There are *three* methods in which atoms can achieve a stable configuration.

(1) The ***transfer*** of one or more electrons from one atom to the other to form an ***electrovalent*** (or an ***ionic***) ***bond.***

(2) ***Sharing*** of one, two or three pairs of electrons between two atoms to form *a* ***covalent*** (or a ***molecular***) ***bond.***

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(3) When the shared electron pairs are contributed by only one of the combining atoms, the bond formed is known as **coordinate** (or **dative**) bond.

2.2 ELECTROVALENT (OR IONIC) BOND

Atoms of metallic elements that have 1, 2 or 3 valence electrons can lose electron(s) to atoms of non-metallic elements, which have 7, 6 or 5 electrons respectively in their outermost shell and thereby forming an electrovalent compound.

After the transfer of electron(s), both the combining atoms acquire the electronic configuration of the nearest inert gas.

A metallic atom, which loses electron(s), becomes a positively charged ion and is known as a **cation** and a non-metallic atom, which gains electron(s), becomes a negatively charged ion and is known as an **anion.**

An **ion** is a charged particle which is formed due to the gain or the loss of one or more electrons by an atom.

A metallic element, whose one atom readily loses electron(s) to form a positively charged ion, is an ***electropositive element***.

$$Na - e^- \rightarrow Na^+ \text{ (cation)}$$

A non-metallic element, whose atom readily accepts electron(s) to form a negatively charged ion, is an ***electronegative element.***

$$Cl + e^- \rightarrow Cl^- \text{ (anion)}$$

The **cation** and the **anion** being oppositely charged attract each other and form a chemical bond. Since this chemical bond formation is due to the electrostatic force of attraction between a cation and an anion, it is called an ***electrovalent*** (or an ***ionic*) *bond.***

Electrovalent (or ionic) compounds : *The chemical compounds formed as a result of the transfer of electrons from one atom of an element to one atom of another element are called* ***ionic*** *(or electrovalent)* ***compounds.***

Electrovalency : *The number of electrons that an atom of an element loses or gains to form a electrovalent bond is called its* ***electrovalency.***

2.2.1 Conditions for the formation of an electrovalent (or ionic) bond

When an ionic compound is formed, the neutral atom is changed to a cation or an anion. The formation of cations and anions depends on the following factors :

1. **Low ionisation potential :** If the ionisation potential of a particular atom is **low,** it will lose electron(s) easily, *i.e., a cation is formed easily.*
2. **High electron affinity :** If the electron affinity value is **high,** anion will be formed easily, *i.e., a higher electron affinity value favours ionic bonding.*
3. **Large electronegativity difference :** If the difference in the electronegativities of two elements is **higher,** then the transfer of electrons will be easier. Therefore, *more the difference in electronegativity, more will be the ionic nature of the resulting compound.*

The **metals** of groups 1, 2 and 13 have a tendency to lose their valence electrons. So they combine with the **non-metals** of Groups 15, 16 and 17, which have a tendency to gain electron(s) and form ionic bonds.

Note : Group 1 elements are most electropositive, *i.e.,* they are metallic in nature. Their metallic nature increases down the group.

Group 17 elements are most electronegative. Fluorine is the most electronegative element. Thus, *caesium fluoride CsF is the most ionic compound.*

Bonds formed between metals and non-metals are ionic or electrovalent.

Why are Ionic Compounds Stable ?

Ionic compounds are formed by ions but there also exists a repulsive force between ions for like charges. Since the electrostatic force of attraction between opposite charges is much higher, it makes the ionic compounds stable.

Examples of electrovalent (ionic) compounds.

NaCl	(Sodium chloride)
$MgCl_2$	(Magnesium chloride)
CaO	(Calcium oxide)
KBr	(Potassium Bromide)
$CaCl_2$	(Calcium chloride)

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2.2.2 Structures of some electrovalent compounds

1. Sodium chloride (NaCl)

The electronic configuration of a **sodium atom is 2, 8, 1.** It has one electron in excess of the stable electronic configuration of the nearest noble gas, neon, (2, 8). Therefore, an atom of sodium shows a tendency to give up the electron from its outermost shell, so as to acquire a stable electronic configuration of neon.

$$\underset{(2,8,1)\text{ atom}}{Na} - 1e^- \rightarrow \underset{(2,8)\text{ cation}}{Na^+}$$

However, *after giving up one electron, the sodium atom is no more electrically neutral. It has eleven protons in its nucleus but only ten electrons revolving around it. Therefore, it has a net positive charge of* +1. *This positively charged atom is called sodium ion and is written as* Na^+ and its electronic configuration resembles that of the noble gas neon.

Table 2.2 Comparison of the properties of sodium atom and sodium ion

Property	*Sodium atom* (Na)	*Sodium cation* (Na^+)
Colour	Silvery white	Colourless
Toxicity	Poisonous	Non-poisonous
Chemical action	Very active	Inactive
Valence shell	Incomplete outermost shell	Complete outermost shell
Electrical state	Neutral	Positively charged
Existence	Combined state	Independent existence

The electronic configuration of **chlorine is 2, 8, 7.** It has an electronic configuration with one electron less than that of the nearest noble gas, argon (2, 8, 8). Therefore, the chlorine atom shows a tendency to acquire an electron to attain octet in its outermost shell.

$$\underset{(2,8,7)\text{ atom}}{Cl} + 1e^- \rightarrow \underset{(2,8,8)\text{ anion}}{Cl^-}$$

An atom of chlorine is electrically neutral, as it contains 17 protons in its nucleus and 17 electrons revolving around it. But, *after acquiring an electron from the sodium atom, the chlorine atom does not remain electrically neutral. It has one electron more than the number of protons in its nucleus and therefore has charge of –1 represented as* Cl^- *i.e.* **chloride ion.**

Table 2.3 Comparison of the properties of chlorine atom and chlorine ion

Property	*Chlorine atom* (Cl)	*Chloride anion* (Cl^-)
Colour	Yellowish green (as Cl_2 gas)	Colourless
Toxicity	Poisonous	Non-poisonous
Odour	Suffocating	Odourless
Chemical action	Very active	Inactive
Valence shell	Incomplete outermost shell	Complete outermost shell
Electrical state	Neutral	Negatively charged
Existence	Not independent	Independent

Chloride ion has an octet of electrons in its outermost shell, and its electronic configuration resembles that of the noble gas argon (Fig. 2.1).

Thus, when an atom of sodium combines with an atom of chlorine (electronic configuration 2, 8, 7), one electron is transferred from the sodium atom to the chlorine atom, resulting in the formation of a sodium chloride molecule.

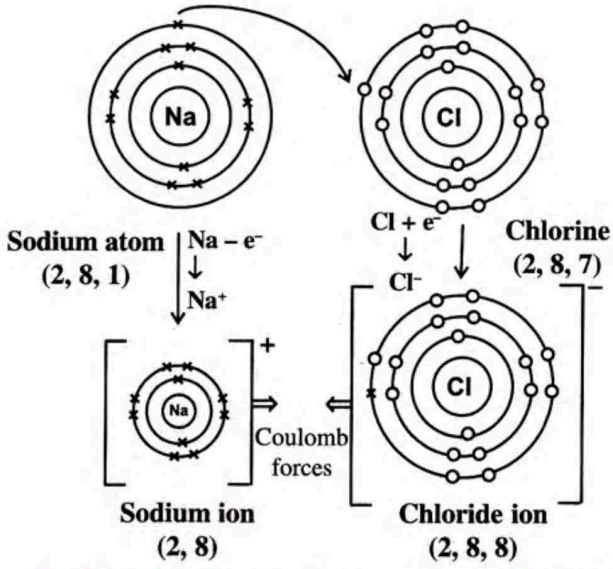

Fig. 2.1 Orbit structure of electrovalent bonding in sodium chloride.

The cation Na^+ and anion Cl^- are attracted towards each other, due to electrical charge or

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coulomb force existing between them and form an ionic compound.

Electron dot symbol (Lewis symbol)

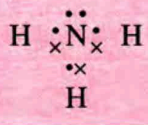
The electron dot symbol for an atom consists of the symbol of the element surrounded by dots representing the outermost shell electrons. The paired electrons are represented by a pair of dots, whereas the unpaired electron in the outermost orbit is represented by a single dot.

Example : Electron dot symbol of Hydrogen is H•

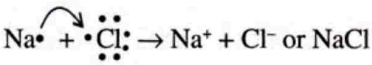
and of Oxygen is :Ö:

Symbols other than dots, such as circles and crosses can be used to distinguish between the electrons of different atoms in a molecule, *for example* :

Ammonia (NH_3) can be represented as



Electron dot structure of NaCl



2. Magnesium chloride ($MgCl_2$)

The number of valence electrons of magnesium (atomic number 12) is 2 and that of chlorine (atomic number 17) is 7. Magnesium atom acquires a stable configuration of 8 electrons by losing two electrons from its outermost shell (one each to each atom of chlorine) and thus becomes a positive magnesium ion, Mg^{2+}.

Mg	–	$2e^-$	→	Mg^{2+}
(2, 8, 2)				(2, 8)
atom				cation

However, each chlorine atom, which contains 7 electrons in its outermost shell, can accept only 1 of the 2 electrons donated by a magnesium atom. Therefore, for each magnesium atom forming a magnesium ion, there must be two chlorine atoms to form two chloride ions.

2Cl	+	$2e^-$	→	$2Cl^-$
(2, 8, 7)				(2, 8, 8)
2 Cl atoms				2 chloride anions

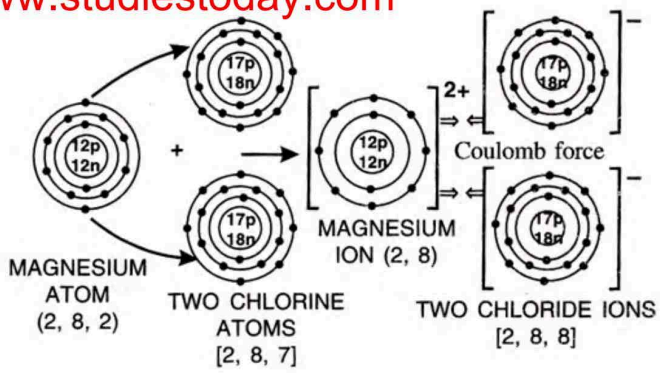
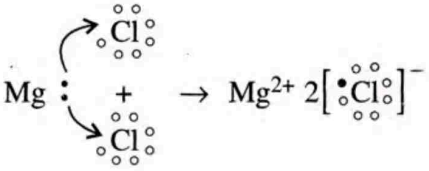


Fig. 2.2 Orbit structure of formation of magnesium chloride ($MgCl_2$) molecules.

Thus, the ratio of magnesium to chloride ions in magnesium chloride must be 1 : 2, so the molecular formula of the compound magnesium chloride is $MgCl_2$ (Fig. 2.2).

Electron dot structure of magnesium chloride



3. Calcium oxide (CaO)

The number of valence electrons of a calcium atom (atomic number 20) is 2, and that of an oxygen atom is 6, *i.e.*, oxygen requires 2 electrons to attain octet. In the presence of oxygen, each calcium atom loses its 2 valence electrons to one oxygen atom (Fig. 2.3). As a result, the calcium atom forms a calcium ion with charge +2 (Ca^{2+}), and the oxygen atom forms an oxide ion with charge –2 (O^{2-}). Since only one oxygen atom is needed to accept the 2 valence electrons donated by a calcium atom, the formula of calcium oxide is CaO and not Ca_2O_2.

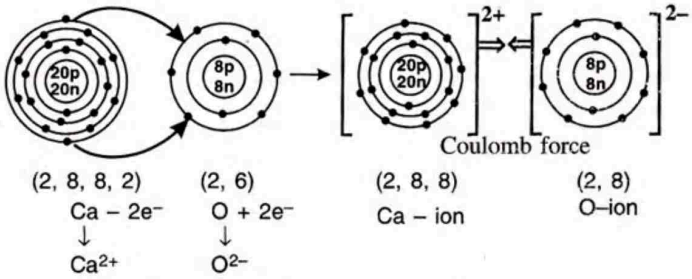


Fig. 2.3 Orbit structure of formation of calcium oxide (CaO).

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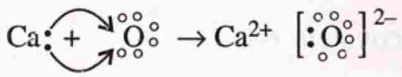
$$Ca - 2e^- \rightarrow Ca^{2+}$$
(2, 8, 8, 2) atom → (2, 8, 8) cation

$$O + 2e^- \rightarrow O^{2-}$$
(2, 6) atom → (2, 8) anion

Electron dot structure of calcium oxide



In the formation of an electrovalent bond, the transfer of electron(s) is involved. The **electropositive atom** undergoes **oxidation**, while the ***electronegative atom*** undergoes ***reduction***. This is known as REDOX PROCESS.

For example :

Formation of sodium chloride : Sodium chloride is formed by the combination of sodium and chlorine.

$$2Na + Cl_2 \rightarrow 2Na^+ + 2Cl^- \text{ (or 2 NaCl)}$$

The reaction can be written as two half reaction:

$$2Na \rightarrow 2Na^+ + 2e^- \text{ (Oxidation)}$$
$$Cl_2 + 2e^- \rightarrow 2Cl^- \text{ (Reduction)}$$

Oxidation

$$2Na + Cl_2 \rightarrow 2Na^+ + 2Cl^- \text{ (Redox Reaction)}$$

Reduction

Oxidation and reduction always occur simultaneously because the electron(s) lost by the reducing agent must be gained by the oxidising agent.

For example :

Oxidised

$$\underset{\text{Oxidising agent}}{CuO} + \underset{\text{Reducing agent}}{H_2} \rightarrow Cu + H_2O$$

Reduced

In this reaction, hydrogen acting as a reducing agent reduces Cu(II) oxide to copper. This is a reduction reaction.

$$Cu^{2+} + 2e^- \rightarrow Cu \text{ (Reduction)}$$

At the same time, hydrogen is oxidised to water by the oxidising agent Cu(II) oxide, and this is an oxidation reaction.

$$2H - 2e^- \rightarrow 2H^+ \text{ (Oxidation)}$$

Thus, the net reaction is a redox reaction.

It can be inferred from the above example that an oxidising agent is an acceptor of electron(s) and a reducing agent is a donor of electron(s).

[Also refer chapter 6, article 6.2 (viii) & (ix)]

Intext Questions

1. How do atoms attain noble gas configuration ?
2. Define :
   (a) a chemical bond, (b) an electrovalent bond,
   (c) a covalent bond.
3. What are the conditions for the formation of an electrovalent bond ?
4. An atom X has three electrons more than the noble gas configuration. What type of ion will it form ? Write the formula of its (i) sulphate (ii) nitrate (iii) phosphate (iv) carbonate (v) hydroxide.
5. Mention the basic tendency of an atom which makes it to combine with other atoms.
6. A solid is crystalline, has a high melting point and is water soluble. Describe the nature of the solid.
7. In the formation of the compound XY_2, an atom X gives one electron to each Y atom. What is the nature of bond in XY_2 ? Draw the electron dot structure of this compound.
8. An atom X has 2,8,7 electrons in its shell. It combines with Y having 1 electron in its outermost shell.
   (a) What type of bond will be formed between X and Y ?
   (b) Write the formula of the compound formed.
9. Draw orbit structure and electron dot diagrams of
   (i) NaCl (ii) $MgCl_2$ (iii) CaO
10. Compare :
    (a) sodium atom and sodium ion
    (b) chlorine atom and chloride ion, with respect to
    (i) atomic structure, (ii) electrical state,
    (iii) chemical action, (iv) toxicity.
11. The electronic configuration of Fluoride ion is the same as that of a neon atom. What is the difference between the two ?

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12. (a) What do you understand by redox reactions? Explain oxidation and reduction in terms of loss or gain of electrons.

(b) Divide the following redox reactions into oxidation and reduction half reactions.

(i) $Zn + Pb^{2+} \rightarrow Zn^{2+} + Pb$

(ii) $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

(iii) $Cl_2 + 2Br^- \rightarrow Br_2 + 2Cl^-$

(iv) $Sn^{2+} + 2Hg^{2+} \rightarrow Sn^{4+} + Hg_2^{2+}$

(v) $2Cu^+ \rightarrow Cu + Cu^{2+}$

(c) Potassium (at No. 19) and chlorine (at No. 17) react to form a compound. Explain on the basis of electronic concept :

(i) oxidation (ii) reduction

(iii) oxidising agent (iv) reducing agent.

2.3 COVALENT (MOLECULAR) BOND

The chemical bond that is formed between two combining atoms by mutual sharing of one or more pairs of electrons is called a ***covalent*** *(or a molecular)* ***bond*** *and the compound formed due to this bond is called a* ***covalent compound.***

The molecule formed due to the sharing of electrons (covalent bond) is called a **covalent molecule.**

The atoms of non-metals usually have 5, 6 or 7 electrons in their outermost shell (except carbon which has 4 and hydrogen which has just 1 electron in the outermost shell). *The atoms of such elements do not favour the loss of its electrons due to energy considerations and thus the transfer of electrons is not possible*. Therefore, this atom can complete its octet only by mutually sharing one or more pairs of electrons. **Each atom contributes equal number of electron(s).** So, whenever a non-metal combines with another non-metal (to attain stable configuration), the sharing of electrons takes place between their atoms and a covalent bond is formed.

For example, hydrogen is a non-metal and chlorine is also a non-metal. When hydrogen combines with chlorine to form hydrogen chloride (HCl), the sharing of electrons takes place between hydrogen and chlorine atoms and a covalent bond is formed. It should be noted that a covalent bond can also be formed between two atoms of the same non-metal. For example, two chlorine atoms combine together by the sharing of electrons to form a chlorine molecule (Cl_2) and a covalent bond is formed between the two chlorine atoms.

Covalent bonds are of following three types:

1. Single covalent bond
2. Double covalent bond
3. Triple covalent bond

A **single covalent** bond is formed by the sharing of one pair of electrons between the atoms, each atom contributing one electron.

A single covalent bond is denoted by putting a short line (—) between the two atoms. So, a hydrogen molecule can be written as H—H.

Likewise, molecules of chlorine, hydrogen chloride, water, ammonia, methane and carbon tetrachloride are examples of single covalent bonds.

Cl—Cl	H—Cl	H—O—H
Chlorine	Hydrogen chloride	Water

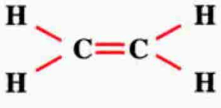
H—N—H | H	H | H—C—H | H	Cl | Cl—C—Cl | Cl
Ammonia	Methane	Carbon tetrachloride

Similarly, a **double bond** is formed by the sharing of two pairs of electrons between two atoms.

A double bond is actually a combination of two single bonds, so it is represented by putting two short lines (=) between the two atoms. *For example*, oxygen molecule, O_2, contains a double bond between two atoms and it can be written as O=O.

A **triple bond** is a combination of three single bonds, so it is represented by putting three short lines (°) between the two atoms. Nitrogen molecule, N_2, contains a triple bond, so it is written as N ≡ N.

Some molecules have a combination of single bond as well as a double or a triple bond. *For example*, ethene (C_2H_4) molecule has one double covalent bond and four single covalent bonds.



Ethene (C_2H_4)

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Similarly, **ethyne** (C_2H_2) molecule has one triple covalent bond and two single covalent bonds.

Ethyne (C_2H_2) H – C ≡ C – H

The covalency *of an atom is the number of its electrons taking part in the formation of shared pairs.* Thus, the covalency of hydrogen is 1, oxygen 2, nitrogen 3 and carbon 4.

Non-polar and polar covalent compounds

Covalent compounds are non polar when shared pair of electron(s) are equally distributed between the two atoms. No charge separation takes place. The molecule is symmetrical and electrically neutral.

If the two covalently bonded atoms are identical the shared electron pair(s) is at equal distance from the combining atoms *i.e.*, the shared electron pair(s) is equally attracted by the nuclei of the two type of charge, such molecules are non-polar. For example hydrogen (H_2), chlorine (Cl_2), oxygen (O_2), *etc.*, are perfectly non-polar compounds.

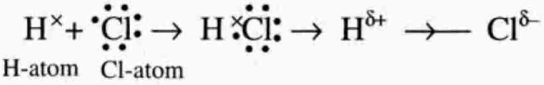
The bond formed between dissimilar atoms can be non-polar if their electronegativity difference is little and their structure permits the shared pair of electrons to attract equally the linked atoms and thus the molecule becomes symmetrical. For example methane (see structure 7, article 2.4.2), carbon tetrachloride, etc.

The shared electron pairs in methane are at equal distance from the carbon and the hydrogen atoms, because the two have nearly equal electronegativities (carbon = 2·5, hydrogen = 2·1).

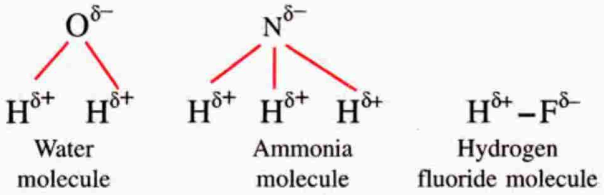
These compounds do not ionise in water due to lack of charge separation.

The covalent compounds are said to be polar when the shared pair of electrons are not at equal distance between the two atoms. *This results in the development of fractional positive and negative charges on them. They ionise in water. For example, hydrogen chloride.*

In hydrogen chloride, the strong nuclear charge of the chlorine atom (the electro-negativity of chlorine is 3) attracts the shared electron pair towards itself, *i.e.*, negative charge shifts towards the chlorine atom thereby developing a slight negative charge (δ^-) on it. The hydrogen atom (electronegativity 2.1) develops a slight positive charge (δ^+). *Therefore, a polar covalent bond is formed.* This is shown below.



Other examples of polar covalent compounds that ionise are water, ammonia and hydrogen fluoride.



The process by which covalent compounds are converted into ions is called **ionisation**.

Note :

(i) The more the electronegativity difference between two atoms forming a bond, the more is the polar nature of the molecule.

The bond formed between two atoms (1) with **same** electronegativity is **non-polar** (2) with **slightly different** electronegativity is **polar** and (3) with **much electronegativity** difference is **ionic**.

(ii) Since a polar covalent molecule has both positive and negative poles, it is also known as a ***'dipole molecule'***.

(A molecule that has both, slight positive and slight negative charge is called a **Dipole molecule**)

2.3.1 Conditions for the formation of a covalent bond

1. Both atoms should have four or more electrons in their outermost shells, *i.e.*, non-metals (exceptions are H, Be, B, Al, *etc.*).
2. Both the atoms should have high electronegativity.
3. Both the atoms should have high electron affinity.
4. Both the atoms should have high ionization energy.
5. The electronegativity difference between the combining atoms should either be zero or negligible.

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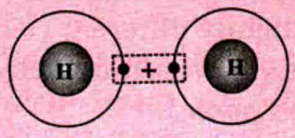
Group →	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII
Chloride →	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_3/PCl_5	S_2Cl_2	
Bonding →	Ionic solid	Ionic solid	Partially ionic-partially covalent solid	Covalent liquid	Covalent liquid/solid	Covalent liquid	
Oxide →	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_2/SO_3	Cl_2O_7
Bonding →	Ionic Solid	Ionic Solid	Ionic Solid	Covalent Solid	Covalent Solid	Covalent Gas/Solid	Covalent Gas

Hydrogen can combine with all non-metals of Group IVA to VIIA with the help of covalent bonds.

Note : As we move across a period, the electrovalent character of chlorides, oxides, etc., decreases. An example of the decreasing electrovalent character of chlorides of the elements of 3rd period is shown below.

Formation of a hydrogen molecule

Before combination



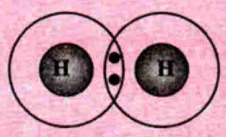
H atom H atom

Electron dot structure

H· + ·H

H-atom H-atom

After combination



Hydrogen molecule

H:H or H—H → H_2

One shared pair of electrons — **Single covalent bond**

[A hydrogen molecule contains two atoms of hydrogen]

2. Chlorine molecule (Non-polar compound)

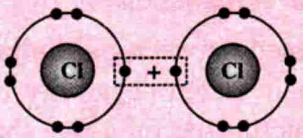
Electronic configuration	*Nearest noble gas*
$_{17}Cl$ [2, 8, 7]	Argon ($_{18}Ar$) [2, 8, 8]

To attain the stable electronic configuration of the nearest noble gas, chlorine needs one electron.

When two chlorine atoms come closer, each contributes one electron and form ***one shared pair of electrons*** between them. Both the atoms of chlorine thus attain an octet. A single covalent bond [Cl — Cl] is formed between them.

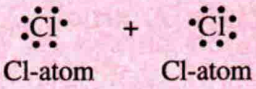
Formation of a chlorine molecule

Before combination



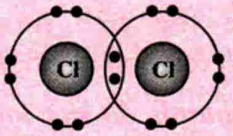
Cl-atom Cl-atom

Electron dot structure



Cl-atom Cl-atom

After combination



Chlorine molecule

:Cl : Cl: or [Cl—Cl] → Cl_2

One shared pair of electrons — **Single covalent bond**

[A chlorine molecule contains two atoms of chlorine]

3. Nitrogen molecule (Non-polar compound)

Electronic configuration	*Nearest noble gas*
Nitrogen ($_7N$) [2, 5]	Neon ($_{10}Ne$) [2, 8]

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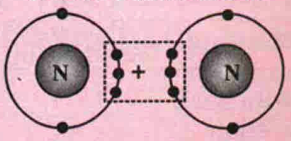
To attain the stable electronic configuration of the nearest noble gas, nitrogen needs three electrons.

When two nitrogen atoms come closer, each contributes three electrons and so they have ***three shared pairs of electrons*** between them. Both atoms attain an octet, resulting in the formation of a triple covalent bond [N ≡ N] between them.

Formation of a nitrogen molecule

Before combination



N-atom N-atom

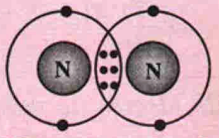
Electron dot structure

:N: + :N:

N-atom N-atom

After combination

Mutual sharing of three pairs of elements



Nitrogen molecule

:N: :N: or [N≡N] → N_2

Three shared pairs of electrons — **Triple covalent bond**

[A nitrogen molecule contains two atoms of nitrogen]

4. Water molecule (Polar compound)

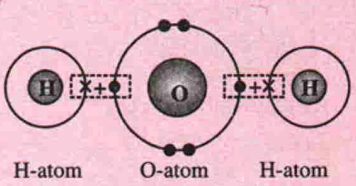
Atoms involved	*Electronic configuration*	*Nearest noble gas*
Hydrogen	$_1$H [1]	Helium [2]
Oxygen	$_8$O [2, 6]	Neon [2, 8]

To attain the stable electronic configuration of the nearest noble gas, hydrogen needs one electron and oxygen needs two electrons.

In the case of a water molecule, *each of the two hydrogen atoms share an electron pair with the oxygen atom* such that hydrogen acquires a duplet configuration and oxygen an octet, resulting in the formation of two single covalent bonds.

Formation of a water molecule

Before combination



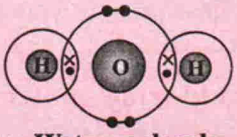
H-atom O-atom H-atom

Electron dot structure

H× + ·O· + ×H

H-atom O-atom H-atom

After combination



Water molecule

H ×O× H or H—O—H → H_2O

One shared pair of electrons — **Two single covalent bond**

[One molecule of water contains a total of three atoms, *i.e.* one atom of oxygen and two atoms of hydrogen]

5. Ammonia molecule (Polar compound)

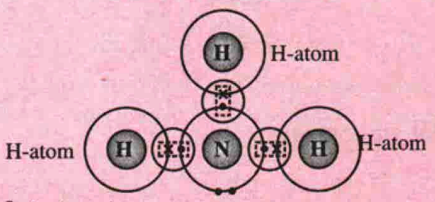
Atoms involved	*Electronic configuration*	*Nearest noble gas*
Nitrogen	$_7$N [2, 5]	Neon [2, 8]
Hydrogen	$_1$H [1]	Helium [2]

To attain the electronic configuration of the nearest noble gas, nitrogen needs three electrons and hydrogen needs one electron.

When a molecule of ammonia is to be formed, one atom of nitrogen *shares three electron pairs, one with each of the three atoms of hydrogen.*

Formation of an ammonia molecule

Before combination



Electron dot structure

H
+
H• + •N• +•H

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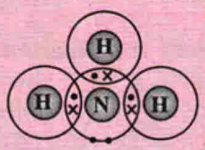
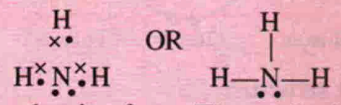
After combination



Ammonia molecule



One shared pair of electrons with each hydrogen atom

Three single covalent bonds

[The ammonia molecule contain a total of four atoms, *i.e.*, one atom of nitrogen and three atoms of hydrogen]

6. Carbon tetrachloride molecule (Non-polar compound)

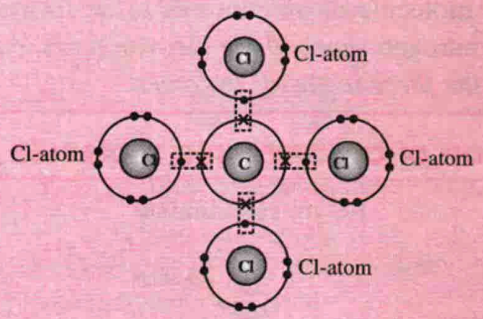
Atoms involved	*Electronic configuration*	*Nearest noble gas*
Carbon	$_6$C [2, 4]	Neon [2, 8]
Chlorine	$_{17}$C1 [2, 8, 7]	Argon [2, 8, 8]

To attain the stable electronic configuration of the nearest noble gas, carbon needs four electrons and chlorine needs one electron.

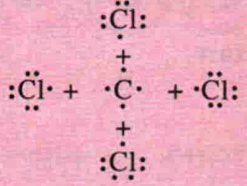
When a molecule of carbon tetrachloride is to be formed, one atom of carbon *shares four electron pairs, one with each of the four atoms of chlorine.*

Formation of a carbon tetrachloride molecule

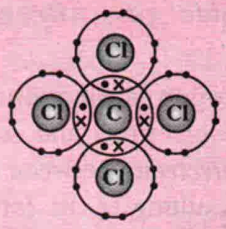
Before combination



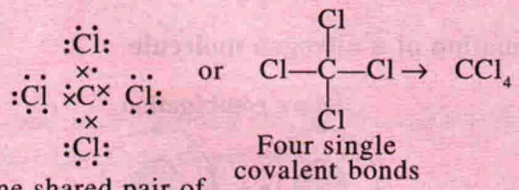
Electron dot structure



After combination



Carbon tetrachloride molecule



One shared pair of electrons with each chlorine atom

Four single covalent bonds

[One molecule of carbon tetrachloride contains five atoms in all, *i.e.* one atom of carbon and four atoms of chlorine]

7. Methane molecule (Non-polar compound)

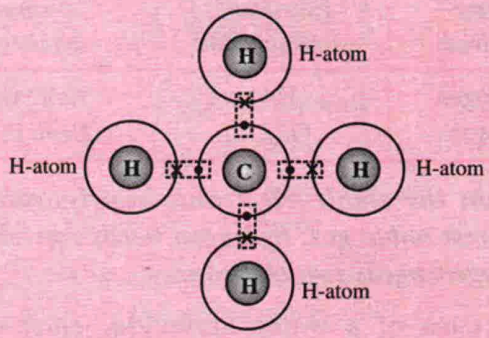
Atoms involved	*Electronic configuration*	*Nearest noble gas*
Carbon	$_6$C [2, 4]	Neon [2, 8]
Hydrogen	$_1$H[1]	Helium [2]

To attain the stable electronic configuration of the nearest noble gas, carbon needs four electrons and hydrogen needs one electron.

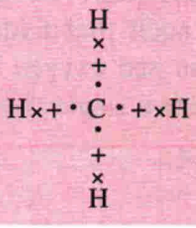
When a molecule of methane is to be formed, one atom of carbon *shares four electron pairs, one with each of the four atoms of hydrogen.*

Formation of a methane molecule

Before combination



Electron dot diagram



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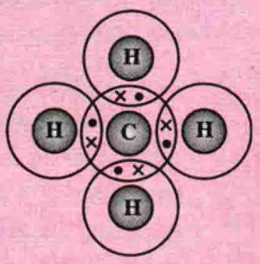
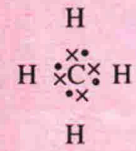
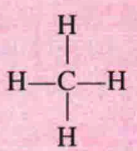
After combination



Methane molecule



or



Four single covalent bonds

[A methane molecule contains a total of five atoms, *i.e.* one atom of carbon and four atoms of hydrogen]

2.4 PROPERTIES AND COMPARISON OF ELECTROVALENT AND COVALENT COMPOUNDS

Electrovalent compounds		Covalent compounds	
Property	***Reason***	***Property***	***Reason***
1. Nature (i) Their constituent particles are ions. (ii) They are hard solids consisting of **ions**.	These have strong electrostatic forces of attraction between their ions, which cannot be separated easily.	**1. Nature** (i) Their constituent particles are molecules. (ii) These are gases or liquids or soft solids.	They have weak forces of attraction between their molecules.
2. Boiling point and melting point These are non-volatile, with high boiling and high melting points.	There exists a strong force of attraction between the oppositely charged ions, so a large amount of energy is required to break the strong bonding force between ions.	**2. Boiling point and melting point** These are volatile, with low boiling and low melting points.	They have weak forces of attraction between the binding molecules, thus less energy is required to break the force of bonding.
3. Electricity conducting nature (i) They do not conduct electricity in the solid state. (ii) They are good conductors of electricity in the fused or in aqueous state.	Electrostatic forces of attraction between ions in the solid state are very strong. These forces weaken in fused state or in solution state. Hence, ions become mobile.	**3. Electricity conducting nature** They are non-conductors of electricity in solid, molten or aqueous state.	Due to the absence of free ions.
4. Ionisation in solution Electrovalent compounds are composed of ions. In solution, these ions become mobile. They are electrolytes.	Water being a polar covalent compound decreases the electrostatic forces of attraction, resulting in free ions in aqueous solution. $NaCl \rightleftharpoons Na^+(aq) + Cl^-(aq)$	**4. Ionisation in solution** On passing electric current, non-polar covalent compounds do not ionise. ***Some of the covalent compounds are polar in nature.*** *They ionize in their solutions and can act as an electrolyte.* e.g. $HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$	Covalent compounds do not have ions. Polar covalent molecules form ions in their solutions.
5. Dissociation Their **ions dissociate and migrate** when an electric current passes through them in their molten or aqueous solution state. *e.g.* $NaCl \rightleftharpoons Na^+ + Cl^-$	Ions dissociate in water or in molten state.	**5. Dissociation** The dissociation of molecules into ions does not take place.	Covalent compounds do not have ions, so they do not dissociate.
6. Solubility These are soluble in water but insoluble in organic solvents.	As water is a polar compound, it decreases the electrostatic forces of attraction, resulting in free ions in aqueous solution. Hence they dissolve.	**6. Solubility** These are insoluble in water but dissolve in organic solvents.	As organic solvents are non-polar, hence, these dissolve in non-polar covalent compounds.

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7. Rate of reaction They show **rapid speed** of chemical reactions in aqueous solutions.	Since free ions are easily formed in different solutions, they unite very fast forming compounds.	**7. Rate of reaction** They show **slow speed** of chemical reactions in aqueous solutions.	In covalent molecules, old bonds are broken and new bonds are formed, thus the reaction is slow between covalent compounds.

2.3.2 Some covalent molecules and their structures

1. Hydrogen molecule (Non-polar compound)

A hydrogen atom has one electron in its only shell. It needs one more electron to attain **duplet**. To meet this need, hydrogen atom shares its electron with another hydrogen atom. Thus, one electron each is contributed by the two atoms of hydrogen, and the resulting pair of electrons is mutually shared by both the atoms to form a hydrogen molecule.

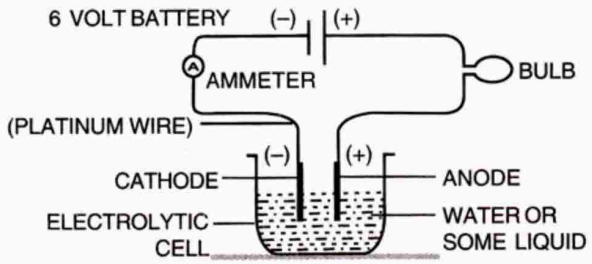

Fig. 2.4 Effects of electric current in different solutions.

2.5 EFFECT OF ELECTRICITY ON ELECTROVALENT AND COVALENT COMPOUNDS

Experiment : Arrange an electrolytic cell as shown in Fig. 2.4. The electric circuit contains a 6-volt battery, an ammeter bulb and platinum electrodes connected in series.

Take the following electrovalent and covalent compounds in solution in separate beakers and dip the platinum electrodes in them for a moment, one by one, and note the change in ammeter reading.

Observations :

Electrovalent compounds : When the current is passed in the solutions of electrovalent compounds, *i.e.*, magnesium chloride, sodium chloride, sodium hydr-oxide and copper sulphate the bulb glows. This shows that they *allow electric current to pass through them.*

Thus, ***ionic compounds are good conductors of electricity in molten or aqueous states due to free mobile ions.***

Covalent compounds : When the current is passed through the solutions of covalent compounds, *i.e.* distilled water, sugar solution, alcohol, chloroform, benzene and petrol the bulb does not glow. This shows that they *do not conduct electricity.* This happens because solutions of covalent compounds contain only molecules and no ions.

Intext Questions

1. What are the conditions necessary for the formation of covalent molecules ?
2. Elements A, B and C have atomic numbers 17,19 and 10 respectively.
   (a) State which one is : (i) a non-metal (ii) a metal, (iii) chemically inert ?
   (b) Write down the formula of the compound formed by two of the above elements.
3. Draw electron dot diagram and structure of :
   (a) nitrogen molecule (b) magnesium chloride
   (c) methane
4. What is the difference between :
   (a) ionic compounds and polar covalent compounds,
   (b) ionic compounds and covalent compounds,
   (c) a polar covalent compound and a non-polar covalent compound ?
5. The element X has the electronic configuration 2, 8, 18, 8, 1. Without identifying X,
   (a) predict the sign and charge on a simple ion of X.
   (b) write if X will be an oxidising agent or reducing agent and why.
6. What do you understand by dipole molecule ? Explain it by taking hydrogen chloride as an example.
7. (a) Explain the bonding in methane molecule using electron dot structure.
   (b) Methane molecule is non-polar molecule. Explain.

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8. Give the characteristic properties of :
   (a) electrovalent compounds,
   (b) covalent compounds.
9. (a) Which type of bond is formed when the combining atoms have :
   (i) zero E.N. difference
   (ii) small E.N. difference
   (iii) large E.N. difference
10. Explain the following :
   (a) Electrovalent compounds conduct electricity.
   (b) Electrovalent compounds have a high melting point and boiling point while covalent compounds have low melting and boiling points.
   (c) Electrovalent compounds dissolve in water whereas covalent compounds do not.
   (d) Electrovalent compounds are usually hard crystals yet brittle.
   (e) Polar covalent compounds conduct electricity.
11. Elements X, Y and Z have atomic numbers 6, 9 and 12 respectively. Which one :
   (a) forms an anion ,
   (b) forms a cation,
   (c) state type of bond between Y and Z and give its molecular formula.
12. Taking $MgCl_2$ as an electrovalent compound, CCl_4 as a covalent compound, give four differences between electrovalent and covalent compounds.
13. Potassium chloride is an electrovalent compound, while hydrogen chloride is a covalent compound. But, both conducts electricity in their aqueous solutions. Explain.
14. (a) Name two compounds that are covalent when taken pure but produce ions when dissolved in water.
   (b) For each compound mentioned above give the formulae of ions formed in aqueous solution.
15. An element M burns in oxygen to form an ionic bond MO. Write the formula of the compounds formed if this element is made to combine with chlorine and sulphur separately.

2.6 COORDINATE BOND

The bond formed between two atoms by sharing a pair of electrons, provided entirely by one of the combining atoms but shared by both is called a ***coordinate bond*** or ***dative bond,*** e.g., Ammonium ion (NH_4^+), Hydronium ion (H_3O^+).

2.6.1 Conditions for the formation of coordinate bond

(i) One of the two atoms must have at least one lone pair of electrons, *e.g.,* ammonia (NH_3), water (H_2O).

A pair of electrons which is not shared with any other atom is known as the **lone pair** of electrons but it is provided to the other atom for the formation of coordinate bond.

(ii) Another atom should be short of at least a lone pair of electrons, *e.g.,* Hydrogen ion (H^+).

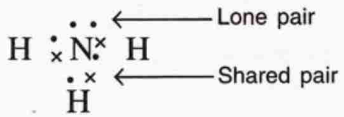
A coordinate bond can also be defined as a bond formed between an ion and an atom of a polar covalent molecule with one or more lone pair of electrons.

A coordinate bond has properties of both covalent and ionic bonds. Therefore, it is also called **co-ionic bond**.

The atom which provides the electron pair for the formation of a coordinate bond is known as the **DONOR** and the atom or ion sharing the donated electron pair is known as the **ACCEPTOR**.

2.6.2 Formation of Ammonium ion (NH_4^+)

Nitrogen atom has five valence electrons and hydrogen atom has one valence electron. In ammonia, each of the three hydrogen atoms are bonded to the nitrogen atom by a pair of shared electrons. (One electron from the hydrogen atom and one from the nitrogen).



Thus, the nitrogen atom in ammonia is left with a **lone pair** of electrons.

This *ammonia molecule* combines with a hydrogen ion H^+, which has no electrons in its outermost orbit in such a way that the lone pair of electrons on the nitrogen atom is shared by the hydrogen ion. The latter thus acquires the helium

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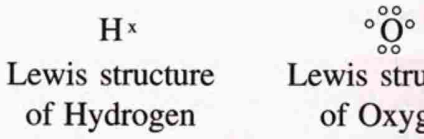
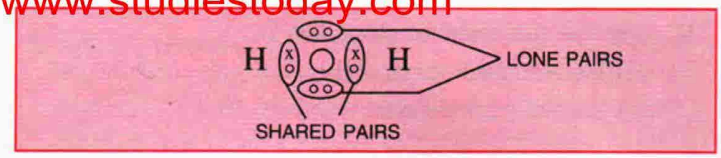
configuration. *This bond which is formed between the nitrogen atom in ammonia and the hydrogen ion is a coordinate bond and is represented by "→" whereas a covalent bond is represented by "—".*

2.6.3 Formation of H_3O^+ ion and lone pair effect of oxygen atom

Water contains two hydrogen atoms and one oxygen atom. The hydrogen atom has one electron in its valence shell while the oxygen atom has six electrons in its valence shell.



Two hydrogen atoms share one electron each with oxygen atom to form a water molecule.



Dot diagram of water molecule

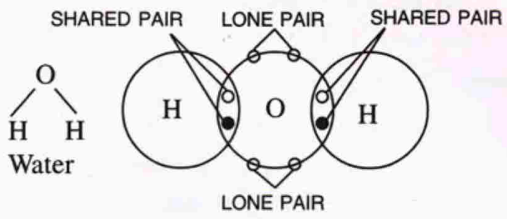
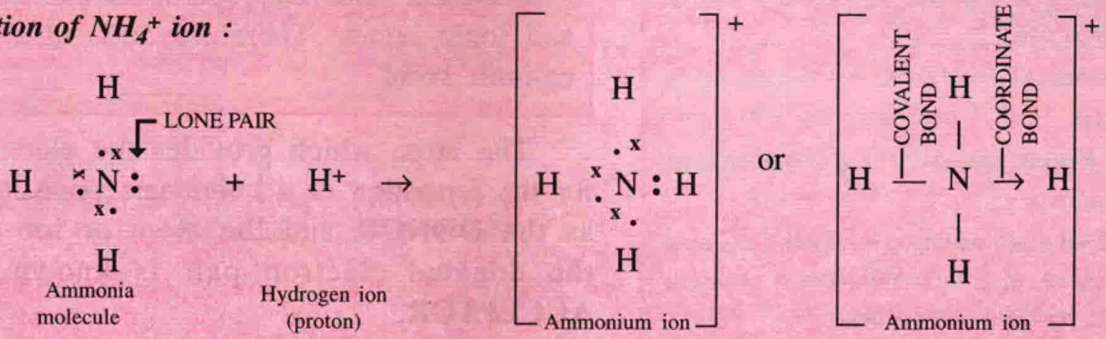


Fig. 2.5 Covalent bonding in water

The two unshared pairs of electrons known as **lone pairs** of electrons, do not participate in the bond formation in water molecule.

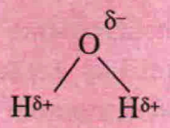
Formation of NH_4^+ ion :



The nitrogen atom in an ammonia molecule contains a ***lone pair of electrons.*** It is a DONOR. The hydrogen ion (H^+) ***accepts that lone pair of electrons*** (ACCEPTOR) and forms a coordinate bond. Once this bond is formed, all four N – H bonds become identical. The linkage forms an ***ammonium ion*** having a ***single positive charge.***

When ammonium chloride NH_4Cl is formed cation NH_4^+ (having 3 covalent and one coordinate bond) and anion Cl^- are attracted towards each other, due to electrical charge existing between them ionic bond is formed. Thus ammonium chloride is a good example of compound having all the three types of bonds *i.e.*, covalent, coordinate and ionic bond.

The O–H bonds in a water molecule are polarised *due to the large difference in the electronegativity values of the oxygen and the hydrogen atoms.* (The oxygen atom has an electronegativity value of 3·5, while that of the hydrogen atom is 2·1). Oxygen has higher electroneg-ativity so it attains slight negative (δ^-) charge while hydrogen atoms attain slight positive charge (δ^+). Therefore, a molecule of water due to these charges is a polar molecule and is represented as shown alongside.



When an acid is added to water, strong dipole interaction (attraction between slight positive and slight negative charge) occurs between the polar water molecules and the polar acid molecules.

As a result of such a dipole interaction, the hydrogen ion gets released from the polar bonds of the acid molecules.

$$HCl \xrightarrow{H_2O} H^+ + Cl^-$$

The hydrogen ion spontaneously adds on to one of the two lone pairs of electrons of the oxygen atom of the water molecule.

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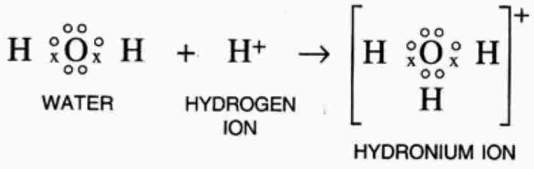
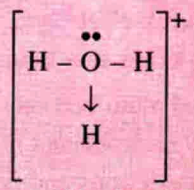


Addition of released H^+ ion to a **lone pair of electrons** of the oxygen atom of the polar water molecule leads to the formation of a *hydronium ion*

Hydronium ion is hydrated proton

$$H^+ + H_2O \rightarrow H_3O^+$$

Proton Water Hydronium ion



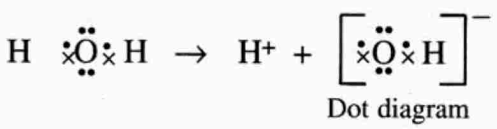
Structure of hydronium ion

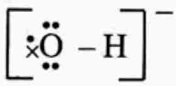
2.6.4 Formation of hydroxyl ion (OH^-) :

The hydroxyl ion or hydroxide ion is OH^-. It is formed when one hydrogen ion (H^+) is removed from the water molecule.

$$H_2O \rightarrow H^+ + OH^-$$

When H^+ is removed from water molecule, the shared pair of electron remains with oxygen as oxygen is more electronegative and thus hydroxyl ion has negative charge.





Structure of hydroxyl ion

Self-ionisation of water

Water molecule contains two hydrogen atoms and one oxygen atom. A positive hydrogen ion (H^+) is formed when water ionises.

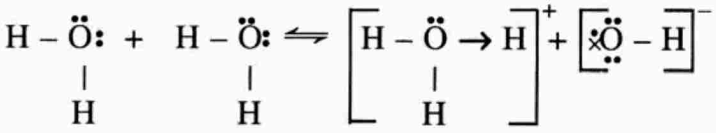
$$H_2O \rightleftharpoons H^+ + OH^-$$

This H^+ ion is transferred from one water molecule to the oxygen atom of another water molecule, forming hydronium ion (H_3O^+).

$$H^+ + H_2O \rightleftharpoons H_3O^+$$

Thus, H_3O^+ and OH^- ions are formed by water molecule.

$$H_2O + H_2O \rightleftharpoons \underset{\text{hydronium ion}}{H_3O^+} + \underset{\text{hydroxide ion}}{OH^-}$$



EXERCISE

1. Define a coordinate bond and give the conditions for its formation.
2. What do you understand by lone pair and shared pair ?
3. State the type of bonding in the following molecules.
   (a) water, (b) calcium oxide,
   (c) hydroxyl ion, (d) methane,
   (e) ammonium ion, (f) ammonium chloride
4. (a) Draw an electron dot diagram to show the structure of each of the following :
   (i) Hydronium ion, (ii) Ammonium ion,
   (iii) Hydroxyl ion.
   State the type of bonding present in them.
   (b) Give two examples in each case :
   (i) Co-ordinate bonds compounds,
   (ii) Solid covalent compounds,
   (iii) Gaseous polar compounds,
   (iv) Gaseous non polar compounds,
   (v) Liquid non polar compounds.
5. Element M forms a chloride with the formula MCl_2 which is a solid with high melting point. M would most likely be in the group in which ..... is placed.
   [(a) Na (b) Mg (c) Al (d) Si.]
6. Complete the following

	Sodium	Phosphorus	Carbon
Formula of chloride			
Nature of bonding			
Physical state of chloride			

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7. Element A has 2 electron in its M shell. Element B has atomic number 7.

(a) Write equations to show how A and B form ions.

(b) If B is a diatomic gas, write the equation for the direct combination of A and B to form a compound.

(c) If the compound formed between A and B is melted and an electric current is passed through the molten compound, the element A will be obtained at the ..... and B at the ..... of the electrolytic cell.

8. (a) How many atoms of each kind are present in the following molecules : calcium oxide, chlorine, water, carbon tetrachloride ?

(b) How many electrons are required by each atom mentioned in (a) to attain the nearest noble gas configuration?

9. Complete the following :

(a) When the nuclei of two different reacting atoms are of ..... mass, then a bond so formed is called ..... covalent bond (equal, unequal, polar, non-polar).

(b) In case of non-polar covalent bond, the covalent bond is formed in the ..... of atoms and shared electrons are ..... distributed (corner, middle, equally, unequally).

(c) The ions in ..... compounds are held very strongly due to strong ..... forces (electrovalent, covalent, electromagnetic, electrostatic).

10. (a) Compound X consists of molecules.

Choose the letter corresponding to the correct answer from the options A, B, C and D given below :

(i) The type of bonding in X will be :

A. ionic B. electrovalent

C. covalent D. molecular

(ii) X is likely to have a :

A. low melting point and high boiling point,

B. high melting point and low boiling point,

C. low melting point and low boiling point,

D. high melting point and high boiling point.

(iii) In the liquid state, X will :

A. become ionic,

B. be an electrolyte,

C. conduct electricity,

D. not conduct electricity.

(b) Electrons are getting added to an element Y :

(i) is Y getting oxidized or reduced ?

(ii) what charge will Y migrate to during the process of electrolysis ?

11. (a) Acids dissolve in water and produce positively charged ions. Draw the structure of these positive ions.

(b) Explain why Carbon tetrachloride does not dissolve in water.

(c) Elements Q and S react together to form an ionic compound. Under normal conditions, which physical state will the compound QS exist in ?

(d) Can Q and S, both be metals? Justify your answer.

(e) The property which is characteristic of an electrovalent compound is that :

A. it is easily vaporized,

B. it has a high melting point,

C. it is a weak electrolyte,

D. it often exists as a liquid.

(f) When a metal atom becomes an ion :

A. it loses electrons and is oxidized,

B. it gains electrons and is reduced,

C. it gains electrons and is oxidized,

D. it loses electrons and is reduced.

2007

(a) (i) Name the charged particles which attract one another to form electrovalent compounds.

(ii) In the formation of electrovalent compounds, electrons are transferred from one element to another. How are electrons involved in the formation of a covalent compound ?

(iii) The electronic configuration of nitrogen is (2, 5). How many electrons in the outer shell of a nitrogen atom are not involved in the formation of a nitrogen molecule ?

(iv) In the formation of magnesium chloride (by direct combination between magnesium and chlorine), name the substance that is oxidized and the substance that is reduced.

2008

(a) Which of the following is **not** a common characteristic of an electrovalent compound ?

A. High melting point.

B. Conducts electricity when molten.

C. Consists of oppositely charged ions.

D. Ionizes when dissolved in water.

(b) What are the terms defined below ?

(i) A bond formed by a shared pair of electrons, each bonding atom contributing one electron to the pair.

(ii) A bond formed by a shared pair of electrons with both electrons coming from the same atom.

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2009

(a) The one which is composed of all the three kinds of bond [ionic, convalent and coordinate bond]

A. Sodium chloride B. Ammonia
C. Carbon tetrachloride D. Ammonium chloride

(b) Draw the structural formula of carbon tetrachloride and state the type of bond present in it.

2010

(a) Select the correct answer from A, B, C and D — Metals lose electrons during ionization — this change is called

A. Oxidation B. Reduction
C. Redox D. Displacement

(b) Select the right answer —

(i) *Sodium chloride* — convlent bond / ionic bond / convalent and coordinate bond.

(ii) *Ammonium ion* — covalent bond / ionic bond / covalent and coordinate bond.

(iii) *Carbon tetrachloride* — covalent bond / ionic bond / covalent and coordinate bond.

2011

(a) (i) In covalent compounds, the bond is formed due to ..... [sharing/transfer] of electrons.

(ii) Electrovalent compounds have a ..... [low/high] boiling point.

(iii) A molecule of ..... contains a triple bond. [hydrogen, ammonia, nitrogen].

(b) By drawing an electron dot diagram, show the lone pair effect leading to the formation of — ammonium ion from ammonia gas and hydrogen ion.

(c) Give reasons — Hydrogen chloride can be termed as a polar covalent compound.

2012

(a) Draw an electron dot diagram of the structure of — hydronium ion. State the type of bonding present in it.

(b) There are three elements E, F, G with atomic number 19, 8 and 17 respectively. Give the molecular formula of the compound formed between E and G and state the type of chemical bond in this compound.

2013

(a) *A chemical term for.* A bond formed by a shared pair of electrons with both electrons coming from the same atom.

(b) Among the compounds identify the compound that has all three bonds [ionic, covalent and coordinate bond].

A. Ammonia B. Ammonium chloride
C. Sodium hydroxide D. Calcium chloride

(c) State which is not a typical property of an ionic compound.

A. High m.p.
B. Conducts electricity in molten and the aqueous state
C. Are insoluble in water
D. Exist as oppositely charged ions even in the solid state.

(d) Compare carbon tetrachloride and sodium chloride with regard to solubility in water and electrical conductivity.

2014

(a) Compound 'X' consists of only molecules. 'X' will have —

A. Crystalline hard structure
B. A low m.p. and low b.p
C. An ionic bond
D. A strong force of attraction between its molecules

(b) The molecule which contains a triple covalent bond is :

A. ammonia B. methane
C. water D. nitrogen

(c) Give one word or phrase for the following : Formation of ions from molecules.

(d) Give a reason why covalent compounds exist as gases, liquids or soft solids.

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3 Acids, Bases and Salts

SYLLABUS

Study of Acids, Bases and Salts

(i) Simple definitions in terms of the molecules and their characteristic properties.

Self-explanatory.

(ii) Ions present in mineral acids, alkalis and salts and their solutions: use of litmus and pH paper to test for acidity and alkalinity.

Examples with equation for the ionisation/dissociation of ions of acids, bases and salts: acids form hydronium ions (only positive ions) which turn blue litmus red, alkalis form hydroxyl ions (only negative ions) with water which turns red litmus blue. Salts are formed by partial or complete replacement of the hydrogen ion of an acid by a metal, should be explained with suitable examples. Introduction to pH scale to test for acidity, neutrality and alkalinity by using; pH paper or Universal indicator.

(iii) Definition of salt; types of salts.

Types of salts: normal salts, acid salt, basic salt, definition and examples.

(iv) General properties of salts:

- *Deliquescence, Efflorescence, water of crystallization.*

Definition and example of each of the above.

- *Decomposition of hydrogen carbonates, carbonates, chlorides and nitrates by appropriate acids with heating if necessary. (relevant laboratory work must be done).*

Action of dilute acids on carbonates, hydrogen carbonates and action of concentrated acid equation of formation of acid rain (Sulphuric acid) on chlorides and nitrates, to obtain carbon dioxide, hydrogen chloride and nitric acid respectively should be taught. This will assist the students in their practical work.

(v) Preparation : laboratory preparation of salts (normal and acid salts) – relevant laboratory work is essential.

(no apparatus details are required). *Laboratory preparation of salts (normal and acid salts) : Direct combination; decomposition; displacement; double decomposition; neutralization.*

3A. ACIDS AND BASES

3.1 INTRODUCTION

Lavoisier and *Davy* studied the terms acids, bases and salts on more scientific lines. Their findings say that

— all **acids** compulsorily have hydrogen as their constituent element in the form of proton (H^+).

Examples : Hydrochloric acid, sulphuric acid and nitric acid.

— all **bases** are either metallic oxide (O^{2-}) or metallic hydroxide (OH^-) or ammonia solution.

Examples : Copper oxide, sodium hydroxide and ammonium hydroxide.

— all salts are chemical compounds. They are formed by the interaction of acid and base replacing hydrogen ions present in an acid, by metallic ion [or non-metallic ammonium ions (NH_4^+)].

Examples : Sodium chloride, calcium carbonate and ammonium phosphate.

3.2 ACIDS

The word 'acid' comes from the Latin word *acidus,* meaning 'sour'. In fact, the sour taste of some foods is due to the acids they contain.

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Some naturally occurring acids	
Acetic acid	Vinegar
Citric acid	Citrus fruits (oranges and lemons)
Butyric acid	Rancid butter
Hydrochloric acid	Gastric juice
Formic acid	Sting of bees and ants
Lactic acid	Sour milk
Maleic acid	Apples
Oleic acid	Olive oil
Oxalic acid	Tomatoes
Stearic acid	Fats
Tartaric acid	Grapes, tamarind and apples
Uric acid	Urine

Acids are defined as ***compounds which contain one or more hydrogen atoms and when dissolved in water, produce hydronium ions*** (H_3O^+) ***the only positively charged ions.*** At first, the acid molecule furnishes hydrogen ion (or proton), *i.e.*, H^+ ion in aqueous solution. But this H^+ ion cannot exist independently. Therefore, it combines with a water molecule to form hydronium ion (H_3O^+). This hydrated ion can now exist independently in solution.

Hydrogen is common in all acids.

For example : Hydrochloric acid, sulphuric acid, nitric acid and acetic acid when dissolved in water produce hydronium ions.

$$HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$$
$$H_2SO_4 + 2H_2O \rightleftharpoons 2H_3O^+ + SO_4^{2-}$$
$$HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$$
$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

The hydrated hydrogen ion that exists in the solutions of acids is called a hydronium ion.

H^+	+	H_2O	→	H_3O^+
Hydrogen ion		Water molecule		Hydronium ion (or simply H^+)

However, for simplicity, we shall use the simplified form, H^+ instead of H_3O^+.

The acidic properties of an acid are actually the properties of hydronium ions present in it.

Note : In fact, the hydronium ion, H_3O^+, can also be written as OH_3^+, since the positive charge is due to the presence of extra hydrogen ion (H^+) with water, which is neutral.

3.3 CLASSIFICATION OF ACIDS

Acids can be classified in different ways.

1. Depending on their sources

(i) Organic acids : *Acids which are obtained usually from plants are called organic acids. They contain carbon atom also alongwith hydrogen atom.*

Examples : Oxalic acid $(COOH)_2$
Acetic acid (CH_3COOH).

They are weak acids and they do not ionise completely in solution. So *they contain ions as well as molecules.*

(ii) Inorganic acids (mineral acids) **:** *Acids which are obtained usually from minerals are known as inorganic acids. They do not contain carbon* ***(except carbonic acid H_2CO_3).***

Examples : Hydrochloric acid (HCl),
Sulphuric acid (H_2SO_4),
Nitric acid (HNO_3).

They are strong acids. They ionise completely in solution producing a high concentration of hydronium ions $[H_3O]^+$.

Carbonic acid (H_2CO_3) is a weak mineral acid. It turns blue litmus pink. It is non-corrosive, and so used in soft drinks.

Acids which contain oxygen along with hydrogen and some other element are ***oxy-acids.*** *Examples :* Nitric acid (HNO_3), sulphuric acid (H_2SO_4).

Hydracids *contain hydrogen and a non-metallic element,* ***and no oxygen.*** *Examples :* Hydrochloric acid (HCl), hydrobromic acid (HBr).

2. Depending on their concentration

Concentration of an acid means the amount of acid present in a definite amount of its aqueous solution.

Concentrated acid : An acid which contains a very small amount of water or no water is called concentrated acid.

Dilute acid : An acid which contains far more

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amount of water than its own mass is known as dilute acid.

In order to dilute an acid, pour acid into water in small amounts and stir constantly. Water is not added to acid as it is an exothermic process and in this process so much heat is produced that splashing of acidic solution may occur, also the container may break which can be fatal to the person.

Note : Concentration of an acid simply tells us the amount of water present in the acid and not at all the strength of the acid. Strength of an acid is the measure of concentration of hydronium ions it produces in its aqueous solution. Thus **dil. HCl is stronger acid than highly concentrated acetic acid.**

Note : The strength of an acid depends on the degree of ionisation [α] and concentration of hydronium ions $[H_3O]^+$ produced by that acid in aqueous solution.

1. Degree of ionisation (α)

$$= \frac{\text{No. of acid molecules ionised}}{\text{(Total no. of acid molecules present in aqueous solution)}} \times 100$$

2. If the degree of ionisation (α) for an acid, base or salt in aqueous solution is greater than 30%, it is strong and if it is less than 30%, it is weak.

3. Depending on their basicity

The basicity of an acid is defined as *the number of hydronium ions $(H_3O)^+$ that can be produced by the ionisation of one molecule of that acid in aqueous solution.*

Therefore we have,

Examples : Hydrochloric acid (HCl),

(i) Monobasic acids : *Acids which on ionisation in water produce one hydronium ion per molecule of the acid are known as monobasic acids.*

For example : Hydrochloric acid

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

Other similar examples are : HBr, HNO_3, HI, CH_3COOH, *etc.*

It must be emphasized here that the basicity of an acid depends not on the number of hydrogen atoms in one molecule of that acid, but on the number of ionisable hydrogen atoms that it has per molecule.

For example, **acetic acid** (CH_3COOH or $C_2H_4O_2$) contains four hydrogen atoms in its molecule, but it is a monobasic acid because **its molecule ionises by liberating only one hydronium ion.**

$$\underset{\text{acetic acid}}{CH_3COOH} + H_2O \rightleftharpoons H_3O^+ + \underset{\text{acetate ion}}{CH_3COO^-}$$

Monobasic acids ionise in one step and so form only one normal salt.

$$NaOH + HCl \rightarrow NaCl + H_2O$$

(ii) Dibasic acids : *Acids which, on ionisation in water, produce two hydronium ions $(H_3O)^+$ per molecule of the acid, are known as dibasic acids.*

Examples : Sulphuric acid (H_2SO_4)
Oxalic acid ($H_2C_2O_4$)
Sulphurous acid (H_2SO_3)
Carbonic acid (H_2CO_3)

Dibasic acids ionise in ***two steps,*** as shown below:

(1) $H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-$
$HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}$

(2) $H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$
$HCO_3^- + H_2O \rightleftharpoons H_3O^+ + CO_3^{2-}$

Dibasic acids have two replaceable hydrogen ions, therefore they form *one acid salt or one normal salt.*

$$NaOH + H_2SO_4 \rightarrow \underset{\textbf{(acid salt)}}{NaHSO_4} + H_2O$$

$$2NaOH + H_2SO_4 \rightarrow \underset{\textbf{(normal salt)}}{Na_2SO_4} + 2H_2O$$

(iii) Tribasic acids : *Acids which, on ionisation in water, produce three hydronium ions per molecule of the acid, are called tribasic acids.*

Examples : Phosphoric acid (H_3PO_4)
Citric acid ($C_6H_8O_7$)

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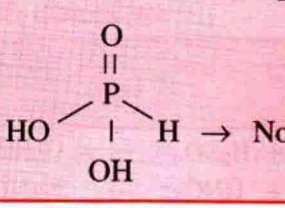
They ionise in ***three steps,*** as shown below :

(a) $H_3PO_4 + H_2O \rightleftharpoons H_3O^+ + H_2PO_4^-$

$H_2PO_4^- + H_2O \rightleftharpoons H_3O^+ + HPO_4^{2-}$

$HPO_4^{2-} + H_2O \rightleftharpoons H_3O^+ + PO_4^{3-}$

H_3PO_3 is a **dibasic acid** because in oxyacids of phosphorus, hydrogen atoms which are attached to oxygen atoms are replacable. Hydrogen atoms directly bonded to phosphorus atoms are not replaceable.



Tribasic acids can form **two** different **acid salts** and **one normal salt,** because they have three replaceable hydrogen ions.

$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$ (monosodium dihydrogen phosphate)

$2NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + 2H_2O$ (disodium hydrogen phosphate)

$3NaOH + H_3PO_4 \rightarrow Na_3PO_4 + 3H_2O$ (trisodium phosphate)

Silicic acid H_4SiO_4 has basicity four.

3.4 PREPARATION OF ACIDS

(1) *By synthesis*

Binary acids (acids containing two elements) are prepared by this method.

Hydrogen		*Non-metal*	→	*Acid*
H_2	+	Cl_2	→	2HCl
H_2	+	Br_2	→	2HBr
H_2	+	S	→	H_2S

(2) *By the action of water on non-metallic or acidic oxides*

Acidic oxides (also called acidic anhydrides) dissolve in water to give acids.

Acidic oxide	+	*Water*	→	*Acid*
SO_3	+	H_2O	→	H_2SO_4 (Sulphuric acid)
SO_2	+	H_2O	→	H_2SO_3 (Sulphurous acid)
CO_2	+	H_2O	→	H_2CO_3 (Carbonic acid)
P_2O_5	+	$3H_2O$	→	$2H_3PO_4$ (Phosphoric acid)
N_2O_5	+	H_2O	→	$2HNO_3$ (Nitric acid)

Note : (i) Only oxy-acids can be prepared by this method.

(ii) NO_2 is called mixed or double acid anhydride because two acids, nitrous acid and nitric acid, are formed when it react with water.

$$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$$

(3) *By the oxidation of non-metals*

A non-metal, *e.g.,* sulphur or phosphorus is oxidised by conc. nitric acid to form sulphuric acid or phosphoric acid respectively.

Non-metal	+	*Acid*	→	*Acid*	+	*Water*	+	*Oxide*
S	+	$6HNO_3$	→	H_2SO_4	+	$2H_2O$	+	$6NO_2$
P	+	$5HNO_3$	→	H_3PO_4	+	H_2O	+	$5NO_2$

(4) *By displacement*

Normal salts of more volatile acids are displaced by a less or non-volatile acid.

Examples : Both hydrochloric and nitric acids are formed by sulphuric acid.

Salt	+	*Less volatile acid*	→	*Salt*	+	*Acid (more volatile)*
NaCl	+	H_2SO_4	→	$NaHSO_4$	+	HCl
$NaNO_3$	+	H_2SO_4	→	$NaHSO_4$	+	HNO_3

3.5 PROPERTIES OF ACIDS

Physical properties

(1) **Taste :** Acids have a sour taste.

Mineral acids like H_2SO_4 and HNO_3 are highly corrosive in nature. Therefore, they should not be tasted.

(2) **Physical state :** Some acids are solids and some are liquids at room temperature.

For example :

Boric acid	H_3BO_3	*Solid Acids*	
Oxalic acid	$(COOH)_2$		
Tartaric acid	$C_4H_6O_6$		
Citric acid	$C_6H_8O_7$		
Phosphoric acid	H_3PO_4		
Acetic acid	CH_3COOH	*Volatile acids* vaporise easily, *i.e.,* at room temperature or on heating at about 100°C.	LIQUID ACIDS
Formic acid	HCOOH		
Carbonic acid	H_2CO_3		
Hydrochloric acid	HCl		
Nitric acid	HNO_3		
Sulphurous acid	H_2SO_3		
Sulphuric acid	H_2SO_4 (conc.)	*Non-volatile acids*	

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(3) **Effect on skin :** All strong mineral acids have corrosive action on the skin and cause painful burns. Conc. H_2SO_4 stains the skin black, conc. HNO_3 makes it yellow and conc. HCl amber colour. Carbonic acid and organic acids are not corrosive.

(4) **They change the colours of indicators.** *Indicators are complex substances that acquire separate colours in acidic and basic mediums. For example, litmus indicator is a natural colouring matter, obtained from certain lichens on dissolving in distilled water.*

The red cabbage extract (natural indicator) remains red in acidic solutions but turns green in basic solutions.

Those substances whose smell (or odour) changes in acidic or basic solutions are called OLFACTORY INDICATORS. *For example,* onion, vanila and clove oil. Onion has a characteristic smell. When a basic solution like sodium hydroxide solution is added to a cloth strip treated with onion (or onion extract), then the onion smell cannot be detected. An acidic solution does not destroy the smell of the onion.

Indicator	*Colour change in acidic medium*
Litmus	Blue to red
Methyl orange	Orange to pink
Phenolphthalein	Remains colourless

(5) They are electrolytes, *i.e.*, they conduct electricity in the aqueous state.

Chemical properties

(1) Reaction with active metals

Both dilute HCl and dilute H_2SO_4 react explosively with active metals like potassium, sodium and calcium, and moderately with less active metals like magnesium, zinc, *etc.*, producing metallic salts and hydrogen.

Active metal	+	*Acid*	→	*Salt*	+	*Hydrogen*
Mg	+	2HCl	→	$MgCl_2$	+	$H_2\uparrow$
Zn	+	2HCl	→	$ZnCl_2$	+	$H_2\uparrow$
Fe	+	H_2SO_4	→	$FeSO_4$	+	$H_2\uparrow$

Note : *Nitric acid is a very strong oxidising agent so it is not used in the preparation of hydrogen. It oxidises hydrogen and form water, thus defeats the purpose of the reaction. Only magnesium and manganese can produce hydrogen with very dilute nitric acid (1% acid).*

$$Mn + \underset{\text{(Dilute 1\%)}}{2HNO_3} \rightarrow Mn(NO_3)_2 + H_2\uparrow$$

(2) Reaction with bases – Neutralisation

Acids neutralise bases to form salt and water only.

Basic oxide/ hydroxide	+	*Acid*	→	*Salt*	+	*Water*
CuO	+	H_2SO_4	→	$CuSO_4$	+	H_2O
NaOH	+	HNO_3	→	$NaNO_3$	+	H_2O

(3) Reaction with carbonates and bicarbonates (hydrogen carbonates)

Acids liberate carbon dioxide on reaction with metallic carbonates and bicarbonates.

Carbonate/ bicarbonate	+	*Acid*	→	*Salt*	+	*Water*	+	*Carbon dioxide*
$CaCO_3$ (carbonate)	+	2HCl	→	$CaCl_2$	+	H_2O	+	$CO_2\uparrow$
$Ca(HCO_3)_2$ (bicarbonate)	+	2HCl	→	$CaCl_2$	+	$2H_2O$	+	$2CO_2\uparrow$

Note : (i) **If the salt produced is insoluble, then the reaction does not proceed. So, we do not expect lead carbonate to react with hydrochloric acid or sulphuric acid. Nor does calcium carbonate react with sulphuric acid.**

(ii) **The hydrogen carbonates of sodium and potassium** are the only common ones, which exist in the solid state.

(4) Reaction with sulphites and bisulphites

Acids react with sulphites and bisulphites of metals to liberate sulphur dioxide.

Sulphite/ bisulphite	+	*Acid*	→	*Salt*	+	*Water*	+	*Sulphur dioxide*
$CaSO_3$ (sulphite)	+	2HCl	→	$CaCl_2$	+	H_2O	+	$SO_2\uparrow$
$NaHSO_3$ (bisulphite)	+	HCl	→	NaCl	+	H_2O	+	$SO_2\uparrow$

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(5) Reaction with sulphides

Acids react with metal sulphides to liberate hydrogen sulphide.

Metal sulphide	+	*Acid*	→	*Salt*	+	*Hydrogen sulphide*
ZnS	+	2HCl	→	$ZnCl_2$	+	$H_2S\uparrow$
FeS	+	H_2SO_4	→	$FeSO_4$	+	$H_2S\uparrow$

(6) Reaction with chlorides

Chlorides do not react with any of the dilute acids. In general, chlorides react with concentrated sulphuric acid on warming to liberate hydrogen chloride.

The reaction takes place in two steps since H_2SO_4 contains two replaceable hydrogen ions.

$$NaCl + H_2SO_4\ (conc.) \xrightarrow{\text{below } 200^\circ C} NaHSO_4 + HCl\uparrow$$

$$2NaCl + H_2SO_4 \xrightarrow{\text{above } 200^\circ C} Na_2SO_4 + 2HCl$$

(7) Reaction with nitrates

Nitrates do not react with dilute acids. However, lead nitrate solution reacts with both dilute HCl and dil. H_2SO_4 to form insoluble lead salts.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 + 2HNO_3$$

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4 + 2HNO_3$$

Nitrates when heated with *conc. sulphuric acid*, produce more volatile nitric acid.

$$KNO_3 + H_2SO_4 \xrightarrow{\text{below } 200^\circ C} KHSO_4 + HNO_3$$

$$2KNO_3 + H_2SO_4 \xrightarrow{\text{above } 200^\circ C} K_2SO_4 + 2HNO_3$$

ACID RAIN

You have studied in class IX (Atmospheric Pollution) that acid rain is a by-product of a variety of human activities that releases oxides of sulphur and nitrogen in the atmosphere. Burning of fossil fuels, coal, oil, petrol and diesel produce sulphur dioxide and nitrogen oxide. They all pollute air. Polluted air also contain many oxidising agents which produces oxygen due to excessive heat. This oxygen futher combines with the oxides of sulphur and nitrogen. When it rains, oxides of sulphur and nitrogen react with rain water and form acids.

$$2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$$

$$4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$$

Sulphuric acid and nitric acid formed are washed down to the earth with water. This is **Acid rain**. The highly acidic rain water has harmful effects.

- Acid rain damages the leaves of plants and trees.
- It removes the basic nutrients such as calcium from the soil.
- It causes respiratory ailments in human beings and animals.
- It affects plant and animal life in aquatic ecosystem.
- It corrodes water pipes resulting in leaching of heavy metals, such as iron, lead and copper, into the drinking water.
- Acid rain damages buildings as it reacts with stone or metals.

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$$

3.6 GENERAL USES OF SOME ACIDS

Acid	*Use*
1. Boric acid	Eye-wash/antiseptic.
2. Citric acid	Food preservation. Vitamin C preparation.
3. Oxalic acid	Ink stain remover.
4. Carbonic acid	Flavoured drinks.
5. Tartaric acid	Baking powder.
6. Acetic acid	Table vinegar and cooking.
7. Hydrochloric acid	Cleaning of metal items
8. Benzoic acid	Preservation of food and making of perfumes and medicines.
9. Nitric acid	Explosives.
10. Phosphoric acid	Fertilizers.

Intext Questions

1. (a) What do you understand by the term, acid ?
   (b) Name the positive ion formed when an acid is dissolved in water.
   (c) Draw the structure of this ion.
2. Write the ionisation of sulphuric acid showing the formation of hydronium ion.
3. Water is never added to acid in order to dilute it. Why ?
4. Define the term 'basicity' of an acid. Give the basicity of : nitric acid, sulphuric acid and phosphoric acid.
5. Give two examples of each of the following :
   (a) oxy-acid, (b) hydracid,
   (c) monobasic acid, (d) dibasic acid,
   (e) tribasic acid.

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6. Name the :
   (a) acidic anhydride of the following acids :
   (i) sulphurous acid, (ii) nitric acid,
   (iii) phosphoric acid, (iv) carbonic acid.
   (b) acids present in vinegar, grapes and lemon.
7. What do you understand by the statement '*acetic acid is a monobasic acid*'?
8. Give a balanced equation for reaction of nitrogen dioxide with water.
9. What do you understand by the strength of an acid ? On which factor does the strength of an acid depend ?
10. Explain the following :
    (a) Carbonic acid gives an acid salt but hydrochloric acid does not.
    (b) Dil. HCl acid is stronger than highly concentrated acetic acid.
    (c) H_3PO_3 is not a tribasic acid.
    (d) Lead carbonate does not react with dilute HCl.
    (e) Nitrogen dioxide is a double acid anhydride.
11. What is acid rain ? How is it formed ?
12. How is an acid prepared from a
    (a) Non-metal (b) salt ? Give an equation for each.
13. Give equations to show how the following are made from their corresponding anhydrides.
    (a) sulphurous acid, (b) phosphoric acid,
    (c) carbonic acid, (d) sulphuric acid.
14. Name an acid used :
    (a) to flavour and preserve food,
    (b) in a drink,
    (c) to remove ink spots,
    (d) as an eyewash.

3.7 BASES

A base is either a metallic oxide or a metallic hydroxide or ammonium hydroxide which reacts with hydronium ions of an acid to form salt and water only.

For example, CuO (metallic oxide), $Mg(OH)_2$ (metallic hydroxide), etc.

Base	+	Acid	→	Salt	+	Water
CuO	+	2HCl	→	$CuCl_2$	+	H_2O
$Mg(OH)_2$	+	H_2SO_4	→	$MgSO_4$	+	$2H_2O$
$Al(OH)_3$	+	$3HNO_3$	→	$Al(NO_3)_3$	+	$3H_2O$
NH_4OH	+	HCl	→	NH_4Cl	+	H_2O

Basic oxide

A basic oxide is a metallic oxide which contains the ion O^{2-} and reacts with an acid to form salt and water only.

Note the importance of the word *only* in this definition. If it is omitted, certain compounds, which are quite different from basic metallic oxides and hydroxides, would be included in the definition of the base.

For example : Lead (IV) oxide is a metallic oxide which reacts with hydrochloric acid to produce lead (II) chloride (a salt) and water, but the word *only* excludes it from the class of bases, because chlorine is also produced.

$$PbO_2(s) + 4HCl(aq) \rightarrow PbCl_2(aq) + Cl_2(g) + 2H_2O$$

Thus Lead (IV) oxide PbO_2 is not a base.

Basic hydroxide

It is a metallic hydroxide which contains OH^- and will react with an acid to give salt and water only.

For example : Sodium hydroxide (NaOH) and Aluminium hydroxide ($Al(OH)_3$).

Basic hydroxide	+	Acid	→	Salt	+	Water
NaOH	+	HCl	→	NaCl	+	H_2O

3.8 ALKALIS

An alkali is a basic hydroxide which when dissolved in water produces hydroxyl (OH^-) ions as the only negatively charged ions.

$$NaOH\ [aq.] \rightleftharpoons Na^+ + OH^-$$

An alkali is a base soluble in water.
All alkalis furnish OH^- ions in aqueous solution

All metallic oxides and hydroxide are insoluble except of sodium, potassium, calcium (slightly soluble). Thus

The most common soluble bases *i.e.* alkalis are :

Sodium hydroxide (NaOH, *caustic soda*), *Potassium hydroxide* (KOH, *caustic potash*),

Calcium hydroxide [$Ca(OH)_2$, *slaked lime*] *and Ammonium hydroxide* (NH_4OH).

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The first two are *strong alkalis* because they are completely ionised in their aqueous solutions and the last two are *weak alkalis*, because they are feebly ionised in their aqueous solutions.

All alkalis are bases but all bases are not alkalis.
For example : Ferric hydroxide [$Fe(OH)_3$] *and Cupric hydroxide* [$Cu(OH)_2$] *are bases, but not alkalis because they are* **insoluble** in water.

3.9 CLASSIFICATION OF BASES

3.9.1 On the basis of their strength

(a) ***A strong base :*** *It undergoes almost complete dissociation in aqueous solution to produce a high concentration of* OH^-.

Examples : Sodium hydroxide (NaOH) and Potassium hydroxide (KOH).

$NaOH\ (aq) \rightleftharpoons Na^+\ (aq) + OH^-\ (aq)$

$KOH\ (aq) \rightleftharpoons K^+\ (aq) + OH^-\ (aq)$

(b) ***A weak base :*** *It undergoes only partial dissociation or ionisation in aqueous solution to produce a low concentration of* OH^- *in solution.*

Examples : Ammonium hydroxide (NH_4OH) and

Calcium hydroxide ($Ca(OH)_2$).

$NH_4OH\ (aq) \rightleftharpoons NH_4^+\ (aq) + OH^-\ (aq)$

$Ca(OH)_2\ (aq) \rightleftharpoons Ca^{2+}\ (aq) + 2OH^-\ (aq)$

2. On the basis of their acidity

The acidity of a base

The **number of hydroxyl ions [OH]⁻ which can be produced per molecule of the base in aqueous solution** or the number of hydrogen ions (of an acid) with which a molecule of that base will react to produce salt and water only is known as acidity of the base.

(a) ***Monoacidic base :*** *It is a base that dissociates in molten (fused) state or in aqueous solutions to produce one* OH^- *ion per molecule of that base.*

For example : NaOH, KOH and NH_4OH.

$NaOH \rightarrow Na^+ + OH^-$

(b) ***Diacidic base :*** *It is a base that dissociates in molten state or in aqueous solutions to produce two* OH^- *ions per molecule of that base.*

For example : $Ca(OH)_2$ and $Cu(OH)_2$.

$Ca(OH)_2\ (aq) \rightleftharpoons Ca^{2+} + 2OH^-$

$Cu(OH)_2\ (fused) \rightleftharpoons Cu^{2+} + 2OH^-$

For water-insoluble hydroxides, acidity is equal to the number of replaceable OH^- *ions in one molecule of that base.*

(c) ***Triacidic base :*** *It is a base that dissociates in aqueous solutions to produce three* OH^- *ions per molecule of that base (or contains three replaceable hydroxyl ions per molecule of the base).*

For example : $Al(OH)_3$ and $Fe(OH)_3$.

3.9.2 Preparation of bases

The common methods of preparation are :

(1) ***From metals :*** Metals when react with oxygen give bases.

Metal	+	Oxygen	→	Base
$4Na$	+	O_2	→	$2Na_2O$
$2Mg$	+	O_2	→	$2MgO$

(2) ***By the action of water on reactive metals*** like sodium, potassium and calcium.

Reactive metal	+	Water	→	Base/alkali	+	Hydrogen
$2Na$	+	$2H_2O$	→	$2NaOH$	+	$H_2\uparrow$
$2K$	+	$2H_2O$	→	$2KOH$	+	$H_2\uparrow$
Ca	+	$2H_2O$	→	$Ca(OH)_2$	+	$H_2\uparrow$

(3) ***By the action of water on soluble metallic oxides*** like oxides of sodium, potassium and calcium .

Metal oxide	+	Water	→	Base (Alkali)
Na_2O	+	H_2O	→	$2NaOH$
K_2O	+	H_2O	→	$2KOH$
CaO	+	H_2O	→	$Ca(OH)_2$

(4) ***By double decomposition :*** Aqueous solutions of salts with a base (alkali) precipitate the respective metallic hydroxide.

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Salt solution	+	Base (Alkali)	→	Basic hydroxide	+	Normal salt
$FeCl_3$	+	$3NaOH$	→	$Fe(OH)_3 \downarrow$	+	$3NaCl$
$CuSO_4$	+	$2NaOH$	→	$Cu(OH)_2 \downarrow$	+	Na_2SO_4
$AlCl_3$	+	$3NH_4OH$	→	$Al(OH)_3 \downarrow$	+	$3NH_4Cl$

(5) ***By the action of oxygen on metal sulphides :***

Metallic sulphide	+	Oxygen	$\xrightarrow{\Delta}$	Metallic oxide	+	Sulphur dioxide
$2ZnS$	+	$3O_2$	→	$2ZnO$	+	$2SO_2$
$2PbS$	+	$3O_2$	→	$2PbO$	+	$2SO_2$

(6) ***By decomposition of salts :***

(i)

Metal carbonate	$\xrightarrow{\Delta}$	Basic oxide	+	Carbon dioxide
$CaCO_3$	→	CaO	+	$CO_2\uparrow$
$CuCO_3$	→	CuO	+	$CO_2\uparrow$

Note : Sodium and potassium carbonate do not decompose on heating.

(ii)

Metal nitrate	$\xrightarrow{\Delta}$	Basic oxide	+	Nitrogen dioxide	+	Oxygen
$2Ca(NO_3)_2$	→	$2CaO$	+	$4NO_2$	+	O_2
$2Zn(NO_3)_2$	→	$2ZnO$	+	$4NO_2$	+	O_2

Note : Sodium and potassium nitrate do not give metal oxide when decomposed.

(7) ***Ammonium hydroxide*** can be prepared by simply dissolving ammonia gas in water. Ammonia is extremely soluble in water.

$NH_3 + H_2O \longrightarrow NH_4OH$ (a weak alkali)

3.9.3 Properties of bases/alkalis

Physical properties

(1) They have a *sharp and bitter taste.*

(2) *They change the colour of indicators.*

Indicator	Colour change
Litmus	Red to blue
Methyl orange	Orange to yellow
Phenolphthalein	Colourless to pink

Insoluble base do not affect indicators.

(3) They are soapy substances, *i.e.*, they are slippery to touch.

Caustic soda and caustic potash are strong alkalis. We know that alkalis react with oil to form soap. Since our skin contains oil (in the form of fat), hence when we touch caustic soda or caustic potash, a reaction takes place and soapy solutions are formed.

(4) They are strong electrolytes.

(5) They show a mild corrosive action (slight burn) on the skin.

Chemical properties

(1) Strong alkalis absorb carbon dioxide from the air to form carbonates.

Strong alkali	+	Carbon dioxide	→	Carbonate	+	Water
$2NaOH$	+	CO_2	→	Na_2CO_3	+	H_2O
$2KOH$	+	CO_2	→	K_2CO_3	+	H_2O

(2) They neutralise acids to form salt and water.

Base/Alkali	+	Acid	→	Salt	+	Water
$Ca(OH)_2$	+	$2HCl$	→	$CaCl_2$	+	$2H_2O$
$Fe(OH)_2$	+	$2HCl$	→	$FeCl_2$	+	$2H_2O$

(3) They precipitate as insoluble metallic hydroxides when added to the solutions of the salts of the heavy metals, *viz.* copper, iron, zinc, *etc.*

Metallic salt	+	Base/ alkali	→	Salt	+	Insoluble hydroxide
$CuSO_4$ (aq.)	+	$2NH_4OH$ (aq.)	→	$(NH_4)_2SO_4$ (aq.)	+	$Cu(OH)_2 \downarrow$ (pale blue)
$ZnSO_4$ (aq.)	+	$2\ NaOH$ (aq.)	→	Na_2SO_4 (aq.)	+	$Zn(OH)_2\downarrow$ (gelatinous white)
$FeSO_4$ (aq.)	+	$2NaOH$ (aq.)	→	Na_2SO_4 (aq.)	+	$Fe(OH)_2 \downarrow$ (dirty green)

Note : The hydroxides of zinc, aluminium and lead, being amphoteric by nature, dissolve in excess of NaOH or KOH, but other hydroxides do not.

$Zn(OH)_2 + 2NaOH \rightarrow \underset{\text{sodium zincate}}{Na_2ZnO_2} + 2H_2O$

(4) When alkalies are warmed with an ammonium salt, ammonia gas is given out.

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Ammonium salt	+	Alkali	$\xrightarrow{\Delta}$	Salt	+ Water	+ Ammonia
NH_4Cl	+	$NaOH$	$\xrightarrow{\Delta}$	$NaCl$	+ H_2O +	$NH_3\uparrow$
$2NH_4Cl$	+	$Ca(OH)_2$	$\xrightarrow{\Delta}$	$CaCl_2$	+ $2H_2O$ +	$2NH_3\uparrow$
$(NH_4)_2SO_4$	+	$2KOH$	$\xrightarrow{\Delta}$	K_2SO_4	+ $2H_2O$ +	$2NH_3\uparrow$

3.9.4 General uses of some bases

Base	*Use*
1. Sodium hydroxide	Manufacture of soap.
2. Potassium hydroxide	Manufacture of salts and soaps and used in batteries.
3. Calcium hydroxide (slaked lime)	Manufacture of bleaching powder and softening of hard water.
4. Magnesium hydroxide	As an antacid [neutralizes acidity].
5. Aluminium hydroxide	As a foaming agent in fire extinguishers.
6. Ammonium hydroxide	To remove grease stains from clothes.
7. Magnesia	In making refractory bricks.
8. Calcium oxide	Drying agent, basic flux.

3.10 STRENGTH OF ACIDIC AND BASIC SOLUTIONS

The acidic, basic or neutral solution can be ascertained on the basis of ionisation. Water ionises to a very small extent. The concentration of $[H_3O^+]$ (aq.) and OH^- (aq.) ions in pure water is found to be 1×10^{-7} mol litre^{-1}. This means that all aqueous solutions contain both $[H_3O^+]$ (aq.) and OH^- (aq.) ions and the product of concentration of these ions is $(1 \times 10^{-7})\ (1 \times 10^{-7}) = 1 \times 10^{-14}$ mol^2 litre^{-2} at 25°C. **This is known as ionic product of water (Kw).**

On adding acid to water the concentration of H_3O^+ ion increases and that of OH^- decreases so that Kw remains the same and if an alkali is added the reverse is noticed. So the **nature of solution depends upon the concentration of H_3O^+ ion in solution.**

A Danish biochemist (1909) devised a scale known as pH scale which represents the $[H_3O^+]$ ion concentration of the given aqueous solution.

The 'p' in pH stands for 'Potenz' in German meaning *power*, so pH means the power of hydrogen.

The pH of a solution is the negative logarithm to the base 10 of the hydrogen ion concentration expressed in moles per litre.

$$pH = \log_{10}\frac{1}{[H^+]} = -\log_{10}[H^+]$$

A solution of H^+ *ions of concentration equal to* 10^{-12} *moles/L will have* pH = 12.

i.e. $$pH = \log_{10}\frac{1}{[H^+]} = \log_{10}\frac{1}{10^{-12}}$$

$\log_{10}10^{12} = 12 \log_{10}10 = 12$. (Since $\log_{10}10 = 1$)

The normal pH scale ranges from 0 to 14, as given in Fig. 3.1.

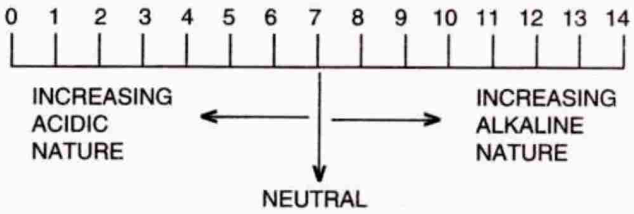

Fig. 3.1 The pH *scale*

The number 7 on the pH scale represents the neutrality of the solution. Numbers less than 7, *i.e.* pH 6, 5, 4, ... indicate that *acidic nature is increasing* as the numbers *decrease.* On the other hand, numbers greater than 7, *i.e.,* pH 8, 9, 10, ... show that *alkalinity increases* as the numbers *increase.*

In a colourless liquid, a reasonably accurate pH value can be obtained by adding a *universal indicator.*

The **universal indicator** is a mixture of indicator dyes that gives a spectrum of colours depending on how acidic or alkaline a solution is. Universal indicators gives different colours at different concentrations of hydrogen ions in a solution.

For example, a universal indicator produces green colour in a neutral solution, i.e., when pH = 7. *It changes in a basic solution progressively from blue to indigo to violet as* pH *increases progressively from 7 to 14.* The colour change in acidic solution is from yellow to pink, and then to red, as pH *progressively* decreases from 7 to 1 (Fig. 3.2).

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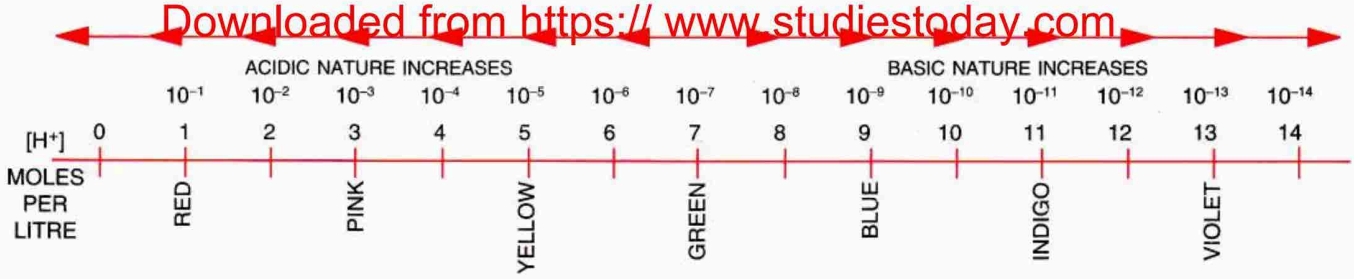

Fig. 3.2 Colour changes in the universal indicator (for different pH *values).*

Importance of pH in everyday life

1. Our body works within a narrow pH range of 7·0 to 7·8. Plants and animals also survive in a narrow range.
2. **In agriculture :** Every crop grows better in a particular pH range. *For example*, rice grows better in slightly acidic soil, sugarcane in neutral and citrus fruits in alkaline soil.
3. **In medicines :** Certain diseases are diagnosed only on the basis of the pH value of blood and urine.
4. **In digestive system :** Hydrochloric acid is produced in the stomach which helps in the digestion of food but if it becomes excess, the pH falls, and pain and irritation occurs. To get rid of this ANTACIDS like milk of magnesia ($Mg(OH)_2$) is generally used to adjust the pH.
5. In saving tooth decay : Substances like chocolates and sweets are degraded by bacteria present in our mouth. When the pH falls to 5·5 tooth decay starts. Tooth enamel (calcium phosphate) is the hardest substance in our body and it gets corroded. The saliva produced by salivary glands is slightly alkaline, it helps to increase the pH, to some extent, but tooth paste is used to neutralise excess acid in the mouth.
6. Bee sting leaves acid in the body. If baking soda, a base, is applied on the stung area, it gives relief.

Intext Questions

1. What do you understand by an alkali ?
   Give two examples of :
   (a) strong alkalis, (b) weak alkalis.
2. What is the difference between :
   (a) an alkali and a base,
   (b) the chemical nature of an aqueous solution of HCl and an aqueous solution of NH_3.
3. Name the ions furnished by :
   (a) bases in solution, (b) an acid.
4. Give one example in each case :
   (a) A basic oxide which is soluble in water,
   (b) A hydroxide which is highly soluble in water,
   (c) A basic oxide which is insoluble in water,
   (d) A hydroxide which is insoluble in water,
   (e) A weak mineral acid,
   (f) A base which is not an alkali,
   (g) An oxide which is a base,
   (h) A hydrogen containing compound which is not an acid,
   (i) A base which does not contain a metal ion.
5. You have been provided with three test tubes. One of them contains distilled water and the other two have an acidic solution and a basic solution respectively. If you are given only red litmus paper, how will you identify the contents of each test tube?
6. HCl, HNO_3, C_2H_5OH, $C_6H_{12}O_6$ all contain H atoms but only HCl and HNO_3 show acidic character. Why ?
7. (a) Dry HCl gas does not change the colour of dry litmus paper. Why ?
   (b) Is PbO_2 a base or not ? Comment.
   (c) Do basic solutions also have $H^+(aq)$? Explain why are they basic by taking an example?

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8. How would you obtain :
   (a) a base from other base,
   (b) an alkali from a base,
   (c) salt from another salt ?
9. Write balanced equations to satisfy each statement.
   (a) Acid + Active metal → Salt + Hydrogen
   (b) Acid + Base → Salt + Water
   (c) Acid + Carbonate or bicarbonate → Salt + Water + Carbon dioxide
   (d) Acid + Sulphite or bisulphite → Salt + Water + Sulphur dioxide
   (e) Acid + Sulphide → Salt + Hydrogen Sulphide
10. The skin *has and needs* natural oils. Why is it advisable to wear gloves while working with strong alkalis ?
11. Complete the table :

Indicator	Neutral	Acidic	Alkaline
Litmus	Purple	.....	.....
Phenolphthalein	Colourless	.....	.....

12. What do you understand by pH value ? Two solutions X and Y have pH values of 4 and 10 respectively. Which one of these two will give a pink colour with phenolphthalein indicator ?
13. You are supplied with five solutions : A, B, C, D and E with pH values as follows:
    A = 1·8, B = 7, C = 8·5, D = 13, and E = 5
    Classify these solutions as *neutral,* slightly or strongly *acidic* and slightly or strongly *alkaline.*
    Which solution would be most likely to liberate hydrogen with :
    (a) magnesium powder,
    (b) powdered zinc metal. Give a word equation for each reaction.
14. Distinguish between :
    (a) a common acid base indicator and a universal indicator,
    (b) acidity of bases and basicity of acids,
    (c) acid and alkali (other than indicators).
15. How does tooth enamal get damaged ? What should be done to prevent it ?
16. When you use universal indicator, you see that solutions of different acids produce different colours. Indeed, solutions of the same acid with different concentrations will also give different colours. Why ?
17. (a) A solution has a pH of 7. Explain how you would : (i) increase its pH; (ii) decrease its pH;
    (b) If a solution changes the colour of litmus from red to blue, what can you say about its pH ?
    (c) What can you say about the pH of a solution, that liberates carbon dioxide from sodium carbonate ?
18. Solution P has a pH of 13, solution Q has a pH of 6 and solution R has a pH of 2.
    Which solution :
    (a) will liberate ammonia from ammonium sulphate on heating ?
    (b) is a strong acid ?
    (c) contains molecules as well as ions ?

3B. SALTS AND THEIR LABORATORY PREPARATIONS

3.11 SALT

Salt is a compound formed by the partial or total replacement of the ionisable hydrogen atoms of an acid by a metallic ion or an ammonium ion.

Partial replacement

$$NaNO_3 + H_2SO_4 \xrightarrow{<200°C} NaHSO_4 + HNO_3$$
(conc.) — salt

Complete replacement

$$2NaNO_3 + H_2SO_4 \xrightarrow{>200°C} Na_2SO_4 + 2HNO_3$$
(conc.) — salt

Ionic definition : Salt is an ionic compound, which dissociates in water to yield a positive ion other than hydrogen ion (H^+) and a negative ion other than hydroxyl ion (OH^-).

For example, sodium chloride is a salt consisting of Na^+ and Cl^-.

$$NaCl \xrightarrow{Water} Na^+ + Cl^-$$
positive ion but not hydrogen — negative ion but not hydroxyl

It is prepared by neutralising sodium hydroxide with hydrochloric acid.

$$NaOH + HCl \rightarrow NaCl + H_2O$$
(base) (acid) (salt) (water)

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3.11.1 Classification of salts

There are *six* kinds of salts.

1. Normal salts
2. Acid salts
3. Basic salts
4. Double salts
5. Mixed salts
6. Complex salts

1. Normal salts : *Normal salts are the salts formed by the complete replacement of the ionisable hydrogen atoms of an acid by a metallic or an ammonium ion.*

For example :

Hydrochloric acid yields sodium chloride.

Sulphuric acid gives sodium sulphate.

Acid +	Base	→	Salt	+ Water
HCl +	$NaOH$	→	$NaCl$	+ H_2O
H_2SO_4 +	$2NaOH$	→	Na_2SO_4	+ $2H_2O$

Normal salts have no ionisable hydrogen atoms.

2. Acid salts : *Acid salts are formed by the partial replacement of the ionisable hydrogen atoms of a polybasic acid by a metal or an ammonium ion.*

For example :

Dibasic sulphuric acid yields sodium hydrogen sulphate ($NaHSO_4$).

$$H_2SO_4 + NaCl \xrightarrow{\text{below } 200^\circ C} NaHSO_4 + HCl$$

OR

$$H_2SO_4 + NaOH \rightarrow \underset{\text{Acid Salt}}{NaHSO_4} + H_2O$$

Other examples of acid salts are :

$NaHSO_3$, Na_2HPO_4 and NaH_2PO_4.

Acid salts are usually formed when the amount of *base taken is insufficient* for the complete neutralisation of the acid.

Acid salts ionise in water solution to give hydronium ions, and therefore, they **show all the properties of an acid.**

3. Basic salts : *Basic salts are formed by the partial replacement of the hydroxyl group of a di- or a tri- acidic base by an acid radical.*

Examples :

Basic lead chloride [$Pb(OH)Cl$],

Basic Magnesium chloride [$Mg(OH)Cl$],

Basic copper chloride [$Cu(OH)Cl$], and

Basic copper nitrate [$Cu(OH)NO_3$]

$$\underset{\text{(diacidic base)}}{Pb(OH)_2} + HCl \rightarrow \underset{\text{(basic salt)}}{Pb(OH)Cl} + H_2O$$

$$Mg(OH)_2 + HCl \rightarrow Mg(OH)Cl + H_2O$$

The basic salt contains a metallic cation, a hydroxyl ion of the base and an anion of the acid.

***4. Double salts :** *Double salts are formed by the union of two simple salts that dissolve in water and crystallise.*

Examples :

Potash alum : $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

Mohr's salt : $FeSO_4.(NH_4)_2SO_4.6H_2O$

Dolomite : $CaCO_3.MgCO_3$

$$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \rightarrow K_2SO_4Al_2(SO_4)_3{\cdot}24H_2O$$

***5. Mixed salts :** *Mixed salts are those salts that contain more than one basic or acid radical.*

Examples :

Sodium potassium carbonate : $NaKCO_3$;
Bleaching powder : $CaOCl_2$

***6. Complex salts :** *Complex salts are those which on dissociation give one simple ion and one complex ion.*

Complex salt	⇌	Simple ion +	Complex ion
$Na[Ag(CN)_2]$	⇌	Na^+ +	$[Ag(CN)_2]^-$

Sodium argentocyanide

Other examples are :

Silver amino chloride : $[Ag(NH_3)_2]Cl$

Tetrammine copper (II) sulphate : $[Cu(NH_3)_4]SO_4$

3.11.2 General properties of salts

(i) Salts are electrovalent compounds. They conduct electricity in their molten state as well as in their aqueous solutions [due to their dissociation into ions].

** Not in syllabus*

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(ii) Salts are non-volatile solids that form crystals.

(iii) *Solubility of salts* : Most of the salts are soluble in water. Their degree of solubility in water varies with temperature (Fig. 3.3).

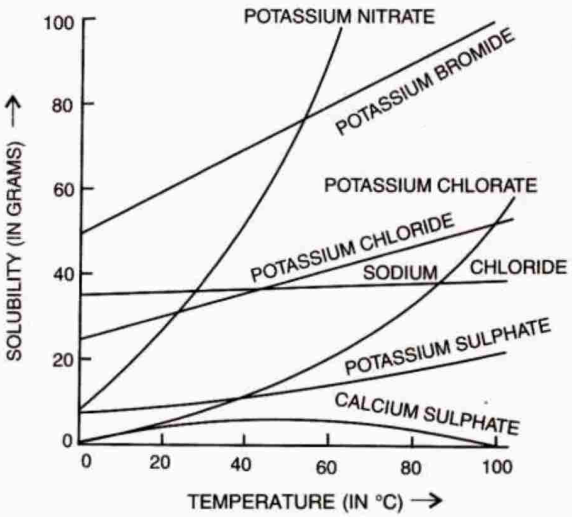

Fig. 3.3 Variation in solubility of some salts with change in temperature

If at least 1g of a substance can be dissolved in 100 mL of water at 298 K, it is called a **soluble salt**.

If 0·1 g to 1g of a substance can be dissolved in 100 mL water at 298 K, it is called a partially **soluble salt**.

If less than 0·1 g of a substance can be dissolved in 100 mL water at 298 K, it is called an **insoluble salt**.

Table 3.1 : Solubility of salts

Salts	*Exceptions*
1. All the compounds of ammonium, sodium and potassium are *soluble* in water.	———
2. All nitrates and nitrites are *soluble.*	———
3. All chlorides, bromides and iodides are *soluble* in water.	Hg_2Cl_2; AgCl; $PbCl_2$ ($PbCl_2$ is soluble in hot water, not in cold water).
4. All sulphates are *soluble.*	$CaSO_4$; $PbSO_4$; $BaSO_4$.
5. All **carbonates sulphides, sulphites** and **phosphates** are insoluble.	Except of ammonium, sodium and potassium.

***Note :* All metallic oxides and hydroxides are insoluble except of sodium, potassium and ammonium. Calcium hydroxide is slightly soluble.**

$NaHCO_3$ and $KHCO_3$ are sparingly soluble.

(iv) *Hydrolysis of salts :*

The phenomenon due to which salt formed by a weak acid and a strong base, or by a strong acid and a weak base, reacts with water to give an acidic or an alkaline solution, is known as ***hydrolysis.***

(a) *Salts formed from strong bases* like NaOH and KOH and weak acids like H_2CO_3, CH_3CO_2H, *etc.*, ***hydrolyse in water to give alkaline solutions. They have pH more than 7.***

Examples : Sodium carbonate (Na_2CO_3), potassium acetate (CH_3COOK),

$$Na_2CO_3 + 2H_2O \rightleftharpoons \underset{\text{weak acid}}{H_2CO_3} + \underset{\text{strong alkali}}{2NaOH}$$

$$CH_3COOK + H_2O \rightleftharpoons \underset{\text{weak acid}}{CH_3COOH} + \underset{\text{strong alkali}}{KOH}$$

(b) *The salts of strong acids and weak bases hydrolyse in water to give an* ***acidic solution*** (*p*H less than 7).

Examples :

Iron (III) chloride ($FeCl_3$),

Copper sulphate ($CuSO_4$),

Aluminium chloride ($AlCl_3$), and

Ammonium sulphate [$(NH_4)_2SO_4$].

$$FeCl_3 + 3H_2O \rightleftharpoons \underset{\text{strong acid}}{3HCl} + \underset{\text{weak base}}{Fe(OH)_3}$$

$$CuSO_4 + 2H_2O \rightleftharpoons \underset{\text{strong acid}}{H_2SO_4} + \underset{\text{weak base}}{Cu(OH)_2}$$

(c) *Salts derived from strong acids* and *strong bases, such as* NaCl, Na_2SO_4, K_2SO_4 and KNO_3 gives a ***neutral solution*** in water.

(d) Salts formed from weak bases like ammonium hydroxide and weak acids like acetic acid may be acidic, alkaline or neutral depending upon the dissociated ions and undissociated molecules.

Examples :

Ammonium acetate [CH_3COONH_4] and

Ammonium carbonate [$(NH_4)_2\ CO_3$].

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3.11.3 Preparation of soluble salts

METHOD	REACTION
1. **DIRECT COMBINATION** (*i.e.*, by heating two elements together)	Metal + Non metal → Salt 2Na(molten) + Cl_2 → 2NaCl 2Fe + $3Cl_2$ → $2FeCl_3$
2. **SIMPLE DISPLACEMENT*** (Action of dilute acids on active metals)	Active metal + Acid → Salt + Hydrogen Zn + H_2SO_4 → $ZnSO_4$ + $H_2\uparrow$ Fe + 2HCl → $FeCl_2$ + $H_2\uparrow$
3. **DECOMPOSITION**** (a) Decomposition of **bicarbonates** by acids.	(i) $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2\uparrow$ sodium bicarbonate — salt (ii) $Ca(HCO_3)_2 + 2HNO_3 \rightarrow Ca(NO_3)_2 + 2H_2O + 2CO_2$ calcium bicarbonate — salt
(b) Decomposition of **carbonates** by acids.	(i) $CuCO_3 + 2HCl \rightarrow CuCl_2 + H_2O + CO_2\uparrow$ copper carbonate — salt (ii) $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + H_2O + CO_2\uparrow$ magnesium carbonate — salt
(c) Decomposition of **chlorides** by conc. H_2SO_4.	(i) $NaCl(s) + H_2SO_4(l) \xrightarrow{\text{below } 200^\circ C} NaHSO_4(aq) + HCl(g)$ sodium chloride; (conc.); sodium bisulphate (acid salt) (ii) $2NaCl(s) + H_2SO_4(l) \xrightarrow{\text{above } 200^\circ C} Na_2SO_4(aq) + 2HCl(g)$ sodium chloride; sulphuric acid (conc.); sodium sulphate; hydrogen chloride gas (iii) $ZnCl_2(s) + H_2SO_4(l) \rightarrow ZnSO_4(aq) + 2HCl(g)$ zinc chloride; zinc sulphate
(d) Decomposition of **nitrates** by conc. H_2SO_4	$KNO_3 + H_2SO_4 \xrightarrow{\text{below } 200^\circ C} KHSO_4 + HNO_3$ acid salt
NEUTRALISATION 4. (i) **Neutralising** an acid with a soluble base is also known as **titration**. (ii) Neutralising an acid with an insoluble base.	$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$ $HCl + NaOH \rightarrow NaCl + H_2O$ $2HCl + MgO \rightarrow MgCl_2 + H_2O$
5. By the action of Alkalis (NaOH or KOH) on certain metals (Al, Zn) their oxides and their hydroxides.	$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$ $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$ $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ $Al(OH)_3 + NaOH \rightarrow NaAlO_2 + 2H_2O$ $ZnO + 2KOH \rightarrow K_2ZnO_2 + 2H_2O$ $Zn(OH)_2 + 2KOH \rightarrow K_2ZnO_2 + 2H_2O$

***Note* : Soluble salts are prepared in solution only.**
Soluble salts are obtained by the evaporation of water, followed by crystallisation.
Some soluble salts, like zincates and aluminates, are found only in solution. These salts cannot be obtained in their pure forms by crystallisation.

* Displacement is a chemical change in which a more active element displaces a less active element from its salt solution
** Decomposition is a chemical change in which a compound breaks up into its elements or simpler compounds

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3.11.4 Preparation of insoluble salts

1. By direct combination

Reaction :

(i) $Pb + S \xrightarrow{\Delta} PbS$ (ii) $Fe + S \xrightarrow{\Delta} FeS$

2. By the combination of an acidic oxide with a basic oxide.

Reaction :

	Acidic oxide	+	Basic oxide	→	Salt
(i)	SO_2	+	CaO	$\xrightarrow{\Delta}$	$CaSO_3$
(ii)	CO_2	+	CaO	$\xrightarrow{\Delta}$	$CaCO_3$

3. By the precipitation reaction (DOUBLE DECOMPOSITION)*

Reaction :

	Soluble salt	+	Acid or soluble salt	→	Insoluble salt	+	Acid or salt
(i)	$BaCl_2$	+	H_2SO_4	→	$BaSO_4\downarrow$	+	2HCl
(ii)	$AgNO_3$	+	HCl	→	$AgCl\downarrow$	+	HNO_3
(iii)	$Pb(NO_3)_2$	+	Na_2CO_3	→	$PbCO_3\downarrow$	+	$2NaNO_3$
(iv)	Na_2SO_4	+	$BaCl_2$	→	$BaSO_4\downarrow$	+	2NaCl
(v)	$CuSO_4$	+	H_2S	→	$CuS\downarrow$	+	H_2SO_4
(vi)	$Pb(NO_3)_2$	+	2HCl	→	$PbCl_2\downarrow$	+	$2HNO_3$

***Note :** An insoluble salt can also be obtained from another insoluble salt, by double decomposition. Insoluble salt is first to be converted into a soluble salt, which can then be used for preparing the desired salt.*

For example :

(i) Insoluble **lead sulphate** is prepared from insoluble lead carbonate or insoluble lead oxide, by first converting either of them into soluble lead nitrate (with the help of dilute nitric acid).

$$PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O$$
$$PbCO_3 + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O + CO_2\uparrow$$

The resulting solution is then treated with sulphuric acid or sodium sulphate solution and a **white precipitate** of lead sulphate is obtained.

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4\downarrow + 2HNO_3$$
$$Pb(NO_3)_2 + Na_2SO_4 \rightarrow PbSO_4\downarrow + 2NaNO_3$$

Adding dilute H_2SO_4 to $PbCO_3$ in order to prepare $PbSO_4$ is an **impractical method** because an insoluble crust of lead sulphate is formed on lead carbonate which prevents further reaction.

$PbCl_2$ is prepared from insoluble lead oxide or lead carbonate by the same method.

(ii) ***Silver chloride and barium sulphate***

silver chloride is usually prepared by reacting of aqueous silver nitrate with dilute hydrochloric acid and precipitate of silver chloride is obtained.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

Barium sulphate is prepared by reacting aqueous barium chloride with dilute sulphuric acid.

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HCl$$
white precipitate

3.12 LABORATORY PREPARATION OF SOME SALTS (NORMAL AND ACID SALTS)

1. Iron (III) chloride [$FeCl_3$]

Method of preparation : Synthesis, *i.e., direct combination* of elements.

Procedure (Fig. 3.4) :

Powdered iron is taken in a combustion tube. Dry chlorine gas (obtained by passing the gas through concentrated sulphuric acid) is passed through the combustion tube. The air inside the apparatus is expelled.

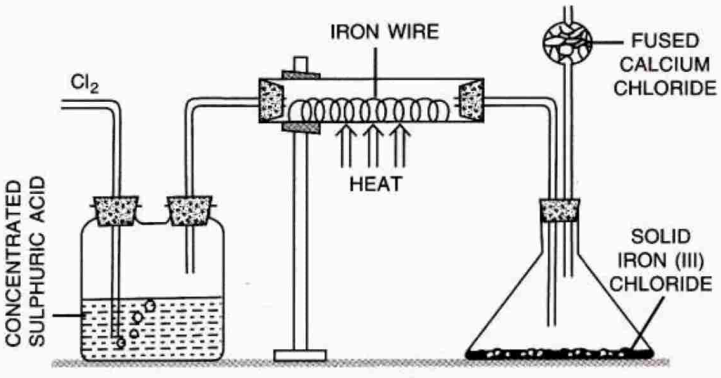

Fig. 3.4. Preparation of Iron (III) chloride.

The iron turns red hot since the reaction is exothermic. Now, external heating is suspended.

Iron (III) chloride is formed, which volatilizes and condenses in the receiver as **brown scale**.

**A chemical change, in which two compounds in solution react to form two other compounds by the mutual exchange of radicals. A solid precepitate is formed as a result of the reaction.*

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Reaction :

$$2Fe + 3Cl_2 \rightarrow 2FeCl_3$$
(heated) (dry)

Iron (III) chloride is highly deliquescent, so it is kept dry with the help of fused calcium chloride (drying agent).

Note : Anhydrous $FeCl_3$ cannot be prepared by simply heating the hydrated ferric chloride $FeCl_3 \cdot 6H_2O$, because on heating, $FeCl_3 \cdot 6H_2O$ produces $Fe_2O_3 \cdot H_2O$ and HCl.

$$2[FeCl_3 \cdot 6H_2O] \xrightarrow{\Delta} Fe_2O_3 + 9H_2O + 6HCl$$

Anhydrous Iron (III) chloride is used as a catalyst in certain reactions.

2. Copper (II) sulphate (or blue vitriol)

Method of preparation : By the action of dilute acid (dil H_2SO_4) on an insoluble base copper hydroxide, copper oxide or copper carbonate.

Reaction :

$$Cu(OH)_2 + H_2SO_4 \rightarrow CuSO_4 + 2H_2O$$
$$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$$
$$CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2\uparrow$$
$$CuSO_4 + 5H_2O \rightarrow CuSO_4.5H_2O$$
(blue vitriol)

Procedure :

Step 1. Take dilute sulphuric acid in a beaker and heat it on a wire gauze.

Step 2. Add copper hydroxide or black cupric oxide or copper carbonate, in small quantities at a time and keep on stirring the solution till no more of it dissolves and the excess compound settles to the bottom.

Step 3. Filter it hot, and collect the filtrate in a china dish. Evaporate the filtrate by heating to the point of crystallisation, and then allow it to cool.

Step 4. *Collect the bright blue crystals of copper* (II) *sulphate penta-hydrate* $CuSO_4.5H_2O$ *(blue vitriol), and dry the crystals.*

Note : Other salts which can be prepared by the same method are : zinc sulphate, lead nitrate, calcium chloride, calcium nitrate and magnesium sulphate.

3. Zinc sulphate and Iron (II) sulphate

Method of preparation : By the action of dilute acid (dil H_2SO_4) on an active metal (Zinc, Iron).

(a) Zinc sulphate (or white vitriol)

Reaction :

$$Zn(s) + H_2SO_4\,(aq) \rightarrow ZnSO_4 + H_2\uparrow$$
$$ZnSO_4 + 7H_2O \rightarrow ZnSO_4 \cdot 7H_2O$$
(white vitriol)

Procedure :

Step 1. Take dilute sulphuric acid (1 volume of acid : 5 volumes of water) in a beaker and heat it on a wire gauze. Add some granulated zinc pieces, a little at a time, with constant stirring.

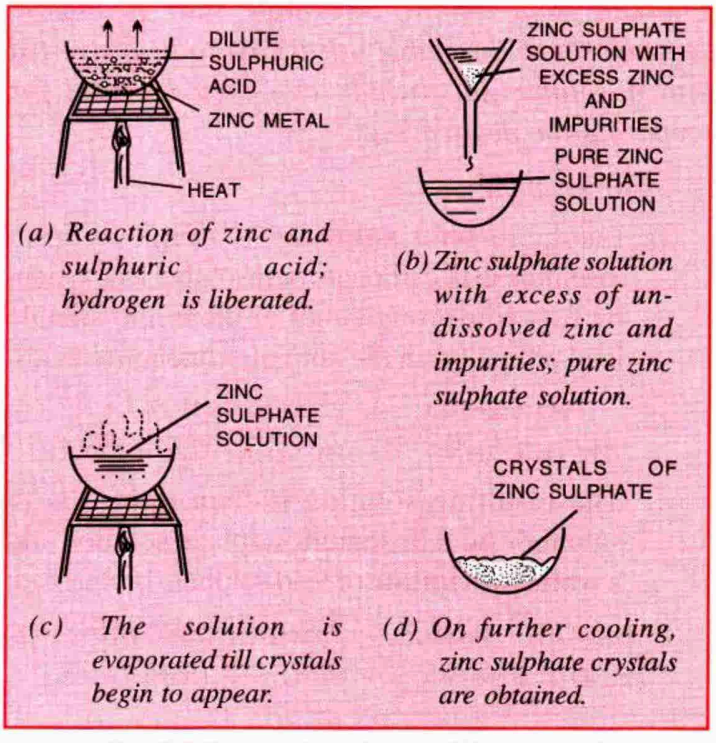
Step 2. Effervescence takes place because of the liberation of hydrogen gas.

Add zinc till the metal settles at the base of the beaker.

When effervescence stops, it indicates that all the acid has been used up.

Step 3. The excess of zinc is filtered off.

Step 4. Collect the solution in a china dish. Evaporate the solution to get crystals. Filter, wash them with water and dry them between the folds of a filter paper.



(a) Reaction of zinc and sulphuric acid; hydrogen is liberated.

(b) Zinc sulphate solution with excess of un-dissolved zinc and impurities; pure zinc sulphate solution.

(c) The solution is evaporated till crystals begin to appear.

(d) On further cooling, zinc sulphate crystals are obtained.

Fig. 3.5 Preparation of zinc sulphate crystals from zinc and sulphuric acid.

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The white, needle-shaped crystals are of hydrated zinc sulphate, commonly known as ***white vitriol***.

(b) Iron (II) sulphate (or green vitriol)

Iron (II) sulphate is prepared by the same method as white vitriol.

Reaction :

$$Fe(s) + H_2SO_4 \rightarrow FeSO_4 + H_2\uparrow$$
$$FeSO_4 + 7H_2O \rightarrow FeSO_4.7H_2O$$
(green vitriol)

Filter the hot solution formed and allow the filtrate to evaporate at room temperature. Pale green crystals of hydrated iron (II) sulphate are formed.

4. Lead Chloride ($PbCl_2$) and Calcium Carbonate ($CaCO_3$)

Method of preparation : **Precipitation or double decomposition**

An insoluble salt can be prepared by mixing the solutions of two appropriate salts, each containing one of its ions. This mixing results in the formation of an insoluble solid precipitate, which can be filtered, washed and dried.

(a) Lead chloride ($PbCl_2$)

It is prepared by adding either dilute hydrochloric acid or sodium chloride solution to a solution of lead nitrate (Fig 3.6).

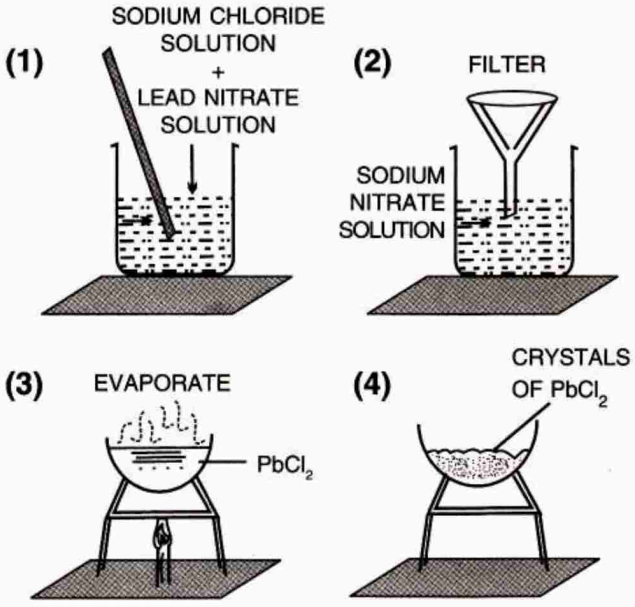

Fig. 3.6 Preparation of lead chloride

Reaction :

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2\downarrow + 2HNO_3$$
or
$$Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2\downarrow + 2NaNO_3$$

The solution of lead nitrate and sodium chloride is then filtered. The heavy, white precipitate of lead chloride accumulate as residue on the filter paper. It is then washed repeatedly with cold water. The precipitate is taken in a china dish with some water and is heated till the precipitate of lead chloride dissolves. The solution is then cooled.

Pure, needle shaped crystals of lead chloride are thus obtained.

Note : lead chloride dissolves in hot water.

The chlorides of Pb, Ag and Hg (ous), and the sulphates of Ba, Pb and Ca are also prepared using this method.

(b) Calcium Carbonate ($CaCO_3$)

It is prepared by adding sodium carbonate solution to a hot solution of calcium chloride in a beaker, till the former is in excess.

Reaction :

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$$

The white precipitate formed by the interchange of radicals is filtered and washed repeatedly with cold water and collected. The washed precipitate is dried to obtain an amorphous powder, *i.e.*, calcium carbonate.

The carbonates of all the metals (except Na, K *and* NH_4^+*) are prepared by the above method.*

5. Preparation of an acid salt sodium bicarbonate ($NaHCO_3$)

Principle :

By passing carbon dioxide gas into a cold solution of sodium carbonate.

Reaction :

$$Na_2CO_3 + CO_2 + H_2O \rightarrow 2NaHCO_3$$

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Procedure :

Step 1. Dissolve nearly 5 grams of anhydrous sodium carbonate in about 25 cm^3 of distilled water in a flask.

Step 2. Cool the solution by keeping the flask in a freezing mixture.

Step 3. Pass carbon dioxide gas in the solution.

Step 4. Crystals of sodium bicarbonate will precipitate out after sometime.

Step 5. Filter the crystals and dry it in the folds of filter paper.

The procedure is shown in Fig. 3.7.

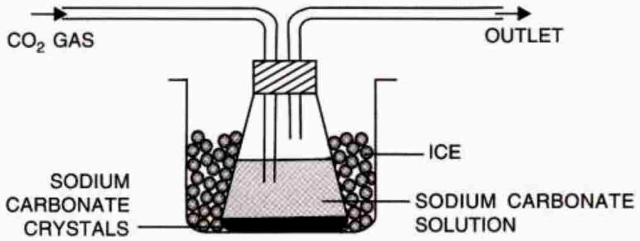

Fig. 3.7 Preparation of sodium bicarbonate $NaHCO_3$.

Note : Sodium hydrogen carbonate is used in **health salts**. Health salts are the mixture of dry citric acid or tartaric acid and sodium hydrogen carbonate. On adding water, acid reacts with $NaHCO_3$ to liberate CO_2 which forms an effervescence drink and releases gas pressure in the stomach.

6. Neutralisation

Neutralisation is the process by which H^+ ions of an acid react completely with the $[OH^-]$ ions of a base to give salt and water only.

For example, in aqueous solutions :

Base + Acid → Salt + Water

$$NaOH + HCl \rightarrow NaCl + H_2O$$

or $$Na^+ + OH^- + H^+ + Cl^- \rightarrow Na^+ + Cl^- + H_2O$$

Na^+ and Cl^- are present in the reactants as well as in the principal product. Thus, by cancelling out the ions present in the reactants as well as in one of the products, we get the following equation :

$$H^+ (aq) + OH^- (aq) \rightleftharpoons H_2O\ (l)$$

OR

$$H_3O^+ (aq) + OH^- (aq) \rightleftharpoons 2H_2O\ (l)$$

Similarly,

$$2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$$

or $$2K^+ + 2OH^- + 2H^+ + SO_4^{2-} \rightarrow 2K^+ + SO_4^{2-} + 2H_2O$$

Heat of neutralisation : *It is the amount of heat liberated when one gram equivalent of an acid or a base is completely neutralised.*

7. Laboratory preparation of sodium sulphate crystals ($Na_2SO_4.10H_2O$)

Principle :

Neutralisation of caustic soda with dilute sulphuric acid.

Reaction :

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

This method is useful for preparing the soluble salts of only sodium, potassium and ammonium.

Since the reactants as well as the products are soluble, a *titration* is conducted to determine the completion of the neutralisation reaction (Fig. 3.8).

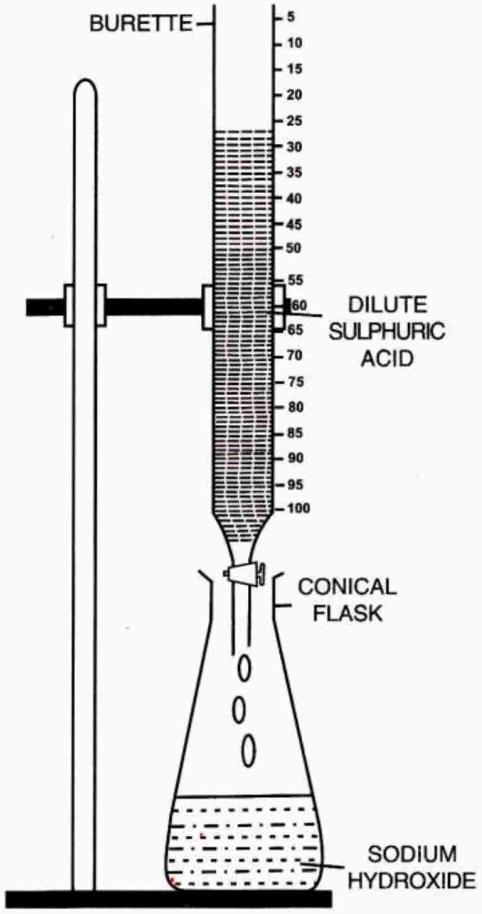

Fig. 3.8 Neutralization of an acid with a base.

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Procedure :

Step 1. 25 cm^3 of sodium hydroxide solution is measured and transferred to a conical flask (see Fig. 3.8). A drop of indicator **phenolphthalein** is added. The solution will turn pink.

Step 2. A burette is filled with dilute sulphuric acid, and the reading on the burette is noted (Zero cc). The acid is run in drops into the conical flask (alkali) till the last drop of the acid turns the solution colourless. This is known as the end point.

Step 3. Note the burette reading at the end point. The difference between the two readings is the volume of the acid required.

The acid and the alkali (in the same ratio as found above) are now mixed in an evaporating basin. The resulting solution contains sodium sulphate and water.

On evaporating this solution, crystals of $Na_2SO_4.10H_2O$ are obtained.

Note :

(i) In place of sodium hydroxide, sodium carbonate can be used in this reaction. However, methyl orange is then used as the indicator.

(ii) Since sodium hydroxide and sulphuric acid are both soluble, an excess of either of them cannot be removed by filtration. Therefore, it is necessary to find out on a small scale, the ratio of the solutions of the two reactants required for complete neutralisation before preparation.

In order to prepare acid salt $NaHSO_4$ add the same volume of sulphuric acid again to the product formed in preparing the normal salt Na_2SO_4. Crystals of sodium hydrogen sulphate are obtained from the resulting solution. *i.e.* for a given amount of alkali, the addition of twice the normal amount of acid will result in the formation of the **acid salt.**

$$2NaOH + 2H_2SO_4 \rightarrow 2NaHSO_4 + 2H_2O$$

or

$$NaOH + H_2SO_4 \rightarrow NaHSO_4 + H_2O$$

Intext Questions

1. Define the following and give two examples in each case : (a) a normal salt, (b) an acid salt, (c) a mixed salt.
2. Answer the following questions related to salts and their preparations :
   (a) What is a 'salt' ?
   (b) What kind of salt is prepared by precipitation?
   (c) Name a salt prepared by direct combination. Write an equation for the reaction that takes place in preparing the salt you have named.
   (d) Name the procedure used to prepare a sodium salt such as sodium sulphate.
3. Describe giving all practical details, how would you prepare :
   (a) copper sulphate crystals from a mixture of charcoal and black copper oxide,
   (b) zinc sulphate crystals from zinc dust (powdered zinc and zinc oxide),
   (c) sodium hydrogen carbonate crystals.
   (d) Calcium sulphate from calcium carbonate,
4. The following is a list of methods for the preparation of salts.
   A – direct combination of two elements.
   B – reaction of a dilute acid with a metal.
   C – reaction of a dilute acid with an insoluble base.
   D – titration of a dilute acid with a solution of soluble base.
   E – reaction of two solutions of salts to form a precipitate.
   Choose from the above list A to E, the best method of preparing the following salts by giving a suitable equation in each case :
   1. Anhydrous ferric chloride, 2. Lead chloride,
   3. Sodium sulphate, 4. Copper sulphate.
5. Name :
   (a) a chloride which is insoluble in cold water but dissolves in hot water,
   (b) a chloride which is insoluble,
   (c) two sulphates which are insoluble,
   (d) a basic salt, (e) an acidic salt,
   (f) a mixed salt, (g) a complex salt,
   (h) a double salt,
   (i) salt whose solubility increases with temperature,
   (j) a salt whose solubility decreases with temperature.

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6. Fill in the blanks with suitable words :

An acid is a compound which when dissolved in water forms hydronium ions as the only ..... ions. A base is a compound which is soluble in water contains ..... ions. A base reacts with an acid to form a ..... and water only. This type of reaction is known as .....

7. What would you observe when :
   (a) blue litmus is introduced into a solution of ferric chloride,
   (b) red litmus paper is introduced into a solution of sodium sulphate,
   (c) red litmus paper is introduced in sodium carbonate solution ?

8. Explain why :
   (a) it is necessary to find out the ratio of reactants required in the preparation of sodium sulphate,
   (b) fused calcium chloride is used in the preparation of $FeCl_3$?
   (c) Anhydrous $FeCl_3$ cannot be prepared by heating hydrated Iron (III) chloride.

9. Give the preparation of the salt shown in the left column by matching with the methods given in the right column. Write a balanced equation for each preparation.

Salt	***Method of preparation***
Zinc sulphate	Precipitation
Ferrous sulphide	Oxidation
Barium sulphate	Displacement
Ferric sulphate	Neutralisation
Sodium sulphate	Synthesis.

10. (a) Give the pH value of pure water. Does it change if common salt is added to it ?
    (b) Classify the following solutions as acids, bases or salts.
    Ammonium hydroxide, barium chloride, sodium chloride, sodium hydroxide, H_2SO_4 and HNO_3.

11. Define the term neutralisation :
    (a) Give a reaction, mentioning clearly acid and base used in the reaction,
    (b) If one mole of a strong acid reacts with one mole of a strong base, the heat produced is always the same. Why ?

12. Write the balanced equations for the preparation of the following salts in the laboratory :
    (a) A soluble sulphate by the action of an acid on an insoluble base,
    (b) An insoluble salt by the action of an acid on another salt,
    (c) An insoluble base by the action of a soluble base on a soluble salt,
    (d) A soluble sulphate by the action of an acid on a metal.

13. You are provided with the following chemicals :
    $NaOH$, Na_2CO_3, H_2O, $Zn(OH)_2$, CO_2, HCl,
    Fe, H_2SO_4, Cl_2 , Zn.
    Using the suitable chemicals from the given list only, state briefly how you would prepare :
    (a) iron (III) chloride, (b) sodium sulphate,
    (c) sodium zincate, (d) iron (II) sulphate,
    (e) sodium chloride ?

14. For each of the salt : A, B, C and D, suggest a suitable method of its preparation.
    (a) A is a sodium salt.
    (b) B is an insoluble salt.
    (c) C is a soluble salt of copper.
    (d) D is a soluble salt of zinc.

15. Choosing only substances from the list given in the box below, write equations for the reactions which you would use in the laboratory to obtain :
    (a) Sodium sulphate, (b) Copper sulphate,
    (c) Iron(II) sulphate, (d) Zinc carbonate,

Dilute sulphuric acid	Copper	Copper carbonate
	Iron	Sodium carbonate
	Sodium	
	Zinc	

16. From the formula listed below, choose one, in each case, corresponding to the salt having the given description :-
    $AgCl$, $CuCO_3$, $CuSO_4 \cdot 5H_2O$, KNO_3, $NaCl$, $NaHSO_4$, $Pb(NO_3)_2$. $ZnCO_3$, $ZnSO_4 \cdot 7H_2O$.
    (a) an acid salt.
    (b) an insoluble chloride.
    (c) on treating with concentrated sulphuric acid, this salt changes from blue to white.
    (d) on heating, this salt changes from green to black.
    (e) this salt gives nitrogen dioxide on heating.

17. (a) $Ca(H_2PO_4)_2$ is an example of a compound called _____ (acid salt / basic salt / normal salt).
    (b) Write the balanced equation for the reaction of :
    A named acid and a named alkali.

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16. State the terms defined by the following sentences :
    (a) A soluble base.
    (b) The insoluble solid formed when two solutions are mixed together.
    (c) An acidic solution in which there is only partial ionization of the solute molecules.

17. Which of the following methods, A, B, C, D or E is generally used for preparing the chlorides listed below from (i) to (v). Answer by writing down the chloride and the letter pertaining to the corresponding method. Each letter is to be used only once.

    A Action of an acid on a metal,

    B Action of an acid on an oxide or carbonate,

    C Direct combination,

    D Neutralization of an alkali by an acid

    E Precipitation (double decomposition)

    (i) Copper (II) chloride

    (ii) Iron (II) chloride

    (iii) Iron (III) chloride

    (iv) Lead (II) chloride

    (v) Sodium chloride

18. Complete te following table :

Reactants	Products	Method
Soluble base + Acid (dil)	Salt + water	Neutralisation Titration
Metal + Non-metal	Salt (soluble/insoluble)	.....
Insoluble base + ...	Salt (soluble) + water	.....
Active metal + Acid (dil)	..... + ...	.....
Soluble salt solution (A) + Soluble salt solution (B)	Precipitated salt + Soluble salt	.....
Carbonate/bicarbonate + Acid (dil)	Salt + ... + ...	Decompostion of carbonate
Chlorides/nitrates + Acid (conc.)	..... + ...	Decompostion of chlorides and nitrates

3C. WATER OF CRYSTALLISATION; EFFLORESCENCE; DELIQUESCENCE

3.13 WATER OF CRYSTALLISATION

Some salts, *while crystallising out from* their solutions, unite with a definite quantity of water, which is *known as* the **water of crystallisation**.

This water of crystallisation is in loose chemical combination with the salts, and it can be driven out by heating the powdered crystals of these salts above 100°C.

Experiment : *To show that copper sulphate crystals contain water of crystallisation.*

Procedure : Take some powdered copper sulphate crystals in a clean, dry test tube, and clamp it in a tilted position (Fig. 3.9) *so that the condensed water may not slip back and crack the test tube at its hot end.* Heat the crystals. Drops of a colourless liquid will condense on the cooler parts of the test tube, and finally a little of it can be collected in a dish.

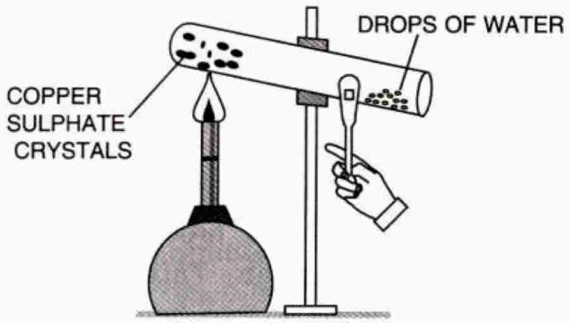

Fig. 3.9 Removal of water of crystallisation by heating.

Reaction :

$$CuSO_4.5H_2O \underset{\text{cool}}{\overset{\text{heat}}{\rightleftharpoons}} CuSO_4 + 5H_2O$$

(blue crystal) (white powder)

Test for condensed liquid :

(i) It is neutral to litmus.

(ii) Spread some anhydrous copper sulphate (white) on a glass slide in a thin layer and add a few drops of the liquid in the dish. ***A blue patch will be formed, thereby showing that the liquid is water.***

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Table 3.2 Some salts containing water of crystallisation

Common name	*Chemical name*	*Formula*
Washing soda crystals	Sodium carbonate decahydrate	$Na_2CO_3.10H_2O$
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4.7H_2O$
Potash alum	Hydrated potassium aluminium sulphate	$K_2SO_4.Al_2(SO_4)_3.24H_2O$
Glauber's salt	Sodium sulphate decahydrate	$Na_2SO_4.10H_2O$
Blue vitriol	Copper (II) sulphate pentahydrate	$CuSO_4.5H_2O$
Plaster of Paris	Calcium sulphate semihydrate	$CaSO_4.\frac{1}{2}H_2O$
White vitriol	Zinc sulphate heptahydrate	$ZnSO_4.7H_2O$
Gypsum	Hydrated calcium sulphate	$CaSO_4.2H_2O$

(iii) The liquid (water) turns blue $CoCl_2$ paper pale pink (almost colourless).

$$\underset{\text{blue}}{CoCl_2} + 6H_2O \rightarrow \underset{\text{pale pink}}{CoCl_2.6H_2O}$$

Crystalline solids not containing water of crystallisation.

Common salt (NaCl),

Nitre (KNO_3),

Sugar ($C_{12}H_{22}O_{11}$),

Potassium permanganate ($KMnO_4$), and

Ammonium chloride (NH_4Cl), etc.

Experiment : Heat gently a few crystals of pure sodium chloride and potassium nitrate separately, in different test tubes.

Observation for sodium chloride : On heating, crystals of sodium chloride produce a crackling sound, called **decrepitation**. *This is due to the bigger crystals breaking up into smaller ones.*

Observation for potassium nitrate : The crystals of potassium nitrate melt into a colourless liquid and, on further heating, produce oxygen which rekindles a glowing splinter. Potassium nitrite, *a pale yellow solid residue, is left behind.*

$$\underset{\substack{\text{potassium}\\\text{nitrate}}}{2KNO_3} \rightarrow \underset{\substack{\text{potassium}\\\text{nitrite}}}{2KNO_2} + O_2$$

Note : These crystalline salts do not produce water vapour on heating.

Conclusion : *All crystalline salts do not contain water of crystallisation.*

3.14 EFFLORESCENCE

Efflorescence is the property of some substances to lose wholly, or partly their water of crystallisation when their crystals are exposed to dry air even for a short time. They become powdery. Such substances are called ***efflorescent substances.***

Efflorescent substances lose their water of crystallisation and thus their crystalline shape becomes powdery, when exposed to air.

Examples of efflorescent substances :

(a) **Washing soda** [hydrated sodium carbonate], when exposed to dry air, becomes a monohydrate.

$$Na_2CO_3.10H_2O \xrightarrow{\text{dry air}} Na_2CO_3.H_2O + 9H_2O$$

(b) **Glauber's salt** [hydrated sodium sulphate, $Na_2SO_4.10H_2O$] becomes a powdery anhydrous sodium sulphate when exposed to air.

$$Na_2SO_4.10H_2O \xrightarrow{\text{dry air}} Na_2SO_4 + 10H_2O$$

(c) **Epsom salt** [magnesium sulphate heptahydrate], when exposed to dry air, becomes a monohydrate.

$$MgSO_4.7H_2O \xrightarrow{\text{dry air}} MgSO_4.H_2O + 6H_2O$$

Note : The higher the temperature of the air, the higher the efflorescence. This is because the air absorbs more water with rising temperature and decreasing moisture.

3.15 HYGROSCOPIC SUBSTANCES

Certain substances absorb moisture (water vapour) from the atmosphere without dissolving in it. Such substances are called

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hygroscopic substances *and the phenomenon is called hygroscopy.*

Some examples of hygroscopic substances are :

(i) Conc. sulphuric acid (H_2SO_4)
(ii) Phosphorus pentoxide (P_2O_5)
(iii) Quicklime (CaO)
(iv) Silica gel

In the laboratory, these substances are used generally for drying gases, *i.e.*, for removal of moisture from gases.

3.16 DELIQUESCENT SUBSTANCES

Certain water-soluble substances, when exposed to the atmosphere at ordinary temperature, absorb moisture from the atmospheric air to become moist, ***and ultimately dissolve in the absorbed water, forming a saturated solution.*** Such a substance is called a ***deliquescent*** substance, and the phenomenon is called ***deliquescence.***

Deliquescent substances

Examples : Caustic soda (NaOH),
Caustic potash (KOH),
Magnesium chloride ($MgCl_2$),
Zinc chloride ($ZnCl_2$),
Calcium chloride ($CaCl_2$),
Ferric chloride ($FeCl_3$),
Zinc nitrate [$Zn(NO_3)_2$], and
Copper nitrate [$Cu(NO_3)_2$].

Deliquescence occurs when the *vapour pressure of the salt* is much *lower* compared to **atmospheric** vapour pressure. Thus, *deliquescence is minimized during dry conditions, whereas efflorescence is maximized in dry conditions.*

Note : Efflorescence substances looses their weight while hygroscopic and deliquescence substances gains weight when exposed to atmosphere.

Note : *Table salt [sodium chloride] turns moist and ultimately forms a solution, on exposure to air* [especially during the rainy season]. Though pure sodium chloride is not deliquescent, the commercial version of the salt contains **impurities,** like **magnesium chloride** and **calcium chloride,** which are deliquescent substances.

This kind of impurity can be removed by passing a current of dry hydrogen chloride gas through a saturated solution of the affected salt. Pure sodium chloride is produced as a precipitate, which can be recovered by filtering and washing first with a little water and finally with alcohol.

3.17 DRYING AND DEHYDRATING AGENTS

Drying or desiccating agents

Certain substances remove moisture from other substances and are, therefore, called ***desiccants*** or ***desiccating agents*** or ***drying agents.*** Almost all hygroscopic substances are desiccating agents.

Examples : Conc. sulphuric acid, phosphorus pentoxide, silica gel and quicklime.

Dehydrating agents are substances that can remove water molecules even from compounds. Concentrated sulphuric acid can remove water molecules from blue vitriol ($CuSO_4.5H_2O$), so it is a dehydrating agent as well.

Differences between drying agent and dehydrating agent

Drying agent	*Dehydrating agent*
(i) They remove moisture from other substances.	(i) They remove chemically combined elements of water in the ratio of 2 : 1 (hydrogen : oxygen) from a compound.
(ii) They are used to dry gases like chlorine, sulphur dioxide, hydrogen chloride, etc. They are also used in desiccators to keep substances dry.	(ii) They prepare substances like carbon monoxide, sugar charcoal, *etc.*
(iii) They represent physical change. ***Examples :*** Phosphorus pentoxide P_2O_5, fused calcium chloride $CaCl_2$, calcium oxide CaO, conc. sulphuric acid H_2SO_4.	(iii) They represent chemical change. ***For example :*** Conc. Sulphuric acid H_2SO_4.

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Intext Questions

1. What do you understand by water of crystallisation?
   Give four substances which contain water of crystallisation and write their common names.
2. (a) Define efflorescence. Give examples.
   (b) Define deliquescence. Give examples.
3. Distinguish between drying and dehydrating agent.
4. Explain clearly how conc. H_2SO_4 is used as dehydrating as well as drying agent.
5. M is an element in the form of a powder. M burns in oxygen and the product obtained is soluble in water. The solution is tested with litmus. Write down *only the* word which will correctly complete each of the following sentences.
   (i) If M is a metal, then the litmus will turn _____.
   (ii) If M is a non-metal, then the litmus will turn _____.
   (iii) If M is a reactive metal, then _____ will be evolved when M reacts with dilute sulphuric acid.
   (iv) If M is a metal, it will form _____ oxide, which will form _____ solution with water.
   (v) If M is a non-metal, it will not conduct electricity in the form of _____.
6. Give reasons for the following :
   (a) Sodium hydrogen sulphate is not an acid but it dissolves in water to give hydrogen ions, according to the equation
   $$NaHSO_4 \rightleftharpoons H^+ + Na^+ + SO_4^{2-}.$$
   (b) Anhydrous calcium chloride is used in a desiccator.
7. State whether a sample of each of the following would increase or decrease in mass if exposed to air.
   (a) Solid NaOH, (b) Solid $CaCl_2$, (c) Solid $Na_2CO_3 \cdot 10H_2O$,
   (d) Conc. Sulphuric acid, (e) Iron (III) Chloride
8. (a) Why does common salt get wet during the rainy season?
   (b) How can this impurity be removed ?
   (c) Name a substance which changes the blue colour of copper sulphate crystals to white.
   (d) Name two crystalline substances which do not contain water of crystallisation.
7. Answer the questions below, relating your answers only to salts in the following list : Sodium chloride, anhydrous calcium chloride, copper sulphate-5-water ?
   (a) What name is given to the water in the compound copper sulphate-5-water ?
   (b) If copper sulphate-5-water is heated, anhydrous copper sulphate is formed. What is its colour ?
   (c) By what means, other than heating, could you dehydrate copper sulphate-5-water and obtain anhydrous copper sulphate ?
   (d) Which one of the salts in the given list is deliquescent ?

EXERCISE

MISCELLANEOUS QUESTIONS BASED ON ICSE EXAMINATIONS

1. Write the balanced equations for the preparation of the following compounds (as the major product) starting from iron and using only one other substance :
(a) Iron (II) chloride, (b) Iron (III) chloride,
(c) Iron (II) sulphate, (d) Iron (II) sulphide.

2. Write balanced reactions for the following conversions (A, B, C, D). **(2015)**

$$Fe \xrightarrow{A} FeCl_2 \xrightarrow{B} FeCO_3 \xrightarrow{C} Fe(NO_3)_2 \xrightarrow{D} Fe(OH)_2$$

3. The preparation of lead sulphate from lead carbonate is a two-step process. (lead sulphate cannot be prepared by adding dilute sulphuric acid to lead carbonate.)
(a) What is the first step that is required to prepare lead sulphate from lead carbonate ?
(b) Write the equation for the reaction that will take place when this first step is carried out.
(c) Why is the direct addition of dilute sulphuric acid to lead carbonate an impractical method of preparing lead sulphate ?

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4. Match the following :

Column A	Column B
(a) Acid salt	A. Sodium potassium carbonate
(b) Mixed salt	B. Alum
(c) Complex salt	C. Sodium carbonate
(d) Double salt	D. Sodium zincate
(e) Normal salt	E. Sodium hydrogen carbonate

5. Write balanced equations for the following reactions :

(a) Lead sulphate from lead nitrate solution and dilute sulphuric acid,

(b) Copper sulphate from copper and concentrated sulphuric acid,

(c) Lead chloride from lead nitrate solution and sodium chloride solution,

(d) Ammonium sulphate from ammonia and dilute sulphuric acid,

(e) Sodium chloride from sodium carbonate solution and dilute hydrochoric acid.

2008

(a) What are the terms defined by the following ?

(i) A salt containing a metal ion surrounded by other ions or molecules,

(ii) A base which is soluble in water.

(b) Making use only of substances chosen from those given below :

Dilute sulphuric acid	sodium carbonate
Zinc	sodium sulphite
Lead	calcium carbonate

give the equations for the reactions by which you could obtain :

(i) hydrogen,

(ii) sulphur dioxide,

(iii) carbon dioxide,

(iv) zinc carbonate (two steps required).

2009

(a) The acid which contains four hydrogen atoms .....

(i) Formic (ii) Sulphuric

(iii) Nitric (iv) Acetic-acid

(b) A black coloured solid which on reaction with dilute sulphuric acid forms a blue coloured solution is

(i) Carbon (ii) Manganese [IV] oxide

(iii) Lead [II] oxide (iv) Copper [II] oxide

(c) Solution A is strong acid B weak acid C strong alkali

(i) Which solution contains solute molecules in addition to water molecules

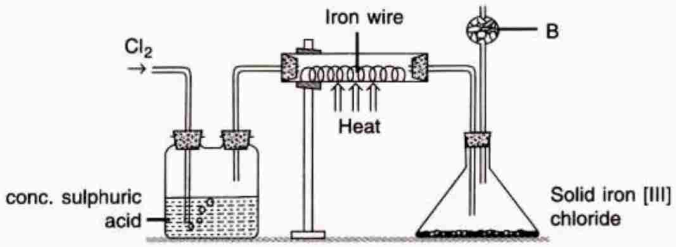
(ii) Which solution gives gelatinous white ppt with zinc sulphate, ppt disappears in excess

(iii) Give example of weak alkali.

(d) Wrrite the equations[s] for the reaction[s] to preapre lead sulphate from lead carbonate.

(e) Define the following term — Neutralization.

(f) The diagram given below is to prepare Iron [III] chloride in the laboratory :



(i) What is substance B.

(ii) What is the purpose of B.

(iii) Why is iron [III] chloride to be stored in a closed container.

(iv) Write the equation for the reaction between iron and chlorine.

2010

(a) Select the correct answer from A, B, C, D and E —

(i) Nitroso iron [II] sulphate

(ii) Iron [III] chloride

(iii) Chromium sulphate

(iv) Lead chloride

(v) Sodium chloride

A. A deliquescent compound

B. A compound soluble in hot water but insoluble in cold water

C. A compound which in the aqueous solution state, is neutral in nature.

(b) Select the correct answer from A, B, C and D —

(i) A weak organic acid is :

A. Formic acid **B.** Sulphuric acid

C. Nitric acid **D.** Hydrochloric acid

(ii) A complex salt is

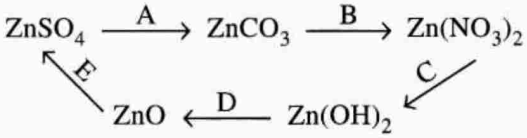
A. Zinc sulphate

B. Sodium hydrogen sulphate

C. Iron [ammonium sulphate]

D. Tetrammine copper [II] sulphate

(c) Give equations for the following conversions A to E —



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(d) For the preparation of the following salts — give a balanced equation in each case.

(i) Copper [II] sulphate from copper [II] oxide

(ii) Iron [III] chloride from the metal iron

(iii) Potassium sulphate from KOH solution

(iv) Lead [II] chloride from lead carbonate [give two equations].

2011

(a) Write the balanced chemical equation — Lead nitrate solution is added to sodium chloride solution.

(b) State what happens to crystals or washing soda when exposed to air. Name the phenomenon exhibited.

(c) Name the method used for the preparation of the following salts from the list given below :

(i) Sodium nitrate (ii) Iron (III) chloride

(iii) Lead chloride (iv) Zinc sulphate

(v) Sodium hydrogen sulphate

List :

A. Simple displacement

B. Neutralisation

C. Decomposition by acid

D. Double decomposition

E. Direct synthesis

2012

(a) Match the following :

Column A	Column B
(i) Acid salt	(A) Ferrous ammonium sulphate
(ii) Double salt	(B) Contains only ions
(iii) Ammonium hydroxide solution	(C) Sodium hydrogen sulphate
(iv) Dilute hydrochloric acid	(D) Contains only molecules
(v) Carbon tetrachloride	(E) Contains ions and molecules

2013

(a) Select the word/s given which are required to correctly complete the blanks —

[ammonia, ammonium carbonate, carbon dioxide, hydrogen, hydronium, hydroxide, precipitate, salt, water]

(i) A solution M turns blue litmus red, so it must contains (1) ..... ions; another solution turns red litmus blue and hence, must contain (2) ..... ions.

(ii) When solution M and O are mixed together, the products will be (3) ..... and (4) .....

(iii) If a piece of magnesium was put into a solution M, (5) ..... gas would be evolved.

(b) Give a suitable chemical term for :

(i) A salt formed by incomplete neutralisation of an acid by a base.

(ii) A definte number of water molecules bound to some salts.

(iii) The process in which a substance absorbs moisture from the atmospheric air to become moist, and ultimately dissolves in the absorbed water.

(c) Choosing the substances from the list given : dil. sulphuric aicd, copper, iron, sodium copper [II] carbonate, sodium carbonate, sodium chloride, zinc nitrate.

Write balanced equations for the reactions which would be used in the laboratory to obtain the following salts

(i) sodium sulphate (ii) zinc carbonate

(iii) copper [II] sulphate (iv) iron [II] sulphate

2014

(a) Fill in the blank from the choices given : The basicity of acetic acid is ..... [3, 1, 4].

(b) Draw the structure of the stable positive ion formed when an acid dissolves in water.

(c) State the inference drawn from the observation :

Salt S is prepared by reacting dilute sulphuric acid with copper oxide. Identify S.

(d) Give balanced chemical equations for the preparation of the following salts :

(i) Lead sulphate – from lead carbonate.

(ii) Sodium sulphate – using dilute sulphuric aicd.

(iii) Copper chloride – using copper carbonate.

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4 Analytical Chemistry : Units of Ammonium Hydroxide and Sodium Hydroxide

SYLLABUS

Analytical Chemistry – Use of Ammonium Hydroxide and Sodium Hydroxide.

(i) On solution of salts: colour of salt and its solution; formation and colour of hydroxide precipitated for solutions of salts of Ca, Fe, Cu, Zn and Pb; special action of ammonium hydroxide on solutions of copper salt and sodium hydroxide on ammonium salts.

On solution of salts :

- *Colour of salt and its solution.*
- *Action on addition of Sodium Hydroxide to solution of Ca, Fe, Cu, Zn and Pb salts drop by drop in excess. Formation and colour of hydroxide precipitated to be highlighted with the help of equations.*
- *Action on addition of Ammonium Hydroxide to solution of Ca, Fe, Cu, Zn and Pb salts drop by drop in excess. Formation and colour of hydroxide precipitated to be highlighted with the help of equations.*
- *Special action of Ammonium Hydroxide on solutions of copper salts and sodium hydroxide on ammonium salts.*

(ii) On certain metals and their oxides (relevant laboratory work is essential).

The metals must include zinc and aluminium, their oxides and their hydroxides, which react with caustic alkalis (NaOH, KOH), showing the amphoteric nature of these substances.

INTRODUCTION

Determination of the chemical components in a given sample is called **Analysis**. In chemistry, we study two types of analysis, ***Qualitative analysis*** which involves the identification of the unknown substances, and ***Quantitative analysis*** which involves the determination of composition of a mixture.

Qualitative analysis, *i.e.,* identification of an unknown substance is done by carrying out chemical tests with the help of *reagents*. A ***reagent*** is a substance that reacts with another substance.

Alkalis are important laboratory reagents. When they are added to certain salt solution characteristic coloured precipitates of metallic hydroxides are formed. Thus a metal ion is identified.

4.1 COLOURS OF THE SALTS AND THEIR SOLUTIONS

The salts of **'representative elements'** of the periodic table, *i.e.,* the elements of the Groups 1, 2, and 13 to 17 are generally **colourless**, while those of the **'transition elements'**, *i.e.,* salts of elements of Groups 3 to 12 are generally coloured.

Different colours of coloured salts help in their identification during qualitative analysis. Some examples of colourless and coloured ions are given below :

Colourless ions		Coloured ions		
Cation	***Symbol***	***Cation***	***Symbol***	***Colour***
Ammonium ion	NH_4^+	Cupric ion	Cu^{2+}	Blue
Sodium ion	Na^+	Ferrous ion	Fe^{2+}	Light green
Potassium ion	K^+			
Calcium ion	Ca^{2+}	Ferric ion	Fe^{3+}	Yellowish brown
Magnesium ion	Mg^{2+}			
Aluminium ion	Al^{3+}	Nickel ion	Ni^{2+}	Green
Lead ion	Pb^{2+}	Chromium ion	Cr^{3+}	Green
Zinc ion	Zn^{2+}	Manganese ion	Mn^{2+}	Pink
Anion	***Symbol***	***Anion***	***Symbol***	***Colour***
Chloride ion	Cl^-	Permanganate ion	MnO_4^-	Pink or Purple
Sulphate ion	SO_4^{2-}			
Carbonate ion	CO_3^{2-}	Dichromate ion	$Cr_2O_7^{2-}$	Orange
Nitrate	NO_3^-			
Hydrogen carbonate ion	HCO_3^-	Chromate ion	CrO_4^{2-}	Yellow
Sulphide ion	S^{2-}			
Bromide ion	Br^-			
Acetate ion	CH_3COO^-			

Action of alkali on metal cations results in the formation of their hydroxides which often appear as precipitates.

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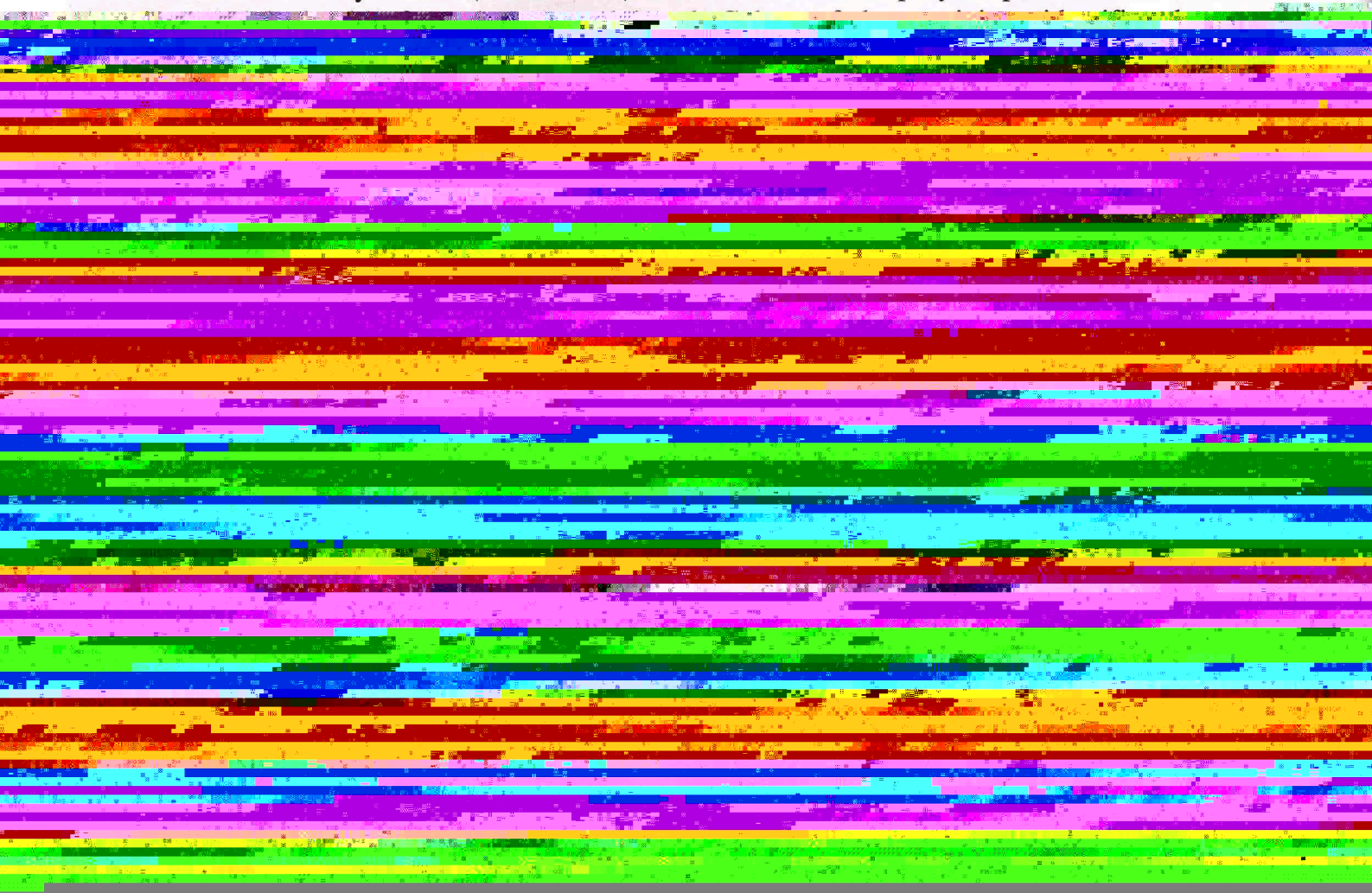
Precipitation is the process of formation of an insoluble solid when solutions are mixed. The solid thus formed is called precipitate. Only those compounds form precipitates which are insoluble in water.

4.2 ACTION OF SODIUM HYDROXIDE SOLUTION ON CERTAIN METALLIC SALT SOLUTIONS

When the sodium hydroxide (caustic soda) solution is added drop by drop to the solution of metallic



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4.3 ACTION OF AMMONIUM HYDROXIDE ON CERTAIN SALT SOLUTIONS

When ammonium hydroxide solution is added dropwise to the solutions of metallic salts, precipitates of their hydroxides are formed, which are identified by their distinct colours. Some precipitated metallic hydroxides are soluble in excess of ammonium hydroxide, because of the formation of soluble amino complexes on further reaction with excess of NH_4OH.

SALT SOLUTION + AMMONIUM HYDROXIDE → METAL HYDROXIDE + SALT FORMED IN SOLUTION	COLOUR OF PPT AND ITS SOLUBILITY IN EXCESS OF AMMONIUM HYDROXIDE
1. CALCIUM SALTS **No precipitation** occurs even with addition of excess of NH_4OH. This is because the concentration of OH^- ions from the ionisation of NH_4OH is so low that it cannot precipitate the hydroxide of calcium.	
2. IRON **(A) IRON (II) SALTS [Fe^{2+} ion]** $FeSO_4$ (green) + $2NH_4OH$ → $Fe(OH)_2\downarrow$ (dirty green ppt.) + $(NH_4)_2SO_4$ [colourless in solution]	DIRTY GREEN PPT INSOLUBLE
(B) IRON (III) SALTS [Fe^{3+} ion] (i) $FeCl_3$ (yellow solution) + $3NH_4OH$ → $Fe(OH)_3\downarrow$ (reddish brown ppt.) + $3NH_4Cl$ [colourless in solution]	REDDISH BROWN PPT INSOLUBLE
(ii) $Fe_2(SO_4)_3$ (yellow solution) + $6NH_4OH$ → $2Fe(OH)_3\downarrow$ (reddish brown ppt.) + $3(NH_4)_2SO_4$ (colourless in solution)	
3. COPPER (II) SALTS [Cu^{2+} ion] $CuSO_4$ (blue) + $2NH_4OH$ → $Cu(OH)_2\downarrow$ (Pale blue ppt.) + $(NH_4)_2SO_4$ (colourless in solution) **With excess of NH_4OH ppt. dissolves** $Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$ Tetrammine copper (II) sulphate or $Cu(OH)_2 + 4NH_4OH \rightarrow [Cu(NH_3)_4](OH)_2 + 4H_2O$ Tetrammine copper hydroxide **This reaction is a characteristic property of Cu^{2+} ion and is used for its detection in qualitative analysis.**	PALE BLUE PPT SOLUBLE in excess of NH_4OH and forms deep blue solution
4. ZINC SALTS [Zn^{2+} ion] $ZnSO_4$ (colourless solution) + $2NH_4OH$ → $Zn(OH)_2\downarrow$ (white, gelatinous ppt.) + $(NH_4)_2SO_4$ (colourless in solution) **With excess of NH_4OH ppt. dissolves.** $Zn(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Zn(NH_3)_4]SO_4 + 4H_2O$ Zinc hydroxide (excess) Tetrammine zinc (II) sulphate (colourless solution) OR $Zn(OH)_2 + 4NH_4OH \rightarrow [Zn(OH_3)_4](OH)_2 + 4H_2O$ Tetrammine zinc hydroxide	GELATINOUS WHITE PPT SOLUBLE in excess of NH_4OH
5. LEAD SALTS [Pb^{2+} ion] $Pb(NO_3)_2$ (colourless) + $2NH_4OH$ → $Pb(OH)_2\downarrow$ (white ppt.) + $2NH_4NO_3$	CHALKY WHITE PPT INSOLUBLE

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Intext Questions

1. What do you understand by the following :
   (i) Analysis, (ii) Qualitative analysis, (iii) Reagent, (iv) Precipitation ?
2. Write the probable colour of the following salts :
   (i) Iron (III) chloride, (ii) Potassium nitrate, (iii) Ferrous sulphate, (iv) Aluminium acetate,
3. Name the probable cation present based on the following observations :
   (i) White precipitate insoluble in NH_4OH but soluble in NaOH (ii)Blue coloured solution,
4. Name the metal hydroxides which are :
   (i) Insoluble, (ii) Soluble,
   in (a) Caustic soda solution (b) Ammonium hydroxide solution.
5. What do you observe when ammonium salt is heated with caustic soda solution ? Write the word equation.
6. How will you distinguish NH_4OH solution from NaOH solution ?
7. Why the alkali is added drop by drop to the salt solution.
   Ans: If alkali is added too quickly, it is easy to miss a precipitate that redissolves in excess.

4.4 ACTION OF ALKALIS ON CERTAIN METALS

Certain metals like zinc, aluminium and lead react with hot concentrated caustic alkalis (NaOH, KOH) to give the corresponding soluble salt and liberate hydrogen.

METAL	+	ALKALI	→	SALT	+	HYDROGEN
1. ZINC						
(i) Zn	+	2NaOH (Hot and conc.)	→	Na_2ZnO_2 Sodium zincate (colourless)	+	H_2
(ii) Zn	+	2KOH (Hot and conc.)	→	K_2ZnO_2 Potassium zincate (colourless)	+	H_2
2. ALUMINIUM						
Aluminium reacts with boiling caustic alkalis solution						
(i) 2Al + 2NaOH +		$2H_2O$	→	$2NaAlO_2$ Sodium meta aluminate (colourless)	+	$3H_2$
(ii) 2Al + 2KOH +		$2H_2O$	→	$2KAlO_2$ Potassium meta aluminate	+	$3H_2$
3. LEAD						
(i) Pb	+	2NaOH	→	Na_2PbO_2 Sodium plumbite (colourless)	+	H_2
(ii) Pb	+	2KOH	→	K_2PbO_2 Potassium plumbite (colourless)	+	H_2

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4.5 ACTION OF ALKALIS ON METAL OXIDES

Oxides of most of the metals are basic in nature. They dissolve in water forming hydroxides (or alkalis). These metal oxides and hydroxides neutralize acids but *do not react with bases.*

For example :

$$\underset{\text{Sodium oxide}}{Na_2O} + H_2O \rightarrow \underset{\text{Sodium hydroxide}}{2NaOH}$$

$$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$$

$$NaOH + HCl \rightarrow NaCl + H_2O$$

A few metallic oxides and hydroxides exhibit dual character, *i.e.,* they show acidic as well as basic character. They are said to be **amphoteric** in nature.

For example :

Zinc oxide and zinc hydroxide react with both acids and concentrated alkalis (NaOH and KOH) forming salt and water.

$$\underset{\text{Zinc oxide}}{ZnO} + \underset{\text{Hydrochloric acid}}{2HCl} \rightarrow \underset{\text{Zinc chloride}}{ZnCl_2} + \underset{\text{Water}}{H_2O}$$

$$\underset{\text{Zinc hydroxide}}{Zn(OH)_2} + 2HCl \rightarrow ZnCl_2 + 2H_2O$$

$$\underset{\text{Zinc oxide}}{ZnO} + \underset{\text{Sodium hydroxide}}{2NaOH} \rightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + \underset{\text{Water}}{H_2O}$$

$$\underset{\text{Zinc hydroxide}}{Zn(OH)_2} + \underset{\text{(Alkali)}}{2NaOH} \rightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + \underset{\text{Water}}{2H_2O}$$

Amphoteric oxides and hydroxides are those compounds which react with both acids and alkalis to form salt and water.

AMPHOTERIC OXIDE / HYDROXIDE	OXIDE / HYDROXIDE	+ ALKALI	⟶	SALT	+	WATER
1. Zinc oxide / Zinc hydroxide	(i) ZnO (white)	$2NaOH$	⟶	Na_2ZnO_2 Sodium zincate (colourless, soluble)	+	H_2O
	$Zn(OH)_2$	$2NaOH$	⟶	Na_2ZnO_2	+	$2H_2O$
	(ii) ZnO (white)	$2KOH$	⟶	K_2ZnO_2 Potassium zincate (colourless, soluble)	+	H_2O
	$Zn(OH)_2$	$2KOH$	⟶	K_2ZnO_2	+	$2H_2O$
2. Aluminium oxide / Aluminium hydroxide	(i) Al_2O_3 (white)	$2NaOH$	⟶	$2NaAlO_2$ Sodium meta aluminate	+	H_2O
	$Al(OH)_3$	$NaOH$	⟶	$NaAlO_2$	+	$2H_2O$
	(ii) Al_2O_3 (white)	$2KOH$	⟶	$2KAlO_2$ Potassium meta aluminate	+	H_2O
	$Al(OH)_3$	KOH	⟶	$KAlO_2$	+	$2H_2O$
3. Lead oxide / Lead hydroxide	(i) PbO (yellow)	$2NaOH$	⟶	Na_2PbO_2 Sodium plumbite (colourless, soluble)	+	H_2O
	$Pb(OH)_2$	$2NaOH$	⟶	Na_2PbO_2	+	$2H_2O$
	(ii) PbO (yellow)	$2KOH$	⟶	K_2PbO_2 Potassium plumbite (colourless, soluble)	+	H_2O
	$Pb(OH)_2$	$2KOH$	⟶	K_2PbO_2	+	$2H_2O$

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EXERCISE

1. Write the probable colour of the following salts :
   (a) Ferrous salts (b) Ammonium salts
   (c) Cupric salts (d) Calcium salts
   (e) Aluminium salts.

2. Name :
   (a) a metallic hydroxide soluble in excess of NH_4OH.
   (b) a metallic oxide soluble in excess of caustic soda solution.
   (c) a strong alkali.
   (d) a weak alkali.
   (e) two colourless metal ions.
   (f) two coloured metal ions.
   (g) a metal that evolves a gas which burns with a pop sound when boiled with alkali solutions.
   (h) two bases which are not alkalis but dissolve in strong alkalis.
   (i) a coloured metallic oxide which dissolves in alkalis to yield colourless solutions.
   (j) a colourless cation not a representative element.

3. Write balanced equations for Q. 2 (g) and (i).

4. What happens when ammonia solution is added first dropwise and then in excess to the following solutions :
   (i) $CuSO_4$ (ii) $ZnSO_4$ (iii) $FeCl_3$.
   Write balanced equations for these reactions.

5. What do you observe when caustic soda solution is added to the following solution, first a little and then in excess:
   (a) $FeCl_3$, (b) $ZnSO_4$,
   (c) $Pb(NO_3)_2$, (d) $CuSO_4$?
   Write balanced equations for these reactions.

6. Name the chloride of a metal which is soluble in excess of ammonium hydroxide. Write equation for the same.

7. On adding dilute ammonia solution to a colourless solution of a salt, a white gelatinous precipitate appears. This precipitate however dissolves on addition of excess of ammonia solution. Identify (choose from Na, Al, Zn, Pb, Fe)
   (a) Which metal salt solution was used ?
   (b) What is the formula of the white gelatinous precipitate obtained ?

8. Name :
   (a) a yellow monoxide that dissolves in hot and concentrated caustic alkali.
   (b) a white, insoluble oxide that dissolves when fused with caustic soda or caustic potash.
   (c) a compound containing zinc in the anion.

9. Select the correct answers :
   (a) Colour of an aqueous solution of copper sulphate is
      (i) Green (ii) Brown
      (iii) Blue (iv) Yellow
   (b) Colour of the precipitate formed on adding NaOH solution to iron (II) sulphate solution is
      (i) White (ii) Brown
      (iii) Green (iv) Pale blue
   (c) A metal which produces hydrogen on reactng with alkali as well as with acid.
      (i) Iron (ii) Magnesium
      (iii) Zinc (iv) Copper

10. What do you observe when freshly precipitated aluminium hydroxide reacts with caustic soda solution ? Give balanced equation.

11. You are provided with two reagent bottles marked A and B. One of which contains NH_4OH solution and the other contains NaOH solution. How will you identify them by a chemical test ?

12. Distinguish by adding :
   (a) Sodium hydroxide solution and
   (b) Ammonium hydroxide solution to
      (i) Calcium salt solution and lead salt solution .
      (ii) Lead salt solution and zinc salt solution.
      (iii) Copper salt solution and ferrous salt solution.
      (iv) Fe(II) salt solution and Fe(III) salt solution.
      (v) Ferrous nitrate and lead nitrate

13. How will you distinguish lead carbonate and zinc carbonate in solution ?

14. What is observed when hot concentrated caustic soda solution is added to (a) Zinc (b) Aluminium. Write balanced equations.

15. (a) What do you understand by amphoteric oxide ?
   (b) Give the balanced equations for the reaction with two different amphoteric oxides with a caustic alkali.
   (c) Name the products formed.

16. Write balanced equations for the following conversions
   (a) $ZnSO_4 \xrightarrow{A} Zn(OH)_2 \xrightarrow{B} Na_2ZnO_2$.
   (b) $CuSO_4 \xrightarrow{A} Cu(OH)_2 \xrightarrow{B} [Cu(NH_3)_4]SO_4$

For I.C.S.E. questions see Chapter 13

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5 Mole Concept and Stoichimetry

SYLLABUS

Mole Concept and Stoichiometry

(i) Gay-Lussac's Law of Combining Volumes; Avogadro's Law.

Idea of mole – a number just as dozen, a gross; Avogadro's Law – statement and explanation; Gay-Lussac's Law of Combining Volumes – statement and explanation; "the mass of 22·4 litres of any gas at S.T.P. is equal to its molar mass" (Questions will not be set on formal proof but may be taught for clear understanding) – simple calculations based on the molar volume.

(ii) Refer to the atomicity of hydrogen, oxygen, nitrogen and chlorine (proof not required).

The explanation can be given using equations for the formation of HCl, NH_3 and NO.

(iii) Relative atomic masses (atomic weight) and relative molecular masses (molecular weights) : either H = 1 or $^{12}C = 12$ will be accepted; molecular mass = 2 × vapour density (formal proof not required). Deduction of simple (empirical) and molecular formula from the percentage composition of a compound; the molar volume of a gas at S.T.P.; simple calculations based on chemical equations; both reacting weight and volumes.

Idea of relative atomic mass and relative molecular mass — standard H atom, or 1/12th of carbon 12 atom.

Relating mole and atomic mass; arriving at gram atomic mass and then gram atom; atomic mass is a number dealing with one atom; gram atomic mass is the mass of one mole of atoms.

Relating mole and molecular mass arriving at gram molecular mass and gram molecule – molecular mass is a number dealing with a molecule, gram molecular mass is the mass of one mole of molecules.

Molecular mass = 2 × vapour density (questions will not be set on formal proof but may be taught for clear understanding); simple calculations based on the formula.

Deduction of simple (empirical) and molecular formula from the percentage composition of a compound.

5A. GAY-LUSSAC'S LAW AND AVOGADRO'S LAW

5.1 INTRODUCTION

You have learnt in Class IX that all gases behave similarly under similar conditions of temperature and pressure as expressed by gas laws.

Gas laws are certain rules applicable to a gas in respect of change in temperature (T) or pressure (P) or volume (V). The change in any one of these affects the other two.

Pressure-Volume Relationship or Boyle's Law

It states that *the volume of a given mass of dry gas is inversely proportional to its pressure at a constant temperature.*

$$P_1V_1 = P_2V_2 = k \text{ at constant temperature}$$

Temperature-Volume Relationship or Charles's Law :

It states that *volume of a given mass of a dry gas is directly proportional to its absolute (kelvin) temperature, if the pressure is kept constant.*

OR

The pressure remaining constant, the volume of a given mass of a dry gas increases or decreases by 1/273 of its volume for each 1°C increase or decrease in temperature respectively.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = k \text{ at constant pressure}$$

Gas Equation

On combining both the laws we conclude that

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the volume of a given mass of a dry gas varies inversely as the pressure and directly as the absolute temperature.

$$V \propto \frac{1}{P} \times T$$

or $$V = \frac{T}{P} \times \text{constant}$$

or $$\frac{PV}{T} = k \text{ (constant)}$$

If the volume of a given mass of a gas changes from V_1 to V_2, pressure from P_1 to P_2 and temperature from T_1 to T_2 then,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = k \text{ (constant)}$$

Standard Temperature Pressure (STP)

Since the volume of a gas changes remarkably with change of temperature and pressure, it becomes necessary to choose standard values of temperature and pressure to which gas volumes can be referred.

The standard values chosen are : ***0°C or 273 K*** for temperature and *1 atmospheric pressure* or ***760 mm or 76 cm of Hg*** for pressure.

Temperature Scales

Absolute Scale or Kelvin Scale : *A temperature scale with absolute zero (zero kelvin) as the starting point is called the absolute scale or the kelvin scale.*

Absolute zero = 0 K = – 273°C

5.2 GAY-LUSSAC'S LAW OF COMBINING VOLUMES

Gay-Lussac (1805) observed that a simple relation exists between the volumes of hydrogen and oxygen, which react together to form water. Thus, one litre of oxygen requires two litres of hydrogen to form two litres of water vapour. So, he found that oxygen and hydrogen react in the ratio 1 : 2 (by volume). When he extended his study to the volumes of other reacting gases, he noted similar simple relationships. Consequently, he generalised these observations as the ***law of combining volumes of gases.***

Gay-Lussac's Law of Combining Volumes

When gases react, they do so in volumes which bear a simple ratio to one another, and to the volume of the gaseous product, provided that all the volumes are measured at the same temperature and pressure.

Note : Gay-Lussac's Law is valid only for gases. **The volumes of solids and liquids are considered to be zero.**

The law may be illustrated by the following examples involving gases or vapours :

1. **Hydrogen chloride**

   One volume of hydrogen when mixed with one volume of chlorine, gives two volumes of hydrogen chloride gas at the same temperature and pressure.

   $$H_2 + Cl_2 \rightarrow 2HCl$$

   Thus, the ratio 1 : 1 : 2 is simple.

2. **Ammonia**

   One volume of nitrogen combined with three volumes of hydrogen gives two volumes of ammonia at the same temperature and pressure.

   $$N_2 + 3H_2 \rightarrow 2NH_3$$

   Thus, the ratio 1 : 3 : 2 is simple.

3. **Carbon dioxide**

   Two volumes of carbon monoxide on combustion with one volume of oxygen gives two volumes of carbon dioxide at the same temperature and pressure.

   $$2CO + O_2 \rightarrow 2CO_2$$

   Thus, the ratio 2 : 1 : 2 is simple.

Numericals based on Gay-Lussac's Law

Example 1 : *What volume of oxygen would be required to burn completely 200 mL of acetylene (C_2H_2) and what would be the volume of carbon dioxide formed ?*

Solution :

$$C_2H_2 + O_2 \rightarrow CO_2 + H_2O$$

On balancing,

$$\underset{\text{2 vols}}{2C_2H_2} + \underset{\text{5 vols}}{5O_2} \rightarrow \underset{\text{4 vols}}{4CO_2} + \underset{\text{Nil}}{2H_2O}$$

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According to the above equation — 2 volumes of acetylene requires 5 volumes of oxygen for complete combustion to produce 4 volumes of carbon dioxide.

∴ 200 mL of acetylene will require

$= \frac{5}{2} \times 200$ mL $= 500$ mL of oxygen, ***i.e., 500 mL of oxygen is required to produce 200 × 2 = 400 mL of carbon dioxide.***

Answer : 500 mL of oxygen is required to burn completely 200 mL of acetylene thereby producing 400 mL of carbon dioxide.

Example 2 : *If 6 litres of hydrogen and 5·6 litres of chlorine are mixed and exploded, what will be the composition by volume of the resulting gaseous mixture ?*

Solution :

$$\underset{1\text{ vol}}{H_2} + \underset{1\text{ vol}}{Cl_2} = \underset{2\text{ vols}}{2HCl}$$

Since 1 volume of chlorine reacts with 1 volume of hydrogen,

∴ 5·6 litres of chlorine will react with only 5·6 litres of hydrogen.

∴ (6 – 5·6) *i.e.,* 0·4 litres of hydrogen will remain unreacted.

Since the volume of HCl gas formed is twice that of chlorine used,

∴ volume of HCl formed will be

5·6 × 2 = 11·2 litres.

Answer : 11·2 litres of HCl and 0·4 litres of residual hydrogen.

Note : The reactant which is completely used up in a reaction is known as **Limiting reagent** or **Limiting reactant**. In example 2 chlorine is the limiting reagent.

Example 3 : *What volume of propane is burnt for every 100 cm³ of oxygen in the reaction.*

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

(Gas volumes measured under the same conditions).

Solution :

From the above reaction, it is clear that for every 5 volumes of oxygen, 1 volume of propane is burnt.

∴ Volume of propane burnt for every 100 cm³ of oxygen $= \frac{1}{5} \times 100 = 20$ cm³

Answer : 20 cm³ of propane is burnt.

Example 4 : *80 cm³ of methane is mixed with 200 cm³ of pure oxygen at room temperature and pressure. The mixture is then ignited when it burns as illustrated by the equation :*

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

Calculate the composition of the resulting mixture if it is cooled to initial room temperature and pressure.

Solution :

$$\underset{1\text{ vol}}{CH_4\ (g)} + \underset{2\text{ vols}}{2O_2\ (g)} \rightarrow \underset{1\text{ vol}}{CO_2\ (g)} + \underset{\text{Nil}}{2H_2O\ (l)}$$

By Gay-Lussac's law,

(i) 1 vol of methane requires 2 vols of oxygen.

∴ 80 cm³ of methane requires

$= 2 \times 80 = 160$ cm³ oxygen

(ii) 1 vol of methane produces 1 vol of carbon dioxide.

∴ 80 cm³ of methane produces 80 cm³ of carbon dioxide.

Hence, the composition of gaseous mixture after reaction is :

Methane = (80 – 80) = 0
Carbon dioxide = 80 cm³
Oxygen = (200 – 160) = 40 cm³
Water = Negligible.

Example 5 : *40 cm³ of methane reacts with chlorine according to the following equation.*

Equation :

$$CH_4\ [g] + 2Cl_2\ [g] \rightarrow CH_2Cl_2\ [g] + 2HCl\ [g]$$

Calculate the volume of HCl gas formed and chlorine gas required.

Solution :

By Gay-Lussac's law :

1 Vol of CH_4 produces 2 Vols of HCl

∴ 40 cm³ of CH_4 will produce

$2 \times 40 =$ **80 cm³ of HCl.**

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Since 1 Vol of CH_4 require 2 Vols of chlorine.

$\therefore$ 40 cm^3 of methane will require

$2 \times 40 =$ **80 cm^3 of chlorine.**

Answer : 80 cm^3 of HCl is formed.

80 cm^3 of chlorine is required.

Example 6 : *200 cm^3 of ethylene $[C_2H_4]$ is burnt in just sufficient air (containing 20% oxygen) to form carbon dioxide gas and steam. If all measurements are made at constant pressure and 100°C, find the composition of the resulting mixture.*

Solution :

The reaction involved is given by :

$$\underset{1 \text{ vol.}}{C_2H_4(g)} + \underset{3 \text{ vols.}}{3O_2(g)} \xrightarrow{100^\circ C} \underset{2 \text{ vols.}}{2CO_2(g)} + \underset{2 \text{ vols.}}{2H_2O(g)}$$

By Gay-Lussac's Law;

(*i*) 1 vol. of ethylene requires oxygen = 3 vols

$\therefore$ 200 cm^3 of ethylene will require oxygen

$= 3 \times 200 = 600$ cm^3.

(*ii*) 1 vol. of ethylene produces carbon dioxide = 2 vols

$\therefore$ 200 cm^3 of ethylene will produce carbon dioxide $= 2 \times 200 = 400$ cm^3.

(*iii*) 1 vol. of ethylene produces steam = 2 vols

$\therefore$ 200 cm^3 of ethylene will produce steam

$= 2 \times 200 = 400$ cm^3.

When oxygen is 20%, unreacted air is = 80%

When oxygen is 600 cm^3, then unreacted air is $= \frac{80 \times 600}{20} = 2400$ cm^3.

Hence, composition of mixture after reaction :

(*i*) Carbon dioxide = 400 cm^3

(*ii*) Steam = 400 cm^3

(*iii*) Unreacted air = 2400 cm^3.

Example 7 : *20 mL of hydrogen, 10 mL of carbon monoxide and 20 mL of oxygen are exploded in an eudiometer. What will be the volume and composition of the mixture of gases, after cooling to room temperature ?*

Write each reaction separately.

Solution :

$$\underset{\substack{2 \text{ vols} \\ 20 \text{ mL}}}{2H_2} + \underset{\substack{1 \text{ vol.} \\ 10 \text{ mL}}}{O_2} = \underset{\text{Nil}}{2H_2O}$$

$$\underset{\substack{2 \text{ vols} \\ 10 \text{ mL}}}{2CO} + \underset{\substack{1 \text{ vol.} \\ 5 \text{ mL}}}{O_2} = \underset{\substack{2 \text{ vols} \\ 10 \text{ mL}}}{2CO_2}$$

Total volume of oxygen used = (10 + 5) *i.e.* 15 mL.

Volume of oxygen left over = 20 – 15 = 5 mL

Volume of carbon dioxide formed = 10 mL

$\therefore$ Total volume of gases = 15 mL

Answer : Total volume of gaseous mixture after cooling to room temperature would be 15 mL, containing 5 mL oxygen and 10 mL carbon dioxide.

Example 8 : *A sample of coal gas contained 45% H_2, 30% CH_4, 20% CO, and 5% C_2H_2 by volume. 100 mL of this gaseous mixture was mixed with 160 mL of oxygen and exploded. Calculate the volume and the composition of the resulting mixture, when cooled to room temperature and pressure.*

Solution :

100 mL of gaseous mixture will contain 45 mL H_2, 30 mL CH_4, 20 mL CO and 5 mL C_2H_2.

Equations representing the involved combustion are :

(a) Hydrogen

$$\underset{\substack{2 \text{ vols} \\ 45 \text{ mL}}}{2H_2} + \underset{\substack{1 \text{ vol} \\ 22{\cdot}5 \text{ mL}}}{O_2} = \underset{\text{Nil}}{2H_2O}$$

(b) Methane

$$\underset{\substack{1 \text{ vol} \\ 30 \text{ mL}}}{CH_4} + \underset{\substack{2 \text{ vol} \\ 60 \text{ mL}}}{2O_2} = \underset{\substack{1 \text{ vol.} \\ 30 \text{ mL}}}{CO_2} + \underset{\text{Nil}}{2H_2O}$$

(c) Carbon monoxide

$$\underset{\substack{2 \text{ vols} \\ 20 \text{ mL}}}{2CO} + \underset{\substack{1 \text{ vol} \\ 10 \text{ mL}}}{O_2} = \underset{\substack{2 \text{ vols} \\ 20 \text{ mL}}}{2CO_2}$$

(d) Acetylene

$$\underset{\substack{2 \text{ vols} \\ 5 \text{ mL}}}{2C_2H_2} + \underset{\substack{5 \text{ vols} \\ 12{\cdot}5 \text{ mL}}}{5O_2} = \underset{\substack{4 \text{ vols} \\ 10 \text{ mL}}}{4CO_2} + \underset{\text{Nil}}{2H_2O}$$

Total volume of oxygen used

$= (22{\cdot}5 + 60 + 10 + 12{\cdot}5) = 105$ mL

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$\therefore$ Volume of oxygen left = (160 – 105)
= 55 mL

Total volume of carbon dioxide formed
= 30 + 20 + 10 = 60 mL

$\therefore$ Total volume of gases left
= (55 + 60) = 115 mL

Answer : 55 mL residual oxygen and 60 mL carbon dioxide and 115 mL total volume.

5.3 AVOGADRO'S LAW

Amedo Avogadro in 1811 put forward a hypothesis based on the relationship between the number of molecules in equal volumes of different gases under similar conditions.

Avogadro's law states that "equal volumes of all gases under similar conditions of temperature and pressure contain the same number of molecules."

This means that one ***litre*** of hydrogen contain the same number of molecules as are present in one ***litre*** of oxygen or in one litre of chlorine or of any other gas, provided the volumes of all gases are measured at the same temperature and pressure.

Avogadro further suggested that the smallest particle of a gaseous element is the molecule and not the atom though it may contain one, two or more atoms.

Example : A molecule of N_2 is made of two atoms of nitrogen, a molecule of NH_3 is made of one atom of nitrogen and three atoms of hydrogen.

He made the following distinction between atoms and molecules of a gaseous element.

*An **atom** is the smallest particle of an element that can take part in a chemical reaction; however, it may or may not exist independently.*

*A **molecule** is the smallest particle of an element or a compound that can exist by itself; it never breaks up except for taking part in a chemical reaction.*

Atomicity

*The number of atoms in a molecule of an element is called its **atomicity.***

(a) Monoatomic

Monoatomic molecule is composed of only one atom.

Examples : Inert gases like Helium, Neon, Argon, etc.

(b) Diatomic

Diatomic molecule is composed of two similar atoms.

Examples : H_2, O_2, Cl_2, N_2, etc.

(c) Triatomic

Triatomic molecule is composed of three similar atoms.

Example : Ozone gas (O_3).

(d) Tetratomic

Tetratomic molecule is composed of four similar atoms.

Example : Phosphorus (P_4).

(e) Octatomic

Octatomic molecule is composed of eight similar atoms.

Example : Sulphur (S_8).

Molecules made up of same type of atoms are homoatomic molecules, e.g. phosphorus (P_4), ozone (O_3) etc. while molecules made up of different types of atoms are hetero-atomic molecules, e.g. HCl, NH_3 etc.

Examples based on Avogadro's Law

Example 9 : *Under same conditions of temperature and pressure, you collect 2 litres of carbon dioxide, 3 litres of chlorine, 5 litres of hydrogen, 4 litres of nitrogen and 1 litre of sulphur dioxide. In which gas sample will there be*

(i) the greatest number of molecules ?

(ii) the least number of molecules ?

Justify your answer.

Solution :

Equal volumes of all gases under similar conditions of temperature and pressure contain the same number of molecules. So, under the same conditions of temperature and pressure, if volume of the gas is decreased, the number of molecules will also decrease.

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Hence,

(i) 5 litres of hydrogen contain the *greatest number* of molecules.

(ii) 1 litre of sulphur dioxide contains the *least number* of molecules.

Example 10 : If 50 cc of a gas A contains y molecules, how many molecules of gas B will be present in 25 cc of B under same conditions ?

Solution :

50 cc contains y molecules.

$\therefore$ 25 cc will contain $\frac{y}{2}$ molecule.

According to Avogadro's law, 50 cc of A will contain equal molecules as 50 cc of B.

$\therefore$ 25 cc will contain half of what 50 cc contains.

EXERCISE-5A

1. State :

(a) Gay-Lussac's Law of combining volumes, **(2011)**

(b) Avogadro's law.

2. (a) Define atomicity of a gas. State the atomicity of Hydrogen, Phosphorus and Sulphur.

(b) Differentiate between N_2 and 2N.

3. Explain Why ?

(a) "The number of atoms in a certain volume of hydrogen is twice the number of atoms in the same volume of helium at the same temperature and pressure."

(b) "When stating the volume of a gas, the pressure and temperature should also be given."

(c) Inflating a baloon seems to violate Boyle's law.

NUMERICAL PROBLEMS

4. (a) Calculate the volume of oxygen at STP required for the complete combustion of 100 litres of carbon monoxide at the same temperature and pressure.

$$2CO + O_2 \rightarrow 2CO_2$$

(b) 200 cm^3 of hydrogen and 150 cm^3 of oxygen are mixed and ignited, as per following reaction,

$$2H_2 + O_2 \rightarrow 2H_2O$$

What volume of oxygen remains unreacted ?

5. 24 cc Marsh gas (CH_4) was mixed with 106 cc oxygen and then exploded. On cooling, the volume of the mixture became 82 cc, of which, 58 cc was unchanged oxygen. Which law does this experiment support? Explain with calculations.

6. What volume of oxygen would be required to burn completely 400 mL of acetylene [C_2H_2] ? Also, calculate the volume of carbon dioxide formed.

$$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O\ (l)$$

7. 112 cm^3 of $H_2S(g)$ is mixed with 120 cm^3 of $Cl_2(g)$ at STP to produce HCl(g) and sulphur(s). Write a balanced equation for this reaction and calculate (i) the volume of gaseous product formed (ii) composition of the resulting mixture.

8. 2500 cc of oxygen was burnt with 600 cc of ethane [C_2H_6]. Calculate the volume of unused oxygen and the volume of carbon dioxide formed, after writing a balanced equation :

Ethane + Oxygen $\rightarrow$ Carbon dioxide + Water vapour

9. What volume of oxygen at STP is required to affect the combustion of 11 litres of ethylene [C_2H_4] at 273°C and 380 mm of Hg pressure?

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$$

10. Calculate the volume of HCl gas formed and chlorine gas required when 40 mL of methane reacts completely with chlorine at STP.

$$CH_4 + 2Cl_2 \rightarrow CH_2Cl_2 + 2HCl$$

11. What volume of propane is burnt for every 500 cm^3 of air used in the reaction under the same conditions ? (Assuming oxygen is 1/5th of air)

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

12. 450 cm^3 of nitrogen monoxide and 200 cm^3 of oxygen are mixed together and ignited. Calculate the composition of resulting mixture.

$$2NO + O_2 \rightarrow 2NO_2$$

13. If 6 litres of hydrogen and 4 litres of chlorine are mixed and exploded and if water is added to the gases formed, find the volume of the residual gas. **(2015)**

14. Ammonia may be oxidised to nitrogen monoxide in the presence of a catalyst according to the following equation.

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

If 27 litres of reactants are consumed, what volume of nitrogen monoxide is produced at the same temperature and pressure ?

15. A mixture of hydrogen and chlorine occupying 36 cm^3 was exploded. On shaking it with water, 4 cm^3 of hydrogen was left behind. Find the composition of the mixture.

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16. What volume of air (containing 20% O_2 by volume) will be required to burn completely 10 cm^3 each of methane and acetylene.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$
$$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$$

17. LPG has 60% propane and 40% butane : 10 litres of this mixture is burnt calculate the volume of carbondioxide added to atmosphere.

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$
$$2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$$

18. 200 cm^3 of CO_2 is collected at STP when a mixture of acetylene and oxygen is ignited. Calculate the volume of acetylene and oxygen at STP in original mixture

$$2C_2H_2\,(g) + 5O_2\,(g) \rightarrow 4\,CO_2\,(g) + 2H_2O\,(g)$$

19. You have collected (a) 2 litres of CO_2 (b) 3 litres of chlorine (c) 5 litres of hydrogen (d) 4 litres of nitrogen and (e) 1 litre of SO_2, under similar conditions of temperature and pressure. Which gas sample will have :

(a) the greatest number of molecules, and

(b) the least number of molecules ?

Justify your answers.

20. The gases chlorine, nitrogen, ammonia and sulphur dioxide are collected under the same conditions of temperature and pressure. The following table gives the volumes of gases collected and the number of molecules (x) in 20 litres of nitrogen. You are to complete the table giving the number of molecules in the other gases in terms of x.

Gas	Volume (in litres)	Number of molecules
Chlorine	10	
Nitrogen	20	x
Ammonia	20	
Sulphur dioxide	5	

21. If 100 cm^3 of oxygen contains Y molecules, how many molecules of nitrogen will be present in 50 cm^3 of nitrogen under the same conditions of temperature and pressure?

ANSWERS

4. (a) 50 litres (b) 50 cm^3 **6.** 1000 mL of O_2, 800 mL of CO_2 **7.** (i) 224 cm^3 HCl (ii) 224 cm^3 HCl + 8 cm^3 Cl_2 **8.** 400cc, 1200cc. **9.** 8·25 litres **10.** HCl = 80 mL, Cl_2 = 80 mL **11.** 20 cc **12.** Unused NO = 50 cm^3, NO_2 formed = 400 cm^3, Total mixture = 450 cm^3. **13.** 2 litres of hydrogen as HCl form dissolves in water **14.** 12 litres **15.** 20 cm^3 of hydrogen, 16 cm^3 of chlorine **16.** 225 cm^3 **17.** 34 litres **18.** Acetylene 100 cm^3, oxygen 250 cm^3 **19.** (a) H_2 (b) SO_2 **20.** $x/2$, x, $x/4$ **21.** $\frac{y}{2}$ molecules

5B. RELATIVE ATOMIC MASS, RELATIVE MOLECULAR MASS AND MOLE CONCEPT

5.4 RELATIVE ATOMIC MASS (ATOMIC WEIGHT)

Atoms being extremely small, cannot be seen or weighed directly. But indirect methods of physics have enabled us to know the absolute mass of nearly all kinds of atoms. The mass of a hydrogen atom is found to be $1{\cdot}6735 \times 10^{-24}$ g while that of an oxygen atom is $26{\cdot}565 \times 10^{-24}$ g. As these masses are too small, it is not convenient to use kilograms or grams as unit. It has, therefore, been considered appropriate to use the mass of some standard atom as a unit and then relate masses of other atoms to it. The resulting masses of atoms are thus ***Relative Atomic Masses*** *(RAM)* or ***Atomic Weight.***

In the beginning, the mass of the hydrogen atom (hydrogen element being the lightest) was chosen as a unit and masses of other atoms were compared with it. In 1961, *carbon-12 was finally selected, because its adoption least affected the values of the atomic mass of the various elements on the old standard.*

The **relative atomic mass** or ***atomic weight*** *of an element is the number of times one atom of the element is heavier than* $\frac{1}{12}$ *times of the mass of an atom of carbon-12. Thus:*

$$\text{Relative atomic mass} = \frac{\text{Mass of 1 atom of the element}}{\frac{1}{12}\text{th the mass of one C-12 atom}}$$

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Atomic mass is expressed in atomic mass units [a.m.u.]. **Atomic mass unit is defined as 1/12 the mass of carbon atom C-12.**

Fractional atomic masses

It is observed that most atomic masses are not whole numbers. The reason is that most natural elements are a mixture of constant composition containing two or more isotopes.* The relative atomic mass of any element is the weighted average of the relative atomic masses of its natural isotopes. For example, chlorine consists of a mixture of two isotopes of masses 35 and 37 in the ratio of 3:1.

The average relative atomic mass of chlorine $= \left(\frac{35\times3+37\times1}{4} = 35{\cdot}5\right)$

Note : The mass of an ion will be same as that of parent atom. The mass of electron(s) gained or lost in forming the ion can be ignored in comparison to the total mass of the atom.

5.5 RELATIVE MOLECULAR MASS (MOLECULAR WEIGHT)

The relative molecular mass (or molecular weight) of an element or a compound is the number that represents how many times one molecule of the substance is heavier than 1/12 of the mass of an atom of carbon-12.

The Relative Molecular Mass (RMM) is obtained by adding together the relative atomic masses (atomic weights) of all the various atoms present in a molecule.

For example, relative molecular mass of sulphuric acid (H_2SO_4)

$$= 2 \times 1 + 32 + 4 \times 16 = 98$$

Thus, molecular mass of H_2SO_4 is 98 a.m.u. *i.e. one molecule of H_2SO_4 is 98 times as heavy as 1/12 the mass of carbon atom C-12.*

Example 1 : *Calculate the molecular masses (or molecular weights) of the following compounds:*

(a) Copper sulphate crystals, $CuSO_4 \cdot 5H_2O$

(b) Ammonium sulphate, $(NH_4)_2SO_4$

Given that the relative atomic masses of Cu = 63·5, S = 32, O = 16, N = 14 and H = 1.

Solution :

(a) The molecular mass of $CuSO_4 \cdot 5H_2O$

$= 63{\cdot}5 + 32 + (16 \times 4) + 5(1 \times 2 + 16)$

$= 159{\cdot}5 + 90 =$ **249·5 a.m.u.** **Ans.**

(b) The molecular mass of $(NH_4)_2SO_4$

$= 2(14 + 1 \times 4) + 32 + (16 \times 4)$

$= 2 \times 18 + 32 + 64$

$= 36 + 32 + 64 =$ **132 a.m.u.** **Ans.**

5.6 GRAM ATOMIC MASS

The atomic mass of an element expressed in grams is called **gram atomic mass.**

For example, the atomic mass of oxygen is 16 a.m.u., therefore its gram atomic mass is 16 g.

The quantity of the element which weighs equal to its gram atomic mass is called ***one gram atom*** *of that element.*

For example 16 g of oxygen is equal to *one gram atom* of oxygen. Similarly,

1 g atom of sodium = 23 gm of sodium

$\Rightarrow$ 1 g atom of He = 4 gm of helium

5.7 GRAM MOLECULAR MASS

The molecular mass of a substance expressed in grams is called gram molecular mass or molar mass. A sample of a substance with its mass equal to its gram molecular mass is called *one gram molecule of the substance.*

The molecular mass of water (H_2O) is 18·00 a.m.u., and its gram molecular mass is 18·00 g. Similarly, the relative molecular mass of ammonia (NH_3) is 17·00, therefore, its given molecular mass is 17·00 g.

5.8 MOLE CONCEPT

It is not practically possible to find the mass of a minute particle like an atom, a molecule, or an ion etc. Therefore, a collection of $6{\cdot}022 \times 10^{23}$ elementary particles called **Mole** is taken for all practical purposes.

Mole is simply like a dozen or a gross. A dozen is a collection of 12 objects, a gross is a collection

* *Atoms of the same element having same atomic number but different mass number are known as ISOTOPES.*

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of 144 objects, similarly **a mole is a collection of 6·022 × 10^{23} particles** (atoms or molecules or ions).

A **mole** is the amount of pure substance containing the same number of chemical units as there are atoms in exactly 12 grams of carbon –12.

Note : When the particle is not specifically mentioned, then one mole is taken as 6·02 × 10^{23} molecule.

Avogadro's number *is defined as the number of atoms present in 12 g (gram atomic mass) of C-12 isotope, i.e., 6·022 × 10^{23} atoms.*

OR

Avogadro's number is the number of elementary units, i.e., atoms, ions or molecules present in one mole of a substance.

It is denoted by N_A.

5.8.1. Mole of atoms

"One mole of atoms contain 6·022 × 10^{23} atoms having mass equal to its gram atomic mass."

Thus, one mole of oxygen atoms contain 6·022 × 10^{23} atoms of oxygen and weighs 16 g.

5.8.2 Mole of Molecules

One mole contain 6·022 × 10^{23} molecules and is equivalent to the gram molecular mass of any given substance. Thus 1 mole of O_2 contain 6·022 × 10^{23} molecules and weighs 32 g.

1 mole of water contain 6·022 × 10^{23} molecules and weighs 18 g.

Mole and molar volume : It has been noticed that one mole of any gaseous molecule occupy 22·4 dm^3 (litre) or 22400 cm^3 (ml) at S.T.P. This volume is known as **molar volume.**

The molar volume of a gas can be defined as the volume occupied by one mole of a gas at S.T.P.

One G.M.M. of the substance = **1 mole** of that substance = **6·02 × 10^{23} molecules** of that substance = **22·4 *l* at S.T.P.** if substance is a gas

Unit : *1 litre = 1 dm^3 = 1000 cm^3 = 1000 ml*

Substance	Moles	Molar Mass (mass in grams)	Volume at S.T.P. in litres or dm^3.	No. of molecules	No. of atoms
Sulphur dioxide (SO_2)	1	64	22·4	6·022 × 10^{23}	S atoms – 6·022 × 10^{23} O atoms – 2 × 6·022 × 10^{23}
Ammonia (NH_3)	1	17	22·4	6·022 × 10^{23}	N atoms – 6·022 × 10^{23} H atoms – 3 × 6·022 × 10^{23}
Calcium Chloride ($CaCl_2$)	1	111	—	6·022 × 10^{23}	Ca^{2+} ions – 6·022 × 10^{23} Cl^- ions – 2 × 6·022 × 10^{23}

Example 2 : *The number of atoms in one mole (one g-atom of an element) is 6 × 10^{23}. Calculate,*

(a) the number of molecules in 14 g of nitrogen gas.

(b) the total number of atoms in 18 g of water.

(c) the number of chloride ions in 111 g of anhydrous calcium chloride.

Solution :

Molecular weight of any substance contain 6 × 10^{23} molecules.

(a) mass of 1 mole of nitrogen is 28 g.

28 g of nitrogen gas contains 6 × 10^{23} molecules.

$\therefore$ 14 g of nitrogen contains $\frac{6\times10^{23}\times14}{28}$

= 3 × 10^{23} molecules.

(b) Mass of 1 mole of water is 18 g.

$\therefore$ 18 g of water contains 6 × 10^{23} molecules

But H_2O contains 3 atoms.

(2 atoms of hydrogen + 1 atom of oxygen)

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$\therefore$ 18 g of water contains

$3 \times 6 \times 10^{23}$ atoms $= 18 \times 10^{23}$ atoms.

(c) Mass of 1 mole of anhydrous calcium chloride ($CaCl_2$) = 40 + 71 = 111 g

But 1 molecule of $CaCl_2$ contains 2 chloride ions.

$\therefore$ 111 g of anhydrous $CaCl_2$ contains

$2 \times 6 \times 10^{23} = 12 \times 10^{23}$ chloride ions.

Answer : (a) 3×10^{23} molecules.

(b) 18×10^{23} atoms.

(c) 12×10^{23} chloride ions.

Example 3 : *How many moles of NaOH are present in 160 g of it ?*

Solution :

Mass of 1 mole of NaOH = 23 + 16 + 1 = 40 g.

$\therefore$ Moles in 160 g NaOH $= \frac{160}{40} = 4$ moles

Answer : There are 4 moles of NaOH in 160 g of it.

Example 4 : *What is the mass in grams of 2·5 moles of slaked lime $Ca(OH)_2$?*

Solution :

Mass of 1 mole of slaked lime $Ca(OH)_2$

$= 40 + 2 \times (16 + 1)$ g = 74 g.

$\therefore$ 2·5 moles of Ca $(OH)_2 = 2{\cdot}5 \times 74 = 185$ g.

Answer : 2·5 moles of Ca $(OH)_2$ weigh 185 g.

Example 5 : *Calculate the mass of*

(i) *one atom of oxygen and*

(ii) *one molecule of oxygen.*

Solution :

(i) Number of oxygen atoms in 16·0 g of atomic oxygen = $6{\cdot}022 \times 10^{23}$ atoms.

$\therefore$ Mass of one atom of oxygen

$$= \frac{16{\cdot}0}{6{\cdot}022 \times 10^{23}} = 2{\cdot}657 \times 10^{-23} \text{ g}$$

Answer : Mass of one atom of oxygen is $2{\cdot}657 \times 10^{-23}$ g.

(ii) Gram molecular mass of oxygen = 32·0 g

Number of O_2 molecules in 32·0 g of O_2

$= 6{\cdot}022 \times 10^{23}$ molecules

$\therefore$ Mass of one molecule of oxygen

$$= \frac{32{\cdot}0}{6{\cdot}022 \times 10^{23}} = 5{\cdot}314 \times 10^{-23} \text{ g.}$$

Answer : Mass of one molecule of oxygen is $5{\cdot}314 \times 10^{-23}$ g.

Example 6 : *How many moles of sulphur atoms and oxygen atoms are present in one mole each of H_2SO_4, H_2SO_3 and SO_2 ?*

Solution :

One mole of H_2SO_4 contains 1 mole of sulphur atoms and 4 moles of oxygen atoms.

One mole of H_2SO_3 has 1 mole of sulphur atoms and 3 moles of oxygen atoms.

One mole of SO_2 has 1 mole of sulphur atoms and 2 moles of oxygen atoms.

Example 7 : *A silicon chip used in an integrated circuit of a microcomputer has a mass of 5.68 mg. How many silicon (Si) atoms are present in this chip?*

Solution :

Gram atomic mass of silicon is 28 g.

1g atom of silicon contain $6{\cdot}022 \times 10^{23}$ atoms

$\therefore$ 28g of silicon contain $6{\cdot}022 \times 10^{23}$ atoms

or 28000 mg of silicon contain $6{\cdot}022 \times 10^{23}$ atom

5.68 mg of silicon contain $\frac{6{\cdot}022 \times 10^{23}}{28000} \times 5{\cdot}68$

$= 1{\cdot}22 \times 10^{20}$ atoms

Example 8 : *Find the number of sodium and sulphate ions in 14·2 gram of sodium sulphate.*

Solution :

Molecular mass of Na_2SO_4

$= 2 \times 23 + 1 \times 32 + 4 \times 16 = 142$ a.m.u.

$\therefore$ 142 g $Na_2SO_4 = 1$ mole

14·2 g $Na_2SO_4 = 0{\cdot}1$ mole

Total molecules $= 0{\cdot}1 \times 6{\cdot}02 \times 10^{23}$

$= 6{\cdot}02 \times 10^{22}$

1 molecule of Na_2SO_4 contain two Na^+ ions and one SO_4^{2-} ions.

$\therefore$ Na^+ ions present $= 2 \times 6{\cdot}02 \times 10^{22}$

$= \mathbf{1{\cdot}204 \times 10^{23}}$

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SO_4^{2-} ions present = $1 \times 6{\cdot}02 \times 10^{22}$

$= \mathbf{6{\cdot}02 \times 10^{22}}$

Example 9 : *Calculate the total number of electrons present in 1·6 gram of methane.*

Solution :

Molecular mass of CH_4

$= 12 + 1 \times 4 = 16$ a.m.u.

16 g = 1 mole

1·6 g = 0·1 mole

1 mole of CH_4 has $6{\cdot}02 \times 10^{23}$ molecule.

0·1 has $6{\cdot}02 \times 10^{22}$ molecule.

1 molecule of CH_4 contain $(6 + 1 \times 4) = 10$ electrons

$6{\cdot}02 \times 10^{22}$ molecules will contain

$= 10 \times 6{\cdot}02 \times 10^{22}$

$= \mathbf{6{\cdot}02 \times 10^{23}}$ **electrons.**

5.9 APPLICATIONS OF AVOGADRO'S LAW

1. It explains Gay-Lussac's law.
2. It determines atomicity of the gases.
3. It determines the molecular formula of a gas.
4. It determines the relation between molecular mass and vapour density.
5. It gives the relationship between gram molecular mass and gram molar volume.

(1) *Avogadro's Law explains Gay-Lussac's Law.*

According to Avogadro's Law, under the same conditions of temperature and pressure, equal volumes of different gases have the same number of molecules.

Since substances react in simple ratio by the number of molecules, volumes of the gaseous reactants and products will also bear a simple ratio to one another. This is what Gay-Lussac's Law says.

H_2	+	Cl_2	→	2HCl
1 vol		1 vol		2 vol (by Gay-Lussac's Law)
n		n		2n (by Avogadro's Law)
molecules		molecules		molecules

(2) *Avogadro's Law predicts atomicity of gases.*

The application of Avogadro's Law to determine atomicity in elementary gases is illustrated by the following examples :

(i) *Atomicity of molecules of hydrogen and chlorine gases.*

In the reaction between hydrogen and chlorine, the volumes bear the following ratio:

H_2	+	Cl_2	→	2 HCl
1 vol		1 vol		2 vol (Gay-Lussac's Law)

Applying Avogadro's Law, x molecules of hydrogen combine with x molecules of chlorine to give $2x$ molecules of hydrogen chloride.

or 1 molecule of H_2 + 1 molecule of Cl_2 → 2 molecules of HCl

or $\frac{1}{2}$ molecule of H_2 + $\frac{1}{2}$ molecule of Cl_2 → 1 molecule of HCl

Since atoms are indivisible, half a molecule each of hydrogen and chlorine must contain at least one atom each and consequently one molecule each of hydrogen and chlorine must contain two atoms each.

(ii) *Atomicity of nitrogen and oxygen.*

Nitrogen	+	oxygen	→	Nitric oxide
N_2	+	O_2	→	2NO
1 vol		1 vol		2 vol
1 molecule of Nitrogen	+	1 molecule of Oxygen	→	2 molecules of Nitric Oxide
$\frac{1}{2}$ molecule	+	$\frac{1}{2}$ molecule	→	1 molecule

Half molecule of nitrogen is equal to one atom of nitrogen as an atom is indivisible.

So, one molecule of nitrogen and one molecule of oxygen contain two atoms each.

(3) *Determination of the molecular formula of a gaseous compound*

Molecular formula is a chemical formula which gives the actual number of atoms of the elements present in one molecule of a compound.

It is experimentally found that 2 volumes of ammonia decompose to give 1 volume of nitrogen and 3 volumes of hydrogen.

	Ammonia	→	Nitrogen	+	Hydrogen
	2 volumes		1 volume		3 volumes (by experiment)
	2 molecules		1 molecule		3 molecules (by Avogadro's Law)
or,	2 molecules		2 atoms		6 atoms
or,	1 molecule		1 atom		3 atoms

(since the molecules of nitrogen and hydrogen are diatomic).

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Therefore, one molecule of ammonia contains 1 atom of nitrogen and 3 atoms of hydrogen. Hence, its molecular formula is NH_3.

(4) *It establishes a relationship between the relative vapour density of a gas and its relative molecular mass.*

The **Relative Vapour Density** of a gas (or a vapour) is :

The ratio between the masses of equal volumes of gas (or vapour) and hydrogen under the same conditions of temperature and pressure.

$$\text{Relative V.D.} = \frac{\text{Mass of volume 'v' of the gas under similar conditions}}{\text{Mass of volume 'v' of hydrogen gas under similar conditions}}$$

According to Avogadro's Law, volumes at the same temperature and pressure may be substituted by molecules. Hence,

$$\text{Relative V.D.} = \frac{\text{Mass of 1 molecule of gas or vapour}}{\text{Mass of 2 atoms of hydrogen}}$$

(Molecule of hydrogen contains 2 atoms) By multiplying both sides by 2

$$2 \times \text{Rel. V.D.} = \frac{\text{Mass of 1 molecule of gas or vapour}}{\text{Mass of 1 atom of hydrogen}}$$

2 × Rel. V.D. = Rel. molecular mass of a gas or vapour.

The relative molecular mass of a gas or vapour is twice its vapour density.

For example, the relative vapour density of gaseous chlorine is 35·5, hence its relative molecular mass would be 2 × 35·5 or 71·0.

(5) *It gives the value of molar volume of gases at S.T.P., and introduces the concept of Avogadro's number.*

The **molar volume** of a gas is *the volume occupied by one gram-molecular mass or simply, by one mole of the gas at S.T.P. It is equal to 22·4 dm^3.*

The value of molar volume is deduced from Avogadro's Law as follows :

It is found by direct weighing that 1 dm^3 (or 1 L) of hydrogen at S.T.P. weighs 0·089 g. Therefore, 1 dm^3 of any gas, whose vapour density at S.T.P. is D, would weigh D × 0·089 g at S.T.P.

OR

D × 0·089 g of any gas occupies 1 dm^3 (1 L).

∴ D g of any gas will occupy $\frac{1}{0{\cdot}089}$ or 11·2 dm^3 or 11·2 L

or 2D of any gas will occupy 22·4 dm^3 or 22·4 L.

But, 2 × V.D. = Molecular mass.

Thus, gram molecular mass of any gas occupies 22·4 litres at S.T.P., which contains 6·02 × 10^{23} molecules of that gas and is equal to 1 mole of that gas.

Example 10 : *Calculate the atomicity of chlorine, if 35·5 g of it occupies 11,200 cm^3 at S.T.P.*

Solution :

[1 mole of any substance = 1 g molecular mass of it and occupies 22400 cm^3 at S.T.P.].

1 mole of chlorine = 1 *g* mol. mass of chlorine and occupies 22,400 cm^3 at S.T.P.

Given :

1 gm atom of chlorine occupies 11,200 cm^3.

How many grams of chlorine will occupy 22,400 cm^3 at S.T.P.

$$= \frac{22400 \times 35{\cdot}5}{11200} = 71$$

∴ gram molecular mass = 71 g

$$\therefore \text{Atomicity} = \frac{\text{Mol. mass}}{\text{At. mass}} = \frac{71}{35{\cdot}5} = 2$$

∴ Atomicity of Chlorine = 2

Example 11 : *Calculate the volume occupied by 2·8 g of N_2 at S.T.P. ?*

Solution :

Gram molecular mass of $N_2 = 2 \times 14 = 28$ g

28 g of N_2 at S.T.P. occupies 22·4 litres

∴ 2·8 g of N_2 at S.T.P. will occupy

$$\frac{22{\cdot}4 \times 28}{28} = 2{\cdot}24 \text{ litres}$$

Example 12 : *Calculate*

(a) *the gram molecular mass of N_2, if 360 cm^3 at S.T.P. weighs 0·45 g;*

(b) *the number of gram molecule of water in 4·5 g of water.*

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Solution :

The mass of 22·4 L of a gas at S.T.P. is equal to its gram molecular mass.

(a) 360 cm^3 of N_2 at S.T.P. weighs 0·45 g

$\therefore$ 22,400 cm^3 of N_2 will weigh

$$\frac{0{\cdot}45}{360} \times 22{,}400 = 28 \text{ g.}$$

Answer : Gram Molecular Mass of N_2 = 28 g

(b) No. of gram molecules $= \dfrac{\text{Mass in grams of water}}{\text{Gram molecular mass}}$

$$= \frac{4{\cdot}5\ g}{18\ g} = 0{\cdot}25 \text{ g molecules.}$$

Answer : 0·25 g molecules are present in 4·5 g of water.

Example 13 : *What volume of oxygen is required to completely burn a mixture of 224 cm^3 of methane and 112 cm^3 of hydrogen at S.T.P. to produce steam and carbon dioxide ? If the steam formed is condensed to water, find the mass of water so formed.*

Solution :

For combustion of methane :

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

According to Gay-Lussac's Law, 1 volume of methane requires 2 volumes of oxygen to burn it completely, giving 1 volume of carbon dioxide and 2 volumes of steam.

Hence, 224 cm^3 of methane will require 448 cm^3 of oxygen for complete combustion and give 224 cm^3 of carbon dioxide and 448 cm^3 of steam.

For combustion of hydrogen :

$$2H_2 + O_2 \rightarrow 2H_2O$$

According to Gay-Lussac's Law, 2 volumes of hydrogen requires 1 volume of oxygen for complete combustion to give 2 volumes of steam.

$\therefore$ 112 cm^3 of hydrogen will require 56 cm^3 of oxygen for complete combustion and thereby form 112 cm^3 of steam.

$\therefore$ Total volume of oxygen needed for burning CH_4 and H_2 = 448 + 56 = 504 cm^3 and total volume of steam formed = 448 + 112 = 560 cm^3.

Molecular mass of H_2O = 2 + 16 = 18 amu

22400 cm^3 of steam at S.T.P. weighs 18 g.

$\therefore$ 560 cm^3 of steam at S.T.P. weighs

$$= \frac{18 \times 560}{22400} = 0{\cdot}45 \text{ g}$$

Answer : Volume of oxygen required is 504 cm^3 and the mass of water formed would be 0·45 g.

Example 14 : *A gas cylinder filled with hydrogen holds 50 g of the gas. The same cylinder holds 200 g of a gas X and 500 g of gas Y. Considering the same conditions of temperature and pressure in the cylinder, calculate the relative molecular masses of gases X and Y.*

Solution :

Vapour Density of a gas

$$= \frac{\text{Mass of a certain volume of a gas at S.T.P.}}{\text{Mass of an equal volume of } H_2 \text{ at S.T.P.}}$$

For Gas X

$$\text{Vapour Density} = \frac{200}{50} = 4$$

$\therefore$ Relative molecular mass of gas X = 2 × V.D. = 2 × 4 = 8

For Gas Y

$$\text{Vapour Density} = \frac{500}{50} = 10$$

$\therefore$ Relative molecular mass of gas Y = 2×V.D. = 2 × 10 = 20

Answer : Relative molecular mass of gas X = 8 and Gas Y = 20

Example 15 : 0·583 g of acetone when vaporized formed 225 cm^3 of vapours at S.T.P. Calculate the gram molecular mass of the gas. [22·4 L. of any gas at stp = 1 g. mol. mass of gas]

Solution :

225 cm^3 of acetone at S.T.P. weighs = 0·583g

$\therefore$ 22400 cm^3 of acetone at stp will weigh

$$= \frac{0{\cdot}583 \times 22400}{225} = 58{\cdot}04 \text{ g.} \qquad \textbf{Ans.}$$

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Example 16 : *Calculate the V.D. and molecular mass of CO_2 if 200 mL of the gas at S.T.P. weighs 0·40 g.*

[1 L of H_2 at S.T.P. weighs 0·09 g.]

Solution :

200 mL of CO_2 at S.T.P. weighs 0·40 g.

$\therefore$ 1 L of CO_2 at S.T.P. will weigh

$$= \frac{0{\cdot}40 \times 1000}{200} = 2 \text{ g.}$$

$$\text{Vapour density} = \frac{\text{Wt. of 1000 mL of } CO_2 \text{ at S.T.P.}}{\text{Wt. of 1000 mL of } H_2 \text{ at S.T.P.}}$$

$$= \frac{2}{009} = \mathbf{22{\cdot}22} \qquad \textbf{\textit{Ans.}}$$

Molecular mass = 2 × V.D.

$$= 2 \times 22{\cdot}22 = \mathbf{44{\cdot}44 \text{ g.}} \qquad \textbf{\textit{Ans.}}$$

Example 17 : *A gas at a pressure of 700 mm of Hg and a temperature of 57°C occupies 700 mL. If at S.T.P. the mass of the gas is 1.5 g, find the vapour density and the molecular mass of the gas (1 L of H*

$\therefore$ Molecular mass = 2 × V.D. = 2 × 31·24

$$= \mathbf{62{\cdot}50}$$

Answer : Vapour density of the gas is 31·25

Molecular weight is 62·50.

Example 18 : *Calculate the mass of a substance A which in gaseous form occupies 10 dm^3 at 27°C and 700 mm pressure. The molecular mass of A is 60g.*

Solution :

$P_1 = 700$ mm of Hg $\qquad P_2 = 760$ mm of Hg

$V_1 = 10$ dm^3 $\qquad V_2 = \chi$ dm^3

$T_1 = 27 + 273 = 300$ K $\qquad T_2 = 273$ K

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{700 \times 10}{300} = \frac{760 \times \chi}{273}$$

$$\therefore V_2 = \frac{700 \times 10 \times 273}{\quad} = \mathbf{8{\cdot}38 \text{ dm}^3}$$

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Important Formulae for Revision

Mole and Gram Atomic Mass :	One mole of atoms	= $6{\cdot}022 \times 10^{23}$ atoms = Gram atomic mass of an element = 1 g atom of the element.
Mole and Gram Molecular Mass :	One mole of molecules	= $6{\cdot}022 \times 10^{23}$ molecules = Gram molecular mass = 1 g molecule of the compound.
Mole in terms of Volume :	One mole of a gas	= 22·4 litres at S.T.P.

Moles of an element = $\frac{\text{Mass of the element}}{\text{Atomic mass or G.A.W.}}$	**Moles of a compound** = $\frac{\text{Mass of the compound}}{\text{Molecular mass or G.M.W}}$
Mass of one atom = $\frac{\text{Atomic Mass or G.A.W.}}{6{\cdot}022 \times 10^{23}}$	**Mass of one molecule** = $\frac{\text{Molecular Mass or G.A.W.}}{6{\cdot}022 \times 10^{23}}$
Number of molecules = Moles × $6{\cdot}022 \times 10^{23}$	**Number of atoms** = Moles × $6{\cdot}022 \times 10^{23}$.

[a] **A mole of atom** = Avogadro's number of atoms = Gram atomic mass of that element.

Examples

- 1 mole of hydrogen atom [H] = $6{\cdot}022 \times 10^{23}$ atoms of hydrogen = One gram of hydrogen
- 1 mole of sodium atom [Na] = $6{\cdot}022 \times 10^{23}$ atoms of sodium = 23g of sodium

[b] **A mole of molecule** = Avogadro's number of molecules = Gram molecular mass of that substance.

Examples

- 1 mole of hydrogen molecule = $6{\cdot}022 \times 10^{23}$ molecules of hydrogen = 2 grams of hydrogen [$H_2 = 1 \times 2$]
- 1 mole of water molecule = $6{\cdot}022 \times 10^{23}$ molecules of water = 18 grams of water [$H_2O = 1 \times 2 + 16$]

[c] **A mole of an ionic compound** = $6{\cdot}022 \times 10^{23}$ formula units of that compound = Mass equal to the formula mass of that substance.

Example

- 1 mole of $CaCl_2$ = $6{\cdot}022 \times 10^{23}$ $CaCl_2$ units = 111 g of $CaCl_2$ [$CaCl_2 = 40 + 2 \times 35{\cdot}5$]
  i.e. $6{\cdot}022 \times 10^{23}$ Ca^{2+} ions
  i.e. 1 mole of Ca^{2+} ions
  and $2 \times 6{\cdot}022 \times 10^{23}$ Cl^- ions
  i.e. 2 moles of Cl^- ions

EXERCISE-5B

1. (a) The relative atomic mass of Cl atom is 35·5 a.m.u. Explain this statement.
(b) What is the value of Avogadro's number ?
(c) What is the value of molar volume of a gas at S.T.P.?

2. Define or explain the terms :
(a) vapour density, (b) molar volume,
(c) relative atomic mass, (d) relative molecular mass,
(e) Avogadro's number, (f) Gram atom,
(g) Mole.

3. (a) What are the main applications of Avogadro's Law ?
(b) How does Avogadro's Law explain Gay-Lussac's Law of combining volumes?

4. Calculate the relative molecular masses of :
(a) Ammonium chloroplatinate $(NH_4)_2\ PtCl_6$,
(b) Potassium chlorate, (c) $CuSO_4 \cdot 5H_2O$,
(d) $(NH_4)_2SO_4$ (e) $CH_3\ COONa$,
(f) $CHCl_3$, (g) $(NH_4)_2\ Cr_2O_7$.

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5. Find the :
(a) number of molecules in 73 g of HCl,
(b) weight of 0·5 mole of O_2,
(c) number of molecules in 1·8 g of H_2O,
(d) number of moles in 10 g of $CaCO_3$,
(e) weight of 0·2 mole of H_2 gas,
(f) number of molecules in 3·2 g of SO_2.

6. Which of the following would weigh most :
(a) 1 mole of H_2O, (b) 1 mole of CO_2,
(c) 1 mole of NH_3, (d) 1 mole of CO ?

7. Which of the following contains maximum number of molecules :
(a) 4 g of O_2, (b) 4 g of NH_3,
(c) 4 g of CO_2, (d) 4 g of SO_2 ?

8. Calculate the number of :
(a) particles in 0·1 mole of any substance,
(b) hydrogen atoms in 0·1 mole of H_2SO_4,
(c) molecules in one kg of calcium chloride.

9. How many grams of :
(a) Al are present in 0·2 mole of it,
(b) HCl are present in 0·1 mole of it ?
(c) H_2O are present in 0·2 mole of it,
(d) CO_2 is present in 0·1 mole of it ?

10. (a) The mass of 5·6 litres of a certain gas at STP is 12 g. What is the relative molecular mass or molar mass of the gas ?
(b) Calculate the volume occupied at S.T.P. by 2 moles of SO_2.

11. Calculate the number of moles of :
(a) CO_2 which contain 8·00 g of O_2,
(b) methane in 0·80 g of methane.

12. Calculate the actual mass of :
(a) an atom of oxygen, (b) an atom of hydrogen,
(c) a molecule of NH_3, (d) the atom of silver,
(e) the molecule of oxygen,
(f) 0·25 gram atom of calcium.

13. Calculate the mass of 0·1 mole of each of the following:
(a) $CaCO_3$, (b) $Na_2SO_4 \cdot 10H_2O$,
(c) $CaCl_2$, (d) Mg.
(Ca = 40, Na = 23, Mg = 24, S = 32, C = 12, Cl = 35·5, O = 16, H = 1)

14. Calculate the number of oxygen atoms in 0·10 mole of $Na_2CO_3 \cdot 10H_2O$.

15. What mass of Ca will contain the same number of atoms as are present in 3·2 g of S ? **(2015)**

16. Calculate the number of atoms in each of the following :
(a) 52 moles of He, (b) 52 amu of He,
(c) 52 g of He.

17. Calculate the number of atoms of each kind in 5·3 grams of sodium carbonate.

18. (a) Calculate the mass of nitrogen supplied to soil by 5 kg of urea $[CO(NH_2)_2]$.
[O = 16; N = 14; C =12; H = 1]
(b) Calculate the volume occupied by 320 g of sulphur dioxide at STP [S = 32; O = 16]

19. (a) What do you understand by the statement that 'vapour density of carbon dioxide is 22' ?
(b) Atomic mass of chlorine is 35·5. What is its vapour density ?

20. What is the mass of 56 cm^3 of carbon monoxide at STP? [C = 12; O = 16]

21. Determine the no. of molecules in a drop of water which weighs ·09 g.

22. The molecular formula for elemental sulphur is S_8. In a sample of 5·12 g of sulphur :
(a) How many moles of sulphur are present,
(b) How many molecules and atoms are present ?

23. If phosphorus is considered to contain P_4 molecules, then calculate the number of moles in 100 g of phosphorus?

24. Calculate :
(a) the gram molecular mass of chlorine if 308 cm^3 of it at STP weighs 0·979 g,
(b) the volume of 4 g of H_2 at 4 atmosphere,
(c) the mass of oxygen in 2·2 litres of CO_2 at STP.

25. A student puts his signature with graphite pencil. If the mass of carbon in the signature is 10^{-12}g, calculate the number of carbon atoms in the signature.

26. An unknown gas shows a density of 3 g per litre at 273°C and 1140 mm Hg pressure. What is the gram molecular mass of this gas ?

27. Cost of Sugar ($C_{12}H_{22}O_{11}$) is ₹ 40 per kg; calculate its cost per mole.

28. Which of the following weighs the least ?
(a) 2 g atom of N (b) 3×10^{25} atoms of carbon
(c) 1 mole of sulphur (d) 7 g silver

29. Four grams of caustic soda contains :
(a) $6·02 \times 10^{23}$ atoms of it,

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(b) 4 g atom of sodium,

(c) $6{\cdot}02 \times 10^{22}$ molecules (d) 4 moles of NaOH.

30. The number of molecules in 4·25 g of ammonia is :

(a) $1{\cdot}0 \times 10^{23}$ (b) $1{\cdot}5 \times 10^{23}$,

(c) $2{\cdot}0 \times 10^{23}$ (d) $3{\cdot}5 \times 10^{23}$.

31. Correct the statements, if required.

(a) One mole of chlorine contains $6{\cdot}023 \times 10^{23}$ atoms of chlorine,

(b) Under similar conditions of temperature and pressure, two volumes of hydrogen combined with two volumes of oxygen will give two volumes of water vapour,

(c) Relative atomic mass of an element is the number of times one molecule of an element is heavier than 1/12 the mass of an atom of carbon [C^{12}],

(d) Under the same conditions of the temperature and pressure, equal volumes of all gases contain the same number of atoms.

ANSWERS

2. (a) 444 (b) 122.5 (c) 249.5 (d) 132 (e) 82, (f) 119·5 (g) 252 **5.** (a) $1{\cdot}2 \times 10^{24}$ (b) 16 g (c) $6{\cdot}02 \times 10^{22}$ (d) 0·1 mole (e) 0·4 g (f) 3×10^{22} **6.** (b) **7.** (b) **8.** (a) $6{\cdot}02 \times 10^{22}$ (b) $1{\cdot}2 \times 10^{23}$ (c) $5{\cdot}42 \times 10^{24}$ **9.** (a) 5·4 g (b) 3·65 g (c) 3·6 g (d) 4·4 g **10.** (a) 48 g (b) 44·8 lts **11.** (a) 0·25 moles (b) ·05 moles **12.** (a) $2{\cdot}657 \times 10^{-23}$ g (b) $1{\cdot}666 \times 10^{-24}$ g (c) $2{\cdot}823 \times 10^{-23}$ g, (d) $1{\cdot}7934 \times 10^{-22}$ g (e) $5{\cdot}314 \times 10^{-23}$ g (f) 10 g of calcium, **13.** (a) 10 g, (b) 32·2 g, (c) 11·1g, (d) 2·4 g, **14.** $7{\cdot}8 \times 10^{23}$ **15.** 4 g **16.** (a) $3{\cdot}131 \times 10^{25}$ (b) 13 atoms (c) $7{\cdot}828 \times 10^{24}$ **17.** Na = $6{\cdot}02 \times 10^{22}$, C = $3{\cdot}01 \times 10^{22}$, O = $9{\cdot}03 \times 10^{22}$ **18.** (a) 2·33 kg (b) 112 *l* **19.** (b) 35·5 **20.** 0·07 g **21.** $3{\cdot}01 \times 10^{21}$ molecules **22.** (a) 0·02 moles (b) $1{\cdot}2 \times 10^{22}$ molecules; $9{\cdot}635 \times 10^{22}$ atoms **23.** 0·805 moles **24.** (a) 71·2 g (b) 11·2 dm^3 (c) 3·14 g **25.** $5{\cdot}019 \times 10^{10}$ atoms **26.** 89·6 g **27.** ₹ 13.68 per mole **28.** (d) **29.** (c) **30.** (b)

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5C. PERCENTAGE COMPOSITION, EMPIRICAL AND MOLECULAR FORMULAE

5.10 Percentage composition

Percentage composition of a compound, is the percentage by weight of each element present in it.

Percentage of an element in a compound

$$= \frac{\text{Total wt. of the element in one molecule}}{\text{Gram molecular weight of the compound}} \times 100$$

***Example 1** : Calculate the percentage composition of various elements in :*

Sodium carbonate, Na_2CO_3

Given that the relative atomic masses of O = 16, Na = 23 and C = 12.

Solution :

Relative molecular mass of Na_2CO_3

$= 23 \times 2 + 12 + 16 \times 3$

$= 46 + 12 + 48 = 106$

Since 106 g of Na_2CO_3 contains 46 g of sodium,

$\therefore$ 100 g of Na_2CO_3 contains $\frac{46 \times 100}{106}$ of sodium

$= \frac{4600}{106} = 43{\cdot}4$ g of Sodium

Similarly, 106 g Na_2CO_3 contains 12 g of carbon.

$\therefore$ 100 " " $\frac{12 \times 100}{106}$

$= \frac{1200}{106} = 11{\cdot}3$ g of C

Again, 106 g of Na_2CO_3 contains 48 g of oxygen.

$\therefore$ 100 " " $\frac{48 \times 100}{106}$

$= \frac{4800}{106} = 45{\cdot}3$ g of O

Answer : In Na_2CO_3 : Na = 43·4%, C = 11·3% and O = 45·3%

***Example 2** : Find the percentage mass of water in washing soda crystals $Na_2CO_3 \cdot 10H_2O$.*

Solution :

Relative molecular mass of

$Na_2CO_3 \cdot 10H_2O$

$= 23 \times 2 + 12 + 16 \times 3 + 10\,(2 + 16)$

$= 106 + 180 = 286$

286 g of washing soda contains 180 g of water of crystallisation.

$\therefore$ 100 " " $\frac{180 \times 100}{286}$

$= \frac{18000}{286} = 62{\cdot}9$ g of H_2O

Answer : The % of H_2O in $Na_2CO_3 \cdot 10H_2O = 62{\cdot}9$

***Example 3** : A given sample has 30% ethyl alcohol by mass, the rest being water. Determine the percentage of C, H and O in the given sample?*

Solution :

Molecular mass of ethyl alcohol

$C_2H_5OH = 2 \times 12 + 5 \times 1 + 16 + 1 = 46$

Carbon present in 46 parts by mass of C_2H_5OH $= 2 \times 12 = 24$ parts by mass.

Carbon present in 30 parts by mass of C_2H_5OH

$= \frac{24}{46} \times 30 = \frac{720}{46} = 15{\cdot}65$

Hydrogen present $= \frac{6}{46} \times 30 = 3{\cdot}91$

Oxygen present $= \frac{16}{46} \times 30 = 10{\cdot}43$

In water (70%)

Hydrogen present $= \frac{2}{18} \times 70 = 7{\cdot}78$

Oxygen present $= \frac{16}{18} \times 70 = 62{\cdot}22$

Percentage of C = 15·65%

Percentage of H = 3·91 + 7·78 = 11·69%

Percentage of O = 10·43 + 62·22 = 72·65%

5.11 EMPIRICAL FORMULA OF A COMPOUND

The empirical formula of a compound is the simplest formula, which gives the simplest ratio in whole numbers of atoms of different elements present in one molecule of the compound.

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For example, the empirical formula of hydrogen peroxide is HO. It indicates the simplest ratio (1 : 1) between the hydrogen and oxygen atoms in its molecule whereas its actual formula is H_2O_2.

Similarly, the empirical formula of glucose ($C_6H_{12}O_6$) is CH_2O. It indicates that the ratio of C, H and O atoms in a molecule of glucose is 1 : 2 : 1.

***The empirical formula mass** is the sum of atomic masses of various elements present in the empirical formula.*

Thus, for hydrogen peroxide (H_2O_2), the empirical formula is HO and its empirical formula mass is 1 + 16 = 17.

5.12 DETERMINATION OF EMPIRICAL FORMULA

The empirical formula can be calculated from the percentage composition of the compound as follows :

(1) The percentage weight of oxygen, carbon or hydrogen (necessary elements in every organic compound) is sometimes not given. Hence, it is first determined by subtracting the sum of other elemental percentages from 100.

(2) The percentage weight of each element is divided by its atomic weight. This gives the ratio of the number of atoms in a molecule of the compound.

(3) If the molecule contains water of crystallisation, its percentage weight is divided by the molecular weight of water, *i.e.*, 18. This gives the ratio of the number of water molecules in one molecule of the compound.

(4) *To get a whole number ratio of atoms of different elements and water molecules,* the numbers obtained in steps 2 and 3 are divided by the smallest number.

(5) The empirical formula is now derived by writing the symbols of various elements side by side with the number of atoms of each one as the subscript to the lower right of its symbol.

***Example 4** : The molecular formula of an organic acid is CH_3COOH. What is its empirical formula?*

Solution :

Molecular formula = CH_3COOH, *i.e.*, $C_2H_4O_2$

$\therefore$ Ratio of C, H and O is 2 : 4 : 2

Simple ratio is, *i.e.* 1 : 2 : 1

$\therefore$ Empirical Formula = CH_2O

***Example 5** : Find the empirical formulae of the compounds with the following percentage compositions :*

(a) *Zn = 47·8; Cl = 52·2*

(b) *Mg = 9·76%, S = 13·01%, O = 26·01%, H_2O = 51·22%*

Solution :

(a) *Zn = 47·8; Cl = 52·2*

Element	*Percentage composition*	*Atomic mass*	*Atomic ratio*	*Simplest ratio*
Zn	47·8	65	$\frac{47·8}{65} = 0·73$	$\frac{0·73}{0·73} = 1$
Cl	52·2	35·5	$\frac{52·2}{35·5} = 1·46$	$\frac{1·46}{0·73} = 2$

Thus, the ratio of Zn : Cl atoms = 1 : 2

Answer : The empirical formula of the compound is $ZnCl_2$.

(b) *Mg = 9·76%, S = 13·01%, O = 26·01%, H_2O = 51·22%*

Solution :

Element compound	*Percentage composition*	*Atomic or molecular mass*	*Ratio of atoms/ molecules*	*Simplest ratio*
Mg	9·76	24	$\frac{9·7}{24} = 0·406$	$\frac{0·406}{0·406} = 1$
S	13·01	32	$\frac{13·01}{32} = 0·406$	$\frac{0·406}{0·406} = 1$
O	26·01	16	$\frac{26·01}{16} = 1·625$	$\frac{1·625}{0·406} = 4$
H_2O	51·22	18	$\frac{51·22}{18} = 2·84$	$\frac{2·84}{0·406} = 7$

Answer : The empirical formula of the compound is $MgSO_4 . 7H_2O$.

***Example 6** : An organic compound has oxygen 26·24% and hydrogen 4·92%. Find its empirical formula.*

Solution :

Every organic compound contains carbon. Find the percentage of carbon :

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Percentage of carbon = 100 – (26·24 + 4·92)

= 68·84

Element	Percentage composition	Atomic mass	Atomic ratio	Simplest ratio
C	68·84	12	$\frac{68·84}{12} = 5·74$	$\frac{5·74}{1·64} = 3·5$
H	4·92	1	$\frac{4·92}{1} = 4·92$	$\frac{4·92}{1·64} = 3$
O	26·24	16	$\frac{26·24}{16} = 1·64$	$\frac{1·64}{1·64} = 1$

Multiply 3·5, 3 and 1 by 2 in order to change them in whole number. Thus, the ratio of C : H : O atoms = 7 : 6 : 2.

Answer : The empirical formula is $C_7H_6O_2$.

5.13 DETERMINATION OF MOLECULAR FORMULA

The molecular formula of a compound denotes *the actual number of atoms of different elements present in one molecule of the compound.*

Molecular formula of Blue vitriol is $CuSO_4.5H_2O$.

It gives the information that a molecule of blue vitriol is made of –

1. One atom of copper,
2. One atom of sulphur,
3. Four atoms of oxygen,
4. Five molecules of water of crystallization.

Steps to find the *Molecular formula* of a compound :

(1) Calculate the empirical weight of the compound from its empirical formula,

(2) Divide its molecular weight by empirical weight which gives the number (n).

(3) Multiply the empirical formula by this number to get the molecular formula.

Molecular formula = empirical formula × n

where $n = \frac{\text{Molecular weight}}{\text{Empirical formula weight}}$

For example, the empirical formula of hydrogen peroxide is HO and its empirical formula weight is 1 + 16 = 17, but its molecular weight is 34. Hence its molecular formula is 34/17 = 2 times the empirical formula, *i.e.*, H_2O_2.

Note : The value of n must be rounded off to its nearest whole number.

Example 7 : *Calculate the empirical formula of a compound whose molecular formula is $C_8H_6O_4$ and empirical formula weight is 83.*

Solution :

Molecular weight of $C_8H_6O_4$ is

= 12 × 8 + 1 × 6 + 16 × 4

= 96 + 6 + 64

= 166

n(Empirical formula weight) = Molecular weight

$\therefore \quad n = \frac{\text{Molecular Weight}}{\text{Empirical Formula Weight}}$

$= \frac{166}{83} = 2$

i.e. $C_8H_6O_4$ = empirical formula × n [$n = 2$]

∴ Empirical Formula = $C_4H_3O_2$

Example 8 : *A compound is found to possess C = 40%, H = 6.7% and O = 53.3%. Its molecular mass is 60. Find the molecular formula of the compound?*

Solution :

Element	Percentage composition	Atomic mass	Atomic ratio	Simplest ratio
C	40	12	$\frac{40}{12} = 3·3$	$\frac{3·3}{3·3} = 1$
H	6·7	1	$\frac{6·7}{1} = 6·7$	$\frac{6·7}{3·3} = 2$
O	53·3	16	$\frac{53·3}{16} = 3·3$	$\frac{3·3}{3·3} = 1$

∴ The empirical formula of the compound is CH_2O.

Let the molecular formula of the compound be n (CH_2O).

∴ Molecular mass = n (12 + 2 + 16) = 30n

30n = 60 or $n = 2$

Answer : The molecular formula of the compound is $C_2H_4O_2$.

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Example 9 : *Calculate the empirical and molecular formula of the compound having the following percentage composition :*

C = 26·59%; H = 2·22%; O = 71·19%

Its vapour density is 45.

Solution :

Element	*Percentage composition*	*Atomic mass*	*Atomic ratio*	*Simplest ratio*
C	26·59	12	$\frac{26·59}{12} = 2·216$	$\frac{2·216}{2·216} = 1$
H	2·22	1	$\frac{2·22}{1} = 2·22$	$\frac{2·22}{2·216} = 1$
O	71·19	16	$\frac{71·19}{16} = 4·45$	$\frac{4·45}{2·216} = 2$

Hence, the empirical formula is CHO_2

Mol. wt. = n(Empirical formula weight)

$2 \times$ V.D. = $n(CHO_2)$

$\therefore \quad 2 \times 45 = n(12 + 1 + 32)$

$90 = n \times 45$

$\frac{90}{45} = n \Rightarrow 2 = n$

Answer : The molecular formula is $C_2H_2O_4$.

Example 10 : *0·5 gram of an organic compound contains 0·062 g of hydrogen and 0·25 g of oxygen. In the vapour state this compound weighs32 times as heavy as the same volume of hydrogen. Determine the molecular formula.*

Solution :

We know, every organic compound contains carbon.

$\therefore$ Mass of carbon = 0·5 – (0·25 + 0·062) = 0·188

Element	*Mass in grams*	*Atomic mass*	*Number of atoms*	*Simplest ratio*
C	0·188	12	$\frac{0·188}{12} = 0·016$	$\frac{0·016}{0·016} = 1$
H	0·062	1	$\frac{0·062}{1} = 0·062$	$\frac{0·062}{0·016} = 4$
O	0·25	16	$\frac{0·25}{16} = 0·016$	$\frac{0·016}{0·016} = 1$

Thus empirical formula is CH_4O

Molecular mass = $2 \times$ V.D.

$= 2 \times 32 = 64$

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$

$$= \frac{64}{32} = 2$$

Hence, the molecular formula is $C_2H_8O_2$.

Example 11 : *3·5 g of nitrogen combines with 2 g of oxygen to form an oxide of nitrogen. What is the empirical formula of this oxide?*

Solution :

Percentage of N_2 is $\frac{3·5}{5·5} \times 100 = \frac{700}{11}$

Percentage of $O_2 = \frac{2}{5·5} \times 100 = \frac{400}{11}$

Element	*Percentage*	*Atomic mass*	*Number of atoms*	*Simplest ratio*
N	$\frac{700}{11}$	14	$\frac{700}{11 \times 14} = \frac{50}{11}$	$\frac{\frac{50}{11}}{\frac{25}{11}} = 2$
O	$\frac{400}{11}$	16	$\frac{400}{11 \times 16} = \frac{25}{11}$	$\frac{\frac{25}{11}}{\frac{25}{11}} = 1$

Answer : Hence the empirical formula is N_2O.

Example 12 : *On heating copper sulphate crystals, $CuSO_4 . xH_2O$ (10g), anhydrous sulphate (6·4g) was left. Write the formula of the crystalline sulphate.*

Solution :

	Mass	Molar mass	Moles	Simple ratio
$CuSO_4$	6·4g	159·5g	$\frac{6·4}{159·5} = 0·04$	$\frac{0·04}{0·04} = 1$
H_2O	3·6g	18g	$\frac{3·6}{18} = 0·2$	$\frac{0·2}{0·04} = 5$

Therefore formula is $CuSO_4 \cdot 5H_2O$.

Example 13 : *0·29 grams of a hydrocarbon with vapour density 29 when burnt completely in oxygen produce 448 ml of carbon dioxide at S.T.P. From the given information, calculate the*

(i) mass of carbon dioxide formed.

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(ii) mass of element carbon in carbon dioxide.

(iii) mass of hydrogen in hydrocarbon.

(iv) empirical formula of hydrocarbon.

(v) molecular formula of hydrocarbon.

Solution :

(i) 1 mole of $CO_2 = 12 + 2 \times 16 = 44$ g.

22400 ml of CO_2 at S.T.P. weighs = 44g

$\therefore$ 448 ml of CO_2 at S.T.P. weighs

$\frac{44}{22400} \times 448 = 0{\cdot}88$ g

(ii) 44 g of CO_2 contains 12 g carbon.

0·88 g of CO_2 contains $\frac{12}{44} \times 0{\cdot}88 = 0{\cdot}24$ g

(iii) Weight of hydrogen

= wt of hydrocarbon – wt. of carbon

= 0·29 – 0·24 = 0·05 g .

(iv)

Element	Mass	At. mass	Gram atom	Ratio	
Carbon	0·24g	12	$\frac{0{\cdot}24}{12} = 0{\cdot}02$	$\frac{0{\cdot}02}{{\cdot}02} = 1$	2
Hydrogen	0·05g	1	$\frac{0{\cdot}05}{1} = 0{\cdot}05$	$\frac{{\cdot}05}{0{\cdot}02} = 2{\cdot}5$	5

Empirical formula is C_2H_5.

(v) Molecular weight = 2 × Vapour density

= 2 × 29 = 58

Molecular wt. = n(Empirical formula wt.)

$58 = n(12 \times 2 + 1 \times 5)$

$\frac{58}{29} = n$

$2 = n$

$\therefore$ Molecular formula = C_4H_{10}

EXERCISE-5C

1. Give three kinds of information conveyed by the formula H_2O.
2. Explain the terms, empirical formula and molecular formula.
3. Give the empirical formula of :
   (a) C_6H_6, (b) $C_6H_{12}O_6$,
   (c) C_2H_2, (d) CH_3COOH.
4. Find the percentage mass of water in the Epsom salt $MgSO_4{\cdot}7H_2O$.
5. Calculate the percentage of phosphorus in :
   (a) Calcium hydrogen phosphate $Ca(H_2PO_4)_2$,
   (b) Calcium phosphate $Ca_3(PO_4)_2$.
6. Calculate the percentage composition of :
   Potassium chlorate, $KClO_3$.
7. Find the empirical formula of the compounds with the following percentage composition. :
   Pb = 62·5%; N = 8·5%; O = 29·0%.
8. Calculate the mass of iron in 10 kg of iron ore which contains 80% of pure ferric oxide.
9. If the empirical formula of two compounds is CH and their vapour densities are 13 and 39 respectively. Find their molecular formula. **(2015)**
10. Find the empirical formula of a compound containing 17·7% hydrogen and 82·3% nitrogen.
11. On analysis, a substance was found to contain :
    C = 54·54%, H = 9·09%, O = 36·36%.
    The vapour density of the substance is 44, calculate :
    (a) its empirical formula,
    (b) its molecular formula.
12. An organic compound, whose vapour density is 45, has the following percentage composition,
    H = 2·22%; O = 71·19%; and remaining carbon.
    Calculate :
    (a) its empirical formula, (b) its molecular formula.
13. An organic compound contains 4·07% hydrogen, 71·65% chlorine and remaining carbon. Its molar mass is 98·96. Find its,
    (a) Empirical formula (b) Molecular formula
14. A hydrocarbon contains 4·8 g of carbon per gram of hydrogen. Calculate :
    (a) the gram atom of each,
    (b) find the empirical formula,
    (c) find molecular formula, if its vapour density is 29.
15. Combine 0·2 g atom of silicon with 21·3 g of chlorine. Find the empirical formula of the compound formed.
16. A gaseous hydrocarbon contains 82·76% of carbon. Given that its vapour density is 29, find its molecular formula. **(2016)**
17. In a compound of magnesium (Mg = 24) and nitrogen (N = 14), 18 g of magnesium combines with 7g of nitrogen.
    Deduce the simplest formula by answering the following questions :

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(a) How many gram-atoms of magnesium are equal to 18g ?

(b) How many gram-atoms of nitrogen are equal to 7g of nitrogen ?

(c) Calculate simple ratio of gram-atoms of magnesium to gram-atoms of nitrogen and hence the simplest formula of the compound formed.

18. Barium chloride crystals contain 14·8% water of crystallisation. Find the number of molecules of water of crystallisation per molecule.

19. Urea is a very important nitrogenous fertilizer. Its formula is CON_2H_4. Calculate the percentage of nitrogen in urea (C = 12, O = 16, N = 14 and H = 1).

20. Determine the formula of the organic compound if its molecule contains 12 atoms of carbon. The percentage compositions of hydrogen and oxygen are 6·48 and 51·42 respectively.

21. (a) A compound with empirical formula AB_2, has the vapour density equal to its empirical formula weight. Find its molecular formula.

(b) A compound with empirical formula AB has vapour density three times its empirical formula weight. Find the molecular formula.

22. A hydride of nitrogen contains 87·5 per cent by mass of nitrogen. Determine the empirical formula of this compound.

23. A compound has O = 61·32%, S = 11·15%, H = 4·88% and Zn = 22·65%. The relative molecular mass of the compound is 287 a.m.u. Find the molecular formula of the compound, assuming that all the hydrogen is present as water of crystallisation.

ANSWERS

4. 51·2% **5.** (a) 26·5%, (b) 20% **6.** K = 31·83%, Cl = 28·98%, O_2 = 39·18% **7.** $Pb(NO_3)_2$, **8.** 5·6 kg **9.** C_2H_2, C_6H_6 **10.** NH_3 **11.** (a) C_2H_4O (b) $C_4H_8O_2$ **12.** (a) CHO_2 (b) $C_2H_2O_4$ **13.** (a) CH_2Cl (b) $C_2H_4Cl_2$ **14.** (i) 0·4, 1 (ii) C_2H_5 (iii) C_4H_{10} **15.** $SiCl_3$ **16.** C_4H_{10} **17.** (a) $\frac{3}{4}$ gram atoms (b) $\frac{1}{2}$ gram atoms (c) Mg_3N_2 **18.** 2 **19.** 46·67% **20.** $C_{12}H_{24}O_{12}$ **21.** (a) A_2B_4 (b) A_6B_6 **22.** NH_2. **23.** $ZnSO_4.7H_2O$

5D. CALCULATIONS BASED ON CHEMICAL EQUATIONS

5.14 CHEMICAL EQUATIONS

A chemical equation is a balanced account of a chemical transaction. It is not merely a qualitative statement, but it also gives quantitative information of a chemical reaction.

Chemical equations give information about :

(i) moles and masses of various reactants and products,

(ii) volumes of gaseous reactants and products measured at S.T.P.

Information conveyed by the chemical equation :

Consider the equation :

$$2\,KClO_3(s) \xrightarrow{MnO_2,\ \Delta} 2\,KCl(s) + 3O_2\,(g)$$

(i) ***The molecular proportion of substances:*** Here, 2 molecules of solid potassium chlorate on heating in the presence of manganese dioxide give 2 molecules of solid potassium chloride and 3 molecules of oxygen gas.

(ii) ***The relative masses of substances :*** 2 × 122·5 g = 245 g of potassium chlorate

gives 2 × 74·5 g = 149 g of KCl

and 3 × 32 = 96 g of oxygen.

(iii) ***The volumes of gaseous substances:*** 3 × 22·4 L = 67·2 L of oxygen at S.T.P. is evolved when 245 g of potassium chlorate is heated.

In solving the problems, proceed as follows:

(i) Write a **balanced** equation for the reaction.

(ii) Write down the molecular weights of each substance below its formula.

(iii) *The same units of weight and volume should be used throughout.*

(iv) Write down the product of 22·4 L and the number of molecules below each gaseous substance.

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5.14.1 Problems based on reacting weights

Example 1 : *A solution of common salt when added to silver nitrate solution yields a precipitate of silver chloride (0·28 g). Find the mass of sodium chloride in the solution, and also the mass of sodium nitrate formed.*

Solution :

$$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$$
$$(23 + 35{\cdot}5) \quad (108 + 35{\cdot}5) \quad (23 + 14 + 48)$$
$$58{\cdot}5 \quad 143{\cdot}5 \quad 85$$

Since 143·5 g of silver chloride is formed from 58·5 g of NaCl,

$\therefore$ 0·28 g " " $= \frac{58{\cdot}5 \times 0{\cdot}28}{143{\cdot}5}$

$= \frac{16{\cdot}38}{143{\cdot}5}$

$=$ 0·114 g of NaCl

Also 58·5 g of sodium chloride forms 85 g of $NaNO_3$.

$\therefore$ 0·114 g " " $= \frac{85 \times 0{\cdot}114}{58{\cdot}5}$

$= \frac{9{\cdot}69}{58{\cdot}5}$

$=$ 0·165 g of $NaNO_3$

Answer : 0·114 g NaCl is present in the solution and 0·165 g $NaNO_3$ is formed.

5.14.2 Problems based on mass-volume relationship

Example 2 : *Find the mass of lead formed by the reduction of 342·5 g of red lead (Pb_3O_4) in a current of hydrogen and also find the volume of hydrogen used up at NTP.*

Solution :

$$Pb_3O_4 + 4H_2 \rightarrow 3Pb + 4H_2O$$
$$207{\times}3 + 16{\times}4 \quad 4{\times}22{\cdot}4\ dm^3 \quad 3{\times}207$$
$$= 685\ g \quad = 89{\cdot}6\ dm^3 \quad = 621\ g$$

685 g of red lead yields 621 g of lead.

$\therefore$ 342·5g " " " $= \frac{621 \times 342{\cdot}5}{685}$

$=$ 310·5 g of lead

Also, 685g of red lead uses 89·6 dm^3 of hydrogen at NTP.

$\therefore$ 342·5 " " dm^3

$=$ 44·8 dm^3 of hydrogen.

Answer : 310·5 g of lead is formed and 44·8 dm^3 of hydrogen at NTP is used up.

Example 3 : *If 16·4 grams of calcium nitrate is heated, calculate the volume of (a) nitrogen dioxide obtained at stp and (b) the mass of calcium oxide obtained.*

Solution :

$$2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$$
$$2[40 + 2(14{+}48)] \quad 112\ g \quad 4\ moles$$
$$328\ g$$

(a) 328 g of $Ca(NO_3)_2$ liberates $4 \times 22{\cdot}4$ L NO_2.

$\therefore$ 16·4 g will liberate $\frac{4 \times 22{\cdot}4}{328} \times 16{\cdot}4$

$=$ 4·48 L NO_2 at STP

(b) 328 g of calcium nitrate gives 112 g of CaO $\therefore$

16·4 g will give $= \frac{112 \times 16{\cdot}4}{328}$

$=$ 5·6 g of CaO.

Answer : (a) 4·48 litres of NO_2.

(b) 5·6 g of CaO.

Example 4 : *On heating 12·4 g of copper (II) carbonate in a crucible only 7·0 g of copper (II) oxide was produced. What was the percentage yield of copper (II) oxide ? [at. wt. Cu = 64, C = 12, O = 16]*

Solution :

Equation for the reaction

$$\underbrace{CuCO_3}_{64 + 12 + 48} \longrightarrow \underbrace{CuO}_{64 + 16} + CO_2$$
$$124\ g \quad 80\ g$$

124 g of copper carbonate produces 80 g of copper oxide, on heating.

$\therefore$ 124 g of copper carbonate produces $\frac{80}{124} \times 12{\cdot}4 =$ **8 g**

Thus expected yield is 8·0 g

Actual yield is 7·0 g

$\therefore$ Percentage yield $= \frac{7}{8} \times 100 = 87{\cdot}5\%$

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Some difficult problems, not in syllabus

Example 5 : *Find how many grams of zinc must be dissolved in dilute hydrochloric acid to produce 300 cm^3 of hydrogen at 27°C and 700 mm pressure.*

Solution :

First convert the volume of hydrogen to NTP.

P = 700 mm; V = 300 cm^3;

T = (273 + 27) K = 300 K

P' = 760 mm; V' = ?

T' = (273 + 0) K = 273 K

By using the gas equation,

$$\frac{PV}{T} = \frac{P'V'}{T'}$$

or $\frac{700 \times 300}{300} =$

or $300 \times 760V' = 700 \times 300 \times 273$

$$\therefore V' = \frac{700 \times 300 \times 273}{300 \times 760}$$

$$= \frac{19110}{76} = 251{\cdot}5 \text{ cm}^3$$

$Zn + 2HCl = ZnCl_2 + H_2$

65 g 22·4 dm^3.

22400 cm^3.

Since 22400 cm^3 of hydrogen at NTP are obtained from 65 g of Zn.

$\therefore$ 251·5 cm^3 " " $= \frac{65 \times 251{\cdot}5}{22400}$ g

$= \frac{16347{\cdot}5}{22400} = 0{\cdot}729$ g of zinc.

Answer : 0·729 g of zinc must be dissolved to produce the given volume of hydrogen.

Example 6 : *The given sample of hydrochloric acid contains only 20% by weight of HCl, the rest being water. What mass of the acid will be required for reacting completely with 20 g of magnesium oxide ?*

Solution :

$$\underset{(24+16)}{MgO} + \underset{2(1+35{\cdot}5)}{2HCl} \rightarrow MgCl_2 + H_2O$$

40 g of MgO requires 73 g of HCl.

$\therefore$ 20 g of MgO will require $= \frac{73}{40} \times 20$

$= 36{\cdot}5$ g of HCl.

But the given acid is 20%. *i.e.* 20 g of HCl is contained in 100 g sample.

$\therefore$ 36·5 g of HCl is contained in $\frac{100}{20} \times 36{\cdot}5$ g of HCl. = 182·5 g of HCl

Answer : 182·5 g of HCl.

Example 7 : *Hydrogen is generated by the action of steam with heated magnesium. Calculate the mass of magnesium which will produce just sufficient hydrogen to combine with all the oxygen that can be obtained by the complete decomposition of 24·5 g of $KClO_3$.*

Solution :

$$Mg + H_2O \rightarrow MgO + H_2 \quad (1)$$

$$2H_2 + O_2 \rightarrow 2H_2O \quad (2)$$

$$2KClO_3 \rightarrow 2KCl + 3O_2 \quad (3)$$

i.e., $2KClO_3 = 3O_2 = 6H_2 = 6Mg$

2 moles of $KClO_3$ correspond to 6 moles of Mg

or 1 mole of $KClO_3$ (122·5 g) corresponds to

3 × 24 g of Mg.

$\therefore$ 24·5 g of $KClO_3$ will correspond to

$$\frac{72}{122{\cdot}5} \times 24{\cdot}5 = 14{\cdot}4 \text{ g of Mg.}$$

Answer : 14·4 g of Mg.

Example 8. *Find the mass of quick lime (calcium oxide) formed by the decomposition of 200 g of limestone. What mass of carbon dioxide will be evolved ? If the limestone is only 90% pure, what mass of quicklime will be produced ?*

Solution :

$$\underset{(40+12+48)}{CaCO_3} \rightarrow \underset{(40+16)}{CaO} + \underset{(12+32)}{CO_2}$$

100 g of limestone on decomposition produces 56 g of CaO.

$\therefore$ 200 g " " $= \frac{56 \times 200}{100}$

$= 112$ g

Also, 100 g of limestone evolves 44 g of CO_2.

$\therefore$ 200 g " " = 88 g of CO_2

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If the limestone is only 90% pure, weight of pure calcium carbonate in 200 g will be

$$= \frac{90 \times 200}{100} = 180 \text{ g}$$

Since 100 g of pure calcium carbonate produces 56 g of CaO.

$\therefore$ 180 g " " $= \frac{56 \times 180}{100}$

$= 100{\cdot}8$ g

Answer : 200 g pure limestone would produce 112 g quicklime and 88 g carbon dioxide. Also, 200 g of 90% pure limestone would produce 100·8 g quicklime.

EXERCISE-5D

Note : Atomic mass wherever required should be taken from the table given on the preliminary pages, rounded off to the nearest whole number.

1. Complete the following blanks in the equation as indicated.

$CaH_2\ (s) + 2H_2O\ (aq) \rightarrow Ca(OH)_2\ (s) + 2H_2\ (g)$

(a) Moles : 1 mole + ... → ..... + ...

(b) Grams : 42 g + ... → ..... + ...

(c) Molecules : $6{\cdot}02 \times 10^{23}$ + ... → ..... + ...

2. The reaction between 15 g of marble and nitric acid is given by the following equation :

$$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$$

Calculate :

(a) the mass of anhydrous calcium nitrate formed,

(b) the volume of carbon dioxide evolved at STP.

3. 66 g ammonium sulphate is produced by the action of ammonia on sulphuric acid.

Write a balanced equation and calculate :

(a) mass of ammonia required,

(b) the volume of the gas used at STP,

(c) the mass of acid required.

4. The reaction between red lead and hydrochloric acid is given below :

$$Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + 4H_2O + Cl_2$$

Calculate : (a) the mass of lead chloride formed by the action of 6.85 g of red lead, (b) the mass of chlorine and (c) the volume of chlorine evolved at STP.

5. Find the mass of KNO_3 required to produce 126 kg of nitric acid. Find whether a larger or smaller mass of $NaNO_3$ is required for the same purpose.

$$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$$

$$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

6. Pure calcium carbonate and dilute hydrochloric acid are reacted and 2 litres of carbon dioxide was collected at 27°C and normal pressure.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

Calculate : (a) the mass of salt required,

(b) the mass of the acid required.

7. Calculate the mass and volume of oxygen at STP, which will be evolved on electrolysis of 1 mole (18 g) of water.

8. 1·56 g of sodium peroxide reacts with water according to the following equation :

$$2Na_2O_2 + 2H_2O \rightarrow 4NaOH + O_2$$

Calculate : (a) mass of sodium hydroxide formed,

(b) volume of oxygen liberated at STP,

(c) mass of oxygen liberated.

9. (a) Calculate the mass of ammonia that can be obtained from 21·4 g of NH_4Cl by the reaction :

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$$

(b) What will be the volume of ammonia when measured at STP ?

10. The reaction between aluminium carbide and water takes place according to the following equation :

$$Al_4C_3 + 12\ H_2O \rightarrow 3CH_4 + 4Al\ (OH)_3.$$

Calculate the volume of methane measured at STP released from 14.4 g of aluminium carbide by excess of water.

11. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

0·02 moles of pure MnO_2 is heated strongly with conc. HCl. Calculate :

(a) mass of MnO_2 used,

(b) moles of salt formed,

(c) mass of salt formed,

(d) moles of chlorine gas formed,

(e) mass of chlorine gas formed,

(f) volume of chlorine gas formed at STP,

(g) moles of acid required,

(h) mass of acid required.

12. Nitrogen and hydrogen react to form ammonia.

$$N_2\ (g) + 3H_2\ (g) \rightarrow 2NH_3\ (g)$$

If 1000 g of H_2 react with 2000 g of N_2.

(a) Will any of the two reactants remain unreacted ? If yes, which one and what will be its mass ?

(b) Calculate the mass of ammonia (NH_3) that will be formed.

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ANSWERS

2. (a) 24·6 g (b) 3·36 L **3.** (a) 17 g (b) 22·4 litres (c) 49 g. **4.** (a) 8·34 g, (b) 0·71 g, (c) 0·224 L **5.** 202 Kg KNO_3, 170 Kg $NaNO_3$ *i.e.*, smaller mass of $NaNO_3$ is required **6.** (a) 8·125 g (b) 5·93 g **7.** 16 g, 11·2 L **8.** (a) 1·6 g, (b) 224 cm^3, (c) 0·32 g **9.** (a) 6·8 g, (b) 8·96 L **10.** 6·72 L **11.** (a) 1·74 grams, (b) 0·02 moles, (c) 2·52 g, (d) 0·02 moles, (e) 1·42 g, (f) 0·448 dm^3 at stp, (g) 0·08 moles, (h) 2·92 g **12.** (a) 571·4 g H_2 remains unreacted, (b) 2428·8 g.

MISCELLANEOUS EXERCISES

1. From the equation for burning of hydrogen and oxygen

$$2H_2 + O_2 \rightarrow 2H_2O \text{ (steam)}$$

Write down the number of mole (or moles) of steam obtained from 0·5 moles of oxygen.

2. From the equation,

$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$

Calculate :

(a) mass of copper needed to react with 63 g of HNO_3,

(b) volume of nitric oxide at STP that can be collected.

3. (a) Calculate the number of moles in 7 g of nitrogen.

(b) What is the volume at STP of 7·1 g of chlorine?

(c) What is the mass of 56 cm^3 of carbon monoxide at STP?

4. Some of the fertilizers are sodium nitrate $NaNO_3$, ammonium sulphate $(NH_4)_2SO_4$ and urea $CO(NH_2)_2$. Which of these contains the highest percentage of nitrogen ?

5. Water decomposes to O_2 and H_2 under suitable conditions as represented by the equation below :

$$2H_2O \rightarrow 2H_2 + O_2$$

(a) If 2500 cm^3 of H_2 is produced, what volume of O_2 is liberated at the same time and under the same conditions of temperature and pressure ?

(b) The 2500 cm^3 of H_2 is subjected to $2\frac{1}{2}$ times increase in pressure (temp. remaining constant). What volume of H_2 will now occupy ?

(c) Taking the volume of H_2 calculated in 5(b), what changes must be made in kelvin (absolute) temperature to return the volume to 2500 cm^3 pressure remaining constant.

6. Urea $[CO(NH_2)_2]$ is an important nitrogeneous fertilizer, and is sold in 50 kg sacks. What mass of nitrogen is in one sack of urea ?

7. Find the molecular formula of a hydrocarbon having vapour density 15, which contains 20% of Hydrogen.

8. The following experiment was performed in order to determine the formula of a hydrocarbon. The hydrocarbon X is purified by fractional distillation.

0·145 g of X were heated with dry copper (II) oxide and 224 cm^3 of carbon dioxide was collected at STP.

(a) Which elements does X contain ?

(b) What was the purpose of Copper (II) oxide ?

(c) Calculate the empirical formula of X by the following steps :

(i) Calculate the number of moles of carbon dioxide gas.

(ii) Calculate the mass of carbon contained in this quantity of carbon dioxide and thus the mass of carbon in sample X.

(iii) Calculate the mass of hydrogen in sample X.

(iv) Deduce the ratio of atoms of each element in X (empirical formula).

9. A compound is formed by 24 g of X and 64 g of oxygen. If atomic mass of X = 12 and O = 16, calculate the simplest formula of compound.

10. A gas cylinder filled with hydrogen holds 5 g of the gas. The same cylinder holds 85 g of gas X under same temperature and pressure. Calculate :

(a) vapour density of gas X,

(b) molecular weight of gas X.

11. (a) When carbon dioxide is passed over red hot carbon, carbon monoxide is produced according to the equation :

$$CO_2 + C \rightarrow 2CO$$

What volume of carbon monoxide at STP can be obtained from 3 g of carbon ?

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(b) 60 cm^3 of oxygen was added to 24 cm^3 of carbon monoxide and mixture ignited. Calculate :

(i) volume of oxygen used up and

(ii) volume of carbon dioxide formed.

12. How much calcium oxide is obtained by heating 82 g of calcium nitrate ? Also find the volume of NO_2 evolved :

$$2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$$ **(2016)**

13. The equation for the burning of octane is :

$$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$$

(i) How many moles of carbon dioxide are produced when **one** mole of octane burns?

(ii) What volume, at STP, is occupied by the number of moles determined in (i) ?

(iii) If the relative molecular mass of carbon dioxide is 44, what is the mass of carbon dioxide produced by burning **two** moles of octane ?

(iv) What is the empirical formula of octane ?

14. Ordinary chlorine gas has two isotopes $^{35}_{17}Cl$ and $^{37}_{17}Cl$ in the ratio of 3 : 1. Calculate the relative atomic mass of chlorine.

15. Silicon (Si = 28) forms a compound with chlorine (Cl = 35·5) in which 5·6 g of silicon combines with 21·3 g of chlorine. Calculate the empirical formula of the compound.

16. An acid of phosphorus has the following percentage composition; Phosphorus = 38·27%; hydrogen = 2·47%; oxygen = 59·26%. Find the empirical formula of the acid and its molecular formula, given that its relative molecular mass is 162.

17. Calculate the mass of substance 'A' which in gaseous form occupies 10 litres at 27°C and 700 mm pressure. The molecular mass of 'A' is 60.

18. A gas cylinder can hold 1 kg of hydrogen at room temperature and pressure. (a) What mass of carbon dioxide can it hold under similar conditions of temperature and pressure? (b) If the number of molecules of hydrogen in the cylinder is X, calculate the number of carbon dioxide molecules in the cylinder. Give reasons for your answer.

19. Following questions refer to one mole of chlorine gas.

(a) What is the volume occupied by this gas at STP?

(b) What will happen to volume of gas, if pressure is doubled ?

(c) What volume will it occupy at 273°C ?

(d) If the relative atomic mass of chlorine is 35·5, what will be the mass of 1 mole of chlorine gas?

20. (a) A hydrate of calcium sulphate $CaSO_4 \cdot xH_2O$ contains 21% water of crystallisation. Find the value of x.

(b) What volume of hydrogen and oxygen measured at STP will be required to prepare 1·8 g of water.

(c) How much volume will be occupied by 2 g of dry oxygen at 27°C and 740 mm pressure ?

(d) What would be the mass of CO_2 occupying a volume of 44 litres at 25°C and 750 mm pressure.

(e) 1 g of a mixture of sodium chloride and sodium nitrate is dissolved in water. On adding silver nitrate solution, 1·435 g of AgCl is precipitated.

$AgNO_3$ (aq) + NaCl (aq) $\rightarrow$ AgCl (s) + $NaNO_3$

Calculate the precentage of NaCl in the mixture.

21. (a) From the equation :

$$C + 2H_2SO_4 \rightarrow CO_2 + 2H_2O + 2SO_2$$

Calculate :

(i) the mass of carbon oxidized by 49 g of sulphuric acid.

(ii) The volume of sulphur dioxide measured at STP, liberated at the same time.

(b) (i) A compound has the following percentage composition by mass : carbon 14·4%, hydrogen 1·2% and chlorine 84·5%. Determine the empirical formula of this compound. Work correct to 1 decimal place. (H = 1; C = 12; Cl = 35·5)

(ii) The relative molecular mass of this compound is 168, so what is its molecular formula ?

22. Find the percentage of

(a) oxygen in magnesium nitrate crystals [$Mg(NO_3)_2 \cdot 6H_2O$].

(b) boron in $Na_2B_4O_7.10H_2O$. [H = 1, B = 11, O = 16, Na = 23].

(c) phosphorus in the fertilizer superphosphate $Ca(H_2PO_4)_2$.

23. A gas occupied 360 cm^3 at 87°C and 380 mm Hg pressure. If the mass of gas is 0·546 g, find its relative molecular mass.

24. Solid ammonium dichromate decomposes as under :

$$(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$$

If 63 g of ammonium dichromate decomposes. Calculate:

(a) the quantity in moles of $(NH_4)_2Cr_2O_7$

(b) the quantity in moles of nitrogen formed

(c) the volume of N_2 evolved at STP.

(d) what will be the loss of mass ?

(e) calculate the mass of chromium (III) oxide formed at the same time. **(2015)**

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25. Hydrogen sulphide gas burns in oxygen to yield 12·8 g of sulphur dioxide gas as under :

$$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$$

Calculate the volume of hydrogen sulphide at STP.

Also, calculate the volume of oxygen required at STP. which will complete the combustion of hydrogen sulphide determined in (litres).

26. Ammonia burns in oxygen and the combustion, in the presence of a catalyst, may be represented by; $2NH_3 + 2½\ O_2 \rightarrow 2NO + 3H_2O$ [H = 1; N = 14; O = 16]. What mass of steam is produced when 1·5 g of nitrogen monoxide is formed ?

27. If a crop of wheat removes 20 kg of nitrogen per hectare of soil, what mass of the fertilizer, calcium nitrate $Ca(NO_3)_2$ would be required to replace the nitrogen in a 10 hectare field ?

28. Concentrated nitric acid oxidises phosphorus to phosphoric acid according to the following equation:

$$P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$$

If 6·2 g of phosphorus was used in the reaction calculate:

(a) Number of moles of phosphorus taken and mass of phosphoric acid formed.

(b) What mass of nitirc acid will be consumed at the same time ?

(c) What would be the volume of steam produced at the same time if measured at 760 mm Hg pressure and 273°C ?

29. 112 cm^3 of a gaseous fluoride of phosphorus has a mass of 0·63 g. Calculate the relative molecular mass of the fluoride. If the molecule of the fluoride contains only one atom of phosphorus, then determine the formula of the phosphorus fluoride. [F = 19; P = 31].

30. Washing soda has the formula $Na_2CO_3.10H_2O$. What mass of anhydrous sodium carbonate is left when all the water of crystallization is expelled by heating 57·2 g of washing soda ?

31. A metal M forms a volatile chloride containing 65·5% chlorine. If the density of the chloride relative to hydrogen is 162·5, find the molecular formula of the chloride (M=56).

32. A compound X consists of 4·8% carbon and 95·2% bromine by mass :

(i) Determine the empirical formula of this compound working correct to one decimal place (C = 12; Br = 80).

(ii) If the vapour density of the compound is 252, what is the molecular formula of the compound ?

33. The reaction : $4N_2O + CH_4 \rightarrow CO_2 + 2H_2O + 4N_2$ takes place in the gaseous state. If all volumes are measured at the same temperature and pressure, calculate the volume of dinitrogen oxide (N_2O) required to give 150 cm^3 of steam.

34. Samples of the gases O_2, N_2, CO_2 and CO under the same conditions of temperature and pressure contain the same number of molecules x. The molecules of oxygen occupy V litres and have a mass of 8 g under the same conditions of temperature and pressure.

What is the volume occupied by :

(a) x molecules of N_2,

(b) $3x$ molecules of CO,

(c) What is the mass of CO_2 in grams ?

(d) In answering the above questions, which law have you used ?

35. The percentage composition of sodium phosphate as determined by analysis is 42·1% sodium, 18·9% phosphorus and 39% oxygen. Find the empirical formula of the compound.

36. What volume of oxygen is required to burn completely a mixture of 22·4 dm^3 of methane and 11·2 dm^3 of hydrogen into carbon dioxide and steam ?

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$
$$2H_2 + O_2 \rightarrow 2H_2O$$

37. The gases hydrogen, oxygen, carbon dioxide, sulphur dioxide and chlorine are arranged in order of their increasing relative molecular masses. Given 8 g of each gas at STP, which gas will contain the least number of molecules and which gas the most ?

38. 10 g of a mixture of sodium chloride and anhydrous sodium sulphate is dissolved in water. An excess of barium chloride solution is added and 6·99 g of barium sulphate is precipitated according to the equation given below :

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$$

Calculate the percentage of sodium sulphate in the original mixture.

39. When heated, potassium permanganate decomposes according to the following equation :

$$2KMnO_4 \rightarrow \underset{\text{solid residue}}{K_2MnO_4} + MnO_2 + O_2$$

(a) Some potassium permanganate was heated in the test tube. After collecting one litre of oxygen at room temperature, it was found that the test tube had undergone a loss in mass of 1·32 g. If one litre of hydrogen under the same conditions of temperature and pressure has a mass of 0·0825 g, calculate the relative molecular mass of oxygen.

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(b) Given that the molecular mass of potassium permanganate is 158, what volume of oxygen (measured at room temperature) would be obtained by the complete decomposition of 15·8 g of potassium permanganate ? (Molar volume at room temperature is 24 litres).

40. (a) A flask contains 3·2 g of sulphur dioxide. Calculate the following :

(i) The moles of sulphur dioxide present in the flask.

(ii) The number of molecules of sulphur dioxide present in the flask.

(iii) The volume occupied by 3·2 g of sulphur dioxide at STP.

(S = 32, O = 16)

(b) An experiment showed that in a lead chloride solution, 6·21 g of lead is combined with 4·26 g of chlorine. What is the empirical formula of this chloride ? (Pb = 207; Cl = 35·5).

41. The volumes of gases A, B, C and D are in the ratio, 1 : 2 : 2 : 4 under the same conditions of temperature and pressure.

(i) Which sample of gas contains the maximum number of molecules ?

(ii) If the temperature and the pressure of gas A are kept constant, then what will happen to the volume of A when the number of molecules is doubled ?

(iii) If this ratio of gas volume refers to the reactants and products of a reaction, which gas law is being observed ?

(iv) If the volume of A is actually 5·6 dm^3 at STP, calculate the number of molecules in the actual volume of D at STP (Avogadro's Number is 6 x 10^{23}).

(v) Using your answer from (iv), state the mass of D if the gas is dinitrogen oxide (N_2O).

42. The equation given below relate to the manufacture of sodium carbonate (molecular weight of Na_2CO_3 = 106).

1. $NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$
2. $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Equations (1) and (2) are based on the production of 21·2 g of sodium carbonate.

(a) What mass of sodium hydrogen carbonate must be heated to give 21·2 g of sodium carbonate?

(b) To produce the mass of sodium hydrogen carbonate calculated in (a), what volume of carbon dioxide, measured at STP, would be required ?

43. A sample of ammonium nitrate when heated yields 8·96 litres of steam (measure at STP).

$$NH_4NO_3 \rightarrow N_2O + 2H_2O$$

(i) What volume of dinitrogen oxide is produced at the same time as 8·96 litres of steam ?

(ii) What mass of ammonium nitrate should be heated to produce 8·96 litres of steam ? (Relative molecular mass of ammonium nitrate is 80).

(iii) Determine the percentage of oxygen in ammonium nitrate (O = 16).

44. Given that the relative molecular mass of copper oxide is 80, what volume of ammonia (measured at STP) is required to completely reduce 120g of copper oxide?

The equation for the reaction is :

$$3CuO + 2NH_3 \rightarrow 3Cu + 3H_2O + N_2$$

45. (a) Calculate the number of moles and the number of molecules present in 1·4 g of ethylene gas. What is the volume occupied by the same amount of ethylene?

(b) What is the vapour density of ethylene ?

46. (a) Calculate the percentage of sodium in sodium aluminium fluoride (Na_3AlF_6) correct to the nearest whole number. (F = 19; Na = 23; Al = 27)

(b) 560 ml of carbon monoxide is mixed with 500 ml of oxygen and ignited. The chemical equation for the reaction is as follows :

$$2\ CO + O_2 \rightarrow 2\ CO_2$$

Calculate the volume of oxygen used and carbon dioxide formed in the above reaction.

2009

47. (a) A gas cylinder of capacity of 20 dm^3 is filled with gas X the mass of which is 10 g. When the same cylinder is filled with hydrogen gas at the same temperature and pressure the mass of the hydrogen is 2 g, hence the relative molecular mass of the gas is :

(i) 5 (ii) 10

(iii) 15 (iv) 20

(b) (i) Calcium carbide is used for the artificial ripening of fruits. Actually the fruit ripens because of the heat evolved while calcium carbide reacts with moisture. During this reaction calcium hydroxide and acetylene gas is formed. If 200 cm^3 of acetylene is formed from a certain mass of calcium carbide, find the volume of oxygen required and carbon dioxide formed during the complete combustion. The combustion reaction can be represented as below.

$$2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O_{(g)}$$

(ii) A gaseous compound of nitrogen and hydrogen contains 12·5% hydrogen by mass. Find the

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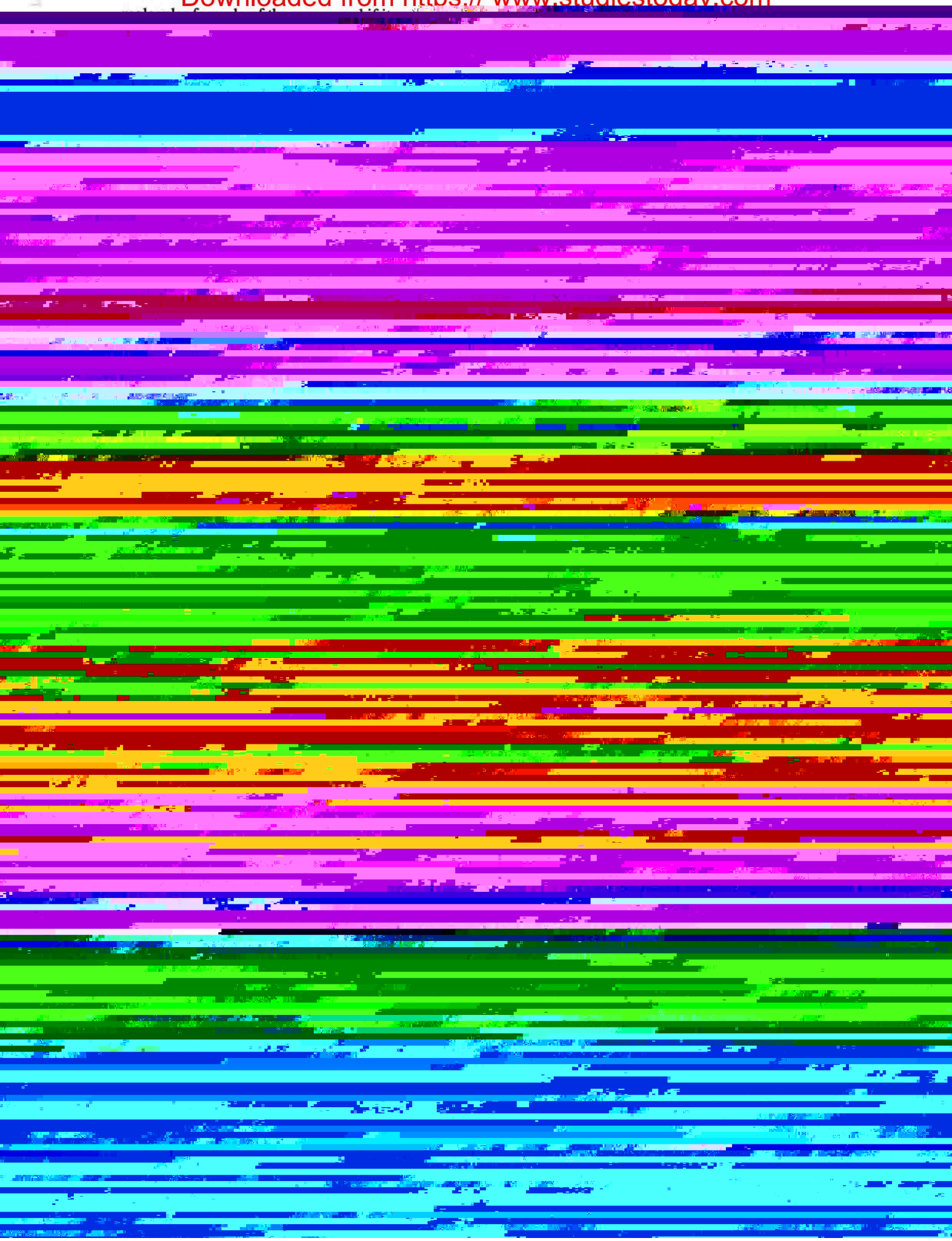

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(iii) A vessel contains **X** number of molecules of hydrogen gas at a certain temperature and pressure. How many molecules of nitrogen gas would be present in the same vessel under the same conditions of temperature and pressure?

(b) O_2 is evolved by heating $KClO_3$ using MnO_2 as a catalyst.

$$2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$$

(i) Calculate the mass of $KClO_3$ required to produce 6·72 litre of O_2 at STP.
[atomic masses of K = 39, Cl = 35·5, O = 16].

(ii) Calculate the number of moles of oxygen present in the above volume and also the number of molecules.

(iii) Calculate the volume occupied by 0·01 mole of CO_2 at STP.

2014

52. (a) (i) Oxygen oxidises ethyne to carbon dioxide and water as shown by the equation:

$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$

What volume of ethyne gas at s.t.p. is required to produce 8·4 dm^3 of carbon dioxide at STP ?
[H = 1, C = 12, O = 16]

(ii) A compound made up of two elements X and Y has an empirical formula X_2Y. If the atomic weight of X is 10 and that of Y is 5 and the compound has a vapour density (V.D.) 25, find its molecular formula.

(b) A cylinder contains 68 g of Ammonia gas at STP

(i) What is the volume occupied by this gas ?

(ii) How many moles of ammonia are present in the cylinder ?

(iii) How many molecules of ammonia are present in the cylinder ?

ANSWERS

1. 1 mole **2.** (a) 24 gm (b) 5·6 L **3.** (a) 0·25 moles (b) 2·24 dm^3 (c) 0·07g **4.** Urea **5.** (a) 1250 cm^3 (b) $\frac{5000}{7}$ cm^3 (c) increased to 3·5 times **6.** 23·3 kg **7.** (a) CH_3 (b) C_2H_6 **8.** (c) (i) 0·01 moles (ii) 0·12 g (iii) 0·025 g (iv) C_2H_5 **9.** XO_2 **10.** (a) 17 (b) 34 a.m.u. **11.** (a) 11·2 L (b) (i) 12 cm^3 (ii) 24 cm^3 **12.** (a) 28g, 22·4 *l* **13.** (a) (i) 3g (ii) 11·2 *l*, (b) (i) $CHCl_2$ (ii) $C_2H_2Cl_4$ **14.** 35·5 **15.** $SiCl_3$ **16.** H_2PO_3 ; $H_4P_2O_6$ **17.** 22·45 g **18.** (a) 22 kg (b) X **19.** (a) 22·4 dm^3 (b) 11·2 dm^3 (c) 44·8 dm^3 (d) 71 g **20.** (a) 2 (b) 2·24 *l*, 1·12 *l* (c) 1·58 *l* (d) 78·14 g (e) 58·5 **21.** (i) 8 moles (ii) 179·2 dm^3 (iii) 704 g (iv) C_4H_9 **22.** (a) 75%, (b) 11·5%, (c) 26·5% **23.** 89·6 a.m.u. **24.** (a) 0·25 moles (b) 0·25 moles (c) 5·6 litres (d) 25 g (e) 38 g **25.** H_2S = 4·48 L O_2 = 6·72 L **26.** 1·35 g **27.** 1171·4 kg **28.** (a) 0·2 mole, 19·6 g (b) 63 g (c) 8·96 L **29.** 126, PF_5 **30.** 21·2 g **31.** M_2Cl_6 **32.** (i) CBr_3 (ii) C_2Br_6 **33.** 300 cc **34.** (a) 1 V litres (b) 3 V litres (c) 11g **35.** Na_3PO_4. **36.** 44·8 dm^3 + 5·6 dm^3 = 50·4 dm^3. **37.** **Hydrogen** : maximum; **chlorine** : minimum. **38.** 42·6% **39.** (a) 32 (b) 1·2 L **40.** (a) (i) 0·05 (ii) 3×10^{22} molecules (iii) 1·12 L (b) $PbCl_4$ **41.** (i) D (ii) 2 V (iii) Gay-Lussac's Law (iv) 6×10^{23} (v) 44 **42.** (a) 33·6g (b) 8·96 dm^3 **43.** (i) 4·48 litres (ii) 16 g (iii) 60% **44.** 22·4 litres **45.** (a) 0·05 moles, 3×10^{22} molecules, 1·12 litres (b) 14 **46.** (a) 32·8% (b) oxygen 280 ml, carbon dioxide 560 ml **47.** (a) 10 (b) (i) 500 cc, 400 cc (ii) N_2H_4 (c) (1) 1120 g (2) 896 dm^3 (ii) 19·33% (iii) 112 g **48.** (a) (i) 18 + 16 = 34 litres (ii) N = 35%, O = 60% (b) (i) $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ (ii) 450 g (iii) 100·8 litres (iv) 499·5 g (v) 9 moles of HCl **49.** (a) (i) 112 litres (ii) 22·4 litres (b) (i) $C_2H_4Br_2$ (ii) (1) 0·533 g (2) 4·4 g **50.** (a) (i) 585 dm^3 (ii) 33·6 litres (iii) X number of molecules (b) (i) 24·5 g (ii) 0·3, $1·806 \times 10^{23}$ (iii) 0·224 litres **52.** (a) (i) 4·2 dm^3 (ii) X_4Y_2 (b) (i) 89·6 dm^3 (ii) 4 moles (iii) $2·4 \times 10^{24}$.

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6 ELECTROLYSIS

SYLLABUS :

(i) Electrolytes and non-electrolytes.
Definitions and examples.

(ii) Substances containing molecules only, ions only, both molecules and ions.
Substances containing molecules only, ions only, both molecules and ions. Examples: relating their composition with their behaviour as electrolyte (strong and weak), non-electrolyte.
Definition and explanation of electrolysis, electrolyte, electrode, anode, cathode, anion, cation, oxidation and reduction (on the basis of loss and gain of electrons).

(iii) An elementary study of the migration of ions, with reference to the factors influencing selective discharge of ions, illustrated by the electrolysis of: molten lead bromide; acidified water with platinum electrodes and aqueous copper (II) sulphate with copper electrodes; electron transfer at the electrodes.
The above electrolytic processes can be studied in terms of electrolyte used, electrodes used, ionization reaction, anode reaction, cathode reaction, use of selective discharge theory wherever applicable.

(iv) Application of electrolysis: electroplating with nickel and silver; purification of copper; choice of electrolyte for electroplating.
Reasons and conditions for electroplating, names of the electrolytes and the electrodes used should be given. Equations for the reactions at the electrodes should be given for electroplating, refining of copper.

(v) Acids, bases and salts as electrolytes: reference should be made to the activity series as indicating the tendency of metals, e.g., Na, Mg, Fe, Cu, to form ions.

6.1 INTRODUCTION

This chapter deals with a process called "Electrolysis" which establishes a relationship between electrical energy and chemical change.

The word electrolysis is made up of two words **'electro'** meaning flow of electrons or electricity and **'lysis'** meaning separating, *i.e.*, bringing about a (chemical) change in a substance by passage of electricity.

Many substances allow electricity to pass through them and many do not. It has also been found that all the substances allowing passage of electricity do not undergo chemical change.

Most metallic **elements** like copper, aluminium, iron and all alloys, whether in solid state or in molten state, allow an electric current to pass through them without undergoing any chemical change. Non-metals (except graphite) like sulphur, phosphorus, etc., do not conduct electricity. Therefore, metals and alloys are called **conductors** and non-metals are called **non-conductors** (insulators) of electricity.

The conduction of a current by a conductor is due to the flow of electrons. Insulators do not possess mobile electrons.

Chemical **compounds** behave differently when an electric current is passed through them. Some compounds conduct electric current in aqueous state or in molten state but not in solid state and are decomposed by the passage of electric current. *For example*, the passage of electric current through copper chloride solution decomposes it to metallic copper and chlorine. Such chemical compounds which conduct electric current in molten or aqueous state are called **electrolytes**. Some compounds do not conduct electric current at all whether in solid state or in aqueous state. They are called **non-electrolytes,** *e.g.*, sugar solution.

Chemical decomposition of a chemical compound can be brought about by using electric current. Such a chemical process is called **electrolysis**.

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Table 6.1 Differences between metallic conductors and electrolytic conductors

Metallic conductors e.g. Cu wire	*Electrolytic conductors* e.g. $CuSO_4$ solution
1. Flow of current consists of a single stream of electrons flowing from the negative pole to the positive pole.	1. Flow of current consists of two streams, one of the positive cations flowing towards the negative cathode and the other of negative anions flowing towards the positive anode.
2. Metals, both in solid and liquid state are good conductors of electricity.	2. Electrolytes are good conductors only in aqueous solution or in molten state.
3. No change of form or in composition occurs during conduction of electricity. This is only a physical process.	3. New products are formed at the electrodes. This involves chemical changes.

6.2 SOME IMPORTANT TERMS

(i) Electrolysis : *It is the process of decomposition of a chemical compound in aqueous solutions or in molten state accompanied by a chemical change by using direct electric current.*

(ii) Electrolytes *are compounds which either in aqueous solution or in molten state allow electric current to pass through them. Electrolytes can be* ***electrovalent compounds****, composed of metals and non-metals, or* ***polar covalent compounds****, that form ions in water.*

Examples of electrolytes

Acids – H_2SO_4, HNO_3, H_3PO_4, HCl;

Bases – NaOH, KOH, NH_4OH, $Ca(OH)_2$;

Salts – NaCl, $CuSO_4$, $PbBr_2$;

Electrolytes are of *two* types, based on their conductivity and ability to dissociate into ions.

1. Strong electrolyte 2. Weak electrolyte

(iii) Non-electrolyte : *It is a compound which neither in solution nor in the molten state allows an electric current to pass through it.*

It also does not decompose at the electrodes.

Non-electrolytes do not have ions even in solution. They contain only ***molecules***.

Table 6.2 Differences between strong and weak electrolytes

Strong electrolytes	*Weak electrolytes*
1. Electrolytes which allow a large amount of electricity to flow through them.	1. Electrolytes which allow small amounts of electricity to flow through them.
2. These are *good conductors of electricity.*	2. These are poor *conductors of electricity.*
3. These are almost, *completely dissociated in fused or aqueous solution state.*	3. These are *partially dissociated* in fused or aqueous solution state.
4. These solutions contain ***(almost) only free mobile ions.***	4. These solutions ***contain ions as well as molecules.***
5. Strong electrolyte allows a bulb to glow brightly.	5. Weak electrolyte allows a bulb to glow dimly.
Examples	*Examples*
Acids – Hydrogen chloride, Sulphuric acid, Nitric acid, etc.	***Acids*** – Carbonic acid, acetic acid, oxalic acid, etc.
Bases – NaOH, KOH (aqueous or molten state)	***Bases*** – Calcium hydroxide, ammonium hydroxide. (aqueous or molten state)
Salts – NaCl (molten or aqueous) $PbBr_2$ (molten), $CuCl_2$ (aq.) $CuSO_4$(aq.)	***Salts*** – Sodium carbonate and bicarbonate.

Examples of non-electrolytes

Distilled water, alcohol, kerosene, carbon disulphide, cane sugar, benzene, glucose and urea.

(iv) Electrolytic cell : A non-conducting vessel in which electrolysis is carried out, is called an **electrolytic cell** or **voltameter**. It converts electrical energy into chemical energy (Fig. 6.1).

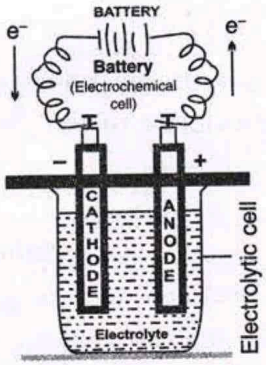

Fig. 6.1 A simple electrolytic cell or voltameter.

(v) Electrochemical cell : It is a device used to convert chemical energy into electrical energy.

Examples : Simple voltaic cell, Daniel cell, *etc.*

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(vi) Electrodes : *Two metal plates or wires or graphite rods or gas carbon rods immersed in the electrolyte through which the current enters and leaves the eletrolytic cell are called* **electrodes.**

Anode : The electrode connected to the positive terminal of the battery is called **anode.**

Cathode : The electrode connected to the negative terminal of the battery is called **cathode.**

Note : When the current is turned on, the *cathode becomes the negative electrode by which electrons enter the electrolyte and the anode becomes the positive electrode by which electrons leave the electrolyte.*

The electrodes ..... Which is which ?
How to remember :
Anode → ADD = + *i.e.* positive
The anode (positive) hence cathode (negative)
Anion (negative) goes to **Anode**
Cation (positive) goes to **Cathode**
A to **A** and **C** to **C**

Table 6.3 Differences between anode and cathode.

Anode	*Cathode*
1. It is the electrode connected to positive terminal of the battery.	1. It is the electrode connected to negative terminal of the battery.
2. Anions migrate to anode.	2. Cations migrate to cathode.
3. It is an oxidising electrode. Oxidation takes place at anode.	3. It is a reducing electrode. Reduction take place at cathode.

(vii) Ions : The atoms or groups of atoms which carry a positive or a negative charge are known as ions.

Atoms which carry positive charge are called ***cations*** and the atoms which carry negative charge are called ***anions***.

The charge on an ion, positive or negative is equal to the valency of the atom or the ion.

Table 6.4 Differences between cations and anions

Cations	*Anions*
1. Are positively charged ions.	1. Are negatively charged ions.
2. Migrate to the cathode during electrolysis.	2. Migrate to the anode during electrolysis.
3. Gain electron(s) from the cathode and get reduced to become a neutral atom.	3. Lose electrons to the anode and get oxidised to become a neutral atom.
Examples :	*Examples :*
Cations : Na^+, Ca^{2+}, Al^{3+}	**Anions:** PO_4^{3-}, Cl^-, SO_4^{2-}, OH^-

(viii) Oxidation

In the electronic concept, **oxidation** is defined as a ***process in which an atom or an ion loses electron(s)***.

Element

$Zn - 2e^- \rightarrow Zn^{2+}$
$Na - e^- \rightarrow Na^+$
$Fe^{2+} - e^- \rightarrow Fe^{3+}$
$S^{2-} - 2e^- \rightarrow S$

Oxidation is also defined as a *chemical process which involves* addition of oxygen or removal of hydrogen.

(a) **Addition of oxygen :**

Element	+	*Oxygen*	→	*Oxide*
C	+	O_2	→	CO_2
2Mg	+	O_2	→	2 MgO

In the above reactions, carbon and magnesium get oxidised to carbon dioxide and magnesium oxide respectively by the gain of oxygen.

(b) **Removal of hydrogen :**

$H_2S + Cl_2 \rightarrow 2HCl + S$

Hydrogen sulphide is oxidised to sulphur due to the loss of hydrogen.

Oxidising Agents : *An oxidising agent is one that oxidises other substances by either accepting electrons or by providing oxygen or an electronegative ion [or by removing hydrogen or an electropositive ion].* Thus, in the above examples, oxygen and chlorine are the oxidising agents.

Some examples of oxidising agents.

1. ***Solids :*** Manganese dioxide, red lead, lead dioxide, etc.
2. ***Liquids :*** Hydrogen peroxide, conc. nitric acid, conc. sulphuric acid, bromine, etc.
3. ***Gaseous :*** Oxygen, ozone, chlorine, sulphur dioxide, etc.

(ix) Reduction

In the electronic concept, ***reduction*** *is defined as a* ***process in which an atom or an ion gains electron(s).***

$Cu^{2+} + 2e^- \rightarrow Cu$
$Fe^{3+} + e^- \rightarrow Fe^{2+}$
$Sn^{2+} + 2e^- \rightarrow Sn$
$S + 2e^- \rightarrow S^{2-}$

Reduction is also defined as a *chemical process which involves removal of oxygen or addition of hydrogen.*

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(a) **Removal of oxygen :**

Oxide	+	*Element*	→	*Metal*
CuO	+	H_2	→	Cu + H_2O
ZnO	+	C	→	Zn + CO

Copper (II) oxide and zinc (II) oxide are reduced to copper and zinc respectively by the loss of oxygen.

(b) **Addition of Hydrogen :**

$Cl_2 + H_2S \rightarrow S + 2HCl$

$3Cl_2 + 2NH_3 \rightarrow N_2 + 6HCl$

In the above reactions, chlorine is reduced to hydrogen chloride due to the gain of hydrogen.

Reducing agents : *A reducing agent is one that reduces other substances by providing electrons, or by providing hydrogen or an electropositive ion, (or by removing oxygen or an electronegative ion).* Thus, in the above examples, hydrogen, carbon, hydrogen sulphide and ammonia are the reducing agents.

Some examples of reducing agents.

1. ***Solids :*** Carbon, metals like Zn, Al, Cu and Na, stannous chloride, glucose, etc.
2. ***Liquids :*** Hydrogen peroxide, hydrogen iodide, hydrogen bromide, etc.
3. ***Gaseous :*** Hydrogen sulphide, carbon monoxide, sulphur dioxide, etc.

6.3 THEORY OF ELECTROLYTIC DISSOCIATION

Svante Arrhenius, in 1887, gave an ionic theory. According to him :

1. An electrolyte on dissolving in water dissociates into free mobile ions *i.e.* cations [+ve ions] and anions [–ve ions] and allows the flow of an electric current through it.
2. All ions carry an electric charge and are responsible for the flow of current through the solution.
3. The conductivity of the electrolyte depends upon the concentration of the ions in the solution.
4. The number of positive charges equals the number of negative charges in the solution and thus the solution is in electrolytic equilibrium.
   The equilibrium is also established between the unionized molecules and the ions produced.
5. In non-electrolytes, such as sugar solution, benzene, etc., there is no ionisation and therefore, **only molecules** are present in solution.
6. The *degree of dissociation* is the extent to which an electrolyte dissociates or breaks up into ions.

The difference between Arrhenius and Modern concept

Arrhenius considered that water ionises electrolytes but Modern concept consider that electrolytes are ionic even in solid state and their ions are held by strong electrostatic forces which make them immobile.

Water renders these ions mobility by breaking the electrostatic forces.

6.4 CONDUCTION IN COMPOUNDS

The electrovalent compounds consist of aggregate of oppositely charged ions held in their positions by the strong electrostatic forces of attraction. Hence, in the *solid state*, an electrovalent compound like sodium chloride cannot conduct electricity despite the presence of ions.

When the electrovalent compounds are melted or dissolved in water, the forces of attraction between their ions are broken and the ions then move about and conduct electricity.

Covalent compounds which are polar in nature also ionise in aqueous solution and conduct electricity *e.g.* HCl, NH_3, H_2O.

6.5 CHARACTERISTICS OF ELECTROLYSIS

1. The passage of electricity through an electrolyte causes the positive ions *i.e.*, cations to migrate towards the cathode and negative ions *i.e.*, anions to migrate towards the anode.
2. The number of electrons gained by the anode is equal to the number of electrons given by the cathode.
3. The products of electrolysis are formed at the anode and cathode itself since the exchange of electrons takes place only at the surface of the electrodes.
4. Only hydrogen gas and metals are liberated at the cathode and hence are called *electro-positive elements.*
5. Only non-metals are liberated at the anode and are called *electronegative elements.*
6. The mass of a substance produced at an electrode during electrolysis is proportional to the quantity of electricity passing through the eletrolyte (Faraday's Law of electrolysis).

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7. **The process of electrolysis is a redox reaction.**

The reaction at the cathode involves reduction of cations as they gain electrons to become neutral atoms while that at anode involves oxidation of anions as they lose electrons to become neutral.

Example : Dissociation of sodium chloride during electrolysis.

$$\underset{\text{(aqueous)}}{NaCl} \rightleftharpoons Na^+ + Cl^-$$

At cathode : $Na^+ + e^- \rightarrow Na$ **(reduction)**

At anode : $Cl^- - e^- \rightarrow Cl$ **(oxidation)**

$$Cl + Cl \rightarrow Cl_2$$

Overall reaction : $2NaCl \rightarrow 2Na + Cl_2$

8. The alternating current (A.C.) does not cause any chemical change when passed through an electrolyte and therefore does not help electrolysis to occur. *Electrolysis is caused by direct current.*

6.5.1 Electrolytic Dissociation

The process due to which an ionic compound dissociates into ions in the fused state or in aqueous solution is called ***electrolytic dissociation.***

Electrolytic dissociation of sodium chloride

1. Solid sodium chloride is an electrovalent compound and its crystal structure contains sodium ions [Na^+] and chloride ions [Cl^-]. **In solid state sodium chloride ions are not free but are held together by an electrostatic force of attraction hence it does not conduct electricity (due to the absence of free ions).** [Fig 6.2].

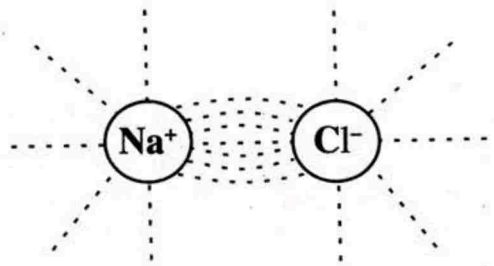

Fig. 6.2 In solid state, ions are held by electrostatic attraction. When electrovalent compounds like NaCl are fused or dissolved in water, the oppositely charged ions become mobile and conduct electricity.

2. When solid sodium chloride is dissolved in water **(aqueous solution)** or melted **(fused state)**, it conducts electricity.

(a) **In the aqueous state,** the slightly negatively charged oxygen atoms of the polar water molecule exerts a pull on the positively charged sodium ions. A similar pull is exerted by the slightly positively charged hydrogen atoms of the water on the negatively charged chloride ions. Thus the ions becomes free in solution. These free ions conduct electricity.

(b) **In the molten state**, the high temperatures required to melt the solid weakens the bond between the particles and the ions are set free. [Fig. 6.3].

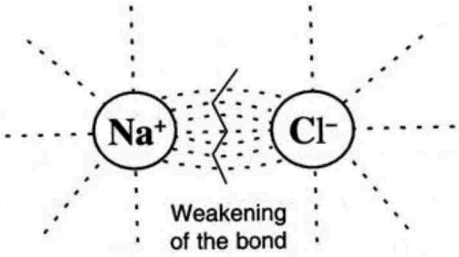

Fig. 6.3 In water solution or in molten state, the attraction is lost and ions are free to move about.

6.5.2 Ionisation

There are *certain* covalent molecules which are not made up of ions but form ions when dissolved in water. These solutions conduct an electric current.

The process by which polar covalent compounds are converted into ions, in water solution, is called ***ionisation.***

For example, hydrogen chloride.

1. In the gaseous state or in the pure liquid state, **HCl** is unionized and does not conduct an electric current.
2. It is, however, polar covalent in nature, *i.e.* shows charge distribution in its molecule such that the hydrogen atom has a slight positive charge and the chlorine atom has a slight negative charge.

$$H^{\delta+} - Cl^{\delta-}$$

3. When hydrogen chloride is added to water [a polar solvent], the slightly negatively charged oxygen atom of the water exerts an electrostatic pull on the slightly positively charged hydrogen ions present in the molecule of **HCl**.

* The quantity of electricity is often expressed in unit of Faraday; 1 Faraday is equal to 96,500 coulombs of electricity.

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$$H^{\delta+} - Cl^{\delta-} + O^{\delta-}\begin{matrix}H^{\delta+}\\H^{\delta+}\end{matrix} \rightleftharpoons (H_3O)^+ + Cl^-$$

Water (Hydronium ion) Chloride ion

Thus, H^+ ions combine with water to form hydronium ions $[H_3O]^+$.

6.5.3 Comparison of ionisation and dissociation

Ionisation	Dissociation
1. Formation of positively or negatively charged ions from molecules which are not initially in the ionic state.	1. Separation of ions which are already present in an ionic compound.
2. Polar covalent compounds show ionisation *e.g.*, HCl, H_2CO_3, NH_4OH, etc. $HCl \xrightarrow{H_2O} H^+ + Cl^-$	2. Electrovalent compounds show dissociation *e.g.*, potassium chloride, lead bromide, etc. $KCl \rightarrow K^+ + Cl^-$

Intext Questions

1. Fill in the blanks.
   (a) Powdered sodium chloride (common salt) does not conduct an electric current, but it does so when ..... or when ..... .
   (b) Molten lead bromide conducts electricity. It is called an ..... It is composed of lead ..... and bromide ..... . The lead ions are ..... charged and are called ..... The bromide ..... are ..... charged and are called .....
   (c) Substances which conduct electricity in the solid state are generally.....
   (d) The electron releasing tendency of zinc is ..... than that of copper.
   (e) A solution of HCl gas in water conducts electricity because ....., but a solution of HCl gas in toluene does not conduct an electric current because .....
2. Define the following terms :
   (a) Electrolysis, (b) Non-electrolyte,
   (c) Cation and anion, (d) Weak electrolytes.
3. What is the difference between :
   (a) Modern explanation and Arrhenius explanation for the theory of electrolysis,
   (b) Electrolytic dissociation and ionisation,
   (c) A cation and an anion,
   (d) Electrolytic dissociation and thermal dissociation.
4. Name :
   (a) a salt which is a weak electrolyte,
   (b) a base which is a weak electrolyte,
   (c) an inert electrode and an active electrode,
   (d) a positively charged non-metallic ion,
   (e) the electrode at which reduction occurs,
   (f) a non-metallic element which is a conductor of electricity.
5. Electrolysis is a redox process. Explain.

6.6 ELECTROCHEMICAL (EC) SERIES

Metal elements lose electrons from the outermost shell of their atoms, forming positive metallic ions.

i.e. forming cations :

$$K \rightarrow K^+ + e^-$$
$$Ca \rightarrow Ca^{2+} + 2e^-$$
$$Al \rightarrow Al^{3+} + 3e^-$$

However, the tendency to lose these valence electrons varies from metal to metal. Consequently, the metallic elements can be arranged in decreasing order of their tendency to lose valence electrons.

6.6.1 Electrochemical Series of metals :

Sodium gets converted to Na^+ much more easily than copper to Cu^{2+}, Zn becomes Zn^{2+} more easily than Ag to Ag^+.

Based on the ease with which atoms of metals lose electrons to form positively charged ions, the metals are arranged in a series known as the ***electrochemical series.***

Table 6.5 : Electrochemical Series of Metals

Metals		Cations	
K	Cations are discharged at cathode by gain of electron(s)	K^+	Increasing ease of discharge at cathode ↓
Ca		Ca^{2+}	
Na		Na^+	
Mg		Mg^{2+}	
Al		Al^{3+}	
Zn		Zn^{2+}	
Fe		Fe^{3+}	
Ni		Ni^{2+}	
Sn		Sn^{4+}	
Pb		Pb^{2+}	
H		H^+	
Cu		Cu^{2+}	
Hg		Hg^{2+}	
Ag		Ag^+	
Au		Au^{3+}	
Pt		Pt^{4+}	

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About Electrochemical Series of Metals

1. A metal with a higher position in the series displaces the one with a lower position.
2. Metals below hydrogen cannot displace hydrogen from acids while those above it can.
3. Elements lower in the series get discharged more easily at the cathode during electrolysis because their cations can easily gain electrons.
4. Metal atoms which form their ions most easily, will accept the electrons back to form the atoms with greatest difficulty.
5. Hydrogen has some properties of a metal, *e.g.* formation of a positive ion. Thus, its position amongst metals confirms its identity with metals. Therefore, it has been placed along with metals in the series.

Electrochemical series of the anions

Anions are negatively charged ions. On electrolysis, they move towards the anode. They lose their extra electrons or the negative charge at the anode and are discharged there. But all anions do not have identical tendency to lose electrons.

Based on their tendency to lose electrons and get discharged at the anode, the anions have been arranged in the increasing order.

ELECTROCHEMICAL SERIES OF ANIONS

Increasing ease of discharge

SO_4^{2-}

NO_3^-

Cl^-

Br^-

I^-

OH^-

Increasing ease of oxidation

ABOUT ELECTROCHEMICAL SERIES OF ANIONS

1. Lower the position of an anion in the series, more easily it gets discharged at anode.
2. Higher the position of an anion in the series, more difficult for it to lose electrons or get oxidised.

SOME SOLVED PROBLEMS

1. Can we store $CuSO_4$ solution in an iron vessel?

Ans. Iron is above copper in the electrochemical series and is thus more electropositive than copper. So, iron displaces copper from copper sulphate solution. This liberated copper appears in the form of a red precipitate. Hence $CuSO_4$ solution cannot be stored in an iron vessel.

$$Fe + CuSO_4\,(aq.) \rightarrow FeSO_4\,(aq.) + Cu$$

2. On electrolysis of dilute copper (II) sulphate solution, copper is deposited at the cathode but no hydrogen gas evolves there. Explain why ?

Ans. Copper is below hydrogen in the electrochemical series. So, Cu^{2+} gets easily reduced to metallic copper than does H^+ to hydrogen gas.

3. When a dilute aqueous solution of sodium chloride is electrolysed between platinum electrodes, hydrogen gas is evolved at the cathode but metallic sodium is not deposited. Why ?

Ans. Position of hydrogen is much below that of sodium in the electrochemical series. So, hydrogen is preferentially discharged at the cathode.

4. Zinc can produce hydrogen on reacting with acids but copper cannot. Explain.

Ans. Since zinc is above hydrogen in the electrochemical series, it is more electropositive and can displace hydrogen from acids like HCl, H_2SO_4, HNO_3.

$$Zn + H_2SO_4\,(dil.) \rightarrow ZnSO_4 + H_2$$

Whereas Cu is below hydrogen and is less electropositive than hydrogen. Therefore, it cannot displace hydrogen from acids.

$$Cu + H_2SO_4\,(dil.) \rightarrow \text{No reaction occurs.}$$

6.7 PREFERENTIAL OR SELECTIVE DISCHARGE OF IONS AT ELECTRODES

When two or more ions of the same charge are present in a solution of an electrolyte, under identical conditions, and are competing for discharge at the same electrode, one of them gets preferentially discharged. This is known as *selective discharge* of ions.

Selective discharge of ions depends on three factors :

(a) *The relative position of ions in the electrochemical (activity) series.*

(b) *The relative concentration of ions.*

(c) *The nature of the electrodes.*

(a) The relative position of ions in the electrochemical series

If all other factors are constant, any ion lower in the electrochemical series gets discharged at the relevant electrode in preference to those above it (Refer Table 6.5). The table given ahead makes the point clear.

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Preferential discharge based on relative positions of ions in the electrochemical series

Electrolyte	Dissociation of electrolytes	Ions migrating to cathode	Ion discharged at cathode	Ions migrating to anode	Ion discharged at anode
1. Dilute $CuSO_4$ solution	$CuSO_4 \rightarrow Cu^{2+} + SO_4^{2-}$ $H_2O \rightleftharpoons H^+ + OH^-$	Cu^{2+}, H^+	Cu^{2+}	SO_4^{2-}, OH^-	OH^-
2. Dilute H_2SO_4 solution	$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{2-}$ $H_2O \rightleftharpoons H^+ + OH^-$	Only H^+	H^+	SO_4^{2-}, OH^-	OH^-
3. Dilute NaCl solution	$NaCl \rightarrow Na^+ + Cl^-$ $H_2O \rightleftharpoons H^+ + OH^-$	Na^+, H^+	H^+	Cl^-, OH^-	OH^-

(b) Concentration of ions in the electrolyte

If an electrolyte has a much higher concentration of a particular ion, then that ion will be discharged at the relevant electrode in preference to those ions which are lower in the electrochemical series but are present alongwith.

Examples :

Electrolysis of concentrated NaCl solution.

Ionic equation :

$$NaCl \rightarrow Na^+ + Cl^-,$$
$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

At cathode :

Both Na^+ and H_3O^+ migrate to cathode. But concentration of Na^+ in this case is much higher than that of H_3O^+ in a highly concentrated solution of NaCl.

So, Na^+ ions discharge in preference to H_3O^+ irrespective of its higher position than H_3O^+ in electrochemical series.

At anode :

Both Cl^- and OH^- migrate to the anode. But the concentration of Cl^- is much higher than that of OH^-, hence Cl^- is discharged, although OH^- has lower position in the electrochemical series.

(c) Nature of the electrodes used

Electrodes used in the process of electrolysis are either **inert** or **active.**

The electrodes used determine the ion preferred for discharge. If **inert electrodes** like graphite and platinum are used, they will not take part in the electrolytic reaction. The electrolysis will thus depend only on the above two factors *i.e.* position in the electrochemical series and the concentration of ions.

The **active electrodes** like Cu, Ni, Ag, if used, take part in the electrolytic reaction and the product formed is different.

Example :

In the electrolysis of copper (II) sulphate solution, copper atom is deposited at the cathode and copper ions are formed at the anode if copper cathode and copper anode are used. Oxygen is formed at the anode if the platinum anode is used. (Section 6.8 III and IV).

6.8 EXAMPLES OF ELECTROLYSIS

I. Electrolysis of molten lead bromide

(i) Electrolyte : Molten lead bromide ($PbBr_2$).

(ii) Temperature : Above 380°C, the melting point of $PbBr_2$ is maintained by continuous heating by a burner.

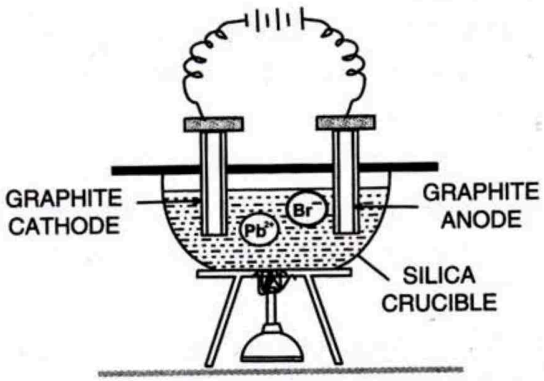

Fig. 6.4 Electrolysis of molten lead bromide.

(iii) Electrolytic cell : Crucible is made of **silica** because silica is non-reactive, withstands high temperature and is almost a non-conductor of electricity.

(iv) Electrodes : The cathode and anode are both made of graphite plates which are inert. They are connected by copper wires to the two ends of a battery.

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(v) Current : 3 amperes.

(vi) Ions present : Pb^{2+} and Br^-, *i.e.*, lead ions and bromide ions.

(vii) Electrode reactions :

At the *cathode* : $Pb^{2+} + 2e^- \rightarrow Pb$

At the *anode* : $Br^- - e^- \rightarrow Br$

$Br + Br \rightarrow Br_2$

The overall reaction is :

$$PbBr_2(l) \rightarrow Pb(s) + Br_2(g)$$

The process is shown in Fig. 6.4.

(viii) Observations :

(a) Dark reddish brown fumes of **bromine** evolve at the **anode**.

(b) *Greyish* white metal lead is formed on the **cathode**.

Explanation : When the current is turned on, the positive lead ions (Pb^{2+}) migrate towards the cathode. They take 2 electrons each and become neutral lead atoms. The negative bromide ions (Br^-) come in contact with the anode, give an electron each and become bromine atoms. These two atoms combine to form bromine molecule (Br_2).

Note :

1. Since silica is non-reactive, hence an electrolytic cell is made of silica . *It can withstand high temperature and is almost a non-conductor of electricity. The silica crucible is heated slowly from outside.*

   (a) Solid lead bromide is a non-conductor of electric current since *its ions are not free but held together by an electrostatic force of attraction.*

   (b) The ions become free when lead bromide is in the fused or molten state. *Hence the crucible is heated from outside to keep lead bromide in the molten state.*

2. Graphite anode is preferred to other inert electrodes such as platinum since *graphite is unaffected by the reactive bromine vapours.*

II. Electrolysis of acidified water using platinum electrodes (Fig. 6.5) :

Electrolytic Cell : Hoffman's Voltameter

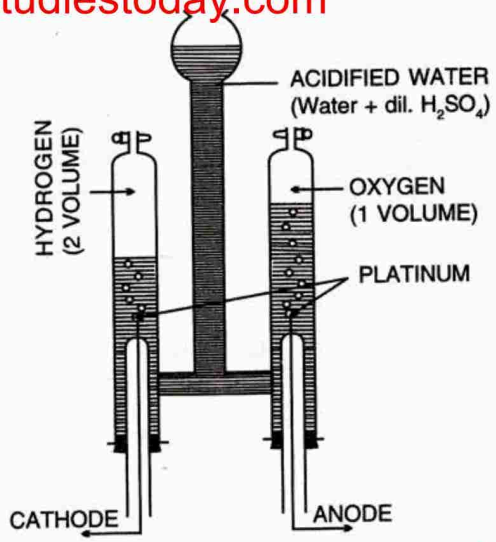

Fig. 6.5 Electrolysis of acidified water using platinum electrodes.

Electrolyte : Acidified water
(Water + dilute H_2SO_4)

Note : Dilute sulphuric acid is preferred because it is non-volatile while dilute nitric acid or hydrochloric acid are volatile acids.

Electrodes : Platinum foils, which are inert, connected by copper wires to the two terminals of a battery.

Current : 3 amperes.

Dissociation of water : Water is a non-conductor of electricity. It consists almost entirely of molecules. It is a polar covalent compound and can form ions, when traces of dilute Sulphuric acid is added. *As dilute sulphuric acid catalyse this dissociation, hence electrolysis of acidified water is considered as an example of catalysis.*

Ionisation of acidified water :

$$H_2O \rightleftharpoons H^+ + OH^-$$

$$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{2-}$$

Ions present solution :

H^+, OH^-, SO_4^{2-}

Particles present in solution :

H^+, OH^-, SO_4^{2-}, H_2O.

Reaction at Cathode :

Hydrogen ions, H^+, being the only positively charged ions, migrate to the cathode. Since the cathode is a reservoir of electrons, H^+ gains an electron and becomes a neutral hydrogen atom.

$H^+ + e^- \rightarrow H$

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Hydrogen atoms combine to form a molecule and this comes out as hydrogen gas.

$$H + H \rightarrow H_2$$

Reaction at Anode :

The SO_4^{2-} and OH^- both migrate to the anode. OH^- being lower in the electro-chemical series is discharged preferentially.

OH^- loses one electron to the anode and becomes neutral OH.

$$OH^- \rightarrow OH + e^-$$

The combination of OH forms water with the liberation of oxygen, which is given off at the anode.

$$OH + OH \rightarrow H_2O + O$$

$$O + O \rightarrow O_2$$

Note : Since SO_4^{2-} ions migrate to the anode, and H^+ ions have been discharged, the concentration of sulphuric acid will decrease at cathode.

At anode : The discharge of OH^- disturbs the ionic equilibrium of water and to maintain it, more water ionises.

$$H_2O \rightarrow H^+ + OH^-$$

The excess of H^+ ions, thus produced and the SO_4^{2-} ions present increase the concentration of sulphuric acid at anode.**

Thus there is *decrease* in acidity at cathode *increase* in acidity at anode.

Ratio of Hydrogen to Oxygen is 2:1 by volume

In the electrolysis of acidified water :

At anode OH^- loses one electron and becomes neutral OH.

$$OH^- \rightarrow OH + e^-$$

The combination of OH forms water with the liberation of oxygen.

$$2OH + 2OH \rightarrow 2H_2O + O_2$$

Since 4 OH neutral particles are involved in the equation so 4 electrons are lost in order to get 4OH neutral particles.

$$4OH^- \rightarrow 4OH + 4e^-$$

Thus, the formation of 1 molecule of oxygen at the anode releases 4 electrons and there is no build-up of electrons in any part of the circuit, the reaction of the cathode must take up 4 electrons, *i.e.*,

$$4H^+ + 4e^- \rightarrow 2H_2$$

This shows that the number of molecules of hydrogen is twice that of oxygen molecules.

According to Avogadro's Law, molecules can be substituted by volumes.

Hence, **electrolysis of water gives 2 volumes of H_2 and 1 volume of O_2.**

$$\underset{\text{acidified water(1 volume)}}{2H_2O} + \text{electrical energy} \rightarrow \underset{\text{hydrogen (2 volumes)}}{2H_2} + \underset{\text{oxygen}}{O_2}$$

III. Electrolysis of copper sulphate solution using platinum anode and copper or platinum cathode

Electrolyte : Saturated solution of copper (II) sulphate, prepared in distilled water with a small amount of conc. sulphuric acid (to increase the electrical conductivity of electrolyte).

Electrolytic cell : Glass or a porcelain jar (Fig. 6.6).

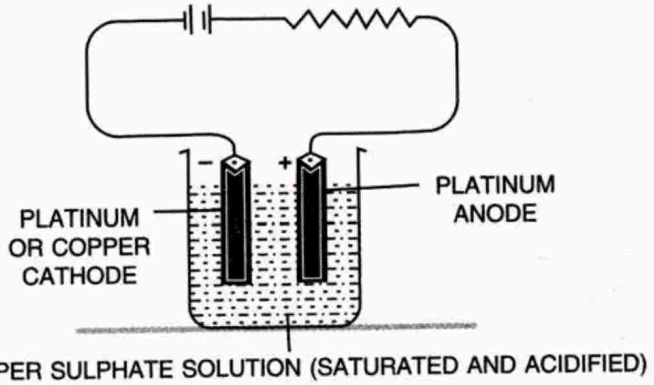

Fig. 6.6 Electrolysis of copper sulphate solution

Dissociation of aqueous copper sulphate :

$$CuSO_4 \rightleftharpoons Cu^{2+} + SO_4^{2-}$$

$$H_2O \rightleftharpoons H^+ + OH^-$$

Particles present :

Cu^{2+}, H^+, SO_4^{2-}, OH^- and H_2O molecules

Ions present :

Cu^{2+}, H^+, SO_4^{2-}, and OH^-

Reaction at cathode : The copper ions are below hydrogen ions in the activity series, thus copper ions discharge in preference to (H^+) ions, to form *neutral copper atoms*.

$$Cu^{2+} + 2e^- \rightarrow Cu$$

The copper atoms deposit themselves on the cathode.

** It may be noted that total acidity at the anode and cathode is unaffected and the overall reaction involves just decomposition of water.

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Reaction at anode : The SO_4^{2-} and OH^- ions both migrate to the anode. The OH^- ions, being lower in electrochemical series as compared to SO_4^{2-} ions, discharge to form neutral (OH) radical. The neutral (OH) radicals reunite to form *water* and *oxygen*.

$$4OH^- - 4e^- \rightarrow 4OH$$
$$2OH + 2OH \rightarrow 2H_2O + O_2$$

Product at anode : Oxygen.

Product at cathode : Reddish brown copper is deposited.

When the deposition of Cu ions is completed, then *electrolysis of water takes place*. Hydrogen gas is liberated at the cathode and oxygen gas is liberated at the anode.

Note : The ***blue colour of Cu^{2+} ions in the electrolyte solution fades*** *due to decrease in Cu^{2+} ions and finally the solution becomes colourless as soon as Cu^{2+} ions are finished.*

IV. Electrolysis of aqueous copper sulphate — using copper electrodes

1. ***Electrolytic cell***	:	Iron crucible or glass voltameter
2. ***Electrolyte***	:	Aqueous copper sulphate
3. ***Electrode***	:	*Cathode* : Copper *Anode* : Copper
4. ***Temperature***	:	Ordinary temperature
5. ***Current***	:	*Current* : 3 ampere

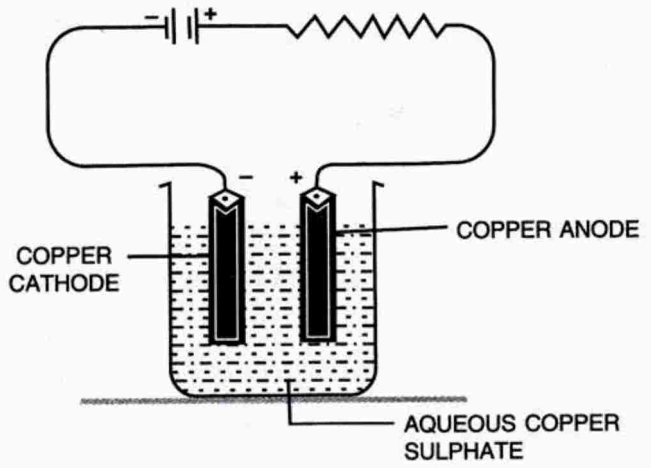

Fig. 6.7 Electrolysis of aqueous copper sulphate

6. ***Dissociation of Copper sulphate (aq.)***	:	$CuSO_4 \rightleftharpoons Cu^{2+} + SO_4^{2-}$ $H_2O \rightleftharpoons H^+ + OH^-$
Ions present	:	Cu^{2+}, H^+, SO_4^{2-}, OH^- [copper ions, hydrogen ions, sulphate ions, hydroxyl ions]
Reaction at cathode	:	**$Cu^{2+} + 2e^- \rightarrow Cu$** Both Cu^{2+} and H^+ ions migrate to the cathode, Cu^{2+} ions discharged as neutral copper atoms, by accepting electrons (as it is lower in the series).
Reaction at anode	:	**$Cu - 2e^- \rightarrow Cu^{2+}$** SO_4^{2-} and OH^- ions migrate to anode but neither of them are discharged **due to the nature of the anode,** (*copper loses electrons more easily than SO_4^{2-} and OH^-*) because copper anode itself ionises to give Cu^{2+} ions.
Product at anode	:	Nil because copper anode keep dissolving during the reaction, as Cu^{2+} ions are formed.
Product at cathode	:	Reddish brown Cu is deposited.

Note :

1. The electrolyte aqueous copper [II] sulphate may be acidified with traces of dilute sulphuric acid.

   This enhances the conductivity of the electrolyte and may prevent its hydrolysis.

2. **The blue colour of aq. copper [II] sulphate solution remains unchanged during its electrolysis due to the copper electrodes.**

Reason : For every copper ion [Cu^{2+}] discharged at the cathode as neutral copper atom, a copper ion [Cu^{2+}] is released or added to the solution by the anode and hence total number of Cu^{2+} ions remains the same. *Therefore, the blue colour does not fade.*

H^+ ions, SO_4^{2-} ions and OH^- ions do not take part in the electrolytic reaction and are known as **spectator ions.**

Intext Questions

1. Name two substances in each case :
   (a) Contain only molecules,
   (b) Contain only ions,
   (c) Contain ions as well as molecules.
2. Explain the following :
   (a) A solution of cane sugar does not conduct electricity, but a solution of sodium chloride is a good conductor,
   (b) Hydrochloric acid is a good conductor of electricity,
   (c) During the electrolysis of an aqueous solution of NaCl, hydrogen ion is reduced at the cathode and not the sodium ion though both Na^+ and H^+ ions are present in the solution.
3. (a) Among Zn and Cu, which would occur more readily in nature as metal and which as ion?
   (b) Why cannot we store $AgNO_3$ solution in copper vessel ?
   (c) Out of Cu and Ag, which is more active ?

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4. (a) How would you change a metal like Cu into its ions?
   (b) How would you change Cu^{2+} ion to Cu ?
5. A solution of caustic soda (NaOH) in water or when fused, conducts an electric current. What is the similarity in these two cases ?
6. During electrolysis of an aqueous solution of sulphuric acid between platinum electrodes, two types of anions migrate towards the anode but only one of them is discharged :
   (a) Name the two anions,
   (b) Name the main product of the discharge of anion at the anode and write the anode reaction,
   (c) Name the product at the cathode and write the reaction.
   (d) Do you notice any change in colour. State why?
   (e) Why this electrolysis, is considered as an example of catalysis ?
7. An electrolytic cell is set up using two platinum electrodes and an aqueous solution of copper (II) sulphate,
   (a) Draw a labelled diagram of the electrolytic cell,
   (b) Name the ions present in the cell,
   (c) Name the ions migrating towards the anode,
   (d) Name the ions migrating towards the cathode,
   (e) Name the ions which will not be discharged at electrodes during electrolysis,
   (f) Write the reaction at the cathode,
   (g) Write the reaction at the anode,
   (h) Name the spectator ion in the solution.
8. State the electrode reaction at the anode during electrolysis of :
   (a) Very dilute sulphuric acid,
   (b) Aqueous copper sulphate solution,
   (c) Sodium chloride solution,
   (d) Fused lead bromide,
   (e) Magnesium chloride (molten).
9. Choosing only words from the following list, write down the appropriate words to fill in the blanks (a) to (e) below: anions, anode, cathode, cations, electrode, electrolyte, nickel, voltameter.

   The electroplating of an article with nickel requires an (a) _____ which must be a solution containing (b) _____ ions. The article to be plated is placed as the (c) _____ of the cell in which the plating is carried out. The (d) _____ of the cell is made from pure nickel. The ions that are attracted to the negative electrode and discharged are called (e) _____

6.9 APPLICATIONS OF ELECTROLYSIS

Electrolysis has several technical and commercial applications. These are as follows :

1. Electroplating with metals,
2. Electrorefining of metals,
3. Extraction of metals (Electrometallurgy).

6.9.1 Electroplating

Electroplating is a process in which a thin film of a metal like gold, silver, nickel, chromium, etc. gets deposited on another metallic article with the help of electricity.

Reasons for electroplating :

(i) *Decoration purposes :* For example, brass objects are frequently plated with silver to give them the shining appearance and beauty of a silver article.

(ii) *To protect from rusting and corrosion :* Iron tools and instruments are often electroplated with nickel or chromium or zinc to protect against rusting.

Conditions for electroplating

Condition	Reason
1. The *article* to be electroplated is always placed at the cathode.	During electrolytic reaction, the metal is always deposited at the cathode by gain of electrons.
2. The metal to be plated on the article is always made the *anode* and has to be periodically replaced.	The metal anode continuously dissolves as ions in solution and is replaced periodically.
3. The electrolyte must contain ions of the metal which is to be plated on the article [*i.e.* be a suitable salt of the plating metal].	The electrolyte dissociates into ions of the metal which migrate towards the cathode and are deposited as neutral metallic atoms on the cathode.
4. A low current for a longer time should be used.	Higher current causes uneven deposition of the metal. Longer time and low current initiates a thicker uniform deposition.
5. A direct current and not A.C. current should be used.	A.C. current causes discharge and ionisation to alternate at the cathode thus giving no effective coating.

I. Electroplating an article with silver

In order to get a perfect deposit of silver on an article, say brass spoon. The spoon is first cleaned and then it is washed in a hot solution of caustic soda followed by dilute hydrochloric acid and water. It is now thoroughly washed with detergent and several times with distilled water.

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The *spoon* is then dried and *made the cathode.*

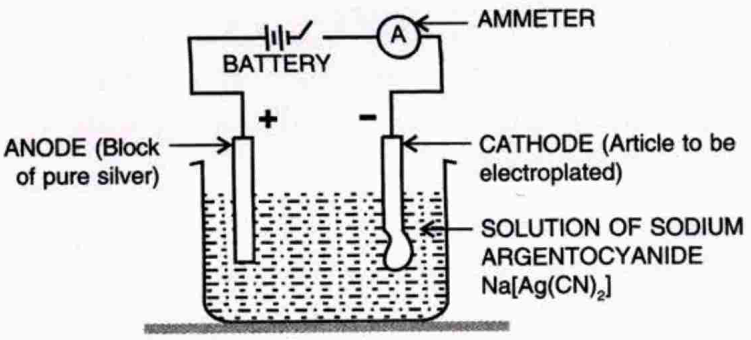

Fig. 6.8 Electroplating of a brass spoon with silver

***Electrolyte** : Sodium argentocyanide* $Na[Ag(CN)_2]$ *or Potassium argentocyanide* $K[Ag(CN)_2]$.

Sodium silver cyanide is prepared by adding sodium cyanide to a solution of silver nitrate until the precipitate formed just dissolves.

$$AgNO_3 + NaCN \rightarrow AgCN\downarrow + NaNO_3$$

$$AgCN + NaCN \rightarrow Na[Ag(CN)_2]$$

(Sodium argentocyanide)

Dissociation :

$$Na[Ag(CN)_2] \rightleftharpoons Na^+ + Ag^+ + 2CN^-$$

$$H_2O \rightleftharpoons H^+ + OH^-$$

Cathode : Article to be electroplated is duly cleaned.

$$Ag^+ + e^- \rightarrow Ag$$

(from electrolyte)

Anode : Plate of pure clean silver.

$$Ag - e^- \rightarrow Ag^+$$

(from anode) (goes into electrolyte)

The negative ions (anions) CN^- and OH^- ions migrate to the anode but none of them are discharged. Instead, the atoms of anode (silver) lose electrons and pass into solution as silver ions (Ag^+) **due to the nature of the anode**. Silver ions are attracted to the brass spoon (cathode). Here, they gain electrons and become atoms of silver which deposit on the spoon as a firm coating.

The thickness of the coating will depend on the duration of the current passed.

The spoon is taken out of the bath repeatedly, washed several times with water, dried and polished.

Note : *If silver nitrate solution is used directly instead of double cyanide of sodium and silver, the deposition of silver will be very fast and hence not very smooth and uniform.*

II. Electroplating with nickel :

Iron gets easily rusted when exposed to air. If iron articles are coated with a metal like nickel, tin or zinc, rusting can be prevented.

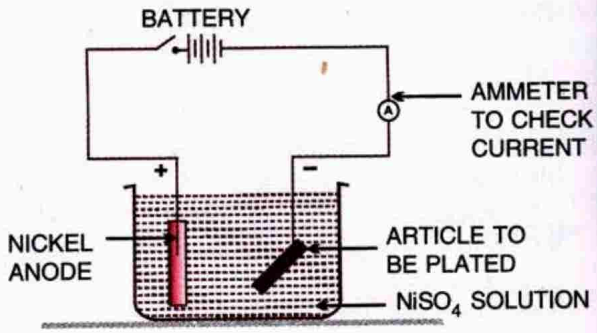

Fig. 6.9. Electroplating of an article with nickel

If nickel plating is to be done, the iron article to be coated is made the cathode, nickel plate is made the anode, and a solution of nickel sulphate containing a small amount of dilute sulphuric acid is used as an electrolyte.

Electrolyte : Aqueous solution of nickel sulphate [$NiSO_4$].

Dissociation : $NiSO_4 \rightleftharpoons Ni^{2+} + SO_4^{2-}$

$$H_2O \rightleftharpoons H^+ + OH^-$$

Cathode : Article to be electroplated is duly cleaned.

Anode : Block of pure nickel metal.

Reaction at cathode :

$$Ni^{2+} + 2e^- \rightarrow Ni \text{ [deposited]}$$

Nickel ions are discharged at the cathode as neutral nickel atoms by gain of electrons [in preference to H^+ ions].

Reaction at anode :

$$Ni - 2e^- \rightarrow Ni^{2+}$$

SO_4^{2-} and OH^- ions both migrate to the anode but neither are discharged due to the nature of electrode. Instead, the nickel anode loses electrons to give Ni^{2+} ions in solution.

Note : *Article to be plated is always kept at the cathode.*

6.9.2 Electrolytic refining of metals

Electrolytic refining is a process by which metals containing impurities are purified electrolytically to give a pure metal.

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Refining of Copper :

The most important use of copper is in the form of wire for electric transmission. For this, *copper must be absolutely pure because very small amounts of impurities reduce its conductivity to a great extent.*

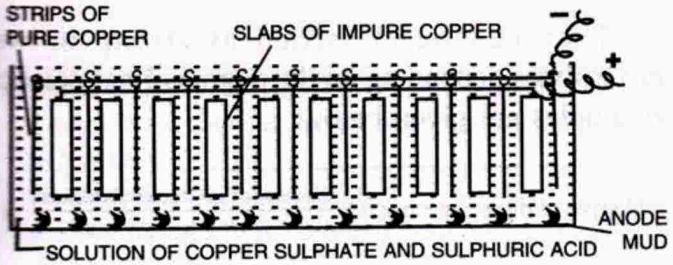

Fig. 6.10 Refining of copper

Electrolyte : *A solution of copper sulphate and dilute sulphuric acid.* (Fig. 6.10).

Cathode : *Thin strip of pure copper.*

Anode : *Impure copper.*

When the current is passed through the electrolyte, the copper ions of the copper sulphate solution are attracted to the cathode where they gain electrons and get deposited on the pure copper strips. The impure copper loses electrons and passes into solution as soluble copper ions.

Reaction :

At cathode : $Cu^{2+} + 2e^- \rightarrow Cu^-$

At anode : $Cu - 2e^- \rightarrow Cu^{2+}$

These copper ions migrate to the cathode in the form of a procession and after gaining electrons from it, get deposited on the cathode. *The net result is that, gradually the impure slabs of copper get finished and thin strips of pure copper become thicker and thicker.* The copper deposited on the cathode is 99·9% pure. After the electrolysis is over, these strips are melted and transformed into the desired shapes.

Some impurities get dissolved in the acid while others, namely silver and gold which are insoluble get collected near the anode (***anode mud***) and are recovered.

Metals like zinc, lead, mercury, silver and copper are refined (purified) by electrolysis.

6.9.3 Electrometallurgy

Electrometallurgy is the process of extraction of metals by electrolysis.

Metals which are higher in the activity series are generally extracted by electrolysis.

Activity series

K, Na, Ca, Mg, Al — They are extracted by electrolysis of their fused salts. These metals have strong affinity for oxygen.

Zn, Fe, Pb, [H], Cu, Hg, Ag — They are generally extracted by reducing agents like carbon, carbon monoxide and hydrogen as their oxides are less stable.

Pt, Au — Metals extracted by thermal decomposition.

Reactive metals are extracted from their halides (chlorides, bromides, etc.) by electrolysis using inert electrodes. Cathode is usually made of iron and anode of graphite.

Reactive metals : They are not extracted from their aqueous salt solutions by electrolysis as they can react with water.

For example :

Potassium

Electrolyte : Fused potassium bromide

Reaction : $KBr \rightleftharpoons K^+ + Br^-$

Reaction at cathode : $K^+ + e^- \rightarrow K$

Reaction at anode $Br^- - e^- \rightarrow Br$

$Br + Br \rightarrow Br_2$

Sodium

Electrolyte : Fused Sodium chloride

Reaction $NaCl \rightleftharpoons Na^+ + Cl^-$

Reaction at cathode : $Na^+ + e^- \rightarrow Na$

Reaction at anode $Cl^- - e^- \rightarrow Cl$

$Cl + Cl \rightarrow Cl_2$

Magnesium

Electrolyte : Fused Magnesium chloride

Reaction $MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-$

Reaction at cathode : $Mg^{2+} + 2e^- \rightarrow Mg$

Reaction at anode $Cl^- - e^- \rightarrow Cl$

$Cl + Cl \rightarrow Cl_2$

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Aluminium* is obtained by electrolysis of its oxide, *i.e.*, Alumina (Al_2O_3), where **gas carbon lining of electrolytic cell act as cathode, and graphite rod act as anode.**

Electrolyte : Pure Alumina (Al_2O_3), cryolite (Na_3AlF_6) and Fluorspar (CaF_2).

Reaction $Al_2O_3 \rightleftharpoons 2Al^{3+} + 3O^{2-}$

Reaction at cathode : $2Al^{3+} + 6e^- \rightarrow 2Al$

Reaction at anode $3O^{2-} - 6e^- \rightarrow 3[O]$

$3[O] + 3[O] \rightarrow 3O_2$

* Refer 7.12

6.10 ACIDS, BASES AND SALTS AS ELECTROLYTES

Acids and alkalis ionise when dissolved in water, whereas bases and salts dissociate in fused (molten) state or in their aqueous solutions into free mobile ions.

They can be classified as strong or weak electrolytes depending on the degree of dissociation#. Examples are given below.

Degree of dissociation (α) = $\frac{\text{Number of molecules dissociated}}{\text{Total number of molecules}} \times 100$

Examples of ionisation or electrolytic dissociation of acids, bases and salts.

ACIDS	BASES	SALTS
(A) STRONG ELECTROLYTES	**STRONG ELECTROLYTES**	**STRONG ELECTROLYTES**
Hydrochloric acid	**Potassium hydroxide**	**Lead bromide**
$HCl \rightleftharpoons H^+ + Cl^-$ [aq]	$KOH \rightleftharpoons K^+ + OH^-$ [aq]	$PbBr_2 \rightleftharpoons Pb^{2+} + 2Br^-$ [molten]
Nitric acid	**Sodium hydroxide**	**Copper chloride**
$HNO_3 \rightleftharpoons H^+ + NO_3^-$ [aq]	$NaOH \rightleftharpoons Na^+ + OH^-$ [aq]	$CuCl_2 \rightleftharpoons Cu^+ + 2Cl^-$ [aq]
Sulphuric acid	**Lithium hydroxide**	**Silver nitrate**
$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{2-}$	$LiOH \rightleftharpoons Li^+ + OH^-$	$AgNO_3 \rightleftharpoons Ag^+ + NO_3^-$
(B) WEAK ELECTROLYTES	**WEAK ELECTROLYTES**	**WEAK ELECTROLYTES**
Acetic acid	**Calcium hydroxide**	**Sodium carbonate**
$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ [aq]	$Ca(OH)_2 \rightleftharpoons Ca^{2+} + 2OH^-$ [aq]	$Na_2CO_3 \rightleftharpoons 2Na^+ + CO_3^{2-}$ [aq]
Formic acid	**Magnesium hydroxide**	**Potassium bicarbonate**
$HCOOH \rightleftharpoons HCOO^- + H^+$ [aq]	$Mg(OH)_2 \rightleftharpoons Mg^{2+} + 2OH^-$ [aq]	$KHCO_3 \rightleftharpoons K^+ + HCO_3^-$ [aq]
Carbonic acid	**Ammonium hydroxide**	**Lead acetate**
$H_2CO_3 \rightleftharpoons 2H^+ + CO_3^{2-}$ [aq]	$NH_4OH \rightleftharpoons NH_4^+ + OH^-$ [aq]	$(CH_3COO)_2Pb \rightleftharpoons 2CH_3COO^- + Pb^{2+}$ [aq]

EXERCISE

1. Give reasons for the following :
   (a) Electrolysis of molten lead bromide is considered to be a reaction in which oxidation and reduction go side by side, *i.e.*, a redox reaction.
   (b) The blue colour of aqueous copper sulphate fades when it is electrolysed using platinum electrodes.
   (c) Lead bromide undergoes electrolytic dissociation in the molten state but is a non-electrolyte in the solid state.
   (d) Aluminium is extracted from its oxide by electrolytic reduction and not by conventional reducing agents.
   (e) The ratio of hydrogen and oxygen formed at the cathode and anode is 2 :1 by volume.
   (f) In the electrolysis of acidified water, dilute sulphuric acid is preferred to dilute nitric acid for acidification.
   (g) Ammonia is unionised in the gaseous state but in the aqeuous solution, it is a weak electrolyte.
   (h) A graphite anode is preferred to other inert electrodes during electrolysis of fused lead bromide.
   (i) For electroplating with silver, silver nitrate is not used as electrolyte.
   (j) Carbon tetrachloride is a liquid but does not conduct electricity.
2. Classify the following substances under three headings : (a) strong electrolytes (b) weak electrolytes (c) non-electrolytes.
   Acetic acid, ammonium chloride, ammonium hydroxide, carbon tetrachloride, dilute hydrochloric acid, sodium acetate, dilute sulphuric acid.
3. Write down the words or phrases from the brackets that will correctly fill in the blanks in the following sentences:
   (a) Pure water consists entirely of _____ (ions/ molecules).
   (b) We can expect that pure water _____ (will/will not) normally conduct electricity.
4. To carry out the so-called "electrolysis of water". sulphuric acid is added to water.
   How does the addition of sulpuric acid produce a conducting solution ?

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5. Copy and complete the following table which refers to two practical applications of electrolysis

	Anode	Electrolyte	Cathode
Silver plating of a spoon		Solution of potassium argentocyanide	
Purification of copper			

6. Complete the sentence by choosing correct words given in brackets.
Electrolysis is the passage of _____ (electricity/electrons) through a liquid or a solution accompanied by a ___ (physical/chemical) change.

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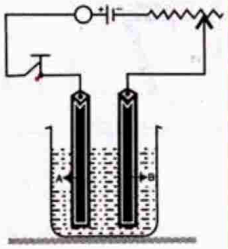
1. Element X is a metal with a valency 2. Element Y is a non-metal with a valency 3.
   (a) Write equations to show how X and Y form ions ?
   (b) If Y is a diatomic gas, write the equation for the direct combination of X and Y to form a compound,
   (c) Write two applications of electrolysis in which the anode diminishes in mass,
   (d) If the compound formed between X and Y is melted and an electric current passed through the molten compound, the element X will be obtained at the ..... and Y at the ..... of the electrolytic cell. (Provide the missing words)
2. (a) What kind of particles will be found in a liquid compound which is a non- electrolyte?
   (b) If HX is a weak acid, what particles will be present in its dilute solution apart from those of water ?
   (c) Cations are formed by ..... (loss/gain) of electrons and anions are formed by ..... (loss/gain) of electrons. (Choose the correct word to fill in the blanks).
   (d) What ions must be present in a solution used for electroplating a particular metal ?
   (e) Explain how electrolysis is an example of *redox reaction.*

2005

1. Explain why :
   (a) Cu, though a good conductor of electricity is a non electrolyte,
   (b) Solid sodium chloride does not allow electricity to pass through ?
2. Name the gas released at cathode when acidulated water is electrolysed.

2006

1. Copper sulphate solution is electrolysed using a platinum anode.



   (a) Study the diagram given alongside and answer the following questions :
   (i) Give the names of the electrodes A and B.
   (ii) Which electrode is the oxidizing electrode ?
   (b) A strip of copper is placed in four different colourless salt solutions. They are KNO_3, $AgNO_3$, $Zn(NO_3)_2$, $Ca(NO_3)_2$. Which one of the solutions will finally turn blue ?

2007

1. Choose A, B, C or D to match the descriptions (i) to (v) below. Some alphabets may be repeated.
   A. non-electrolyte B. strong electrolyte
   C. weak electrolyte D. metallic conductor
   (i) Molten ionic compound,
   (ii) Carbon tetrachloride,
   (iii) An aluminium wire,
   (iv) A solution containing solvent molecules, solute molecules and ions formed by the dissociation of solute molecules.
   (v) A sugar solution with sugar molecules and water molecules.

2008

1. (a) Here is an electrode reaction :
   $Cu \rightarrow Cu^{2+} + 2e^-$.
   At which electrode (anode or cathode) would such a reaction take place? Is this an example of oxidation or reduction ?
   (b) A solution contains magnesium ions (Mg^{2+}), iron (II) ions (Fe^{2+}) and copper ions (Cu^{2+}). On passing an electric current through this solution, which ions will be the first to be discharged at the cathode? Write the equation for the cathode reaction.
   (c) Why is carbon tetrachloride, which is a liquid, a non-electrolyte ?
2. During the electrolysis of molten lead bromide, which of the following takes place?
   A. Bromine is released at the cathode,
   B. Lead is deposited at the anode,
   C. Bromine ions gain electrons,
   D. Lead is deposited at the cathode.

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7 Metallurgy

SYLLABUS

(i) Definition of Metals and Non-metals.
Self-explanatory.

(ii) Position of the metals (Alkali metals and alkaline earth metals) in the Periodic Table and general characteristics applied to these elements with reference to the following — occurrence, nature, bonding, action of air, action of water, action of acids.
Self-explanatory.

(iii) Comparison of Metals and Non-metals.
General properties with special reference to physical properties: state, lustre, melting point, density, ductility, malleability, brittleness, conduction of electricity (exceptions to be specifically noted – e.g. graphite, mercury); chemical properties: a metal forms at least one basic oxide; non-metal, an acidic or neutral oxide; discharge of metallic ions at the cathode from fused metallic chlorides (link with bonding and ion formation); many metals liberate hydrogen from dilute HCl and H_2SO_4.
In the physical properties of metals and non-metals, atomicity and valence electrons should also be included; suitable examples must be given for basic, acidic and neutral oxides; formation and discharge of ions at the cathode (metallic) and anode (non-metallic) should be explained with examples.

(iv) Reduction of metallic oxides; some can be reduced by hydrogen, carbon and carbon monoxide (e.g. copper oxide, lead oxide, iron (II) oxide) and some cannot (e.g. Al_2O_3 MgO - refer to activity series).
Equations with conditions and observations should be given.

(v) Extraction of metals based on the activity series.
Extraction of metals: Principle of extraction of metal from its compounds by reduction – carbon reduction, electrolytic reduction. Active metals by electrolysis e.g. sodium, aluminium (reference only).

(iv) Corrosion of iron and its prevention
Experiment to illustrate that moisture and oxygen in air are responsible for the corrosion : Reaction of corrosion, Prevention by painting and galvanization.

(vii) Metals and their alloys : common ores of iron, aluminium and zinc. Extraction of Aluminium;
Metals and their alloys : Occurrence of metals in nature - mineral and ore. Common ores of iron, aluminium and zinc. Dressing of the ore - hydrolytic method, magnetic separation, froth floatation method, chemical method by using chemical - NaOH for purifying bauxite – Baeyer's process.
Extraction of Aluminium : the constituents in the charge, method of electrolytic extraction (flow chart to be used); structure of electrolytic cell and reason for using cryolite, electrolyte, electrodes, electrode reaction.
Description of the changes occurring, purpose of the substances used and the main reactions with their equations.
(a) Uses of iron, aluminium and zinc and their alloys.
Uses of iron, aluminium and zinc and their alloys. Composition of their alloys – steel, duralumin, brass.
(b) Other important alloys — bronze, fuse metal and solder
Uses only

7A. METALS AND NON-METALS

7.1 INTRODUCTION

There are 118 different elements known to us today. These elements are widely distributed in earths crust in free state as well as in the combined form.

Approximate relative abundance of some elements in the earth crust are :

Oxygen	46·6%	Silicon	27·7%
Aluminium	8%	Iron	5%
Calcium	3·6%	Sodium	2·8%
Potassium	2·8%	Magnesium	2·1%
Hydrogen	0·14%	Phosphorus	0·12%

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These elements have been classified into **metals** and **non-metals** based on their properties. Of these known elements, only 22 are non-metals.

All the metallic elements (except mercury) are solids whereas 11 non-metals are gases, one is a liquid (bromine) and the remaining non-metals are solids.

7.2 METALS

The knowledge of metals is very old. *Copper was the first metal to be used by man* for making utensils, weapons and for other purposes.

Gold, silver, copper, lead, iron, tin, antimony and mercury (earlier named **quick silver**) have been known to mankind from the ancient times.

Metals are widely used in our daily life for a large number of purposes. Some of the metals are used in making *jewellery and coins*; some are employed in making utensils and furniture, and some in the construction of heavy machinery, tools, transport vehicles and many in the manufacturing of other equipments. Metals often find applications as *catalysts* in various industrial processes such as hydrogenation of vegetable oils, manufacture of ammonia, sulphuric acid, nitric acid, dyes, drugs, etc. Metals, though in small quantities, have also been recognized as essentials for various biological processes. *For example, iron is a constituent of blood pigment and magnesium of plant pigment.*

Metals are defined as the *elements which form positive ions by the loss of electrons.*

***For example*:** Sodium, magnesium and aluminium are metals because they form positive ions by the loss of 1, 2 and 3 electrons respectively.

$$Na - e^- \rightarrow Na^+$$
$$Ca - 2e^- \rightarrow Ca^{2+}$$
$$Al - 3e^- \rightarrow Al^{3+}$$

Note : Hydrogen is a **non metal**, which forms positive ion.

$$H - e^- \rightarrow H^+$$

7.3 NON-METALS

Non-metals though small in numbers play a very vital role in our daily life.

Hydrogen, the lightest element known, is used in the hydrogenation of vegetable oils to make ghee, as a fuel and in the manufacture of compounds. It is also the essential part of organic compounds. Proteins, fats, carbohydrates, enzymes, vitamins, *etc.*, are all compounds of carbon, and are essential for the growth and development of living organisms. The presence of oxygen gas in air is essential for respiration, and for combustion. Nitrogen is the most abundant element present in the atmosphere. Its presence in air reduces the rate of combustion. Due to its inertness, it is also used to preserve food. Life would not have been possible in the absence of non-metals like carbon, oxygen, nitrogen and hydrogen.

Non-metals are the elements which form negative ions by the gain of electrons.

For example **:** Chlorine, oxygen and nitrogen are non-metals because they form negative ions by gain of 1, 2 and 3 electrons respectively.

$$Cl + e^- \rightarrow Cl^-$$
$$O + 2e^- \rightarrow O^{2-}$$
$$N + 3e^- \rightarrow N^{3-}$$

The number of electrons lost or gained by an element is equal to its **valency**.

Metals constitute the mineral wealth of a country. The major metals in the earth's crust in the decreasing order of their abundance are aluminium, iron, calcium, sodium, potassium and magnesium. The major non-metals in the earth's crust in the decreasing order of their abundance are oxygen, silicon, phosphorus and sulphur.

7.4 CATEGORISATION OF METALS

The metals have been categorised on the basis of their characteristics as given below.

7.4.1 Alkali Metals

The elements such as lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs) and francium (Fr) have one electron in their outermost orbit and show one valency. So they are placed in IA group (the first column on the left) of the Periodic Table. They are known as alkali metals, as *they react with water to form their hydroxides which are strong alkalies.*

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$$\underset{\text{(Alkali metal)}}{2M} + 2H_2O \rightarrow \underset{\text{(Alkali)}}{2MOH} + H_2$$

Where M is any metal of IA group.

Hydrogen is placed in this group as it has one electron, but it is not an alkali metal.

Characteristics

1. **Occurrence :** All of them are very reactive and do not occur in free state.
2. **Nature :** They are soft (can be cut with a knife) and have low melting and boiling points.
3. **Bonding :** All alkali metal salts (except some salts of lithium) are ionic in nature.
4. **Action of air :** They react rapidly with oxygen and water vapour in the air. *Reactivity of these metals increases down the group.*
5. **Action of water :** They react violently with water and produce hydrogen.

$$\underset{\text{(Any Alkali metal)}}{2M} + 2H_2O \rightarrow 2MOH + H_2$$

6. **Action of acids :** They react violently with dil HCl and dil. H_2SO_4, to produce hydrogen

$$\underset{\text{(Any Alkali metal)}}{2M} + 2HCl \rightarrow 2MCl + H_2$$

7. They have low ionisation energy and form unipositive ion easily.

$$M(g) - e^- \rightarrow M^{1+}(g)$$

They are strong *reducing agents*.

8. When freshly cut, they show typical silvery white metallic lustre but soon they turn dull (tarnish) because of their reaction with air.
9. All alkali metal salts impart colour to the flame, *e.g.*, lithium (*crimson red*), sodium (*golden yellow*), potassium (*pale violet*).
10. Alkali metals are obtained by electrolysis of their molten salts.

7.4.2 Alkaline Earth Metals

The elements such as beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba) and radium (Ra) are known as alkaline earth metals, because their oxides occur in the earth's crust and their hydroxides are alkalies.

All the elements of this group have two electrons in their outermost orbit and show the valency two, so they are placed in **IIA group** of the Periodic Table.

Characterstics

1. **Occurrence :** They are reactive metals (less reactive than alkali metals); hence they do not occur in free state in nature.
2. **Nature :** They are greyish white in colour and harder than the alkali metals.
3. **Bonding :** All the alkaline earth metal salts except beryllium are ionic compounds.
4. **Action of air :** They are less reactive than alkali metals.

   Their reactivity increases on moving down the group.
5. **Action of water :** They react with water to produce hydrogen.

$$\underset{\text{(Any alkaline metal)}}{M} + 2H_2O \rightarrow \underset{\text{(Alkali)}}{M(OH)_2} + H_2$$

6. **Action of acids :** They react with dil HCl and dil H_2SO_4 to produce hydrogen.

$$M + 2HCl \rightarrow MCl_2 + H_2$$

$$M + H_2SO_4 \rightarrow MSO_4 + H_2$$

7. They have low ionisation energy (higher than alkali metals) and form dipositive ion.

$$M - 2e^- \rightarrow M^{2+}$$

They are also strong **reducing agents**.

8. Melting points and boiling points are comparatively low but these are higher than the alkali metals.
9. Except beryllium and magnesium, all other alkaline earth metals impart colour to the flame like calcium-*brick red*, strontium-*crimson*, barium-*apple green* and radium-*crimson*.
10. They are obtained by electrolysis of their molten salts.

7.4.3 Transition metals*

They are placed in the middle of the Periodic Table (group 3 to group 12). Their two outermost

* Not in syllabus

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shells are incomplete. *Examples are iron, zinc, copper, mercury.*

Characteristics

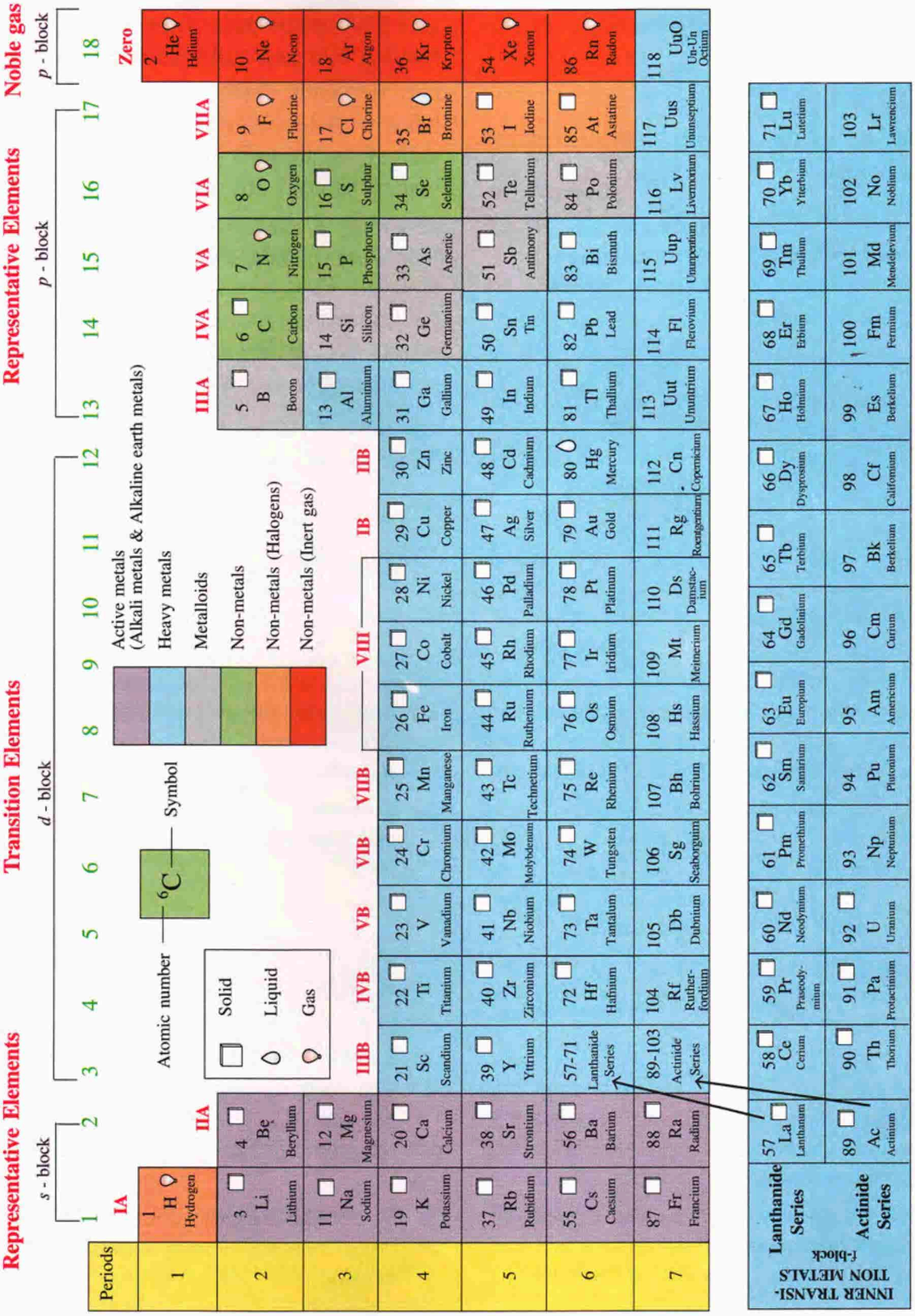
(a) All these elements are metals with high melting and boiling points.

(b) They are good conductors of heat and electricity.

(c) Some of these elements are attracted towards a magnet.

(d) Most of these elements are used as catalysts.

TABLE 7.1 : THE LONG FORM OF PERIODIC TABLE OF ELEMENTS



Elements with atomic number 110 and above have been reported but not yet fully authenticated and named.
As of 2005, the table contains 116 chemical elements whose discoveries have been confirmed;
94 are found naturally on earth and the rest are synthetic elements.

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(e) Most of these elements exhibit variable valencies.

7.4.4 Inner transition metals*

They are placed at the bottom of the Periodic Table. Their actual place is in the third column, *i.e.*, Group III B and the sixth period (called lanthanides) and the seventh period (called actinides) of the Periodic Table.

Characteristics

(a) They are heavy metals with high melting and boiling points.

(b) They show variable valencies.

(c) They form coloured ions.

(d) Some inner transition metals *i.e.* actinides are radioactive by nature.

Certain elements which show properties of both metals and non-metals are called **metalloids** or **semi-metals**.

They are : Boron (B), Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te) and Polonium (Po).

7.5 POSITION OF METALS AND NON-METALS IN THE PERIODIC TABLE

The metals have been placed on the left side (Group IA and IIA) and in the middle of the Periodic Table (Group IB to VII B and Group VIII), whereas non-metals have been placed on the right side of the Periodic Table. Elements in Group IIIA (except Boron) are metals and their metallic nature increases on moving down the group. Elements of Group IVA, VA, VIA and VIIA are mostly non-metallic, though a few elements at the bottom of the group show some metallic nature. Only *one non-metal, i.e., hydrogen* [H] *has been placed on the left side of the Periodic Table.*

The metals and non-metals in the Periodic Table are separated by a zig-zag line of metalloids.

* Not in syllabus

7.6 COMPARISON OF METALS AND NON-METALS (GENERAL PROPERTIES)

Metals	*Non-metals*
A. Physical properties	
1. Physical State : All metals are solids at room temperature **except *mercury*** (*which is a liquid*) and ***gallium*** (which changes to liquid when handled).	**1.** Non-metals exist in all the three states. Eleven non-metals are gases; only ***bromine*** is a ***liquid*** non-metal and others are solids.
2. Lustre : Metals possess a high lustre (called metallic lustre). They can be polished to give a highly reflective surface (in case of sodium and potassium, this lustre lasts for a very short time). **Lead is a metal with dull appearance.**	**2.** Non-metals have a dull appearance, **except iodine and graphite** whose crystals are quite lustrous.
3. Hardness : Most of the metals are hard and strong. (***Exceptions :* sodium and potassium** can be *cut with a knife*).	**3.** Some solid non-metals (like sulphur and phosphorus) are soft; whereas some are hard. Carbon in the form of **diamond is the hardest substance known.**
4. M.P. and B.P. : Usually, metals possess high melting and boiling points and vaporise only at high temperatures (***Exceptions* : gallium, mercury, sodium and potassium**).	**4.** The melting and boiling points of non-metals are low. They vaporise at low temperatures, (***Exceptions* : carbon, boron and silicon**).
5. Density : Metals have a high density (***Exceptions* : sodium and potassium** are less dense than water).	**5.** Non-metals have low densities. Density of iodine is higher than that of many metals. Diamond has a high density.

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6. **Ductility** : Metals are usually ductile. In other **words they can be drawn into thin wires without** breaking. Silver wires of 0·0008 mm diameter have been drawn; tungsten can be drawn into such fine wires that are almost invisible (***Exceptions*** **: zinc, mercury and gallium**).	6. Non-metals are not ductile **except carbon fibre**.
7. **Malleability** : Metals are usually malleable, *i.e.,* can be beaten into thin sheets without cracking (**except zinc and mercury**). *Silver and gold are highly malleable*. Gold has been beaten into foil less than 0·0001 mm in thickness.	7. Non-metals are not malleable in solid form *i.e.* they break into pieces when hammered.
8. **Brittleness** : Metals are hard but not brittle (***Exception*** **: Zinc**).	8. Non-metals are generally brittle, *i.e.,* they break when force is applied.
9. **Conduction of heat and electricity** : Metals are good conductors of heat and electricity ; silver and copper excel in this property. (***Exception*** **: tungsten**).	9. Non-metals are non-conductors of heat and electricity (only carbon in the form of ***gas-carbon*** and ***graphite conducts electricity***).
10. **Alloys and Amalgams** : They form alloys and amalgams.	10. They do not form alloys and amalgams. (***Exceptions:*** **carbon and phosphorus** can be used in alloys).
11. **Solubility :** Usually, metals do not dissolve in liquid solvents.	11. Non-metals dissolve in many liquid solvents, *e.g.,* sulphur dissolves in carbon disulphide and iodine in chloroform.
B. Chemical Properties	
1. **Valence Electrons (electronic configuration) :** Atoms of metals usually have 1, 2 or 3 electrons in their outermost shell. The less is the number, the more active is the metal.	1. Atoms of non-metals usually have 5, 6 or 7 electrons in their outermost shell. The more is the number, the more active is the non-metal. (***Exception is hydrogen***).
2. **Formation of ions** : Metals form +ve ions (or cations) by loss of electrons; they are electropositive. $Na - e^- \rightarrow Na^+$ $Mg - 2e^- \rightarrow Mg^{2+}$ $Al - 3e^- \rightarrow Al^{3+}$ The number of electrons lost is the valency of the metal.	2. Non-metals form –ve ions (or anions) by gain of electrons; they are electronegative. $Cl + e^- \rightarrow Cl^-$ $O + 2e^- \rightarrow O^{2-}$ $N + 3e^- \rightarrow N^{3-}$ The number of electrons gained is the valency of the non-metal (**Exception is hydrogen which generally forms positively charged cation, H^+**).
3. **Discharge of ions** : These are discharged at the cathode during the electrolysis of their compounds. ***At cathode :*** $Na^+ + 1e^- \rightarrow Na$ cation → neutral atom	3. Non-metals are liberated at the anode during electrolysis. (***Exception:*** **Hydrogen as H^+ ions are liberated at the cathode**). ***At anode :*** $Cl^- - 1e^- \rightarrow Cl$ anion → neutral atom $Cl + Cl \rightarrow Cl_2$
4. **Atomicity** : Molecules of metals in vapour state are usually monoatomic.	4. Molecules of non-metals in the vapour state are usually polyatomic. (***Exception :*** **Inert gases**).
5. **Nature of oxides :** Oxides of metals are usually basic. Soluble basic oxides dissolve in water forming an ***alkaline solution***; metallic oxides are ***electrovalent***.	5. Oxides of non-metals are usually **acidic.** Soluble acidic oxides dissolve in water forming an ***acidic solution***; non-metallic oxides are ***covalent***.

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Basic oxides: Na_2O, CaO, MgO, Fe_2O_3, CuO. **Amphoteric oxides :** Al_2O_3, PbO, ZnO **Acidic oxides :** CrO_3, Mn_2O_7	**Acidic oxides:** CO_2, SO_2, SO_3, NO_2, P_2O_5. **Neutral oxides :** CO, NO, N_2O, H_2O.
6. Reaction with acids : Metals above hydrogen in the activity series usually *replace hydrogen from dilute non-oxidising acids* (hydrochloric and sulphuric acids); $M + 2HCl \rightarrow MCl_2 + H_2$ $M + H_2SO_4 \rightarrow MSO_4 + H_2$	**6.** Non-metals do not react with dil. hydrochloric acid or sulphuric acid.
7. Nature of chlorides : Metallic chlorides are electro-valent (ionic) compounds. They are electrolytes. When electricity is passed through fused or aqueous solution of metallic chloride, metals are discharged at the cathode and chlorine at the anode.	**7.** Non-metallic chlorides are covalent compounds and are non-electrolytes. *For example*, CCl_4 and HCl.
8. Oxidising and reducing property : They ionise by loss of electrons and hence are reducing agents. $Na \rightarrow Na^+ + e^-$	**8.** They ionise by gain of electrons and hence are oxidising agents. (***Exception :* Carbon and hydrogen are strong reducing agents**). $Cl + e^- \rightarrow Cl^-$
9. Nature of hydrides : Metals do not form hydrides. But if they form, they are **not very stable,** *e.g.*, – NaH (sodium hydride) and CaH_2 (calcium hydride).	**9.** Non-metals form quite stable hydrides., *e.g.*, CH_4, (methane), H_2O (water) and NH_3 (ammonia).

EXERCISE-7A

1. (a) Name the three classes in which elements are classified. Which was the first metal used by man ?
(b) Name the metal and non-metal present in abundance in the earth crust.

2. Name the metal which is a constituent of :
(a) blood pigment, (b) plant pigment.

3. Give the importance of the following in living beings :
(a) Nitrogen, (b) Hydrogen,
(c) Carbon.

4. State the position of the following in the Periodic Table :
(a) Alkali metals, (b) Alkaline earth metals,
(c) Iron and zinc, (d) Aluminium.

5. Give the general characteristics of :
(a) Alkali metals,
(b) Alkaline earth metals with reference to
(i) bonding (ii) action of air
(iii) action of water (iv) action of acid.

6. Name :
(a) a liquid non-metal,
(b) a metal with dull appearance,
(c) a metal with low melting and boiling points,
(d) a non-metal with very high m.p. & b.p.,
(e) a metal which can float on water,
(f) a metal which can be cut with a knife,
(g) a metal which is a bad conductor of heat and electricity,
(h) a non-metal which is ductile,
(i) a non-metal used in alloys,
(j) a non-malleable metal.
(k) two metalloids

7. Distinguish between metals and non-metals on the basis of :
(i) ion formation, (ii) discharge of ions, (iii) nature of oxide formed, (iv) oxidising and reducing property, (v) reaction with acids.

8. Complete :
(a) $Na - \ldots\ldots \rightarrow Na^+$
(b) $N + \ldots\ldots \rightarrow N^{3-}$
(c) $Cl + e^- \rightarrow \ldots\ldots$
(d) $Mg - \ldots\ldots \rightarrow Mg^{2+}$
(e) $M + HCl \rightarrow MCl_2 + \ldots\ldots$
(f) $Mg + H_2SO_4 \rightarrow \ldots\ldots + \ldots\ldots$

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9. Select from the following list :

Fe_2O_3, NO, PbO, Mn_2O_7.

(a) Basic oxide .....

(b) Amphoteric oxide .....

(c) Acidic oxide .....

(d) Neutral oxide .....

10. Take an element from an alkali metal and one from an alkaline earth metal and write an equation for their action with :

(a) Hydrochloric acid, (b) Oxygen,

(c) Sulphuric acid, (d) Water.

11. (a) The table below compares some properties of metals and non-metals. Write down the missing statements (i) to (iv) :

METALS	NON-METALS
(i)	Poor conductors of heat.
(ii) Malleable	.....
(iii) Form positive ions	.....
(iv)	Form acidic oxides.

(b) How many valence electrons are present in :

(i) metals and (ii) non-metals ?

7B. STUDY OF ACTIVITY SERIES OF METALS

7.7 THE ACTIVITY SERIES OF METALS

Metals differ in tendency to lose valence electrons and can be arranged in a series.

> The arrangement of metals in the decreasing order of their reactivity is called the *activity* or *reactivity* series of the metals.

K
Na
Ca
Mg
Al
Zn
Fe
Pb
[H]
Cu
Hg
Ag
Au

The metal potassium is kept at the top of the series because it is most reactive. Also it is most electropositive as it easily oxidises, *i.e.*, gives up valence electrons. The least active metal is gold which is placed at the bottom of the list.

Hydrogen, though a non-metal, has been included in this series because it can form a positive ion. It would occupy the position based on its formation of the positive ion.

Main Points of the Activity Series

(i) The electropositive character decreases on moving down the series.

Potassium and sodium react with oxygen at normal temperature, and burn when heated, forming superoxide and if oxygen is in limited supply, then their oxide is formed.

$$K + O_2 \rightarrow KO_2$$

$$4K + O_2 \rightarrow 2K_2O$$

Metal	Oxides formed
Ca	$2Ca + O_2 \rightarrow 2CaO$
Mg	$2Mg + O_2 \rightarrow 2MgO$
Al	**Forms oxide at ordinary temperature :** $4Al + 3O_2 \rightarrow 2Al_2O_3$
	The metals listed below form oxides on strong heating :
Zn	$2Zn + O_2 \rightarrow 2ZnO$
Fe	$3Fe + 2O_2 \rightarrow Fe_3O_4$
Pb	$2Pb + O_2 \rightarrow 2PbO$
[H]	
Cu	$2Cu + O_2 \rightarrow 2CuO$ $4Cu + O_2 \rightarrow 2Cu_2O$
Hg	$2Hg + O_2 \rightarrow 2HgO$
Ag, Au	Oxides are not formed even on strong heating.

Note : Metals like calcium and magnesium have no action with air at ordinary temperature. On burning, they form oxides and nitrides.

$$2Ca + O_2 \rightarrow 2CaO$$

$$3Ca + N_2 \rightarrow Ca_3N_2$$

$$2Mg + O_2 \rightarrow 2MgO$$

$$3Mg + N_2 \rightarrow Mg_3N_2$$

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(ii) The reducing power of metals decreases on moving down the series. Thus, potassium is the strongest reducing agent.

(iii) The tendency of metals to lose valence electrons *i.e.*, tendency of metals to get oxidised decreases on moving down the series. Thus, potassium gets most readily oxidised.

(iv) The ability of the metals to reduce water to hydrogen decreases in passing down the series.

Potassium and sodium reacts vigorously with cold water and the hydrogen produced during the reaction gets ignited by the heat evolved during the reaction, producing lilac flame and golden yellow flame respectively.

$$2K + 2H_2O \rightarrow \underset{\text{potassium hydroxide}}{2KOH} + H_2\uparrow$$

$$2Na + 2H_2O \rightarrow \underset{\text{sodium hydroxide}}{2NaOH} + H_2\uparrow$$

Calcium : Reacts fairly vigorously with cold water but the hydrogen does not ignite.

$$Ca + 2H_2O \rightarrow \underset{\text{calcium hydroxide}}{Ca(OH)_2} + H_2\uparrow$$

Magnesium : Reacts very slowly with water, vigorously with warm water and burns in steam to give hydrogen.

$$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2\uparrow$$

$$Mg + \underset{\text{(steam)}}{H_2O} \rightarrow \underset{\text{magnesium oxide}}{MgO} + H_2\uparrow$$

Aluminium, zinc and iron : Reacts with steam when hot.

$$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2\uparrow$$

$$Zn + H_2O \rightarrow ZnO + H_2\uparrow$$

$$\underset{\text{(red hot)}}{3Fe} + 4H_2O \rightleftharpoons \underset{\text{ferrosoferric oxide}}{Fe_3O_4} + 4H_2\uparrow$$

(v) All the metals above hydrogen, in the activity series, **reduce hydrogen ions from dil. HCl or dil. H_2SO_4 and give out hydrogen.**

The rate of reaction decreases on moving from potassium (violent reaction) to lead (hardly any reaction) in the activity series.

The reaction of dilute acids with lead is very slow, because the salts lead chloride and lead sulphate formed are sparingly soluble in dilute acids. During the reaction, the chloride or sulphate gets coated on the metal and prevents reaction to proceed further.

Metals	Reaction with acid
K, Na	Reacts explosively with dilute acid liberating hydrogen. $2Na + 2HCl \rightarrow 2NaCl + H_2$
Ca	Calcium reacts less vigorously $Ca + 2HCl \rightarrow CaCl_2 + H_2$
Mg	Reacts with dilute acids less vigorously, liberating hydrogen. $Mg + 2HCl \rightarrow MgCl_2 + H_2$
Al	Due to great affinity towards oxygen, a coating of oxide is formed, which does not allow any further reaction with acid.
Zn	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$
Fe	$Fe + 2HCl \rightarrow FeCl_2 + H_2$
Pb	Lead reacts with conc. HCl to liberate hydrogen $Pb + 2HCl \rightarrow PbCl_2 + H_2$
[H]	
Cu, Hg, Ag, Au	Do not produce hydrogen on reaction with dilute or conc. acids.

(vi) Any metal higher up in the series displaces the metal below it from the salt solutions of the latter.

The larger the difference in the positions of the metals in the series, more rapidly does the displacement occur.

(vii) **Reduction of metallic oxides :** Oxides of metals like K, Na, Ca, Mg and Al cannot be reduced by common reducing agents like H_2, CO or C.

However, metal oxides below aluminium can be reduced by heating in the presence of **reducing agents**.

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Copper oxide

$$CuO + C \xrightarrow{\Delta} Cu + CO$$
$$CuO + CO \xrightarrow{\Delta} Cu + CO_2$$
$$CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$$
(black) (Pink/brown)

Lead oxide

$$PbO + C \xrightarrow{\Delta} Pb + CO$$
$$PbO + CO \xrightarrow{\Delta} Pb + CO_2$$
$$PbO + H_2 \xrightarrow{\Delta} Pb + H_2O$$
(yellow) (greyish white)

Iron (II) oxide

$$FeO + C \xrightarrow{\Delta} Fe + CO$$
$$FeO + CO \xrightarrow{\Delta} Fe + CO_2$$
$$FeO + H_2 \xrightarrow{\Delta} Fe + H_2O$$

For detailed study of reduction of metallic oxides refer to the topic "Extraction of metals" given ahead in this chapter.

7.7.1 Action of heat on some metallic compounds

Oxides	Hydroxides	Carbonates	Nitrates
K, Na, Ca, Mg, Al: Oxides of metals from potassium to aluminium are stable to heat and so can be reduced only by electrolysis.	K, Na: Metal hydroxides are stable to heat and are soluble in water. Ca, Mg, Al: Decompose on heating to form metal oxide and water vapour.	K, Na: Stable to heat and soluble in water. Ca, Mg, Al: Decompose on heating with decreasing vigour to form the metal oxide and carbon dioxide.	K, Na: On heating metal, nitrite and oxygen are formed. $2KNO_3 \rightarrow 2KNO_2 + O_2$ Ca, Mg, Al: Decompose on heating to form the metal oxide, nitrogen dioxide and oxygen.
Zn: Zinc oxide can be reduced by coke only.	Zn: $Ca(OH)_2 \rightarrow CaO + H_2O$; $Zn(OH)_2 \rightarrow ZnO + H_2O$	Zn: $MgCO_3 \rightarrow MgO + CO_2$	Zn: $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$
Fe, Pb, Cu: Oxides of iron, lead and copper are reduced by C, CO, H_2, NH_3.	Fe: $Pb(OH)_2 \rightarrow PbO + H_2O$; Pb: $Cu(OH)_2 \rightarrow CuO + H_2O$	Fe: $ZnCO_3 \rightarrow ZnO + CO_2$; Pb: $CuCO_3 \rightarrow CuO + CO_2$	Fe: $2Zn(NO_3)_2 \rightarrow 2ZnO + 4NO_2 + O_2$; Pb: $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; Cu: $2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$
Hg, Ag: Oxides of mercury and silver decompose to give metal and oxygen. $2Ag_2O \xrightarrow{\Delta} 4Ag + O_2$	Hg: Yields metal, oxygen and water vapour. $4AgOH \rightarrow 4Ag + O_2 + 2H_2O$	Hg: Forms the metal, oxygen and carbon dioxide. $2Ag_2CO_3 \rightarrow 4Ag + O_2 + 2CO_2$	Hg: Forms metal, nitrogen dioxide and oxygen. $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$

7.8 CORROSION

It has been observed that the surface of many metals, which is in direct contact of the atmosphere, starts reacting with the components of the atmosphere such as air, moisture and gases.

The compounds so formed on the surface of the metal, are generally porous and gradually fall off from the surface of the metal, thus exposing the inner layers of the metal underneath to the attack by atmospheric air and moisture.

When the surface of a metal is attacked by air, moisture or any other substance around it, the metal is said to corrode and the phenomenon is known as ***corrosion****.*

In such a process, mostly metal oxides are formed, but if metals contain impurities like sulphur and carbon, they may lead to the formation of sulphides and carbonates as well.

The effects of corrosion based on the activity series are as follows :

(i) **Alkali metals** like sodium and potassium react with air to form oxides. *These react with water present in air to yield hydroxides which slowly absorb carbon dioxide of the air to form carbonates.*

$$4Na + O_2 \rightarrow 2Na_2O$$
$$Na_2O + H_2O \rightarrow 2NaOH$$
$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$

Reactive metals like sodium and potassium are kept in kerosene oil as they easily react with moisture and air.

(ii) **Aluminium** and **magnesium** form white colour oxide on exposure to atmosphere.

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(iii) **Iron** forms hydrated oxide which is **brown** in colour known as rust ($Fe_2O_3 \cdot xH_2O$).

(iv) A **white deposit** is formed on the surface of **lead** when it comes in contact with moist air. It is a mixture of lead hydroxide and lead carbonate called basic lead carbonate [$2PbCO_3 \cdot Pb(OH)_2$].

(v) **Copper** forms a **green deposit on its surface** when exposed to moist air. It is a mixture of copper hydroxide and copper carbonate (basic copper carbonate) [$2CuCO_3 \cdot Cu(OH)_2$].

(vi) **Silver** gets **tarnished** when exposed to the atmosphere which contains pollutant H_2S and forms a **black** coating of Ag_2S.

The noble metals such as gold and platinum do not corrode easily.

7.8.1 Corrosion of Iron

Corrosion of iron is known as RUSTING

Rusting is the slow oxidation of iron by atmospheric oxygen in the presence of water.

Iron when comes in contact with moist air reacts to give rust which is a **brown powder mainly hydrated iron (III) oxide** ($Fe_2O_3 \cdot xH_2O$).

$$4Fe + 3O_2 + 2xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$$

Rust is a soft powdery material, which easily falls off from the surface of iron and hence it further exposes the inner part of the iron metal to air and moisture for further rusting. Thus, iron gets damaged.

It is a wasteful reaction. It may lead to collapse of bridges, buildings, leakage of roofs and other harmful consequences.

Note : 1. Rusting differs from tarnishing of metals; tarnishing is the oxidation of the metallic surfaces other than iron. The oxide so formed does not flake off, but sticks to the metal surface. Thus after tarnishing further oxidation stops.

2. Cast iron or pig iron contains carbon and other impurities. It gets rusted slower than steel which has less carbon content.

Activity : To prove that air and water are necessary for corrosion (*i.e.*, rusting) of iron.

Polish some iron nails with an emery cloth until they shine. Emery cloth removes rust and grease from metals when it is rubbed on them. Place the nails in test tubes under different conditions, as outlined below :

(a) Put some nails in a test tube and cover them with distilled water. Close the test tube with a cork to stop the water from evaporating. These nails are in contact with both air and water.

(b) Keep some anhydrous calcium chloride at the bottom of the second test tube and cover it with cotton wool. (Calcium chloride is able to absorb all the moisture from the air). Place the bright nails on top and cork the tube. These nails are thus in contact with only dry air.

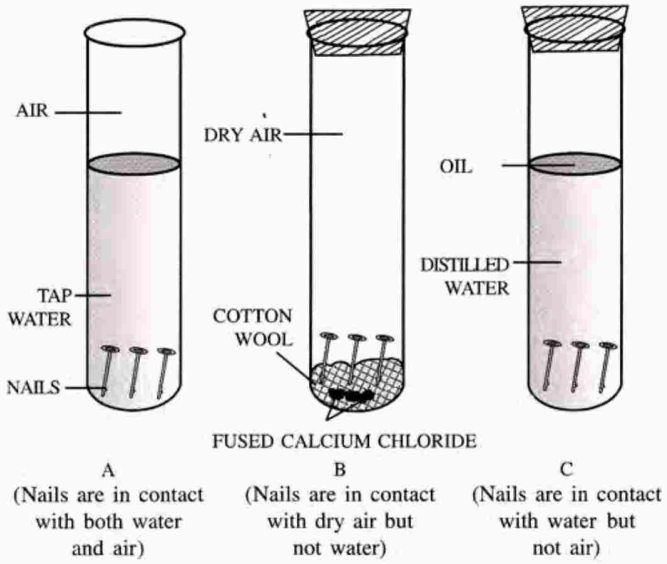

Fig. 7.1

(c) Boil some water to expel the dissolved air from it. Pour some of it on to the three bright nails kept in the third test tube. To prevent the air in the tube from dissolving in the water again as soon as it cools, a layer of vaseline or oil is poured on top of the water. This forms an air-proof layer. These nails are thus in contact with only water.

(d) Leave the test tubes for some days and then find out which nails have rusted. Find out what are the conditions for rusting ?

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7.8.2 Factors favourable for increased corrosion

In addition to the **presence of oxygen** and **moisture** it is noticed that

1. Metals which are placed higher in the activity series, corrode more easily.
2. Dissolved salts in water act as electrolyte and enhance the rate of corrosion (iron corrodes fast in sea water).
3. The presence of a metal which is less reactive than iron increases the rate of rusting of iron, since it will favour the more reactive metal like iron to lose electrons and get oxidised. Due to this reason **galvanisation** (coating of zinc) is preferred to tinning in the prevention of rusting of iron.
4. The presence of pollutants like NO_2 and CO_2 increases rusting.

7.8.3 Prevention of corrosion

The rusting of iron can be prevented by oiling, greasing, painting, galvanising, chrome-plating, anodising or making alloys. These methods can be classified into three categories

1. **Barrier protection :** When metal surface is not allowed to come in contact with atmospheric agents like air, water then process is known as Barrier Protection process. It is done by
   (i) Coating the metal (Iron) with another metal with the help of electricity (**electroplating**).
   (ii) Prevention by painting the metal (Iron) so that the iron do not come in contact with atmospheric reagents.
   (iii). By covering the metal (Iron) by grease or oil or antirust solution.
2. **Sacrificial protection :** The metal to be protected is covered with more electropositive metal like zinc, magnesium, etc., which gets oxidised and saves the metal.

   For example, Iron is coated with zinc and the process is known as GALVANISATION.
3. **Cathode protection :** The object to be saved (iron) from rusting is connected to a piece of more electropositive metal like **zinc**, magnesium or aluminium, by a piece of wire. The anode is made of more reactive element like zinc, which loses electrons and get oxidised and goes on disappearing thus saves the cathode (iron) from rusting (GALVANISATION).

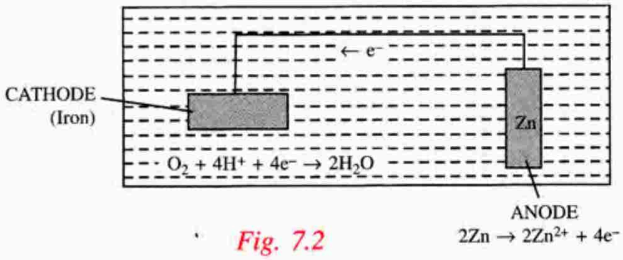

Fig. 7.2

This method is usually used for protecting iron objects like sewer pipes and storage tanks which are buried underground.

7.8.4 Advantage of corrosion

Sometimes corrosion of metals is an advantage as it prevents the metal underneath from further damage.

For example, on exposure to air, the surface of metals like aluminium and zinc forms layers of their oxides which are very sticky and impervious in nature, and hence act as a **protective layer**. This layer protects the metal from further damage. Thus the articles made up of aluminium, do not corrode easily.

EXERCISE-7B

1. Name :
(a) two metals which are liquid at room temperature,
(b) metal which do not corrode easily
(c) a metal which lacks ductility,
(d) a non-metal which is lustrous,
(e) a non-metal which conducts electricity,
(f) a metal which is brittle,
(g) two elements which are monoatomic,

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(h) two metallic oxides which are acidic,

(i) two metallic oxides which are amphoteric,

(j) two metals which react with cold water,

(k) the compound responsible for green deposit on the surface of copper,

(l) a non-metal which can form a positive ion,

(m) a non-metal which shows reducing property,

(n) a metal whose oxide is reduced only by carbon.

2. Explain how the activity series accounts for each of the following :

(a) occurrence of metals,

(b) tendency to corrosion,

(c) reaction with water,

(d) reaction with acids.

3. Give the balanced reactions for the following :

(a) Sodium is dropped in water,

(b) Magnesium reacts with boiling water,

(c) Red hot iron reacts with steam,

(d) Iron reacts with dilute HCl.

4. (a) Why are alkali metals kept in kerosene oil ?

(b) What is :

(i) basic lead carbonate and

(ii) brown powder deposit on iron ?

(c) Why is hydrogen kept in the metal activity series ?

5. Metal A has an electronic configuration of 2, 8, 1 and metal B has 2, 8, 8, 2 which is a more reactive metal.

(a) Identify A and B and give their reactions with dil. HCl and dil. H_2SO_4.

(b) Give the effect of heat on their :

(i) oxides, (ii) hydroxides,

(iii) carbonates, (iv) nitrates.

6. (a) What is corrosion ? What are necessary conditions for corrosion ?

(b) State under what conditions corrosion is faster.

(c) Corrosion can be an advantage in some case. Explain.

7. (a) What is rust ? Give the equation for the formation of rust.

(b) State two conditions necessary for rusting of iron.

(c) How does the painting of an iron object prevent rusting ?

8. What is galvanisation ? How does it protect iron from rusting ?

9. A student has been collecting silver coins and copper coins. One day she observed a black coating on silver coins and a green coating on copper coins. Which chemical phenomenon is responsible for these coatings ? Write the names of black and green coatings.

10. Aluminium is said to be more reactive than iron, towards oxygen (or air). Yet iron undergoes corrosion to a greater extent than aluminium. Explain.

11. Why do gold ornaments look new even after several years of use ?

12. From the metals : copper, iron, magnesium, sodium, and zinc, select a different metal in each case which :

(a) does not react with dilute hydrochloric acid,

(b) can form 2 + and 3 + ions,

(c) has a hydroxide that reacts with both acids and alkalis,

(d) does not react with cold water but reacts with steam when heated.

(e) Arrange the above metals in the decreasing order of reactivity.

7C. EXTRACTION OF METALS BASED ON THE ACTIVITY SERIES

7.9 OCCURRENCE OF METALS

Most of the metals are reactive so they occur in combined state in the form of their oxides, carbonates, halides, sulphides, sulphates, etc. mixed with mud, clay, sand and stone.

Since the reactivity series lists the metals in the decreasing order of electropositivity, it might be expected that those at the top would combine with more electronegative elements in their minerals. *For example*, sodium and potassium occur mainly as their chlorides, calcium and magnesium as carbonates, aluminium as its oxide while zinc, lead, copper, mercury and silver are found as sulphides.

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Metals	Main Ores
K	Carnallite $KCl·MgCl_2·6H_2O$
Na	Rock salt NaCl (in sea water)
Ca	Limestone, Marble ($CaCO_3$), Calcium sulphate dihydrate ($CaSO_4·2H_2O$), Gypsum $CaSO_4$
Mg	Carnallite, magnesium carbonate $MgCO_3$ (Magnesite)
Al	Bauxite $Al_2O_3·2H_2O$ Cryolite Na_3AlF_6
Zn	Zinc blende ZnS Calamine $ZnCO_3$
Fe	Red Haematite Fe_2O_3 Brown haematite $2Fe_2O_3·3H_2O$
Pb	Galena PbS
Cu	Copper pyrites $CuFeS_2$
Hg	Cinnabar HgS
Ag	Argentite Ag_2S Horn silver AgCl

Note : Insoluble ores are found as rocks while soluble ones like alkali chloride are found in sea water.

Metals which have low chemical reactivity are found in free state. Gold and platinum are found in free state (native state).

COMMON TERMS USED IN EXTRACTION

(i) Metallurgy :

The process used for the extraction of metals in their pure form from their ores is referred to as ***Metallurgy***.

Metallurgy deals with the production and purification of metals and manufacture of alloys.

(ii) Minerals :

The naturally occurring compounds of metals which are generally mixed with other matter such as soil, sand, limestone and rocks are known as *minerals.*

(iii) Gangue (Matrix) :

Earthly impurities including silica (SiO_2), mud, etc., associated with the ore are called *gangue* or *matrix*.

(iv) Ores :

Those minerals from which the metals are extracted commercially at a comparatively low cost and with minimum effort are called *ores*.

(v) Flux :

A flux is a substance that is added to the charge in a furnace to remove the gangue.

Flux combines with the gangue to form a fusible mass called slag.

(vi) Slag :

It is the fusible product formed when flux reacts with impurities (gangue) during the extraction of metals.

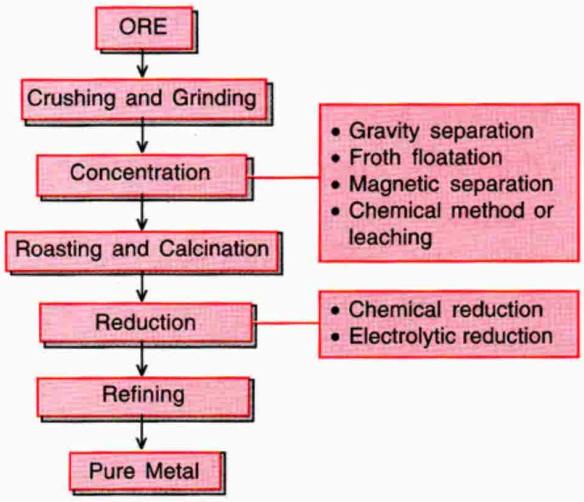
$$\underset{\text{gangue}}{SiO_2} + \underset{\text{flux}}{CaCO_3} \rightarrow \underset{\text{slag}}{CaSiO_3} + CO_2$$

(vii) Smelting :

Smelting is the process of reducing the roasted oxide ore and removing the gangue with the help of an appropriate flux added with the ore. *It reduces the oxide ore to metals in the molten condition.*

7.10 STEPS INVOLVED IN EXTRACTION

Extraction of a metal from its ore consists of the following processes.



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7.10.1 Crushing and Grinding

Ores are crushed into a fine powder in big jaw crushers and ball mills. This process is called **Pulverisation.**

7.10.2 Dressing (Concentration) of ores

The process of removing *gangue, the rocky impurities like* SiO_2 present in an ore, is called *concentration of an ore* or ***ore dressing*** and the purified ore is called ***concentrated ore***.

Concentration of the crushed ore is done mainly by the following methods.

(i) Hydrolytic (Gravity separation) method

Principle : *The difference in the densities of the ore and the gangue is the main criterion.*

Method : The ore is poured over a sloping, vibrating corrugated table with grooves and a jet of water is allowed to flow over it. The denser ore particles settle down in the grooves and lighter gangue particles are washed down by water (Fig. 7.3).

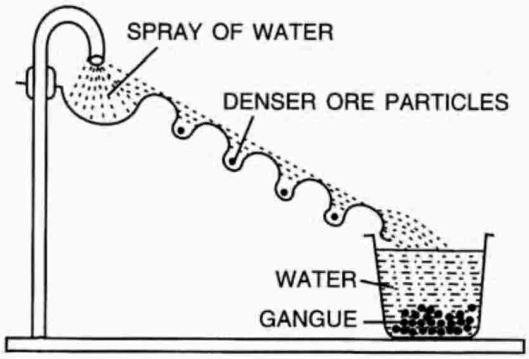

Fig. 7.3 Gravity separation using water and vibrating corrugated table

(ii) Magnetic Separation

Principle : Magnetic properties of the ores.

Method : The crushed ore is placed over a conveyer belt, which rotates around two metal wheels, one of which is magnetic. The magnetic particles are attracted to the magnetic wheel and fall separately apart from the non-magnetic particles (Fig. 7.4).

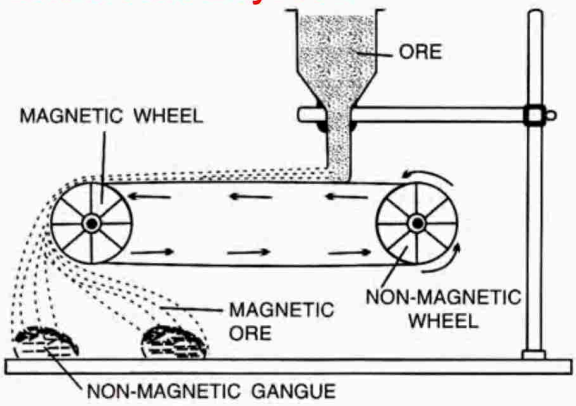

Fig. 7.4 Electromagnetic separation

(iii) Froth floatation

Principle : This process depends on the preferential wettability of the ore with oil (pine oil) and the gangue particles by water.

Method : The crushed ore is taken in a large tank containing oil and water and agitated with a current of compressed air. The ore is wetted by the oil and gets separated from the gangue in the form of froth (Fig. 7.5).

Since the ore is lighter, it comes on the surface with the froth and impurities are left behind.

Sulphide ores like zinc blende (ZnS) Galena (PbS) are lighter than their impurity. They are concentrated by froth floatation.

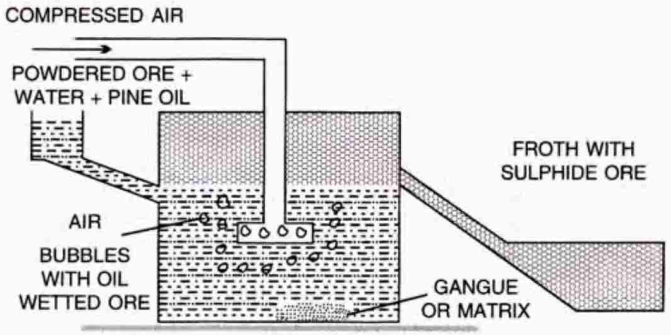

Fig. 7.5 Concentration of ore by froth floatation process

(iv) Chemical Method or Leaching

The ore is treated with a suitable reagent (acid, base or some other reagent) such that the ore is soluble in it but the impurities are not. The impurities are removed by filtration. The solution of the ore, *i.e.*, the filtrate is treated with a suitable

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reagent which precipitates the ore (refer to extraction of aluminium).

7.10.3 Roasting and Calcination

***Roasting** is a process of heating the concentrated ore to a high temperature in presence of air.*

Roasting is generally employed in case of sulphide ores or the ores containing metallic sulphides as impurities. It converts sulphide ores into oxides.

If an ore is a carbonate or a hydrated oxide, it is heated in the *absence of air* to a temperature that is high but insufficient to melt the ore. The process is known as ***Calcination.***

Comparison of Roasting and Calcination

Roasting	Calcination
(i) The ore is heated in the excess of air.	(i) The ore is heated in the absence of air.
(ii) Generally, sulphide ores are roasted, so SO_2 is given off. $2ZnS+3O_2 \xrightarrow{800°-900°C} 2ZnO+2SO_2$	(ii) Carbonate and hydrated ores are calcined and CO_2 or water vapours are given off. $ZnCO_3 \rightarrow ZnO + CO_2$
(iii) Volatile impurities are removed as oxides (SO_2, P_2O_5, As_2O_3) and the ore becomes porous and more reactive.	(iii) Moisture and organic impurities are removed and the ore becomes porous and more reactive.

7.10.4 Reduction of metal oxides

The roasted metal oxide is reduced by reducing agents like carbon, carbon monoxide or hydrogen. If the ore is not an oxide ore, it is first converted into an oxide ore by roasting or by calcination.

Reduction can also be explained based on the activity series :

Oxides of highly active metals like **potassium, sodium, calcium, magnesium** and **aluminium** *have great affinity towards oxygen and so* **cannot be reduced by carbon or carbon monoxide or hydrogen.**

These metals are obtained by electrolytic reduction of fused metallic salts (halides or oxides) (Refer 6.9.3).

The **metals in the middle** of the activity series such as iron, zinc, lead, copper, *etc.*, are moderately reactive. These are usually present as sulphides or carbonates in nature.

Sulphide ores are converted into oxides by heating in excess of air (***roasting***) as oxides are easy to reduce.

$$\underset{\text{(Zinc blende)}}{2ZnS} + 3O_2 \xrightarrow{800°C} 2ZnO + 2SO_2$$

The carbonate ores are heated in a limited supply of air (***calcination***) to convert them into oxides.

$$\underset{\text{(Calamine)}}{ZnCO_3} \xrightarrow{400°C} ZnO + CO_2$$

The metal oxides are then reduced by using suitable reducing agents (***carbon reduction process***).

$$ZnO(s) + C(s) \xrightarrow{400°C} \underset{\text{(Zinc spelter)}}{Zn(s)} + CO(g)$$

ZnO and Coke are heated in retort by producer gas ($CO + N_2$) to get zinc spelter which on distillation or electrolytic refining gives pure zinc.

Zinc oxide is not reduced by hydrogen or carbon monoxide. It is reduced only by heating with carbon.

$$ZnO + C \xrightarrow{\Delta} Zn + CO$$

Similarly, the ore of **iron,** *i.e.*, **haematite (Fe_2O_3)** is reduced in blast furnace by using **coke** and $CaCO_3$ (flux).

Haematite, coke and lime stone are introduced from the top of the blast furnace and hot air is introduced at the lower region. Different reactions that takes place in blast furnace are :

Lower region $C + O_2 \rightarrow CO_2$

Middle region $CO_2 + C \rightarrow 2CO$

$CaCO_3 \rightarrow CaO + CO_2$

$CaO + SiO_2 \rightarrow CaSiO_3$ (slag)

Upper region $\mathbf{Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2}$

Observation : *Red metallic oxide gets reduced to greyish white metal.*

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Molten iron solidifies to give pig iron. It has limited applications, so is converted into steel after removing impurities and adding a calculated amount of alloy SPIEGEL (containing C, Mn and Fe).

Note : The **highly reactive metals can be used as reducing agents** because they can displace metals of lower reactivity from their compounds. *For example*, in order to obtain manganese, its oxide is heated with aluminium powder.

$$3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + Heat$$

These displacement reactions are highly exothermic. The amount of heat evolved produces metals in the molten state.

Lead and copper are obtained by the *reduction of their oxides with carbon*, carbon monoxide and hydrogen.

Lead oxide

$$PbO + C \xrightarrow{\Delta} Pb + CO$$
$$PbO + CO \xrightarrow{\Delta} Pb + CO_2$$
$$PbO + H_2 \xrightarrow{\Delta} Pb + H_2O$$

Observation : Yellow lead monoxide gets reduced to greyish white metal.

Copper oxide

$$CuO + C \xrightarrow{\Delta} Cu + CO$$
$$CuO + CO \xrightarrow{\Delta} Cu + CO_2$$
$$CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$$

Observation : Black copper oxide is reduced to brown/red copper.

Metals low in the activity series are very less reactive. *The oxides of these metals can be reduced to metals by heating alone.*

For example, firstly mercury sulphide (cinnabar), an ore of mercury, gets converted into its oxide by heating in air.

$$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$$

and finally, this mercury oxide, on further heating is reduced to mercury.

$$2HgO \xrightarrow{\text{above } 300^\circ C} 2Hg + O_2$$

Similarly $2Ag_2O \xrightarrow{\text{above } 300^\circ C} 4Ag + O_2$

Copper which is found as Cu_2S in nature can be obtained from its ore by just heating in air.

$$2Cu_2S + 3O_2 \xrightarrow{\Delta} 2Cu_2O + 2SO_2$$
$$2Cu_2O + Cu_2S \xrightarrow{\Delta} 6Cu + SO_2$$

Less active mercury is also obtained by merely heating its sulphide in air.

$$HgS + O_2 \rightarrow Hg + SO_2$$

Silver and gold are obtained by displacement from solutions containing their ions by more electropositive metal zinc.

$$Zn + \underset{\text{(in sol.)}}{2\,Ag^+} \rightarrow Zn^{2+} + \underset{\text{(free)}}{2Ag\downarrow}$$

7.10.5 Refining of the metal

It is the separation of the above extracted metal from the residual impurity such as :

(i) the presence of other metals and non-metals like silicon and phosphorus.

(ii) unreduced oxides and sulphides of the metal.

The method of purification used depends upon the

(i) nature of the metal,

(ii) nature of the impurities present in the metal and

(iii) purpose for which the metal is to be used.

(i) Distillation

Metals like mercury and zinc which are *volatile* distil over in pure form and the non-volatile impurity remains behind.

(ii) Liquation

Metals like lead and tin have *low melting points*, so they are heated on the sloping hearth of a furnace. The molten or fused metal flows away leaving behind the impurities (Fig. 7.6).

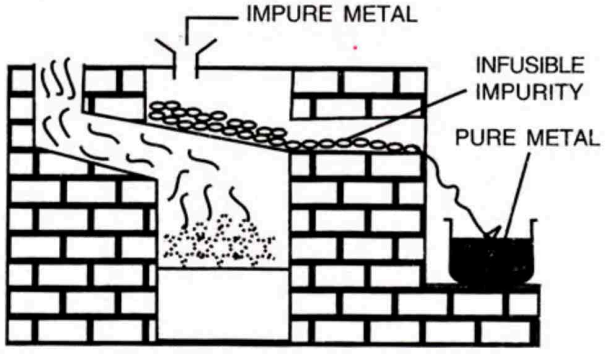

Fig. 7.6 Liquation process

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(iii) Oxidation

Metals like iron which contain volatile impurities and can be easily oxidised are purified by this method. The volatile oxides of phosphorous, sulphur and other impurities rise to the surface and are removed while the molten metal is left behind.

(iv) Electro-refining

The impure metal is made the **anode**, while a thin sheet of pure metal is made the **cathode**.

Electrolyte used is a salt solution of a metal, which is to be refined. Pure metal deposits at the cathode and impurities settle down forming anode mud.

Metals like **copper, aluminium** and **zinc** are refined by this process. (Fig. 7.7).

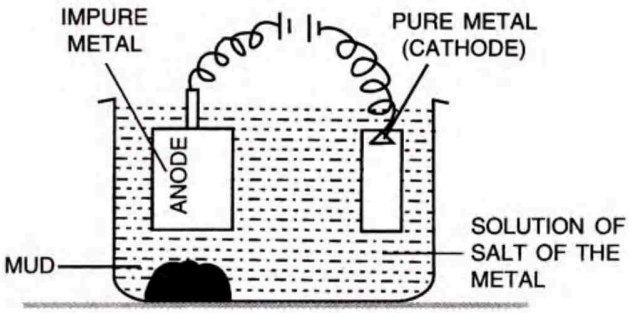

Fig. 7.7 Electro-refining

EXERCISE-7C

1. Define the term 'metallurgy'. State the processes involved in metallurgy.
2. Which metal occurs as :
   (a) a sulphide, (b) a halide,
   (c) a carbonate, (d) an oxide.
   Also give the names of their respective ores.
3. Distinguish between :
   (a) a mineral and an ore,
   (b) an ore and a metallic compound.
4. Which metal can be extracted from each one of the following ores :
   (a) bauxite (b) calamine (c) haematite.
5. State three objectives achieved during the roasting of ores.
6. Give the principles of :
   (a) hydrolytic method, (b) froth floatation,
   (c) electromagnetic separation.
7. Name :
   (a) the processes involved in
   (i) concentration (ii) refining of ores.
   (b) two metallic oxides which cannot be reduced by carbon, carbon monoxide or hydrogen.
8. Explain the following terms :
   (a) flux, (b) gangue,
   (c) slag, (d) smelting.
9. Why does iron or zinc not occur free in nature ?
10. What do you observe when hydrogen is passed over heated copper oxide ?
11. Compare roasting and calcination.
12. (a) Name an ore of zinc.
    (b) Which process is applied to concentrate it ?
    (c) How is concentrated ore changed to oxide ?
13. Some metallic oxides can be reduced by hydrogen, carbon and carbon monoxide and some cannot. Explain.
14. How are the following metallic oxides reduced. Write equations :
    (a) Iron (II) oxide, (b) Zinc oxide.
15. State why aluminium is extracted from its oxide by electrolysis while copper, lead, iron by reducing agents and mercury and silver by thermal decomposition.
16. An ore on being heated in air forms sulphurous anhydride. Write the process used for the concentration of this ore.
17. (a) On which factors does purification of metals depend ?
    (b) Name the methods used for purification.
    (c) How is electro-refining done ?
18. Complete and balance :
    (a) $Ag_2O \xrightarrow{\Delta}$ (b) $MnO_2 + Al \xrightarrow{\Delta}$
    (c) $Cu(OH)_2 \xrightarrow{\Delta}$ (d) $ZnCO_3 \xrightarrow{\Delta}$
    (e) $NaNO_3 \xrightarrow{\Delta}$ (f) $Pb(NO_3)_2 \xrightarrow{\Delta}$
    (g) $AgNO_3 \xrightarrow{\Delta}$ (h) $Cu_2O + Cu_2S \xrightarrow{\Delta}$
    (i) $HgS + O_2 \xrightarrow{\Delta}$

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7.11 COMMON ORES OF ALUMINIUM, IRON AND ZINC

Ore of Al	Chemical name	Formula
1. Bauxite	Hydrated aluminium oxide	$Al_2O_3.2H_2O$
2. Cryolite	Sodium aluminium fluoride	Na_3AlF_6
3. Corundum	Anhydrous aluminium oxide	Al_2O_3

Ore of Fe	Chemical name	Formula
Red haematite	Anhydrous ferric oxide	Fe_2O_3
Brown haematite	Hydrated ferric oxide	$2Fe_2O_3.3H_2O$
Magnetite	Triferric tetraoxide	Fe_3O_4
Iron pyrites	Iron disulphide	FeS_2
Siderite	Ferrous carbonate	$FeCO_3$

Ores of Zn	Chemical name	Formula
Zinc blende	Zinc sulphide	ZnS
Zincite	Zinc oxide	ZnO
Calamine	Zinc carbonate	$ZnCO_3$

7.12 ALUMINIUM

Symbol : Al **Relative atomic mass :** 27

Colour : Silvery white metal **Atomic number :** 13

Specific gravity : 2.2 **Valency :** 3

Electronic configuration : 2, 8, 3

Position in Periodic Table : Period 3, Group IIIA (13)

7.12.1 Discovery and Occurrence

Aluminium was discovered by *Oersted* in 1825. **It is the most abundant metal in the earth crust.** Aluminium is a reactive metal and so does not occur free in nature.

7.12.2 Extraction of Aluminium

Hall and Heroult, in 1885, developed the process for the extraction of aluminium.

Aluminium is extracted from its main ore **bauxite $Al_2O_3 \cdot 2H_2O$.** Bauxite contains 60% Al_2O_3, the rest being sand, ferric oxide and titanium oxide.

Concentration of ore or purification of ore i.e. conversion of bauxite into alumina (Baeyer's process)

Bauxite is grinded finely and heated to remove volatile impurities. It is then heated under pressure with conc. caustic soda solution for 2 to 8 hours at 140°C to150°C.

$$Al_2O_3 \cdot 2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$$

Bauxite dissolves and forms sodium meta aluminate, leaving behind insoluble impurities called **red mud.** Red mud consists of ferric oxide, sand, etc., which are removed by filtration.

On diluting sodium meta aluminate with water and cooling to 50°C, it gets hydrolysed to give aluminium hydroxide as a precipitate. The impurities dissolve in sodium hydroxide.

$$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3 \downarrow$$

The precipitate is filtered, washed, dried and ignited at 1000°C to get alumina (Al_2O_3).

$$2Al(OH)_3 \xrightarrow[1000°C]{\text{heat}} Al_2O_3 + 3H_2O$$

The filtrate containing sodium hydroxide is again used to extract aluminium from bauxite.

Aluminium oxide due to its great affinity for oxygen is a very stable compound. It is not reduced easily by common reducing agents like carbon, carbon monoxide or hydrogen. Hence, *electrolytic reduction was chosen as the method for reducing alumina.*

Difficulty faced in obtaining aluminium from alumina

Electrolytic reduction can be done in molten state only. Alumina melts at approximately 2050°C. At this temperature, aluminium vaporises, so it was not easy to collect aluminium.

This difficulty was removed by Hall who found a suitable solvent in another ore of aluminium, *cryolite* (Na_3AlF_6). When he applied the current to the molten mixture of alumina (20%), cryolite (60%) and calcium fluoride (20%), he saw some shining globules of aluminium appearing at the cathode. He got elated over the success of his

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three years' concerted efforts and immediately ran excitedly to his professor and presented to him the few small pieces of the metal he had collected. It was February 23, 1886. Hall's discovery revolutionised the aluminium industry. *Aluminium which was once a costlier and rarer metal than gold became a cheap metal of common use in every home.*

Electrolytic reduction of fused alumina (Hall-Heroult's process)

Electrolytic cell : Rectangular iron tank with sloping bottom. The sloping bottom of the tank facilitates the removal of the molten aluminium. The tank is lined with gas carbon.

Electrolyte : It is the mixture of molten alumina 20%, cryolite 60% and fluorspar 20%. Powdered coke is sprinkled over the surface of the electrolyte.

Temperature : The molten electrolyte is kept at 950°C by electrical heating.

Voltage used : 5 to 6 volts. A low voltage is used to avoid decomposition of the molten cryolite.

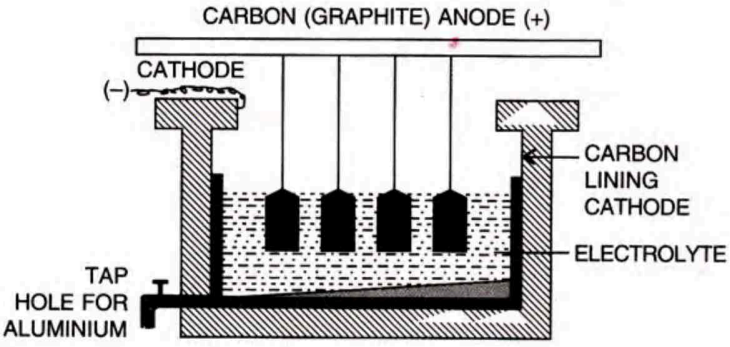

Fig. 7.8 Extraction of aluminium

Reactions

Cryolite : $Na_3AlF_6 \rightleftharpoons 3Na^+ + Al^{3+} + 6F^-$

Fluorspar : $CaF_2 \rightleftharpoons Ca^{2+} + 2F^-$

Alumina : $Al_2O_3 \rightleftharpoons 2Al^{3+} + 3O^{2-}$

Positive ions formed in the solution are Na^+, Ca^{2+} and Al^{3+} while *negative ions* formed in the solution are F^-, O^{2-}. Aluminium being lower in electrochemical series is preferentially discharged at the cathode while oxygen at the anode.

Cathode : (Inner carbon lining of the electrolytic cell)

$$4Al^{3+} + 12e^- \rightarrow 4Al$$

Anode : Thick rods of graphite are suspended into the fused electrolyte.

$$6O^{2-} - 12e^- \rightarrow 6[O]$$

$$3O + 3O \rightarrow 3O_2$$

Anode is oxidised to carbon monoxide, which further forms carbon dioxide.

$$2C + O_2 \rightarrow 2CO$$

$$2CO + O_2 \rightarrow 2CO_2$$

The anode has got to be replaced from time to time, as it gets oxidised by the oxygen evolved at the anode.

When the supply of alumina decreases, the voltage suddenly jumps from 40 to 60 volts. If a bulb is attached to the apparatus, it will glow brightly. Then fresh alumina is added. Aluminium obtained is 99·8% pure.

Further purification can be done by electrolysis.

Note :

1. **Cryolite :** Lowers the fusion temperature from 2050°C to 950°C and enhances conductivity.
2. **Fluorspar and cryolite**
   (i) act as a solvent for the electrolytic mixture.
   (ii) increases its conductivity since pure alumina is almost a non-conductor of electricity.
3. **Powdered coke** is sprinkled over the surface of the electrolytic mixture. It :
   (i) reduces heat loss by radiation.
   (ii) prevents the burning of anode.

Refining of aluminium :

(Hoope's electrolytic process). The process uses an electrolytic cell which contains three layers of molten substance of differing specific gravity.

Molten impure aluminium forms the **bottom layer.** The bottom layer has carbon lining and serves as **anode.** Pure molten aluminium with carbon electrodes serves as **cathode** in the **top layer**.

The **middle layer** consists of a mixture of fluorides of sodium, barium and aluminium.

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On passing current, aluminium from the middle layer passes to the top layer and an equivalent amount of it shifts from the lower layer to the middle one. Pure aluminium is tapped from the top. (Fig. 7.9)

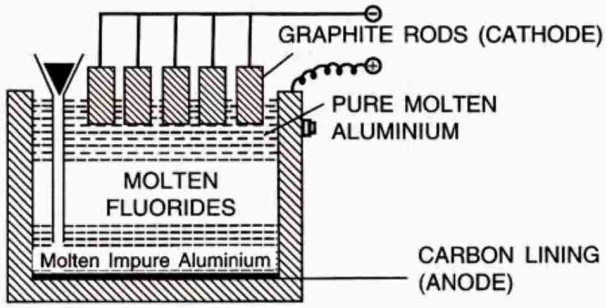

Fig. 7.9 Refining of aluminium

Anode : $\underset{\text{(impure)}}{Al} - 3e^- \rightarrow Al^{3+}$

Cathode : $Al^{3+} + 3e^- \rightarrow \underset{\text{(pure)}}{Al}$

Flow Chart for extraction of aluminium

ORE – Bauxite $Al_2O_3 \cdot 2H_2O$

Concentration by NaOH to obtain Al_2O_3

$Al_2O_3 \cdot 2H_2O + 2NaOH \xrightarrow{140°C-150°C} 2NaAlO_2 + 3H_2O$

$NaAlO_2 + 2H_2O \xrightarrow{50°C} NaOH + Al(OH)_3\downarrow$

$2Al(OH)_3 \xrightarrow{1000°C} Al_2O_3 + 3H_2O$

Electrolytic reduction of alumina

Electrolyte : Alumina, cryolite, fluorspar (1 : 3 : 1)

Cathode : $4Al^{3+} + 12e^- \rightarrow 4Al$

Anode : $6O^{2-} - 12e^- \rightarrow 3O_2$

$C + O_2 \rightarrow CO_2$

Purification by electrolysis

Cathode : $Al^{3+} + 3e^- \rightarrow Al$

Anode : $Al - 3e^- \rightarrow Al^{3+}$

7.12.3 Properties of aluminium

(A) Physical properties

(i) It is a silvery white metal.

(ii) It is a light (*specific gravity* 2.2) but very strong metal, having high tensile strength.

(iii) It is both malleable and ductile.

(iv) It is a very good conductor of heat and electricity.

(v) It can be highly polished to produce an attractive lustre.

(vi) It has a m.p. 660°C and b.p. 2050°C.

(B) Chemical properties

1. *Reaction with air*

Its surface gets *oxidised to oxide at room temperature which forms a thin but tough layer over the surface and protects the rest of the metal.*

Aluminium powder burns in air at about 800°C forming its oxide and nitride *with a bright light.*

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

$$2Al + N_2 \rightarrow 2AlN$$

2. *Reaction with water*

Water also has no action on aluminium due to the layer of oxide on it.

When steam is passed over *pure heated aluminium,* hydrogen is produced.

$$2Al + 3H_2O \hookrightarrow Al_2O_3 + 3H_2$$

3. *Amphoteric nature*

Aluminium reacts with bases (alkalies), as well as with acids.

It reacts with boiling caustic alkalies solution to produce meta aluminate while with fused alkali it produces aluminate.

$$2Al + 2NaOH + 2H_2O \rightarrow \underset{\text{(Sodium meta aluminate)}}{2NaAlO_2} + 3H_2\uparrow$$

$$\underset{\text{(fused)}}{2Al + 6NaOH} \rightarrow \underset{\text{(Sodium aluminate)}}{2Na_3AlO_3} + 3H_2$$

It reacts with acids to produce salt and hydrogen

$$\underset{\text{(dil. or conc. acid)}}{2Al + 6HCl} \rightarrow 2AlCl_3 + 3H_2$$

The reaction is slow in the beginning because the protective layer of aluminium oxide (Al_2O_3) on the surface of aluminium resists the attack of acid. However, after some time, the oxide layer gets dissolved in HCl and the clean surface of aluminium reacts with HCl at an appreciable speed.

Dilute sulphuric acid reacts with the metal to liberate hydrogen.

$$\underset{\text{(dil.)}}{2Al + 3H_2SO_4} \rightarrow Al_2(SO_4)_3 + 3H_2$$

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Concentrated sulphuric acid reacts with aluminium to produce sulphur dioxide.

$$2Al + 6H_2SO_4 \rightarrow Al_2(SO_4)_3 + 6H_2O + 3SO_2\uparrow$$
(conc.)

Dilute or concentrated nitric acid does not attack the metal aluminium. It renders aluminium passive due to the formation of an oxide film on its surface.

4. *Action on non-metals*

Heated aluminium combines directly with non-metals :

$$2Al + N_2 \xrightarrow{\Delta} 2AlN$$
(Nitrogen) (Aluminium nitride)

$$2Al + 3Cl_2 \xrightarrow{\Delta} 2AlCl_3$$
(Chlorine) (Aluminium chloride)

$$2Al + 3S \xrightarrow{\Delta} Al_2S_3$$
(Sulphur) (Aluminium sulphide)

$$4Al + 3C \xrightarrow{\Delta} Al_4C_3$$
(Carbon) (Aluminium carbide)

5. *Reducing action*

Aluminium is a powerful **reducing agent**. *When a mixture of aluminium powder and iron oxide is ignited, the latter is reduced to metal. This process is called* ***aluminothermy.***

$$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$$

EXERCISE-7D

1. State the position of aluminium in the Periodic Table.
2. (a) Give the chemical names and formulae of main ores of (i) aluminium (ii) iron (iii) zinc.
   (b) Which impurities are present in bauxite.
   (c) What is red mud, how is it removed ?
3. In order to obtain 1 tonne of aluminium, the following inputs are required. 4 tonnes of bauxite, 150 kg of sodium hydroxide and 600 kg of graphite. The aluminium compound in bauxite is aluminium oxide and the main impurity is iron (III) oxide. Aluminium is obtained by the electrolysis of aluminium oxide dissolved in cryolite.
   (a) When bauxite is treated with sodium hydroxide solution, what happens to :
      (i) the aluminium oxide,
      (ii) the iron (III) oxide.
   (b) (i) Name the process used for the purification of bauxite.
      (ii) Write the equation for the action of heat on aluminium hydroxide.
   (c) (i) Write the formulae of cryolite.
      (ii) Write down the word which correctly completes the following sentence.
      By dissolving aluminium oxide in cryolite a ..... (conducting/non conducting) solution is produced.
      (iii) Why is so much graphite required for the electrolytic process ?
      (iv) Write the equation for the reaction which takes place at cathode.
4. Aluminium is extracted from its chief ore, bauxite. The ore is first purified and then the metal is extracted from it by electrolytic reduction.
   (a) Write three balanced equations for the purification of bauxite.
   (b) Name a chemical used for dissolving aluminium oxide. In which state of subdivision is the chemical used ?
   (c) Write an equation for the reaction which takes place at the anode during the extraction of aluminium by the electrolytic process.
   (d) Mention one reason for the use of aluminium in *thermite welding.*
5. (a) A to F below relate to the source and extraction of either zinc or aluminium :
      A. Bauxite, B. Coke,
      C. Cryolite, D. Froth floatation,
      E. Sodium hydroxide solution,
      F. Zinc blende.
      (i) Write down the three letters each from the above list which are relevant to :
         1. Zinc, 2. Aluminium.
      (ii) Fill in the blanks using the most appropriate words from A to F.
         1. The ore from which aluminium is extracted must first be treated with ..... so that pure aluminium oxide can be obtained.

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2. Pure aluminium oxide is dissolved in ..... to make a conducting solution.

(iii) Write the formula of cryolite.

6. Explain with reasons :

(a) In the electrolytic reduction of alumina, the graphite anode is gradually consumed.

or

Why the anode has to be replaced in this process ?

(b) Roasting is carried out on sulphide ores and not on carbonate ores.

(c) Carbon can reduce lead oxide but not aluminium oxide.

(d) Electrolytic reduction is done to obtain aluminium ?

(e) Why 'the food containing iron salts' should not be cooked in aluminium utensils ?

7. For each substance listed below, explain its significance in the extraction of aluminium :

(a) Bauxite, (b) Sodium hydroxide,

(c) Cryolite, (d) Graphite.

8. Distinguish between electrolytic methods of reduction and refining.

9. Give *three* ways in which the metal zinc differs from the non-metal carbon. At least one of the differences must be a chemical difference.

10. (a) Aluminium is a more active metal than iron, but suffers less corrosion. Why ?

(b) Explain and give reasons why aluminium vessels should not be cleaned with powders containing alkalis.

11. How does aluminium react with the following :

(a) Air, (b) Water,

(c) Acid, (d) Base.

7E. USES OF METALS AND ALLOYS

7.13 USES OF ALUMINIUM

(1) Being a **strong, light and corrosion resistant metal**, it is used in alloys.

(2) *Aluminium is light, it has high tensile strength, is resistant to corrosion, good conductor of heat, unaffected by organic acids and has attractive appearance. It is used as follows:*

(i) For making cooking utensils.

Note : When aluminium vessels like kettle are exposed to the atmospheric air, aluminium combines with the oxygen of the air and a film of aluminium oxide (Al_2O_3) is formed at the surface of the kettle. This hard, tightly adhering film of aluminium oxide protects the kettle from the corrosive action of water.

(ii) In building and construction work.

(iii) Aluminium is highly malleable so it is used in the form of thin foils for packaging *i.e.* wrapping chocolates, medicines, etc, for capping milk bottles, in making tubes for tooth-paste, etc.

(iv) For painting electric and telegraphic poles, made of iron to prevent them from rusting.

(v) For making furniture and other household gadgets.

(3) *Aluminium has a strong affinity for oxygen.* It is used :

(i) as a deoxidizer in the manufacture of steel.

(ii) for making flash bulbs for photography.

(iii) for making certain explosives like *ammonal* (a mixture of aluminium powder and ammonium nitrate).

(4) Being an **excellent reflector of light,** aluminium is used *for making mirrors* for reflecting telescope.

(5) Aluminium being a *good conductor* of electricity, is used for the manufacture of cables for power transmission.

(6) As aluminium is **unaffected by sea water**, ships are made of alloys of aluminium.

(7) Aluminium is a good reducing agent so it is used in **aluminothermy.** *This process is used in joining (or welding) broken iron girders, rails and machine parts.*

(8) Aluminium amalgams liberate hydrogen when treated with water so it is used for carrying out reduction of organic compounds.

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GOLDSCHMIDT'S ALUMINOTHERMIC PROCESS (Thermite welding) :

In this process, **thermit** (a mixture of 3 parts of ferric oxide and one part of aluminium powder) is taken in a crucible with a hole at the bottom. The crucible is placed over the broken pieces of iron rails.

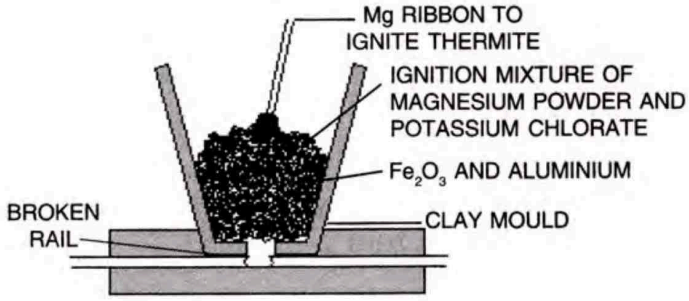

Fig. 7.10 Welding by aluminothermic process

An ignition *mixture of potassium chlorate and magnesium powder is placed on the top of the thermit. To start the process, a fuse of burning magnesium is inserted into the ignition mixture* which catches fire and ignites the thermit. A vigorous reaction occurs with the evolution of a large amount of heat. During this reaction, a temperature of 3000°C is achieved.

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + Heat$$

Iron formed from ferric oxide is allowed to drop over the gap between the broken pieces and they join together.

7.14 FORMS AND USES OF IRON

Depending upon the carbon content, various forms of iron are known such as :

(a) **Cast iron** with carbon content 2·5–5%.

Use : It is used in drain pipes, gutter covers, weights and railings.

(b) **Wrought iron** with carbon content 0·1 to 0·25%,

Use : It is used in chains, horse shoes and electromagnets.

(c) **Steel** has high structural and tensile strength.

Use : It is used in the construction of buildings, overhead structures, machines, automobiles, instruments and in various alloys.

The properties of steel can be changed by heat treatment *i.e.* **tempering.**

When the steel is heated red hot at temperature above 800°C and immediately plunged into cold water or oil, *it becomes harder and more brittle.* This process is called ***quenching*** (hardening) of steel. **Specific machine parts are made by quenching.**

If this quenched steel is reheated carefully to a regulated temperature between 220°C – 300°C and then allowed to cool slowly, *it retains its hardness but its brittleness vanishes.* This *process is known as **annealing** of steel.* **Razor blades, springs, *etc.*, are made by annealing.**

(i) **Mild steel** contains a very low percentage of carbon (0.15 to 0.5%). Used to manufacture **nuts, bolts, *etc***.

(ii) **Hard steel** contains slightly higher carbon content (0.5-1.5%). It is used to make **tools.**

7.15 USES OF ZINC

Zinc has a strong electropositive character. It is used for the following purposes.

1. Zinc is mostly used for coating iron and steel sheets to prevent them from rusting. This process is known as *galvanisation*. Coating is done either by spraying molten zinc on the surface of iron or by immersing the iron object in a bath tub containing molten zinc. *Galvanised iron sheets are used for making pipes, buckets, dustbins, bath tubs, water tanks, wire and roofs.*
2. It is meant for making useful **alloys such as brass, bronze, German silver,** etc.
3. Due to strong electropositive nature, it easily forms Zn^{2+} ions. It is used to make **dry cell** containers which act as negative electrode.
4. It is used for preparing lithographic plates for printing.
5. **Zinc dust** (zinc contains traces of ZnO) is used as a reducing agent for many organic reductions. These reductions are employed in the manufacture of **drugs, dyes, perfumes**, etc.

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6. Zinc compounds are used for paints, electroplast (antiseptic), preservatives for leather and mordant for textiles. *Zinc carbonate is used for making cosmetics, especially face creams.*

7.16 ALLOYS

Alloy is a homogeneous mixture of two or more metals or of one or more metals with certain non-metallic elements.

The properties of alloys are often greatly different from those of the components. Gold is too soft to be used without a small percentage of copper. The corrosion and oxidation resistance of steel is markedly increased by incorporation of 15 to 18% of chromium and often a few per cent of nickel (stainless steel). The presence of carbon up to 1.5%

7.16.1 Purpose of making alloys

Alloys are made to change the property of their major constituents to achieve a specific objective.

For example :

(i) *Strength of iron is increased by making steel.*

(ii) **Brass** is stronger than its components — copper and zinc.

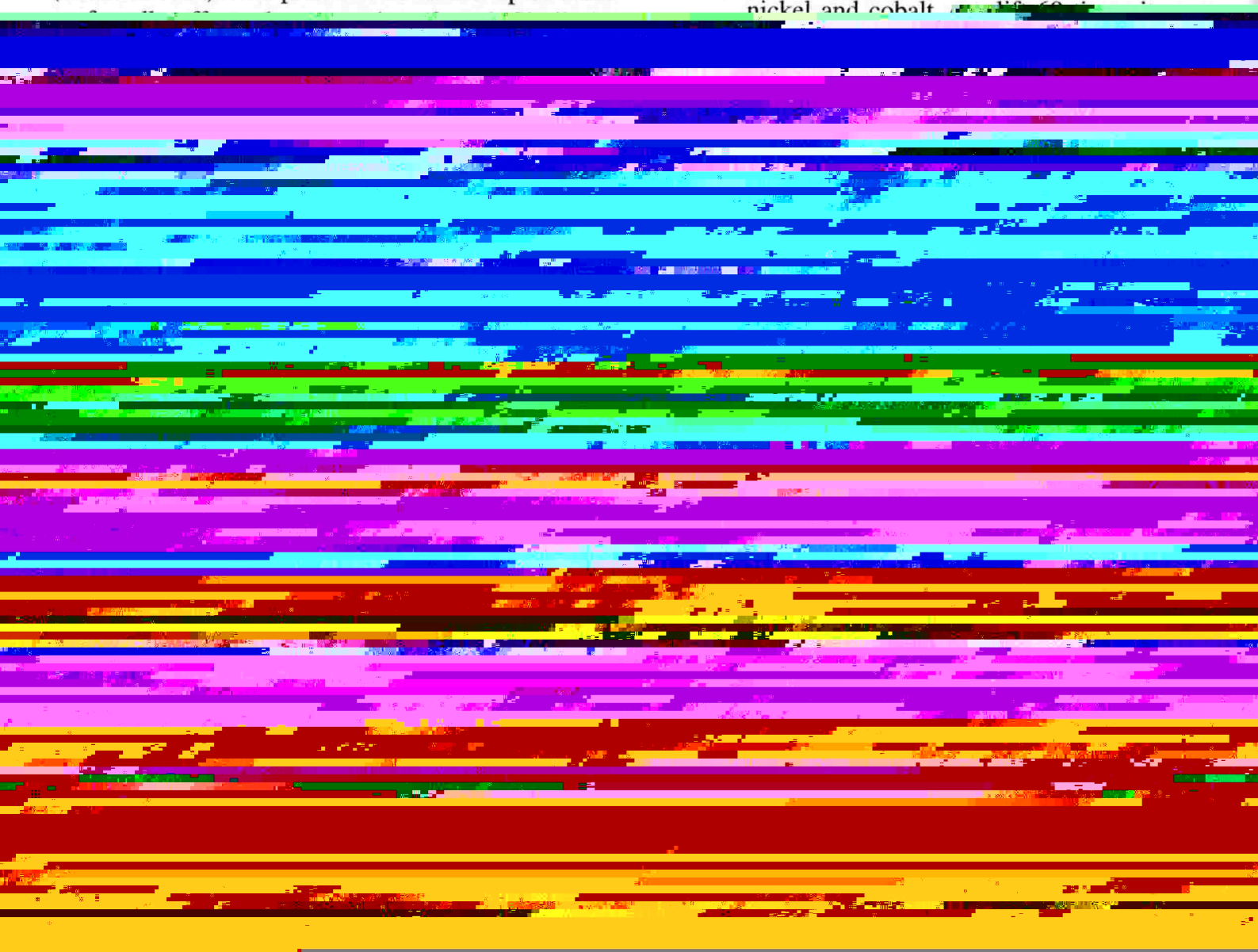
(iii) **Aluminium bronze** looks like gold.

(iv) **Nichrome,** an alloy of Ni, Fe and Cr, is more resistant to electricity.

(v) **Carboloy,** an alloy of tungsten, carbon and cobalt, is as hard as diamond.

(vi) **Bell metal** is more sonorous than copper or tin.

(vii) **Alnico**, an alloy containing aluminium, nickel and cobalt



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3. To *modify casting ability.*
   e.g. Type metal (Pb + Sn + Sb) expands on solidification and is easily cast.
4. To *lower the melting point.*
   e.g. Solder (Pb + Sn) melts at 180°C which is lower than the melting point of lead or tin.
5. To *increase hardness, and tensile strength.*
   e.g. Brass (Cu + Zn) is harder than copper. Duralumin has a strength up to six times greater than pure aluminium.
6. To *increase resistance to electricity.*
   e.g. Nichrome (Ni + Fe + Cr) has more resistance (can produce much heat) to electricity than copper.

7.16.3 Method of making alloys

1. By fusing the metals together. *For example,* brass is made by melting zinc and copper.
2. By compressing finely divided metals. *For example,* **wood metal :** an alloy of lead, tin, bismuth and cadmium powder is a **fusible alloy.** This alloy is used in automatic sprinkler which provides a spray of water to prevent fires from spreading.

7.16.4 Alloys as solid solutions

Alloys can be considered solid solutions in which the metal with a high concentration is solvent and other metals are solute. *For example,* brass is a solid of zinc (solute) in copper (solvent).

Table 7.2 : Common alloys, their compositions, properties and uses

Principal metal	Alloy's name	Composition	Properties	Used for making
Aluminium	1. Duralumin	95% Al 4% Cu 0.5% Mg 0.5% Mn	1. Light but as strong as steel. 2. Hard and resistant to corrosion. 3. Highly ductile.	1. Bodies of aircraft, buses and tube trains. 2. Light tools. 3. Pressure cooker.
"	2. Magnalium	90 - 95% Al 10 - 5% Mg	1. Resists corrosion. 2. Light. 3. Strong.	1. Aircraft 2. Scientific instruments. 3. Metal mirrors. 4. Light tools 5. Beams of balance. 6. Household appliances.
"	3. Alnico	Al, Ni, Co, Fe	1. Light. 2. Shiny 3. Resists corrosion.	– Magnets
Iron	1. Stainless steel	73% Fe, 18% Cr, 8% Ni, 1% C	1. Resists corrosion 2. Lustrous, hard 3. Resistant to acids and alkalis	1. Utensils 2. Cutlery 3. Ornamental pieces 4. Surgical instruments
"	2. Manganese steel	85% Fe, 1% C, 14% Mn	1. Durable, tough and hard	1. Safes. 2. Rock drills and 3. Armour plates
"	3. Tungsten steel	84% Fe, 5% W. 1% C,	1. Very hard	1. Cutting tools for high speed lathes
	4. Nickel steel	95-98% Fe 5-3% Ni	1. Hard and elastic 2. Resistant to corrosion	1. Electric wire cables 2. Automobile parts
	5. Invar	Fe 63%, Ni 36% C 1%	1. Negligible expansion	1. Metre scales 2. Scientific instruments

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Zinc and Copper	1. Brass	60 - 70% Cu 40 - 30% Zn	1. Malleable and ductile. 2. Can be easily cast. 3. Resists corrosion. 4. Yellow/Silvery in colour.	1. Decorative hardware, utensils. 2. Screws and handles. 3. Cartridge, containers. 4. Parts of watches. 5. Musical instruments. 6. Electrical goods.
,,	2. Bronze	80% Cu 18% Sn 2% Zn	1. Hard and easily cast. 2. Can take up polish. 3. Resists corrosion.	1. Medals. 2. Statues. 3. Utensils. 4. Bearing and 5. Coins.
,,	3. German silver	50% Cu, 30% Zn 20% Ni	1. White and light like Silver. 2. Malleable and ductile. 3. High electrical resistance.	1. Decorative articles. 2. Electric heaters, Rheostats. 3. Resistors.
,,	4. Bell metal	78%Cu 22% Sn	1. Sonorous (produces sound). 2. Hard and brittle.	1. Bell, gongs. 2. Statues.
,,	5. Gun metal	Cu 88%, Sn 8% Zn 1%, Pb 1%	1. Hard & brittle 2. Easily cut	1. Barrels of cannons. 2. Bearings, Gears, etc.
Lead	1. Solder or Fuse metal	Pb, Sn	1. Low m.p., high tensile strength	1. Welding 2. Fuse
	2. Type metal	75% Pb, 15% Sb, 10% Sn,	1. Low m.p. easily cast.	1. For printing blocks.

EXERCISE-7E

1. State a reason why zinc is used in :
(a) galvanisation, (b) dry cells, (c) cosmetics ?

2. State on what special properties the use of each of these metals depends :
(a) aluminium, (b) zinc ?

3. Explain the following :
(a) Zinc is used to cover iron so as to prevent rusting of iron. Why ?
(b) A neutral gas other than oxygen which is formed at the anode during electrolysis of fused alumina.
(c) Nitric acid can be stored in aluminium containers.
(d) In construction work, why is the alloy of aluminium-duralumin used rather than pure aluminium ?

4. State the use of :
(a) Cast Iron, (b) Wrought Iron,
(c) Mild steel, (d) Hard steel.

5. Which metal is used for :
(a) making pipes, buckets, water tanks,
(b) lithographic plates for printing,
(c) making face creams.

6. Give reasons, why aluminium is used in :
(a) making alloys,
(b) Wrapping chocolates,
(c) Painting electric and telegraphic poles,
(d) In aluminiothermy,
(e) In making ships.

7. Aluminium is used in thermite welding :
(a) What is thermit ?
(b) What is ignition mixture ?
(c) Write reaction for process ?

8. What is an alloy ? How do the properties of an alloy differ from its constituents ?

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9. Calcium, copper, lead, aluminium, zinc, chromium, magnesium and iron.

Choose the major metals from the list given above to make the following alloys :

(a) Stainless steel, (b) Brass.

10. Both brass and bronze contain copper as major constituents. Name other elements in these alloys.

11. Name an alloy of :

(a) aluminium used in aircraft construction.

(b) lead used in electrical wiring or electrical work in joining metals.

(c) copper in electrical appliances or household vessels.

(d) zinc used in simple voltaic cells.

12. What is an amalgam ? State its use with an example.

13. (a) State two properties of brass that render it more useful for some purposes than its components.

(b) a metal which forms a liquid alloy at ordinary temperature.

14. What is magnalium ? Name the main elements present in it ? Write its one use.

15. Name the constituents of :

(a) Duralumin, (b) Solder,

(c) Bronze, (d) Invar.

SUMMARY

1. Metals : Mostly solids, (Hg is liquid) brilliant lustre, high density, good conductor of heat and electricity, malleable, ductile, tenacious.

They usually form basic oxides, are non-volatile and electrovalent chloride, have electropositive nature, react with dilute acids to liberate hydrogen (except Cu, Ag and Au), form positively charged ions which get deposited on the cathode on electrolysis of fused salts or their aqueous solutions. Some oxides like Al_2O_3 and PbO are amphoteric.

2. Non-Metals : They usually form acidic oxides, are volatile and covalent chlorides, do not react with dilute acids to liberate hydrogen gas, usually form negative ions which get discharged at the anode, and have electronegative nature. Some oxides like CO, NO and H_2O are neutral.

3. Metal Activity series : K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Pt > Au

(i) Metals below hydrogen, cannot displace hydrogen from acids.

(ii) Metals, which are higher in the series, can displace metals below it.

(iii) The higher the position, the more active is the metal.

(iv) Hydrogen has electropositive character and so placed among the metals.

4. Minerals : Naturally occurring materials from which, metals can be extracted.

5. Ores : Minerals from which, the metals may be extracted economically in reasonably pure conditions.

6. Flux : A substance, added in the furnace during smelting to remove the *gangue*.

7. Gangue : Rocky and earthly impurities present in ores. Gangue may be acidic (like SiO_2) or basic (like FeO).

8. Slag : A substance formed when flux reacts with gangue.

$$\underset{\text{gangue}}{SiO_2} + \underset{\text{flux}}{CaCO_3} \rightarrow \underset{\text{slag}}{CaSiO_3} + CO_2$$

9. Roasting : Heating the concentrated ore in the presence of air to remove volatile impurities like S, As, moisture, etc. and to render the ore porous.

10. Smelting : The process of reducing roasted ore and removing the gangue.

11. Methods of Concentration of Ores

(i) Froth floatation : Finely ground sulphide ores + water + pine oil + Air blast. Ore particles come up with the froth and float on surface. Rocky matter settles to the bottom e.g. zinc blende is concentrated by froth floatation.

(ii) Electromagnetic Separation : Magnetic ores are separated by this process like iron ores.

(iii) Gravity Separation : Heavier ores are separated.

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12. Roasting : Concentrated ore heated in air without melting to drive off S, As, C and moisture. It makes the ore porous and even converts the sulphide and carbonate ores to respective oxides.

$$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$$
$$ZnCO_3 \rightarrow ZnO + CO_2$$

13. Reduction of metallic oxides : Less reactive metals like copper, zinc, lead and iron can be reduced by heating the metallic oxide by carbon, carbon monoxide or hydrogen. Reactive metals like K, Na, Ca, Mg and Al are reduced by electrolysis.

14. Refining of metals

(i) Distillation : Easily volatile metals like Zn, Cd and Hg may be removed.

(ii) Liquation : Easily fusible metals like lead and tin may be removed.

(iii) Oxidation : Easily oxidisable impurities like Cu and Sn may be removed by passing air blast through molten impure metals.

(iv) Electrolytic refining : Many metals like Cu, Al, Zn, Ag and Au may be refined using the impure metals as anodes, pure metal strip as cathode and metal salt solution as electrolyte.

15. Extraction of Aluminium :

(a) Purification of bauxite (Baeyer's process) :

STEP 1 : Conversion of impure bauxite into sodium aluminate.

$$\underset{\text{(bauxite)}}{Al_2O_3 . 2H_2O} + 2NaOH(Conc.) \xrightarrow{150°C} \underset{\text{Sodium aluminate}}{2NaAlO_2} + 3H_2O$$

STEP 2 : Conversion of sodium aluminate into aluminium hydroxide.

$$\underset{\substack{\text{Sodium}\\\text{aluminate}}}{NaAlO_2} + 2H_2O \rightarrow \underset{\substack{\text{Aluminium}\\\text{hydroxide}}}{Al(OH)_3\downarrow} + NaOH$$

STEP 3 : Conversion of aluminium hydroxide into pure alumina

$$2Al(OH)_3 \xrightarrow[50°C]{1000°C} \underset{\text{Purified alumina}}{Al_2O_3\downarrow} + 3H_2O\uparrow$$

(b) Electrolytic reduction of purified bauxite (Hall-Heroult's Process) :

Electrolyte : Purified alumina : Cryolite : Fluorspar (1 : 3 : 1)

Electrodes : Carbon lining as **cathode;** graphite rods as **anode.** Temperature : 950°C.

Products formed : Pure Al at the cathode; oxygen gas at the anode.

Role of cryolite :

(i) Melting point of alumina gets lowered from 2050°C to 950°C

(ii) Increases electrical conductivity of the mixture.

(c) Refining or purification (Hoopes electrolytic process) : It is done in an electrolytic cell which consists of three layers of molten substances differing in specific gravity. Molten impure aluminium forms the bottom layer and acts as anode, pure molten aluminium forms the upper layer and acts as cathode. The middle layer is of fluorides.

at Anode : $Al - 3e^- \rightarrow Al^{3+}$

at Cathode : $Al^{3+} + 3e^- \rightarrow Al$

Reactions of Aluminium : (i) With moist air, Al_2O_3 is formed. When ignited both Al_2O_3 and AlN are formed.

(ii) $2Al + 3H_2O$ (steam) $\rightarrow Al_2O_3 + 3H_2\uparrow$

(iii) $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2\uparrow$

$2Al + 6H_2SO_4$ conc. $\rightarrow Al_2(SO_4)_3 + 3SO_2\uparrow + 6H_2O$

(iv) $2Al + 2NaOH + 2H_2O \rightarrow \underset{\text{sodium meta-aluminate}}{2NaAlO_2} + 3H_2\uparrow$

16. Alloy : Alloy is a homogeneous mixture of two or more metals: or of one or more metals with certain non-metallic elements.

17. Galvanisation : It is the process of giving a thin, adherent coating of Zn on iron sheets or wares to protect iron from rusting.

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MISCELLANEOUS EXERCISE

1. Name the following :

(a) A metal which is liquid at room temperature.

(b) The process of heating an ore to a high temperature in the presence of air.

(c) The compound formed by the reaction between calcium oxide and silica.

(d) A compound which is added to lower the fusion temperature of the electrolytic bath in the extraction of aluminium.

(e) Name an allotrope of a non-metal that allows electricity to pass through it.

2. From the list of characteristics given below, select the five which are relevant to non-metals and their compounds :

A. Ductile, B. Conduct electricity,
C. Brittle, D. Acidic oxides,
E. Basic oxides, F. Discharged at anode,
G. Discharged at cathode, H. Ionic chlorides,
I. Covalent chlorides,
J. Reaction with dilute sulphuric acid yields hydrogen,
K. 1, 2 or 3 valence electrons,
L. 5, 6, 7 valence electrons.

(Write the five letters corresponding to the correct characteristics).

3. The following is an extract from 'Metals in the Service of Man, Alexander and Street/Pelican 1976' :

'Alumina (aluminium oxide) has a very high melting point of over 2000°C so that it cannot readily be liquefied. However, conversion of alumina to aluminium and oxygen, by electrolysis, can occur when it is dissolved in some other substance.'

(a) Which solution is used to react with bauxite as a first step in obtaining pure aluminium oxide ?

(b) The aluminium oxide for the electrolytic extraction of aluminium is obtained by heating aluminium hydroxide. Write the equation for this reaction.

(c) Name the element which serves both as the anode and the cathode in the extraction of aluminium.

(d) Write the equation for the reaction that occurs at the cathode during the extraction of aluminium by electrolysis.

(e) Give the equation for the reaction which occurs at the anode when aluminium is purified by electrolysis.

2008

(a) The following is a sketch of an electrolytic cell used in the extraction of aluminium :

(i) What is the substance of which the electrodes A and B are made ?

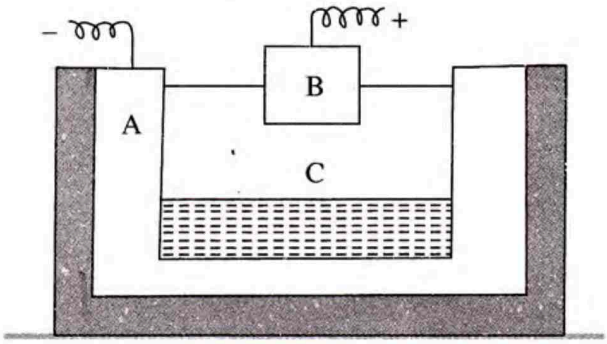
(ii) At which electrode (A or B) is the aluminium formed ?

(iii) What are the two aluminium compounds in the electrolyte C ?

(v) Why is it necessary for electrode B to be continuously replaced ?

(b) Brass is an alloy of :

A. Copper and tin, B. Copper and zinc,
C. Zinc and lead, D. Lead and tin.



2009

(a) The metal oxide which can react with acid as well as alkali is :

(i) Silver oxide (ii) Copper(II) oxide
(iii) Aluminium oxide (iv) Calcium oxide

(b) State the property of the metal being utilized in the following :

Use of metal	Property
Zinc in Galvanization	
Aluminium in Thermite welding	

2010

(a) Which one of the following is **not** true of metals :

(i) Metals are good conductors of electricity

(ii) Metals are malleable and ductile

(iii) Metals form non-polar covalent compounds

(iv) Metal will have 1 or 2 or 3 electrons in their valence shell.

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(b) Name the **main** constituent metal in the following alloys :

(i) Duralumin (ii) Brass

(iii) Stainless steel

2011

(a) Answer the following questions :

(i) Name a metal which is found abundantly in the earth's crust.

(ii) What is the difference between calcination and roasting ?

(iii) Name the process used for the enrichment of sulphide ore.

(iv) Write the chemical formulae of one main ore of iron and aluminium.

(v) Write the constituents of electrolyte for the extraction of aluminium

2012

(a) Name the following metals :

(i) A metal present in cryolite other than sodium.

(ii) A metal which is unaffected by dilute or concentrated acids.

(iii) A metal present in period 3, group 1 of the periodic table.

(b) The following questions are relevant to the extraction of Aluminium :

(i) State the reason for addition of caustic alkali to bauxite ore during purification of bauxite.

(ii) Give a balanced chemical equation for the above reaction.

(iii) Along with cryolite and alumina, another substance is added to the electrolyte mixture. Name the substance and give one reason for the addition.

2013

(a)

	X	Y
Normal electronic Configuration	2, 8, 7	2, 8, 2
Nature of oxide	Dissolves in water and turns blue litmus red	Very low solubility in water. Dissolves in hydrochloric acid
Tendency for oxidising and reducing reactions	Tends to oxidise elements and compounds	Tends to act as a reducing agent
Electrical and Thermal conductivity	Very poor electrical conductor Poor thermal conductivity	Good Electrical conductor Good thermal conductor
Tendency to form alloys and amalgams	No tendency to form alloys	Forms alloys

Using the information above, complete the following:

(i) ..... is the metallic element.

(ii) Metal atoms tend to have a maximum of ..... electrons in the outermost energy level.

(iii) Non-metallic elements tend to form ..... oxides while metals tend to form ..... oxides.

(iv) Non-metallic elements tend to be ..... conductors of heat and electricity.

(v) Metals tend to ..... electrons and act as ..... agents in their reactions with elements and compounds.

2014

(a) The main ore used for the extraction of iron is:

(i) Haematite (ii) Calamine

(iii) Bauxite (iv) Cryolite

(b) Heating an ore in a limited supply of air or in the absence of air at a temperature just below its melting point is known as :

(i) Smelting (ii) Ore dressing

(iii) Calcination (iv) Bessemerisation

(c) State the main components of the following alloys.

(i) Brass (ii) Duralumin

(iii) Bronze

(d) Name the following :

(i) The property possessed by metals by which they can be beaten into sheets.

(ii) A compound added to lower the fusion temperature of electrolytic bath in the extraction of aluminium.

(iii) The ore of zinc containing its sulphide.

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8 Study of Compounds — Hydrogen Chloride

SYLLABUS

(i) Hydrogen chloride: preparation of hydrogen chloride from sodium chloride; refer to the density and solubility of hydrogen chloride (fountain experiment); reaction with ammonia; acidic properties of its solution.

Preparation of hydrogen chloride from sodium chloride; (the laboratory method of preparation can be learnt in terms of reactants, product, condition, equation, diagram or setting of the apparatus, procedure, observation, precaution, collection of the gas and identification).

Simple experiment to show the density of the gas (Hydrogen Chloride) – heavier than air.

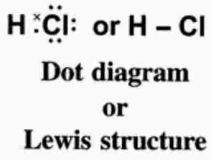
Solubility of hydrogen chloride (fountain experiment); (setting of the apparatus, procedure, observation, inference) – method of preparation of hydrochloric acid by dissolving the gas in water – the special arrangement and the mechanism by which the back suction is avoided should be learnt.

Reaction with ammonia

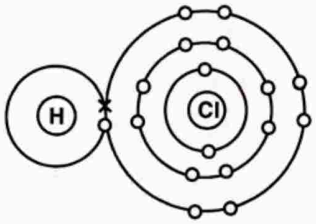
Acidic properties of its solution – (reaction with metals, their oxides, hydroxides and carbonates to give their chlorides, decomposition of carbonates, hydrogen carbonates, sulphides, sulphites, thiosulphates and nitrates).

HYDROGEN CHLORIDE

Molecular formula : HCl
Molecular mass : 36.5
Bond : Covalent



Dot diagram or Lewis structure



Orbital diagram

8.1 OCCURRENCE

Glauber prepared the acid in 1648 by heating common salt (NaCl) with concentrated sulphuric acid. Lavoisier named it ***muriatic acid***. Davy in 1810 named it as hydrochloric acid.

1. Hydrogen chloride gas occurs in free state in volcanic emissions.
2. Hydrochloric acid (0·2 – 0·4 percent) is present in gastric juice of mammals and it helps in digestion.

8.2 GENERAL PREPARATION OF HYDROGEN CHLORIDE GAS

1. *By synthesis (Direct combination).*

(i) Moist hydrogen gas combines with chlorine in the presence of diffused sunlight.

$$\text{Hydrogen} + \text{Chlorine} \rightarrow \text{Hydrogen chloride}$$

$$H_2\,(g) + Cl_2\,(g) \xrightarrow[\text{Sunlight}]{\text{Diffused}} 2HCl\,(g)$$

Note : The reaction is *explosive in direct sunlight* but it is *negligible in the dark.*

However, in the presence of a *catalyst such as activated carbon,* the reaction takes place even in the dark. (Activated carbon absorbs hydrogen which increases reactivity).

(ii) Burning jet of hydrogen also burns in chlorine forming hydrogen chloride.

2. *By heating metallic chloride with conc. sulphuric acid*

Metal chloride	+	Sulphuric acid [conc.]		Salt	+	Hydrogen chloride
$NaCl$	+	H_2SO_4	$\xrightarrow{< 200\ ^\circ C}$	$NaHSO_4$	+	HCl [g]
$2NaCl$	+	H_2SO_4	$\xrightarrow{\text{above } 200\ ^\circ C}$	Na_2SO_4	+	$2HCl$ [g]
$CuCl_2$	+	H_2SO_4	$\xrightarrow{\Delta}$	$CuSO_4$	+	$2HCl$ [g]

8.3 LABORATORY PREPARATION OF HYDROGEN CHLORIDE GAS

In the laboratory, hydrogen chloride gas can be prepared by the action of concentrated sulphuric acid on sodium chloride.

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Reactants : Sodium chloride and concentrated sulphuric acid.

Procedure : Set up the apparatus as shown in Fig. 8.1.

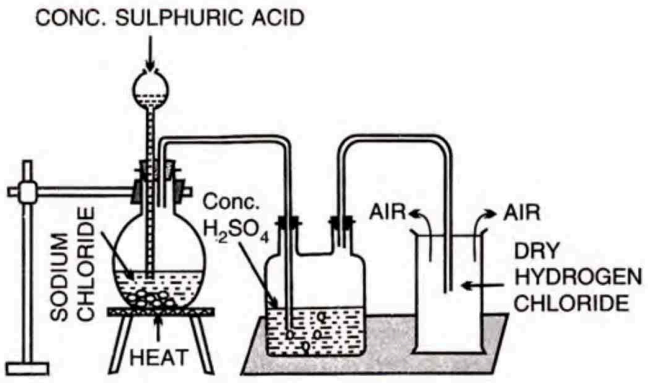

Fig. 8.1 Preparation of hydrogen chloride gas from common salt.

Place some common salt in a flask* and pour concentrated sulphuric acid through the thistle funnel. The reaction proceeds slowly in the cold. On heating the mixture gently there is effervescence and HCl gas is evolved in a controlled manner.

The gas is dried by passing through washer bottle contianing concentrated sulphuric acid.

It is collected by upward displacement of air in a gas jar.

Reaction :

$$NaCl + H_2SO_4 \xrightarrow{<200\,°C} NaHSO_4 + HCl\uparrow$$

Though it is a reversible reaction, yet it goes to completion as hydrogen chloride continuously escapes as a gas.

The reaction can occur upto the stage of formation of sodium sulphate on heating above 200°C.

$$**NaHSO_4 + NaCl \xrightarrow{\text{above } 200\,°C} Na_2SO_4 + HCl\uparrow$$

or

$$2NaCl + \underset{\text{(Conc.)}}{H_2SO_4} \xrightarrow{\text{above } 200\,°C} Na_2SO_4 + 2HCl$$

* Flat bottom flask or round bottom flask can be used. Round bottom flasks are normally used to heat at high temperature.

** This reaction indicates clearly the acid nature of sodium hydrogen sulphate.

Note : (i) Sodium chloride is cheap and therefore it is preferred for preparation of HCl over other metal chlorides.

(ii) **Conc. nitric acid is not used during the preparation of HCl** because it is volatile and may volatilize out alongwith hydrogen chloride.

Purification of HCl gas :

It is dried by passing through conc. sulphuric acid.

The other drying agents like phosphorus pentoxide (P_2O_5) and quick lime (CaO) cannot be used, since they react with hydrogen chloride.

$$2P_2O_5 + 3HCl \rightarrow POCl_3 + 3HPO_3$$

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

Collection :

1. Hydrogen chloride gas is collected by the downward delivery (upward displacement of air) as it is 1·28 times heavier than air.
2. It is not collected over water, since it is highly soluble in water.

Identification

When the jar is completely filled with hydrogen chloride, fumes appear above the jar's mouth.

When hydrogen chloride gas is exposed to air, *it gives white fumes,* due to the formation of hydrochloric acid on reacting with atmospheric water vapour.

In order to know whether the gas jar is full, bring a rod dipped in ammonium hydroxide near its mouth.

$$HCl + NH_4OH \rightarrow NH_4Cl + H_2O$$

Dense white fumes of ammonium chloride will be produced proving thereby that the jar is full of hydrogen chloride gas.

Precautions :

(a) The lower end of the thistle funnel must be dipped in conc. sulphuric acid.

(b) Delivery tube should be dipped in drying agent *i.e.,* conc. H_2SO_4.

(c) Temperature is maintained at nearly 200°C

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At higher temperature *i.e.*, above 200°C :

(i) the apparatus which is made of glass may crack.

(ii) fuel is wasted.

(iii) sodium sulphate formed, forms a hard crust which sticks to the glass and is difficult to remove.

8.4 PHYSICAL PROPERTIES OF HYDROGEN CHLORIDE GAS

1. Colour	*colourless* gas
2. Smell	*pungent* choking smell
3. Taste	sour (*acidic*) in taste
4. Physiological nature	It is corrosive in nature. *It irritates nose, throat and lungs.*
5. Density	It is about one and a quarter times heavier than air (V.D. of HCl is 18·25 and that of air 14·4).
6. Boiling point	–83°C
7. Melting point	–113°C
8. Liquefaction and solidification	When subjected to a high pressure (40 atm.) at low temperature (10°C), it is liquefied to a colourless liquid.
9. Solubility	Hydrogen chloride gas is highly soluble in water (1 volume of water dissolves 452 volumes of the gas at room temperature). HCl being polar covalent compound is soluble in organic non-polar solvents also such as acetone and toluene.

Experiment to demonstrate density : *To show that HCl gas is heavier than air.*

It can be proved by pouring the gas in a jar with a burning candle as shown in Fig. 8.2. The candle gets extinguished because HCl gas being heavier occupies the lower portion of the jar and forces the air out of it.

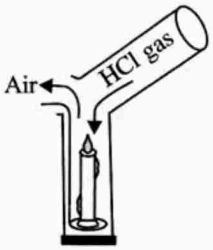

Fig. 8.2 To show that HCl gas is heavier than air

Experiment to demonstrate solubility : *To show that HCl gas is highly soluble.*

The great solubility of the gas can be demonstrated by means of *Fountain Experiment* as illustrated below.

Fountain Experiment : Take a *dry* round bottomed flask filled with dry HCl gas. Arrange the apparatus as shown in Fig. 8.3. On the mouth of the flask, fix a rubber stopper with two holes. Through one hole pass a long jet tube, and through other hole pass a dropper with few drops of water. Put the jet tube in the beaker containing blue litmus solution.

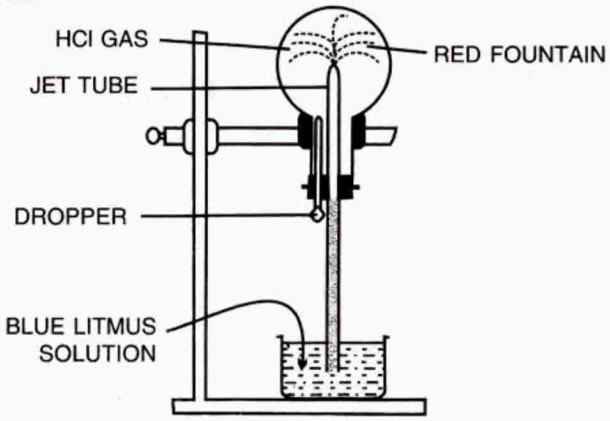

Fig. 8.3 Fountain experiment

Press the dropper. It is seen that the blue litmus solution enters the jet tube with a great force, forming a red fountain.

Reason : As the water goes in the flask from the dropper, HCl gas present in the flask dissolves due to its high solubility, thereby lowering the pressure inside.

The outside pressure being higher, pushes the blue litmus solution inside, through the jet tube.

The blue litmus solution turns red due to the acidic nature of hydrogen chloride gas.

Note : Due to the high solubility, hydrogen chloride gas fumes in moist air forming tiny droplets of hydrochloric acid.

8.5 CHEMICAL PROPERTIES OF HYDROGEN CHLORIDE GAS

1. **Combustibility :** The gas is neither combustible nor a supporter of combustion. *It does not burn, rather it extinguishes a burning splint.*

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2. **Thermal dissociation :** On heating above 500°C, it dissociates into hydrogen and chlorine.

$$2HCl \xrightarrow{>500°C} H_2 + Cl_2$$

3. **With metals :** Metals that come before hydrogen in the electrochemical series form their corresponding chlorides when **heated** with HCl and liberate hydrogen.

Metal	+	Hydrogen chloride (g)	→	Metal chloride	+	Hydrogen
2Na	+	2HCl	→	2NaCl	+	H_2
Ca	+	2HCl	→	$CaCl_2$	+	H_2
Mg	+	2HCl	→	$MgCl_2$	+	H_2
Zn	+	2HCl	→	$ZnCl_2$	+	H_2
Fe	+	2HCl	→	$FeCl_2$	+	H_2

4. **Reaction with ammonia :** It combines with ammonia to form dense white fumes of ammonium chloride.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

i.e., two gases combine to form solid

8.6 HYDROCHLORIC ACID

When hydrogen chloride gas is dissolved in water, hydrochloric acid is formed.

The covalent compound ionises in water due to its polar nature.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

Important : (i) Dry hydrogen chloride gas and liquefied hydrogen chloride do not turn blue litmus paper red, showing non-acidic character of the dry gas.

(ii) *Liquefied hydrogen chloride does not conduct electricity. It shows covalent nature of hydrogen chloride.*

The gas is also soluble in ***toluene****, (organic solvent) but in that case, it neither turns blue litmus red nor conducts electricity.*

This indicates the absence of H_3O^+ in toluene showing thereby that hydrogen chloride is a covalent compound.

8.6.1 Laboratory method of preparation of hydrochloric acid

The aqueous solution of HCl gas is known as Hydrochloric acid.

It is prepared by dissolving hydrogen chloride gas in water.

The gas is passed into water until no more gas is absorbed. The product is concentrated and contains about 36% of hydrogen chloride by mass.

Procedure :

An inverted funnel, connected to the hydrogen chloride gas supply, is placed in the beaker in such a way that it just touches the water taken in the trough.

Since HCl gas is highly soluble in water, the water rises up in the funnel, back-suction occurs and in turn, the level outside the funnel falls creating an air gap between the rim of the funnel and the surface of water.

The pressure outside and inside then becomes equal and the water which had risen in the funnel falls down again.

This process continues till the water in the trough is saturated with hydrogen chloride gas resulting in the formation of hydrochloric acid.

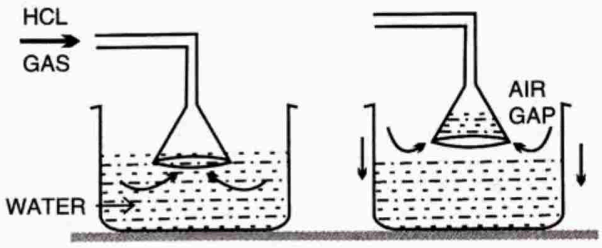

Fig. 8.4 To show funnel arrangement

Mechanism by which back suction is avoided or minimized

The funnel arrangement —

(a) prevents or minimizes back-suction of water.

(b) provides a large surface area for absorption of HCl gas.

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Why is funnel arrangement used ?

Hydrogen chloride gas is sufficiently soluble, so, it is absorbed in water more quickly than it is being generated in the flask. In this case, the pressure in the delivery tube and flask is reduced and the atmospheric pressure from outside forces the water back upto the delivery tube. This effect is called '*back suction*'.

If the tube is made of ordinary narrow glass tubing, the water will quickly fill it and pass over into the generating flask. *This would stop the reaction and might result in an explosion due to the heat produced when water comes in contact with hot concentrated sulphuric acid.*

Precaution : To prevent such an accident, an empty flask (anti-suction device) is put between the generative flask and the water trough. In case, the back suction occurs, the water will collect in it and will not reach the generating flask.

A dilute aqueous solution of hydrochloric acid gets gradually concentrated on distillation, till the *concentration* of the acid reaches 22.2% HCl by weight and 77.8% water by weight which boils at 110 °C. No further increase in concentration is possible on boiling, as molecules of HCl (g) get mixed with water vapours.

A constant boiling mixture or azeotrope is a solution which boils without any change in its composition. **HCl acid forms constant boiling mixture at 110°C.**

8.6.2 Properties of hydrochloric acid

Physical properties

1. Colour	Colourless
2. Smell	Pungent choking smell
3. Taste	Sour (acidic)
4. Physiological action	**Concentrated** acid is corrosive and causes blisters on the skin.
5. Solubility	Readily soluble in water in all proportions.
6. Boiling point	110°C (volatile acid)

Chemical properties

1. ***Nature :*** Aqueous solution is strongly acidic, and shows all properties of acids.

Action on indicators

Indicator	Original colour	Changes to
Moist litmus	Blue	Red
Methyl orange	Orange	Pink
Phenolphthalein	Colourless	Colourless

2. **Action on metals :** Hydrochloric acid reacts with metals above hydrogen in the activity series forming metallic chlorides and evolving hydrogen.

Metal	+	Hydrochloric acid	→	Metallic chloride	+	Hydrogen
Ca	+	2HCl	→	$CaCl_2$	+	$H_2\uparrow$
Mg	+	2HCl	→	$MgCl_2$	+	$H_2\uparrow$
Zn	+	2HCl	→	$ZnCl_2$	+	$H_2\uparrow$

3. **Action on oxides and hydroxides :** It reacts with oxides and hydroxides (bases) to form salt and water only.

Metallic oxide/ hydroxide	+	Hydro-chloric acid	→	Metallic chloride	+	Water
MgO	+	2HCl	→	$MgCl_2$	+	H_2O
ZnO	+	2HCl	→	$ZnCl_2$	+	H_2O
FeO	+	2HCl	→	$FeCl_2$	+	H_2O
Fe_2O_3	+	6HCl	→	$2FeCl_3$	+	$3H_2O$
CuO	+	2HCl	→	$CuCl_2$	+	H_2O
NaOH	+	HCl	→	NaCl	+	H_2O
$Ca(OH)_2$	+	2HCl	→	$CaCl_2$	+	$2H_2O$
$Zn(OH)_2$	+	2HCl	→	$ZnCl_2$	+	$2H_2O$
$Cu(OH)_2$	+	2HCl	→	$CuCl_2$	+	$2H_2O$

4. **With salts of weaker acids :** It decomposes salts of weaker acids *e.g.*, carbonates, hydrogen carbonates, sulphites and sulphides.

Carbonates/ hydrogen carbonates	+	Hydrochloric acid	→	Metal chloride	+	Water	+	Carbon dioxide
Na_2CO_3	+	2HCl	→	2NaCl	+	H_2O	+	$CO_2\uparrow$
$CaCO_3$	+	2HCl	→	$CaCl_2$	+	H_2O	+	$CO_2\uparrow$
$NaHCO_3$	+	HCl	→	NaCl	+	H_2O	+	$CO_2\uparrow$
$Ca(HCO_3)_2$	+	2HCl	→	$CaCl_2$	+	$2H_2O$	+	$2CO_2\uparrow$

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Sulphites/ hydrogen sulphites	+ Hydrochloric acid (dil)	→ Metal chloride	+ Water	+ Sulphur dioxide
Na_2SO_3	+ 2HCl	→ 2NaCl	+ H_2O	+ $SO_2\uparrow$
K_2SO_3	+ 2HCl	→ 2KCl	+ H_2O	+ $SO_2\uparrow$
$NaHSO_3$	+ HCl	→ NaCl	+ H_2O	+ $SO_2\uparrow$

Metallic sulphide	+ Hydrochloric acid (dil)	→ Metal chloride	+ Hydrogen sulphide
Na_2S	+ 2HCl	→ 2NaCl	+ $H_2S\uparrow$
FeS	+ 2HCl	→ $FeCl_2$	+ $H_2S\uparrow$

5. Action on thiosulphates : Dilute hydrochloric acid reacts with thiosulphates to produce sulphur dioxide gas and yellow sulphur is precipitated.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2 + S\downarrow$$

Note : *This reaction can be used as a test for distinguishing thiosulphates and sulphites as sulphur is not precipitated when sulphites are treated with dilute HCl.*

6. Reaction with nitrates : Dil. HCl does not normally react with nitrates. However **lead nitrate and mercury (I) nitrate** react with hydrochloric acid to give white precipitate of lead and mercury (I) chloride.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2\downarrow + 2HNO_3$$
$$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2\downarrow + 2HNO_3$$

Silver nitrate solution react with hydrochloric acid and it gives a thick curdy white precipitate of silver chloride.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

The white precipitate is *insoluble in nitric acid but soluble in ammonium hydroxide* solution, and forms a complex salt called diammine silver (I) chloride.

$$AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]^+ Cl^- + 2H_2O$$

When dilute nitric acid is added to the clear solution of diammine silver (I) chloride, a white precipitate of silver chloride is formed.

$$[Ag(NH_3)_2]^+Cl^-(aq) + \text{Nitric acid} \rightarrow \text{White precipitate of silver chloride}$$

When exposed to light, silver (I) *chloride blackens since it decomposes into chlorine and metallic silver which is liberated as a fine black powder.*

$$2AgCl \rightarrow 2Ag + Cl_2$$

$AgNO_3$ is used as a test for hydrochloric acid and also for chloride ion.

Note : On warming with metallic nitrites, dilute hydrochloric acid gives metallic chlorides and oxides of nitrogen.

$$2KNO_2 + 2HCl \rightarrow 2KCl + H_2O + NO + NO_2$$

7. Oxidation of hydrochloric acid : *Concentrated hydrochloric acid* can readily be oxidised to **chlorine** by strong **oxidising agents such as manganese dioxide, lead dioxide and red lead.** Alternately, we can say that hydrochloric acid readily reduces these compounds.

Oxidising agent	+ Hydrochloric acid	→ Metallic chloride	+ Water	+ Chlorine
MnO_2	+ 4HCl (Conc.) $\xrightarrow{\Delta}$	$MnCl_2$	+ $2H_2O$	+ $Cl_2\uparrow$
PbO_2	+ 4HCl (Conc.) $\xrightarrow{\Delta}$	$PbCl_2$	+ $2H_2O$	+ $Cl_2\uparrow$
Pb_3O_4	+ 8HCl (Conc.) $\xrightarrow{\Delta}$	$3PbCl_2$	+ $4H_2O$	+ $Cl_2\uparrow$
$2KMnO_4$	+ 16HCl (Conc.) ⟶	$2MnCl_2 + 2KCl$	+ $8H_2O$	+ $5Cl_2\uparrow$
$CaOCl_2$ (bleaching powder)	+ 2HCl (dil.) $\xrightarrow{\Delta}$	$CaCl_2$	+ H_2O	+ $Cl_2\uparrow$

8. Formation of aqua regia : A mixture having three parts of conc. hydrochloric acid and one part of conc. nitric acid is called *aqua-regia.*

$$\underset{\text{Conc.}}{3HCl} + \underset{\text{Conc.}}{HNO_3} \rightarrow \underset{\text{Nitrosyl chloride}}{NOCl} + \underset{\text{Nascent chlorine}}{2H_2O + 2[Cl]}$$

It gives nascent chlorine. The nascent chlorine reacts with noble metals like gold and platinum, to give their soluble chlorides.

$$\underset{\text{Gold}}{Au} + 3[Cl] \rightarrow \underset{\text{Gold (III) chloride}}{AuCl_3}$$

$$Pt + 4[Cl] \rightarrow \underset{\text{Platinum (IV) chloride}}{PtCl_4}$$

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8.7 USES OF HYDROCHLORIC ACID

I. General uses

In the laboratory as a reagent and for preparation of aqua regia.

II. Industrial uses :

(1) In the manufacture of
   (a) chlorine and chlorides, *e.g.*, ammonium chloride used in dry cells.
   (b) dyes, drugs, paints and photographic chemicals (silver chloride).
   (c) glucose from starch.

(2) In industry *to pickle* steel* (clean metal surface by using acid), as HCl dissolves the oxides. Steel before being plated with tin or chromium has to be purified by pickling in HCl and some inhibitors.

(3) For purifying bone black, because HCl dissolves the calcium phosphate present in bones.

(4) To remove rust from iron sheets.

(5) For cleaning metal surfaces before painting, electroplating, galvanising, soldering, etc.

(6) In the extraction of glue from bones.

(7) In tanning and calico printing industry.

* Pickling of metals is a process of removal of oxide coating from the surface of metals before they are painted, electroplated or galvanised.

III. Medicine :

Dilute HCl is prescribed to patients with decreased activity of their gastric juices.

HCl acid helps in the digestion of proteins and also in destroying microorganisms that enter the alimentary canal alongwith the food.

8.8 TESTS FOR HYDROGEN CHLORIDE AND HYDROCHLORIC ACID

(1) HCl gas possesses a characteristic irritating smell.

(2) HCl gas gives thick white fumes of ammonium chloride, when a glass rod dipped in ammonia solution is held near the vapours of the acid.

$$NH_3 + HCl \rightarrow NH_4Cl$$

(3) With silver nitrate solution, both the gas and the acid give a *white precipitate of silver chloride.*

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

The precipitate is insoluble in nitric acid but soluble in ammonium hydroxide.

(4) A greenish-yellow gas, *i.e.* chlorine, is liberated when concentrated hydro-chloric acid is heated with oxidising agent like manganese dioxide.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$$

The gas liberated turns moist starch iodide paper blue black.

EXERCISE

1. (a) Draw a labelled diagram for the laboratory preparation of hydrogen chloride gas.
   (b) Give the balanced equation for the reaction.
   (c) Name the drying agent used in drying hydrogen chloride gas.
   (d) Phosphorus pentoxide and calcium oxide are good drying agent but they cannot be used to dry hydrogen chloride gas. Why ?

2. Explain why :
   (a) anhydrous HCl is a poor conductor while aqueous HCl is an excellent conductor.
   (b) when the stopper of a bottle full of hydrogen chloride gas is opened there are fumes in the air.
   (c) a solution of hydrogen chloride in water turns blue litmus red, and conducts electricity, while a solution of the same gas in toluene :
      (i) has no effect on litmus, and
      (ii) does not conduct electricity.
   (d) thick white fumes are formed when a glass rod dipped in NH_4OH is brought near the mouth of a bottle full of HCl gas.
   (e) dry hydrogen chloride gas does not affect a dry strip of blue litmus paper but it turns red in the presence of a drop of water.
   (f) hydrogen chloride gas is not collected over water.

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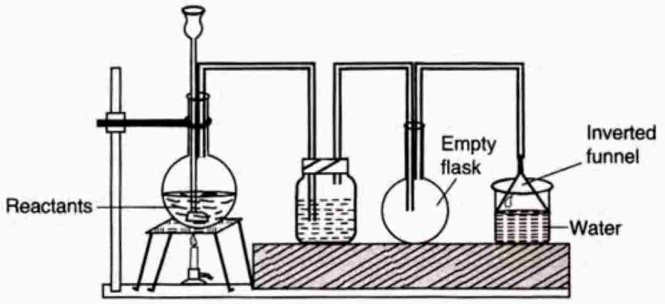
3. The given set up in the figure is for the preparation of an acid.



(a) Name the acid prepared by this method.

(b) Name the reactants used.

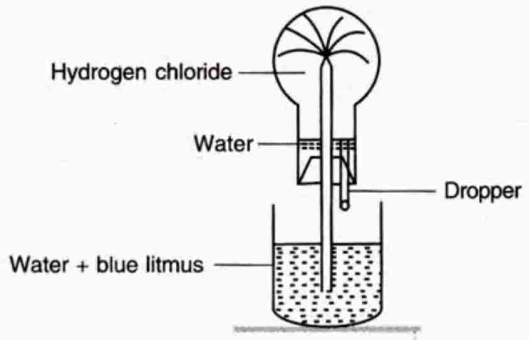
(c) Why empty flask is used.

(d) What is the drying agent used ? Why is this drying agent chosen ?

(e) What is the role of inverted funnel in the arrangement.

4. (a) (i) Name the experiment illustrated below.

(ii) State the colour of the water that has entered the round-bottomed flask.



(b) What property of hydrogen chloride is demonstrated when it is collected by downward delivery (upward displacement) ?

5. (a) Name an element which reacts with hydrogen to form a compound which is strongly acidic in water.

(b) Explain why dilute hydrochloric acid cannot be concentrated by boiling beyond 22.2%.

6. How will you prove that Hydrochloric acid contains (i) hydrogen (ii) chlorine. Write equations for the reactions.

7. Name

(a) black metallic oxide which reacts with hydrochloric acid to give a coloured solution.

(b) two colourless gases, which when mixed produce a white solid.

(c) two gases which chemically combine to form a liquid.

(d) a chloride which is soluble in excess of ammonium hydroxide.

(e) the chemical in which gold can be dissolved.

(f) the experiment which demonstrates that hydrogen chloride is soluble in water.

(g) the gas produced when chlorine water is exposed to sunlight.

8. Solution A reacts with an acid B (which gives greenish yellow gas on reacting with oxidising agents like Pb_3O_4) to give white precipitate C insoluble in nitric acid but soluble in ammonium hydroxide. Name A, B and C.

9. Complete and balance the following reactions, state whether dilute or conc. acid is used.

(a) $NH_4OH + HCl \rightarrow$ (b) $NaHSO_3 + HCl \rightarrow$

(c) $Pb(NO_3)_2 + HCl \rightarrow$ (d) $Pb_3O_4 + HCl \rightarrow$

(e) $Zn + HCl \rightarrow$ (f) $Ca(HCO_3)_2 + HCl \rightarrow$

10. How will the action of dilute hydrochloric acid enable you to distinguish between the following :

(a) Sodium carbonate and sodium sulphite

(b) Sodium thiosulphate and sodium sulphite.

11. Give three distinct tests [apart from using an indicator] you would carry out with solution of HCl to illustrate the typical properties of an acid.

12. MnO_2, PbO_2 and red lead react with conc. HCl acid liberates Cl_2.

What is the common property being shown by these metal oxides ?

13. State which of the two — a solution of HCl in water or in toluene is an electrolyte. Explain.

14. Convert :

(a) Two soluble metallic nitrates to insoluble metallic chlorides using dil. HCl.

(b) Hydrochloric acid to nascent chlorine.

15. A solution of hydrogen chloride in water is prepared. The following substances are added to separate portions of the solution :

S. No.	Substances added	Gas evolved	Odour
1.	Calcium carbonate	.....	.....
2.	Magnesium ribbon	.....	.....
3.	Manganese (IV) oxide with heating	.....	.....
4.	Sodium sulphide	.....	.....

Complete the table by writing the gas evolved in each case and its odour.

16. State the composition of aqua regia. State which component is the oxidising agent in aqua regia.

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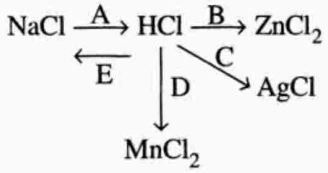
17. Write an equation for the reactions of hydrochloric acid on :

(a) silver nitrate solution, (b) magnesium foil,

(c) caustic soda solution, (d) zinc carbonate,

(e) manganese (IV) oxide, (f) copper oxide.

18. Study the flow chart and give balanced equations with conditions for the conversions A, B, C, D and E.



19. Write balanced equations for the reaction of dilute hydrochloric acid with each of the following :

(a) iron,

(b) sodium hydrogen carbonate,

(c) iron(II) sulphide,

(d) sodium sulphite,

(e) sodium thiosulphate solution.

2010

(a) Aqua regia is a mixture of :

(i) Dilute hydrochloric acid and concentrated nitric acid

(ii) Concentrated hydrochloric acid and dilute nitric acid

(iii) Concentrated hydrochloric acid [1 part] and concentrated nitric acid [3 parts]

(iv) Concentrated hydrochloric acid [3 parts] and concentrated nitric acid [1 part]

(b) How would you distinguish between dilute HCl and dilute HNO_3, by addition of only one solution.

(c) Name two gases which can be used in the study of the fountain experiment. State the common property demonstrated by the fountain experiment.

2011

(a) Choose the correct answer from the choices given :

Hydrogen chloride gas being highly soluble in the water is dried by :

(i) Anhydrous calcium chloride

(ii) Phosphorous pentoxide

(iii) Quick lime

(iv) Conc. Sulphuric acid

(b) Write balanced chemical equation

(i) Sodium thiosulphate is reacted with dilute hydrochloric acid.

(ii) Calcium bicarbonate reacts with dilute hydrochloric acid.

(c) In the laboratory preparation of hydrochloric acid, hydrogen chloride gas is dissolved in water.

(i) Draw a diagram to show the arrangment used for the absorption of HCl gas in water.

(ii) State why such an arrangment is necessary. Give two reasons for the same.

(iii) Write balanced chemical equations for the laboratory preparation of HCl gas when the reaction is :

A. Below 200°C B. Above 200°C

2012

(a) Rewrite the correct statement with the missing word/s :

Aqua regia contains one part by volume of nitric acid and three parts by volume of hydrochloric acid.

(b) Give reason for the following :

Hydrogen chloride gas cannot be dried over quick lime.

(c) Give a balanced equation for the reaction :

Conc. hydrochloric acid and potassium permanganate soluion.

(d) Give balanced equations with conditions, if any, for the following conversion.

(i) Sodium chloride → Hydrogen chloride

(ii) Hydrogen chloride → Iron (II) chloride

(iii) Hydrogen chloride → Ammonium chloride

(iv) Hydrogen chloride → Lead chloride

2013

(a) Identify the gas evolved when :

(i) Potassium sulphite is treated with dilute hydrochloric acid.

(ii) Concentrated hydrochloric acid is made to react with manganese dioxide.

(b) State one appropriate observation for

(i) Copper sulphide is treated with dilute hydrochloric acid.

(ii) A few drops of dil. HCl are added to $AgNO_3$ solution, followed by addition of NH_4OH solution.

2014

(a) Fill in the blank from the choices in the bracket :

Quicklime is not used to dry HCl gas because ..... [CaO is alkaline, CaO is acidic, CaO is neutral].

(b) Write balanced equation for : Action of dilute hydrochloric acid on sodium sulphide.

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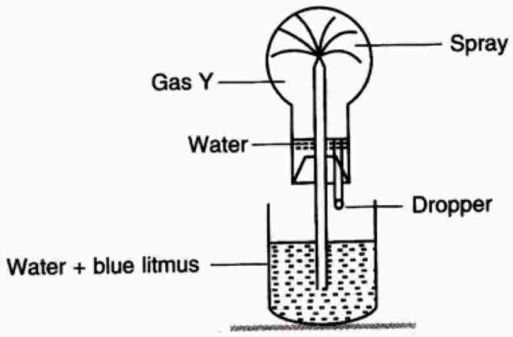
(c) State your observation : Dilute HCl is added to sodium carbonate crystals.

(d) Study the figure given aside and answer the questions that follow :

(i) Identify the gas Y.

(ii) What property of gas Y does this experiment demonstrate.

(iii) Name another gas which has the same property and can be demonstrated through this experiment.



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9 Study of Compounds — Ammonia

SYLLABUS

(i) Ammonia : Its laboratory preparation from ammonium chloride and collection; ammonia from nitrides like Mg_3N_2 and AlN and ammonium salts Manufacture by Haber's Process; density and solubility of ammonia (fountain experiment); aqueous solution of ammonia; its reactions with hydrogen chloride and with hot copper (II) oxide and chlorine; the burning of ammonia in oxygen; uses of ammonia.

Laboratory preparation from ammonium chloride and collection (the preparation can be studied in terms of setting of the apparatus and diagram, procedure, observation, collection and identification).

Manufacture of ammonia on a large scale – reference should be made to Haber Process for the manufacture of ammonia;

Ammonia from nitrides like Mg_3N_2 and AlN and ammonium salts; the reactions can be studied in terms of reactant, product, condition, equation.

Density and solubility of ammonia (fountain experiment); the property can be learnt in terms of setting of the apparatus, procedure and observation and inference.

Aqueous solution of ammonia – reaction with sulphuric acid, nitric acid, hydrochloric acid and solutions of iron (III) chloride, iron (II) sulphate. lead nitrate, zinc nitrate and copper sulphate.

Its reaction with : hydrogen chloride, hot copper (II) oxide, with chlorine in excess and ammonia in excess, burning of ammonia in oxygen; all these reactions may be studied in terms of reactants; products, condition, equation and observation; reference should be made to preparation of nitrogen from air and from ammonium nitrite.

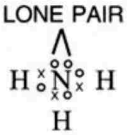
Uses of ammonia – manufacture of fertilizers, explosives, nitric acid, refrigerants gas (Chlorofluro carbon – and its suitable alternatives which are non-ozone depleting), cleansing agents, source of hydrogen.

(ii) The catalytic oxidation of ammonia, as the source of nitric acid (refer to Ostwald process) : simple diagram for a catalytic oxidation of ammonia in the laboratory (with conditions and reactions only).

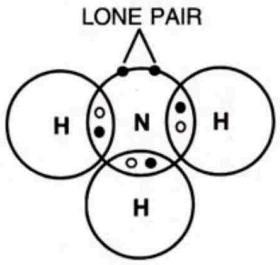
Self-explanatory

Molecular formula : NH_3

Relative molecular mass : 17



Lewis diagram or dot diagram



Covalent bonding in ammonia

9.1 OCCURRENCE

Free state : Ammonia is present in small amounts in air and in traces in natural water.

Combined state : In nature, ammonia occurs in *combined form* in many compounds such as ammonium chloride (Hindi : *Nausadar*), ammonium sulphate, etc.

Ammonia and ammonium compounds being highly soluble in water, do not occur as minerals.

Note : The pungent smell which one encounters near a *decaying heap* of organic matter or a toilet is due to ammonia.

It is produced due to bacterial decomposition of *urea* (NH_2CONH_2), present in the urine.

$$CO(NH_2)_2 + 2H_2O \rightarrow 2NH_3 + H_2O + CO_2$$

Forms of ammonia

Ammonia is used commonly in the following forms :

(i) **Gaseous ammonia** (dry ammonia gas)

(ii) **Liquid ammonia**. When dry ammonia is liquefied by applying high pressure it is called liquid ammonia.

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(iii) **Liquor ammonia fortis** is a saturated solution of ammonia in water. It is called 880 ammonia. It has a relative density of 0·880. It is stored in tightly stoppered bottles in a cold place.

(iv) ***Laboratory bench reagent :*** Dilute solution of liquor ammonia is used as a laboratory reagent.

9.2 PREPARATION OF AMMONIA GAS

9.2.1 General methods of preparation

Ammonia gas can be prepared by warming an ammonium salt with caustic alkali, such as slaked lime or caustic soda or caustic potash.

Ammonium salt +	Alkali $\xrightarrow{\Delta}$	Salt +	Water +	Ammonia
$2NH_4Cl$ (s) +	$Ca(OH)_2$(s) Slaked lime	$CaCl_2$(s) +	$2H_2O$ +	$2NH_3\uparrow$
$(NH_4)_2SO_4$ +	$Ca(OH)_2$	$CaSO_4$ +	$2H_2O$ +	$2NH_3\uparrow$
$(NH_4)_2SO_4$ +	2NaOH Caustic soda	Na_2SO_4 +	$2H_2O$ +	$2NH_3\uparrow$
$(NH_4)_2SO_4$ +	2KOH Caustic potash	K_2SO_4 +	$2H_2O$ +	$2NH_3\uparrow$

9.2.2 Laboratory preparation

(1) *From ammonium chloride*

Reactants : Ammonium chloride (NH_4Cl) and calcium hydroxide [$Ca(OH)_2$] in the ratio of 2 : 3 by weight.

Reaction :

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3.$$

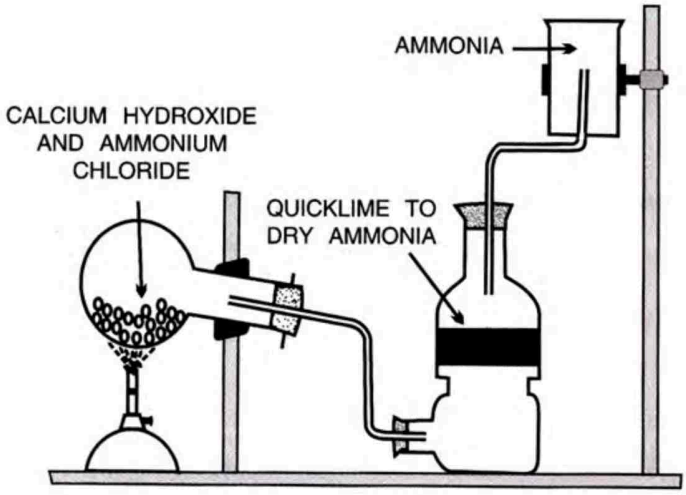

Fig. 9.1 Lab. preparation of dry ammonia

Procedure : Reactants are finely grinded and taken in a round-bottom flask fitted in a slanting position, mouth downwards as shown in Fig. 9.1. On heating the mixture, ammonia gas is evolved.

Precautions : *The flask is tilted in such a way that the water formed in the reaction does not trickle back into the heated flask and thus break it.*

Drying of ammonia gas : In order to get dry ammonia, the gas is passed through a drying tower containing lumps of ***quicklime*** (CaO).

Unsuitability of other drying agents : Other drying agents like conc. sulphuric acid, phosphorous pentoxide and anhydrous calcium chloride are not used, as ammonia being basic, reacts with them.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

$$6NH_3 + P_2O_5 + 3H_2O \rightarrow 2\,(NH_4)_3PO_4$$
ammonium phosphate

$$CaCl_2 + 4NH_3 \rightarrow CaCl_2.4NH_3$$
(addition compound)

Collection :

Ammonia gas is collected in inverted gas jars by the *downward displacement* of air because it is :

(i) lighter than air (V.D. of NH_3, 8·5; that of *air, 14·4*),

(ii) highly soluble in water and therefore, it cannot be collected over water.

Note : 1. The reactants NH_4Cl and $Ca(OH)_2$ are taken in the ratio of 2 : 3 by weight, higher ratio by weight of the alkali may counteract the loss by sublimation of NH_4Cl.

2. Calcium hydroxide is used, as it is cheap and not deliquescent like other alkalies.

3. Though all ammonium salts, on heating with alkalies give NH_3, yet NH_4NO_3 *(explosive in nature)* and NH_4NO_2 on warming with alkalies do not produce ammonia because they decompose on heating.

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

(2) *From metal nitrides*

Ammonia can also be obtained by the action

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of warm water on nitrides of metals like magnesium or Aluminium. (This method is costly)

Reaction :

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3\uparrow$$

or

$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3\uparrow$$

Procedure : Magnesium nitride or Aluminium nitride is kept in a round bottomed flask fitted with a thistle funnel and a delivery tube, as shown in Fig 9.2. Warm water is added very slowly.

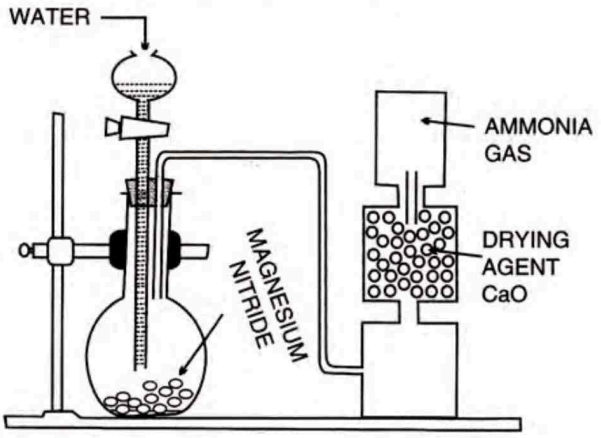

Fig. 9.2 Preparation of ammonia from magnesium nitride

Ammonia gas is liberated and magnesium hydroxide or aluminium hydroxide is also formed, but it remains in the flask and the liberated ammonia gas escapes through the delivery tube.

The ammonia gas formed is collected by holding the jar with its **mouth downwards** over the delivery tube.

The gas can be dried by passing it through **quicklime packed** in a drying tower.

9.3 PREPARATION OF AQUEOUS AMMONIA

An aqueous solution of ammonia is prepared by dissolving ammonia in water. The rate of dissolution of ammonia in water is very high, therefore, **back suction*** of water is possible. To avoid this, a funnel is attached to the outer end of the delivery tube with rubber tubing (the process is similar to the method used to change HCl gas into hydrochloric acid).

* The funnel provides a larger surface area for dissolution of ammonia in water and prevents back suction of water into the hot flask.

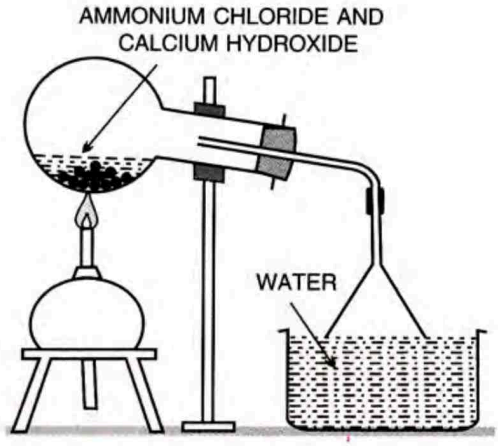

Fig. 9.3 Aqueous solution of ammonia

Procedure : Water is taken in a container and only a small portion of the mouth of the funnel is dipped in water.

As ammonia dissolves in water at a higher rate than its production in the flask, the pressure in the funnel above water level decreases for a moment, and water rushes into the funnel. As a result, the rim of the funnel loses its contact with water. Since, ammonia produced pushes the water down, the funnel comes in contact with water again. In this way, ammonia dissolves in water without back suction of water.

9.4 MANUFACTURE OF AMMONIA (HABER'S PROCESS)

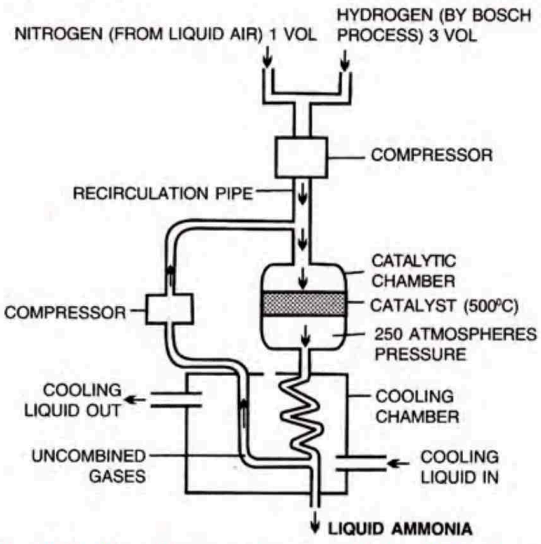

Fig. 9.4. Manufacture of ammonia by Haber's process

Reactants : Nitrogen and hydrogen in the ratio of 1 : 3 by volume.

Sources of reactants :

Nitrogen gas is obtained by fractional distillation of liquid air.

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Hydrogen gas is obtained from water gas (Bosch process) or from natural gas.

Reaction :

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + \text{heat.}$$

The reaction is reversible, exothermic and proceeds with a decrease in volume.

Favourable conditions :

Temperature : Optimum temperature is 450-500 °C.

Pressure : Above 200 atm.

Catalyst : Finely divided iron*.

Promoter : Traces of molybdenum or Al_2O_3.

Rate of conversion : About 15% of the reacting gases get converted to ammonia.

Recovery of ammonia :

Ammonia is separated from the unreacted nitrogen and hydrogen by :

1. *Liquefaction* — Ammonia has a much higher boiling point as compared to N_2 and H_2 so it condenses easily.
2. *Absorbing in water* because NH_3 is highly soluble in water, while N_2 and H_2 are very slightly soluble.

Process : Nitrogen and hydrogen are dried, purified and then mixed in the ratio of 1 : 3 respectively, at a relatively high pressure. This mixture is passed in an electrically heated catalytic chamber containing finely divided iron with aluminium oxide or a little amount of molybdenum at a temperature of 500°C. Mixture of ammonia formed along with residual nitrogen and hydrogen exchanges their heat with the incoming unreacted nitrogen and hydrogen mixture.

The hot mixture of the outgoing gases contains nearly 15% ammonia. The mixture is passed through the cooling pipes (condenser).

Ammonia **liquefies** first, whereas nitrogen and hydrogen do not liquefy easily.

The unchanged nitrogen and hydrogen are recirculated through the plant to get more ammonia. By re-circulating in this way, an eventual yield of 98% can be achieved. The ammonia produced is stored as liquid under pressure.

Note :

(i) The reaction is exothermic, hence low temperature will favour the synthesis. However, at low temperature, the reaction proceeds slowly. In practice, the optimum temperature has been found to be in the range of 450°C to 500°C.

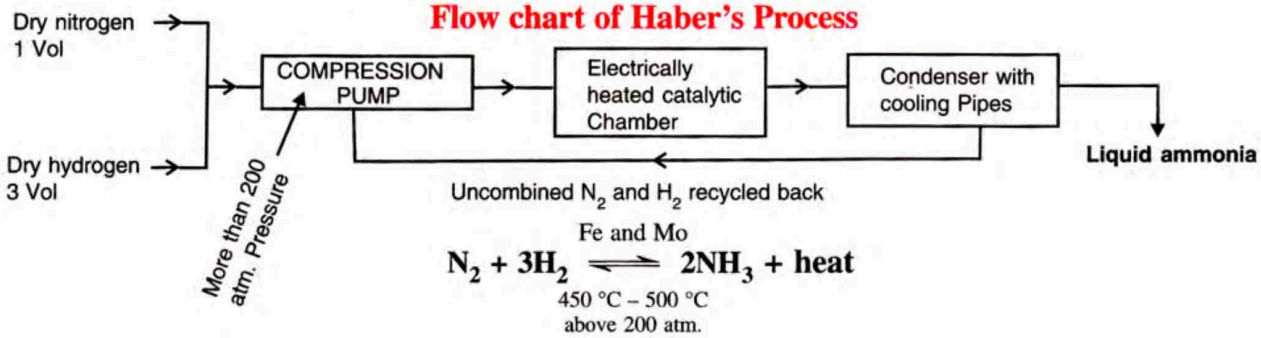
(ii) Four volumes of reactants produce two volumes of product, hence ***high pressure*** favours the formation of ammonia.

The optimum pressure is found to be 200 to 900 atm. Practically, a pressure of about 250 atm. is used.

(iii) Catalyst and promoter : *The speed of the reaction can be improved by using a catalyst, which is finely divided iron,* obtained by the reduction of iron oxide.

A promoter **molybdenum** or Al_2O_3 *is used to increase the efficiency of the catalyst.*

Flow chart of Haber's Process



* Finely divided catalyst has more surface area, this increases the efficiency of the catalyst.

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(iv) Since the reaction is exothermic, the heat evolved further maintains the temperature. External heating is, therefore, not required after the reactants are initially heated.

Note : The purification of nitrogen and hydrogen is necessary as impurities like *carbon dioxide, carbon monoxide and traces of sulphur compounds* (H_2S) *poisons the catalyst.* The removal of these **catalytic poisons** from nitrogen and hydrogen is very essential.

Intext Questions

1. (a) State the type of bonding present in ammonia, show by a diagram.
   (b) What is the formula of liquid ammonia ? Account for the basic nature of this compound.
2. (a) Write a balanced chemical equation for the lab preparation of ammonia.
   (b) How is ammonia dried and collected in the laboratory?
   (c) Ammonia cannot be collected over water. Give reason.
   (d) Explain with a diagram the preparation of aqueous ammonia.
3. Name a drying agent for ammonia. Why are other drying agents such as, P_2O_5 and $CaCl_2$ are not used ?
4. A substance 'A' was heated with slaked lime and a gas 'B' with a pungent smell was obtained. Name the substances A and B and give a balanced equation.
5. Ammonia is manufactured by Haber Process.
   (a) Under what conditions do the reactants combine to form ammonia ? Give a balanced equation for the reaction.
   (b) In what ratio by volume, are the above gases used ?
   (c) State one possible source of each reactant used in Haber Process.
   (d) State whether the formation of ammonia is promoted by the use of high pressure or low pressure?
   (e) Mention two possible ways by which ammonia produced is removed from unchanged gases.
   (f) What is the function of :
   (i) finely divided iron,
   (ii) molybdenum in the above process ?
   (g) What is the percentage formation of ammonia ?
   (h) How can this percentage formation be increased ?
6. Give reasons :
   (a) Ammonium compounds do not occur as minerals.
   (b) Ammonium nitrate is not used in the preparation of ammonia.
   (c) Conc H_2SO_4 is a good drying agent, yet it is not used to dry NH_3.
7. (a) Complete the table.

Name of process	Inputs	Equations	Output
			Ammonia

   (b) State the following conditions required in above process
   (i) Temperature (ii) Pressure (iii) Catalyst
8. Choose the correct word or phrase from the brackets to complete the following sentences and write balanced equations for the same.
   (a) Ammonium chloride is a soluble salt prepared by ..... [precipitation, neutralization].
   (b) When ammonium chloride is heated, it undergoes ..... [thermal decomposition/dissociation].
   (c) Heating ammonium chloride with sodium hydroxide produces ..... [ammonia, nitrogen].
9. Correct the following :
   (a) A reddish brown precipitate is obtained when ammonium hydroxide is added to ferrous sulphate.
   (b) Liquid ammonia is a solution of NH_3.
   (c) Finely divided platinum is used in Haber Process.
   (d) Conc. H_2SO_4 is a drying agent for NH_3.
   (e) Ammonium salts, on heating, decompose to give ammonia.
10. Give reasons for the following :
   (a) Liquid ammonia is used as a refrigerant in ice plants.
   (b) Aqueous solution of ammonia is used for removing grease stains from woollen clothes.
   (c) Aqueous solution of ammonia gives a pungent smell.
   (d) Aqueous solution of ammonia conducts electricity.

9.5 PHYSICAL PROPERTIES OF AMMONIA

1. Colour : Colourless.

2. Odour : Strong, pungent choking smell.

3. Taste : Slightly bitter (alkaline) taste.

4. Physiological action : Non-poisonous.
If inhaled, affects the respiratory system and brings tears to eyes. It is a powerful heart stimulant and may cause death when inhaled in large quantities.

5. Density : Vapour density = 8·5 *[Lighter than air]*

6. Nature : Alkaline nature

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7. Liquefaction : Easily liquefied at 10°C by compressing it at 6 atm. pressure.

8. Boiling point : Liquid ammonia boils at –33·5°C.

9. Freezing point : Solid NH_3 melts at –77·7°C.

10. Solubility : *Highly soluble in water,* 1 vol. of water dissolves about 702 vols. of ammonia at 20°C and 1 atm. pressure.

FOUNTAIN EXPERIMENT

To demonstrate the high solubility of ammonia gas in water.

Apparatus :

1. A round bottomed flask filled with ammonia gas.
2. Mouth of the flask is fitted with a rubber stopper with two holes, one for jet tube and the other for a dropper containing water.
3. A trough containing red litmus solution.
4. A retort stand is used to fix the round bottomed flask in inverted position as shown in Fig. 9.5.

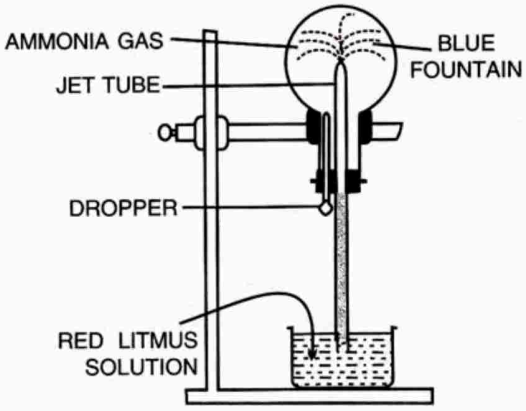


Fig. 9.5. Fountain experiment

Procedure :

1. The dropper containing water is squeezed.
2. As few drops of water enter the flask, ammonia gas present in the flask gets dissolved in water, due to its high solubility. This creates a partial vacuum in the flask.
3. Since, outside pressure is higher, the red litmus solution rushes up through the jet tube and emerges as a blue fountain. [Ammonia gas being basic, changes red litmus solution blue.]

A bottle of liquor ammonia should be opened very carefully, only after cooling it in ice or cold water. There is high pressure inside the bottle of ammonia and on cooling, the pressure drops. This prevents sudden flushing out of the gas.

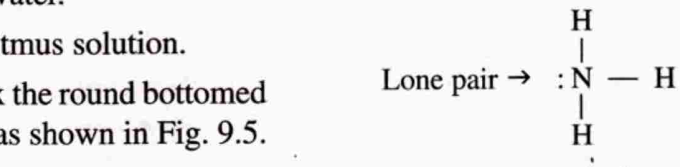
9.6 CHEMICAL PROPERTIES OF AMMONIA

1. ***Thermal dissociation :*** Ammonia gas dissociates into nitrogen and hydrogen at high temperature or by electric sparks.

$$2NH_3 \rightleftharpoons N_2 + 3H_2$$

2. ***Basic properties :*** Dry ammonia is a covalent molecule. So, it is neutral even in liquefied form.

Its aqueous solution is a weak base, *the basicity is due to a lone pair of electrons on* its nitrogen atom.



An aqueous solution of ammonia [NH_4OH] is a weak base. It dissociates partially to give hydroxyl ions.

The alkaline behaviour is due to the hydroxyl ions [OH^-]

$$NH_3 + H_2O \rightleftharpoons NH_4OH$$

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

Action on indicators

Indicator	*Change in colour*	
	Original	*To*
Moist litmus	Red	Blue
Methyl orange	Orange	Yellow
Phenolphthalein	Colourless	Pink

3. ***Reaction with acids :*** Ammonia being a weak base accepts protons [hydrogen ions] to form ammonium ions.

It reacts with acids to form ammonium salts.

Ammonia gas	Acid	Ammonia salt
NH_3 (gas) +	HCl (gas) →	NH_4Cl (solid) Ammonium chloride
NH_3 (gas) +	HNO_3 →	NH_4NO_3 Ammonium nitrate
$2NH_3$ (gas) +	H_2SO_4 →	$(NH_4)_2SO_4$ Ammonium sulphate

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Ammonium chloride is a good example, where *two colourless gases react to form a white solid, directly.*

Note : *Ammonium nitrate and ammonium chloride both show* ***thermal dissociation*** *reaction.*

$$NH_4Cl \rightleftharpoons NH_3 + HCl$$

$$NH_4NO_3 \rightleftharpoons N_2O + 2H_2O$$

The same salts, which are produced by the action of ammonia on acids, are produced if ammonia is first dissolved in water to form ammonium hydroxide.

$$NH_3 + H_2O \rightarrow NH_4OH$$

Ammonia [aq.]		*Acid*		*Ammonium salt*		*Water*
NH_4OH	+	HCl	→	NH_4Cl	+	H_2O
NH_4OH	+	HNO_3	→	NH_4NO_3	+	H_2O
$2NH_4OH$	+	H_2SO_4	→	$(NH_4)_2SO_4$	+	$2H_2O$

4. ***Reaction with soluble metal salts :*** Ammonium hydroxide reacts with soluble salts of metals to give *insoluble precipitates of their respective metallic hydroxides,* (except sodium and potassium).

This behaviour is similar to NaOH solution and KOH solution.

The precipitates formed by different metallic hydroxides differ in *colour* and *solubility.* Hence, ammonium hydroxide is used in qualitative analysis for *identifying cations.*

5. ***Reaction with oxygen :*** Ammonia is not a supporter of combustion. It extinguishes a burning splint and does not burn in air. It burns in oxygen with **yellowish green flame**.

Experiment : Burning of ammonia in oxygen (Fig. 9.6).

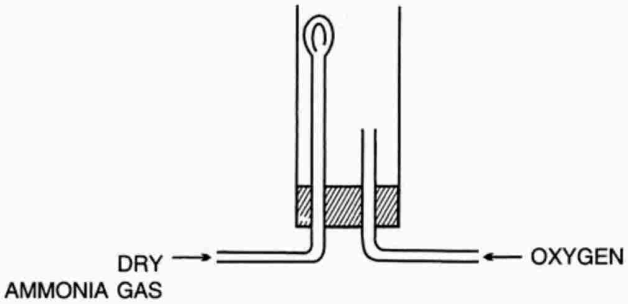

Fig. 9.6 Burning of ammonia in oxygen

Reaction : $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$

Procedure : A wide glass tube is fitted with a cork carrying two tubes, one of which reaches just near the top and the other remains lower. Ammonia is passed through the longer tube, and oxygen through the shorter tube and a lighted match is applied.

Observation : *Ammonia burns with a yellowish green flame* and produces water vapours and nitrogen. It shows that ammonia contains nitrogen and hydrogen.

The reaction is *irreversible and strongly exothermic.*

Mixture of ammonia and oxygen is explosive and therefore dangerous.

Metallic salt solution		*Ammonium hydroxide*		*Ammonium salt*		*Metallic hydroxide [ppt.]*	*Colour of precipitate*	*Solubility in excess of* NH_4OH
$FeSO_4$	+	$2NH_4OH$	→	$(NH_4)_2SO_4$	+	$Fe(OH)_2\downarrow$	Dirty green	Insoluble
$FeCl_3$	+	$3NH_4OH$	→	$3NH_4Cl$	+	$Fe(OH)_3\downarrow$	Reddish brown	Insoluble
$Pb(NO_3)_2$	+	$2NH_4OH$	→	$2NH_4NO_3$	+	$Pb(OH)_2\downarrow$	White	Insoluble
$Zn(NO_3)_2$	+	$2NH_4OH$	→	$2NH_4NO_3$	+	$Zn(OH)_2\downarrow$	White gelatinous	Soluble
$CuSO_4$	+	$2NH_4OH$	→	$(NH_4)_2SO_4$	+	$*Cu(OH)_2\downarrow$	Pale blue	Soluble

* The pale blue precipitate of copper hydroxide dissolves in excess of ammonium hydroxide forming tetraamine copper [II] sulphate, an azure blue (deep blue) soluble complex salt. $Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$

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Catalytic oxidation of ammonia

In the presence of *platinum at 800°C, ammonia reacts with oxygen to give nitric oxide and water vapour.*

Procedure : Pass dry ammonia gas and oxygen gas through inlets over heated platinum placed in the combustion tube, which in the heated state emits a reddish glow.

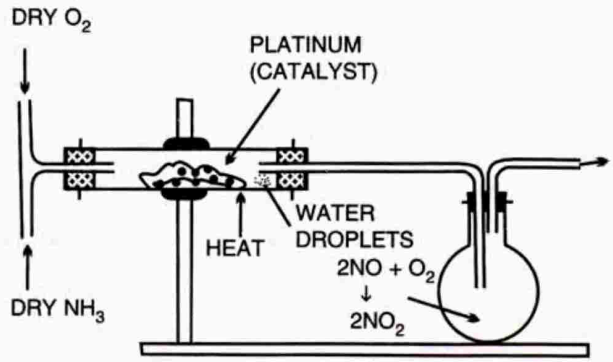

Fig. 9.7 Catalytic oxidation of ammonia

Reaction :

$$4NH_3 + 5O_2 \xrightarrow{Pt,\ 800°C} 6H_2O + 4NO\uparrow + Heat$$

$$2NO + O_2 \longrightarrow 2NO_2 \text{ (brown gas)}$$

Observations :

(i) Reddish brown vapours of *nitrogen dioxide* [NO_2] are seen in the flask due to the oxidation of nitric oxide [NO].

$$2NO + O_2 \rightarrow 2NO_2$$

(ii) The platinum [catalyst] continues to glow even after the heating is discontinued since the catalytic oxidation of ammonia is an exothermic reaction.

6. *Reaction with carbon dioxide :* Ammonia reacts with carbon dioxide at 150°C and 150 atm. pressure to give **urea** which is a valuable nitrogenous fertiliser.

$$2NH_3 + CO_2 \xrightarrow[150\ atm]{150°C} \underset{\text{urea}}{NH_2CONH_2} + H_2O$$

7. *Ammonia as a reducing agent :*

(i) *Reduction of chlorine to hydrogen chloride*

When chlorine reacts with ammonia, it is reduced to HCl, which further combines with excess of ammonia and forms NH_4Cl.

(a) **When *ammonia is in excess*,**

$$2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl$$
$$6NH_3 + 6HCl \rightarrow 6NH_4Cl$$
$$\therefore \quad \underset{\text{(excess)}}{8NH_3} + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$$

Observation :

The yellowish-green colour of chlorine disappears and white fumes of ammonium chloride are formed.

(b) **When chlorine is in excess**, then the products are hydrogen chloride and *yellow coloured highly explosive liquid nitrogen trichloride.*

$$NH_3 + 3Cl_2 \rightarrow 3HCl + NCl_3$$

These reactions show that :

- Chlorine has a great affinity for hydrogen.
- Ammonia is a reducing agent.

(ii) Reduction of metallic oxides

Ammonia reduces heated metallic oxides to give metals, water vapour and nitrogen.

Procedure : Pass ammonia gas over heated copper oxide in a hard glass tube.

Reaction :

$$\underset{\substack{\text{black}\\\text{solid}}}{2NH_3 + 3CuO} \rightarrow \underset{\substack{\text{reddish}\\\text{brown solid}}}{3Cu + 3H_2O} + \underset{\text{gas}}{N_2\uparrow}$$

Observation : *The black copper oxide is reduced to reddish brown copper.*

Similarly, *Ammonia reduces heated yellow lead monoxide to greyish metallic lead.*

$$\underset{\text{(yellow)}}{3PbO} + 2NH_3 \rightarrow \underset{\text{(greyish)}}{3Pb} + 3H_2O + N_2$$

Collection of gas : Nitrogen is collected over water in a gas jar (Fig. 9.8).

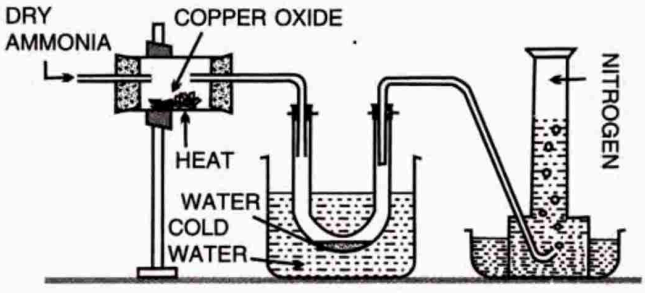

Fig. 9.8 Reducing action of ammonia

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The water formed is collected in the U-tube and is tested with anhydrous copper sulphate which turns blue.

***Note :** This reaction proves that ammonia is made up of nitrogen and hydrogen and is, therefore, a nitrogen hydride.*

9.7 TESTS FOR AMMONIA GAS AND AMMONIUM ION

Ammonia gas can easily be recognised by the following characteristics :

(1) It has a sharp characteristic odour.

(2) It turns
- moist red litmus blue,
- moist turmeric paper brown
- phenolphthalein solution pink.

(3) It gives dense white fumes with conc. hydrochloric acid.

$$NH_3 + HCl \rightarrow NH_4Cl.$$

(4) When NH_3 gas is passed through copper sulphate solution blue precipitate is formed which changes to deep blue solution on passing the gas in excess.

(5) All ammonium salts generally produce ammonia gas when warmed with alkali like sodium hydroxide.

(6) Ammonium salt gives *brown colour* or precipitates with **Nessler's solution** K_2HgI_4 (a slightly alkaline solution of potassium mercuric iodide made by adding KOH).

9.8 USES OF AMMONIA

1. **Liquid ammonia is used as a refrigerant in ice plants.** Liquid ammonia takes a lot of energy to vaporize (5700 calories per mole at –33 °C). This heat is taken from the surrounding bodies which are consequently cooled down.

   Anhydrous ammonia is a clear, colourless liquid under pressure. It evaporates rapidly and produce cooling effect. This makes ammonia a good refrigerant.

   The main refrigerants used are freon chlorofluoro carbon (CFC). They deplete ozone layer and also contribute to global warming. Thus these refrigerants are harmful to life. The chlorofluoro carbons are decomposed by the ultraviolet rays to highly reactive chlorine which is produced in the atomic form.

$$CF_2Cl_2(g) \xrightarrow{\text{Ultraviolet}} CF_2Cl(g) + Cl(g)$$

(free radical)

The free radical [Cl] reacts with ozone and chlorine monoxide is formed.

$$Cl(g) + O_3(g) \rightarrow ClO(g) + O_2(g)$$

This causes depletion of ozone. Chlorine monoxide further reacts with atomic oxygen and produces more free chlorine radicals.

$$ClO(g) + O(g) \rightarrow Cl(g) + O_2(g)$$

Again this free radical [Cl] destroys ozone and the process continues thereby giving rise to **ozone depletion**.

Ozone layer acts as a blanket in the atmosphere above 16 km height from the earths surface. It absorbs the harmful ultraviolet rays coming from the Sun and thus protects life on earth.

Advantages of Ammonia as refrigerant

(i) Ammonia is environmentally compatible. It does not deplete ozone layer and does not contribute the global warming.

(ii) It has superior thermodynamic qualities as a result ammonia refrigeration systems use less electricity.

(iii) Ammonia has a recognizable odour and so leaks are not likely to escape.

Ammonia has some disadvantage.

It is not compatible with copper, so it cannot be used in any system with copper pipes. Ammonia is poisonous in high concentration although it is easily detectable due to its peculiar smell and also being lighter than air goes up in the atmosphere not affecting the life too much on earth.

2. Ammonia solution is used as an important **laboratory reagent in qualitative analysis as it gives characteristic coloured metallic hydroxide precipitates.**

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3. **Aqueous NH_3** emulsifies or dissolves fats, grease, etc. so it is used :
   (i) to remove grease and perspiration stains from clothes,
   (ii) for cleaning tiles, windows, etc.
4. **Ammonia is used in the manufacture of :**
   (a) **nitrogenous fertilisers** like ammonium sulphate, diammonium hydrogen phosphate, ammonium nitrate and urea.
   (b) **explosives** like ammonium nitrate. Ammonium nitrate decomposes explosively leaving no residue behind

$$NH_4NO_3 \xrightarrow{\Delta} N_2O(g) + 2H_2O(vap.)$$

   (c) other ammonium salts such as **ammonium carbonate** (used as smelling salt for reviving a fainted person), **ammonium chloride** – used in dry cells and in the preparation of nitrogen gas.

$$NaNO_2(aq) + NH_4Cl(aq) \rightarrow NaCl(aq) + NH_4NO_2(aq.)$$
Sodium nitrite

$$NH_4NO_2(aq) \xrightarrow{\Delta} N_2(g) + 2H_2O(g)$$
Ammonium nitrite

   (d) nylon and rayon, and sodium cyanamide, plastics, dyes, organic chemicals, explosives and wood pulp.
   (e) sodium carbonate by Solvay process. This process involves the interaction of ammonia, carbon dioxide, and sodium chloride in water solution resulting in the precipitation of sodium hydrogen carbonate.

$$NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3\downarrow + NH_4Cl$$

   The sodium hydrogen carbonate is separated and ignited to obtain sodium carbonate.

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2\uparrow$$

   (f) **nitric acid by the Ostwald process.**

9.9 MANUFACTURE OF NITRIC ACID

A German chemist Ostwald, in 1914, developed **Ostwald process.**

The process is explained in three steps :

Step I : Catalytic oxidation of ammonia

A mixture of dry air (free from carbon dioxide and dust particles) and dry ammonia in the ratio of 10 : 1 by volume, is first compressed and then passed into the **catalytic chamber** containing platinum gauze as catalyst at about 800°C (Fig. 9.9).

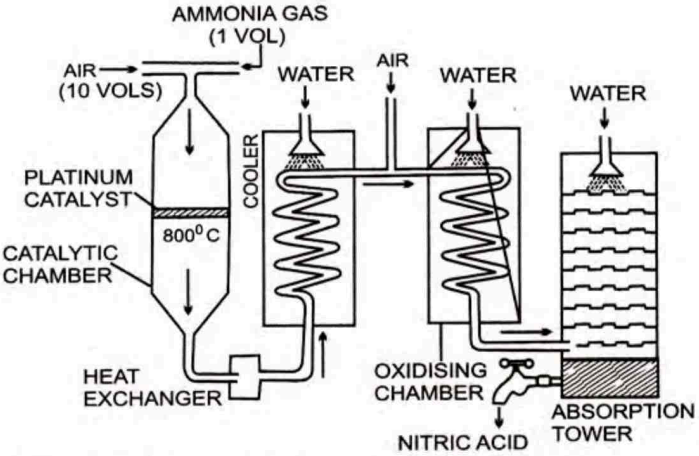

Fig. 9.9 Manufacture of nitric acid by Ostwald process

This results in the oxidation of ammonia into nitrogen monoxide (Nitric oxide).

$$4NH_3 + 5O_2 \xrightarrow[800°C]{Pt} 4NO + 6H_2O + Heat$$

The reaction is **exothermic** and the heat evolved maintains the temperature of the catalytic chamber.

The hot gases are then passed through a heat exchanger, where they are mixed with air and passed into another cooling chamber called **oxidising chamber.**

Step II : Oxidation of nitric oxide in oxidation chamber

In this chamber, nitric oxide combines with oxygen to form nitrogen dioxide at about 50°C.

$$2NO(g) + O_2(g) \xrightarrow{50°C} 2NO_2(g)$$

Step III : Absorption of nitrogen dioxide in water

The nitrogen dioxide together with the remaining air (sometimes more air is added at this stage) is passed through the **absorption tower** (a steel tower filled with quartz stone packed in layers), from the top of which warm water trickles.

The nitrogen dioxide and oxygen present in the air react with water to form nitric acid.

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$4NO_2(g) + 2H_2O(l) + O_2(g) \rightarrow 4HNO_3$ (aq.).

The nitric acid obtained at the bottom of the tower is concentrated above 50%.

The above acid on further distillation gives 68% nitric acid. This acid is **concentrated nitric acid** which is used in laboratories.

Concentration of nitric acid

Nitric acid up to 98% **(fuming nitric acid)** is obtained by distilling 68% HNO_3 over conc. sulphuric acid.

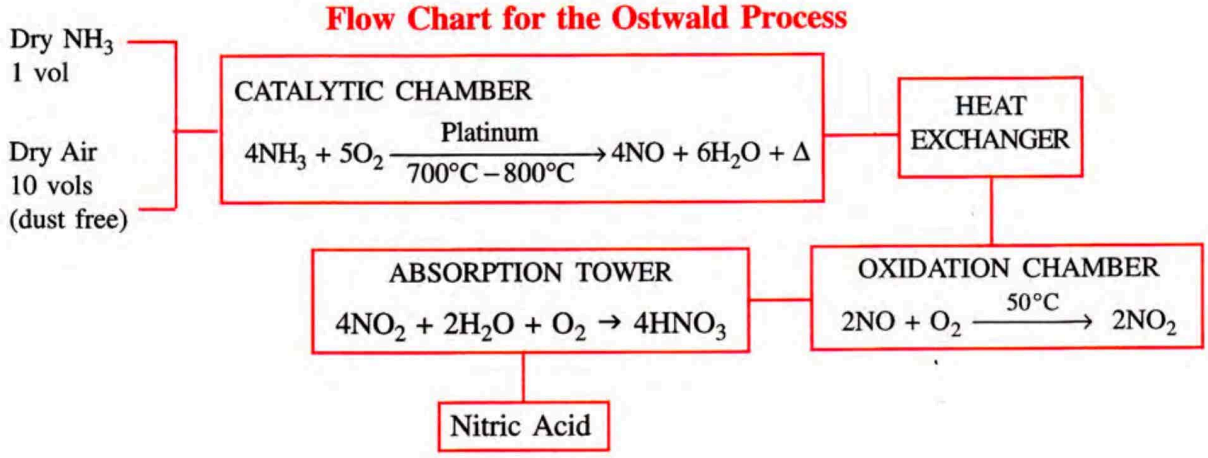
Pure 100% nitric acid is obtained by strongly cooling the 98% acid, when colourless crystals of pure nitric acid get separated at – 42°C.

Note : In the manufacture of nitric acid,

(i) Higher ratio of air is used because oxygen is only 1/5th in air and it is required in all the three chambers.

(ii) Quartz is acid resistant and when packed in layers it helps in dissolving nitrogen dioxide uniformly in water.

Flow Chart for the Ostwald Process



EXERCISE

1. (a) Is ammonia more dense or less dense than air?

(b) What property of ammonia is demonstrated by the Fountain Experiment?

(c) Write the balanced equation for the reaction between ammonia and sulphuric acid.

2. Pick the odd member from the list giving reasons:

(a) Ammonia, sulphur dioxide, hydrogen chloride, carbon dioxide.

(b) Copper oxide, aluminium oxide, sodium oxide, magnesium oxide.

3. The following reactions are carried out :

A : Nitrogen + metal → compound X.

B : X + water → ammonia + another compound.

C : Ammonia + metal oxide → metal + water + N_2.

One metal that can be used for reaction A is magnesium.

(a) Write the formula of the compound X formed.

(b) Write the correctly balanced equation for reaction B where X is the compound formed.

(c) What property of ammonia is demonstrated by reaction C ?

4. A gas 'P' gives dense white fumes with chlorine. Its aqueous solution gives a blue colour with copper (II) hydroxide. (a) Name the gas P. (b) Give its formula. (c) Give three uses of P.

5. Ammonia solution in water gives a blue precipitate when it combines with a solution of copper salt. The blue precipitate further dissolves in excess of ammonia solution to give azure blue solution. Explain with equation.

6. Give chemical equation(s) to prove that NH_3 contains nitrogen and hydrogen ?

7. Copy and complete the following equations.

(a) $AlN + H_2O \rightarrow$ (b) $2NH_3 + 3PbO \rightarrow$

(c) $NH_3 + 3Cl_2 \rightarrow$ (d) $NH_3 + CO_2 \rightarrow$

(i) Which property of ammonia is illustrated by equation (c) ?

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(ii) What important fertiliser is prepared from equation (d) ? State the conditions.

8. What do you observe when ammonium hydroxide is added to the aqueous solution of :

(a) $FeSO_4$, (b) Iron (III) chloride,

(c) Lead nitrate, (d) Zinc nitrate ?

9. Give a chemical test to distinguish between the following:

(a) Ammonium chloride and sodium chloride,

(b) Ferric salt and ferrous salt,

(c) Sodium sulphate and ammonium sulphate.

10. Give balanced equations for the following conversions :

(a) Ammonia to nitrogen using an acidic gas,

(b) Ammonia to brown gas,

(c) Ammonia to nitrogen trichloride,

(d) Ammonia solution to an amphoteric hydroxide,

(e) A nitride of a trivalent metal to ammonia,

(f) Lead oxide to lead.

11. Name :

(a) the gas which is prepared by Haber's Process,

(b) two gases which give dense white fumes with ammonia,

(c) one salt of ammonia in each case which is used in : (i) dry cell (ii) explosives (iii) medicine,

(d) an acidic gas which reacts with a basic gas liberating a neutral gas,

(e) a metallic chloride soluble in ammonium hydroxide,

(f) the gas obtained when ammonia burns in an atmosphere of oxygen without any catalyst,

(g) a nitride of a divalent metal which reacts with warm water liberating ammonia,

(h) an amphoteric oxide reduced by the basic gas,

(i) a white salt produced by an acid gas and a basic gas.

12. When ammonium hydroxide is added to solution B, a pale blue precipitate is formed. This pale blue precipitate dissolves in excess ammonium hydroxide giving an inky blue solution. What is the cation [positive ion] present in solution B? What is the probable colour of solution B.

13. When an ammonium salt is warmed with sodium hydroxide solution, ammonia gas is evolved. State three ways in which you could identify this gas.

14. A gas 'A' reacts with another gas 'B' in the presence of a catalyst to give a colourless gas 'C'. The gas 'C' when comes in contact with air produces a brown gas 'D'. The solution of 'A' in water turns red litmus blue. Explain the observations.

15. (a) Name the common refrigerent. How does it deplete ozone layer ?

(b) What is the alternative of chlorofluoro carbon ?

(c) State the advantages and disadvantages of using ammonia as refrigerent?

16. Name a compound prepared by ammonia and is used as :

(a) Explosive, (b) Fertilizers,

(c) Medicine, (d) Laboratory reagent.

17. Ammonia is used in the Ostwald process,

(a) Give the source of reactants used in this process.

(b) Name the catalyst used in the process.

(c) Name the oxidising agent used in this process.

(d) What is the ratio of ammonia and air taken in this process ?

(e) Why is quartz used in this process ?

18. Write the equation for the action of heat on :

(a) Ammonium chloride, (b) Ammonium nitrate.

State whether each reaction is an example of thermal decomposition or thermal dissociation.

19. (a) Which feature of ammonia molecule leads to the formation of the ammonium ion when ammonia dissolves in water ?

(b) Name the other ion formed when ammonia dissolves in water.

(c) Give one test that can be used to detect the presence of the ion produced in (b).

20. (a) Of the two gases, ammonia and hydrogen chloride, which is more dense ? Name the method of collection of this gas.

(b) Give one example of a reaction between the above two gases which produces a solid compound.

(c) Write a balanced equation for a reaction in which ammonia is oxidized by :

(i) a metal oxide,

(ii) a gas which is not oxygen.

2008

(a) Write equation for the following :

Aluminium nitride and water.

(b) Choose the correct from the following :

Ammonia can be obtained by adding water to

A : Ammonium chloride,

B : Ammonium nitrite,

C : Magnesium nitride,

D : Magnesium nitrate.

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2009

(a) Name the gas evolved in each case [formula is not acceptable]. The gas that burns in oxygen with a green flame.

(b) Write a fully balanced equation for — Magnesium nitride is treated with warm water.

(c) Identify the substance 'O' based on the information given — The white crystalline solid 'Q' is soluble in water. It liberates a pungent smelling gas when heated with sodium hydroxide solution.

2010

(a) Complete the blanks (i) to (v) in the passage given, using the following words. [Ammonium, reddish brown, hydroxyl, nitrogen dioxide, ammonia, dirty green, alkaline, acidic]. In the presence of a catalyst nitrogen and hydrogen combine to give (i) ..... gas. When the same gas is passed through water it foris a solution, which will be (ii) ..... in nature, and will contain the ions (iii) ..... and (iv) ..... A (v) ..... coloured precipitate of iron [II] hydroxide is formed when the above solution is added to iron [II] sulphate solution.

(b) State your observation when — in the absence of catalyst, ammonia is burnt in an atmosphere of oxygen.

(c) Give the equation for the reaction – ammonium chloride is heated with sodium hydroxide.

(d) In the manutacture of ammonia

(i) Name the process

(ii) State the ratio of the reactants take

(iii) State the catalyst used

(iv) Give the equation for the manufacture of the gas – ammonia.

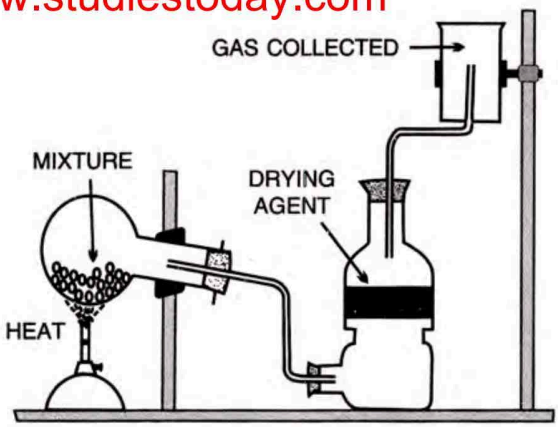
(e) Write a relevant equation to show that ammonia acts as a reducing agent.

(f) Name two gases which can be used to study the fountain experiment. State the common property demonstrated by the fountain experiment.

2011

(a) State what is observed when – Ammonium hydroxide is first added in a small quantity and then in excess to a solution of copper sulphate.

(b) The diagram below shows the set up for the laboratory preparation of a pungent alkaline gas.



(i) Name the gas collected in the jar.

(ii) Give a balanced equation for the above preparation.

(iii) State how the above gas is collected.

(iv) Name the drying agent used.

(v) State how you will find out that the jar is full of the pungent gas.

(c) Write a balanced chemical equation – Chlorine reacts with excess of ammonia.

(d) State your observation – Water is added to the product formed, when Al is burnt in a jar of nitrogen gas.

2012

(a) Name – The gas produced when excess ammonia reacts with chlorine.

(b) Rewrite the correct statement with the missing word/s : Magnesium nitride reacts with water to liberate ammonia.

(c) Give balanced equation for the reaction : Ammonia and oxygen in the presence of a catalyst.

(d) The following question are based on the preparation of ammonia gas in the laboratory :

(i) Explain why ammonium nitrate is not used in the preparation of ammonia.

(ii) Name the compound normally used as a drying agent during the process.

(iii) How is ammonia gas collected. Explain why it is not collected over water.

2013

(a) State one appropriate observation for : Excess of chlorine gas is reacted with ammonia gas.

(b) Nitrogen gas can be obtained by heating :

(i) Ammonium nitrate

(ii) Ammonium nitrite

(iii) Magnesium nitride

(iv) Ammonium chlorde

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(c) State two relevant observations for : Ammonium hydroxide solution is added to zinc nitrate solution slowly and then in excess.

(d) Give a balanced equation for : Reduction of hot copper (II) oxide to copper using ammonia gas.

(e) Copy and complete the following table relating to important industrial process :

Name of the process	Temperature	Catalyst	Equation for the catalyzed reaction
Haber's process			

(f) Identify : *An alkaline gas* which produces dense white fumes when reacted with HCl gas.

2014

(a) Fill in the blank from the choices given in bracket : Ammonia gas is collected by ..... [upward displacement of air, downward displacement of water, downward displacment of air].

(b) Write balanced equation for : Action of warm water on magnesium nitride.

(c) Distinguish between the following pairs of compounds using the test given in bracket :

(i) Iron [II] sulphate and iron [III] sulphate (using ammonium hydroxide).

(ii) A lead salt and a zinc salt (using excess ammonium hydroxide).

(d) State your observation : Calcium hydroxide is heated with ammonium chloride crystals.

(e) Name the other ion formed when ammonia dissolves in water. Give one test that can be used to detect the presence of the ion produced.

(f) State the conditions required for : Catalytic oxidation of ammonia to nitric oxide.

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10 Study of Compounds — Nitric Acid

SYLLABUS

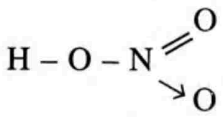
Nitric Acid : One laboratory method of preparation of nitric acid from potassium nitrate or sodium nitrate; Nitric acid as an oxidising agent.

Nitric Acid : Laboratory method of preparation of nitric acid from potassium nitrate or sodium nitrate; the laboratory method can be studied in terms of reactant, product, condition, equation, setting, diagram, precaution, collection, identification.

As an oxidising agent : its reaction with copper, carbon, sulphur.

Molecular formula : HNO_3

Relative molecular mass : 63



Structure of nitric acid

(Nitrogen shows valency of 5)

10.1 INTRODUCTION

The acid was formerly known as **aqua fortis** *meaning strong water. It is so called because it reacts with nearly all metals. It can even dissolve silver which does not dissolve in other acids.*

10.2 DISCOVERY

Glauber, in 1658, obtained nitric acid by distilling nitre, (potassium nitrate, KNO_3) with sulphuric acid.

Lavoisier, in 1776, proved that nitric acid contains oxygen.

Cavendish, in 1784, proved that it also contains hydrogen and nitrogen in addition to oxygen.

10.3 OCCURRENCE

In the ***free*** state, nitric acid is found in rain water, where it occurs in traces after lightning.

In the **combined state**, it is found in the form of metallic nitrates such as Chile saltpetre ($NaNO_3$), nitre (KNO_3), or calcium nitrate [$Ca(NO_3)_2$].

Formation of nitric acid in atmosphere

1. During *lightning discharge,* the nitrogen present in the atmosphere reacts with the oxygen to form nitric oxide.

$$N_2 + O_2 \rightleftharpoons 2NO$$

2. Nitric oxide is further *oxidised* to nitrogen dioxide.

$$2NO + O_2 \rightleftharpoons 2NO_2$$

3. This nitrogen dioxide *dissolves in atmospheric moisture* or *rain water* in the presence of *oxygen* of the air and forms nitric acid in the ***free state***, which is washed down by the rain and combines with the salt present on the surface of the earth.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

The conversion of free atmospheric nitrogen into useful nitrogenous compounds in the soil is known as ***fixation of atmospheric nitrogen.***

10.4 LABORATORY PREPARATION OF NITRIC ACID

Reactants : It is obtained by distilling conc. sulphuric acid with nitrates of potassium KNO_3 (nitre) or sodium $NaNO_3$ (Chile saltpetre).

Procedure : A mixture of equal parts, by weight, of potassium/sodium nitrate and concentrated sulphuric acid is heated gently to 180°C-200°C in a glass retort (Fig 10.1). Sulphuric acid is a non-volatile acid and produces volatile nitric acid on reacting with potassium nitrate or sodium nitrate.

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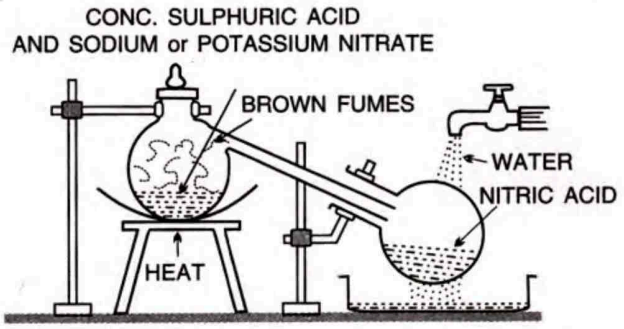

Fig. 10.1 Laboratory preparation of nitric acid

Reactions :

Nitre/Chile saltpetre	+Sulphuric acid	→	Potassium/ Sodium bisulphate	+ Nitric acid
KNO_3	+ H_2SO_4 conc.	$\xrightarrow{<200°C}$	$KHSO_4$	+ HNO_3
$NaNO_3$	+ H_2SO_4 conc.	$\xrightarrow{<200°C}$	$NaHSO_4$	+ HNO_3

Collection : *The vapours of nitric acid are condensed to a* ***light yellow liquid*** *by* chilling the receiver with running cold water.

Note : *Pure acid is colourless but the acid obtained in the laboratory is slightly yellow. The yellow colour is due to dissolution of reddish brown coloured* ***nitrogen dioxide gas*** *in the acid.* This gas is produced due to the thermal decomposition of a portion of nitric acid.

$$4HNO_3 \rightarrow 2H_2O + 4NO_2 + O_2$$

The yellow colour of the acid is removed :

(i) If *dry air* or CO_2 is bubbled through the yellow acid, the latter turns colourless because it drives out NO_2 from warm acid which is further oxidised to nitric acid.

(ii) By addition of excess of water, nitrogen dioxide gas dissolves in water and thus the yellow colour of the acid is removed. The reaction is the reverse of the decomposition of nitric acid.

Precautions :

(1) All glass apparatus is used because nitric acid vapours *attack rubber and cork.*

(2) Conc. HCl is not used in place of conc. H_2SO_4 because HCl *is volatile and hence nitric acid vapours will carry* HCl *vapours.*

(3) *The temperature of the reaction should not exceed* 200°C, because sodium sulphate formed at higher temperature forms a hard crust which sticks to the walls of the retort and is difficult to remove, although the yield of HNO_3 is higher.

$$NaNO_3 + NaHSO_4 \xrightarrow{> 200\ °C} Na_2SO_4 + HNO_3$$

The higher temperature

(i) may damage the glass apparatus,

(ii) decomposition of nitric acid can also occur,

(iii) Wastage of fuel.

10.5 PROPERTIES OF NITRIC ACID

(A) Physical properties

(1) ***Colour :*** It is a colourless liquid (98% concentration) but the commercial nitric acid (68% concentration) is yellowish *brown in colour.*

(2) ***Odour :*** Suffocating smell.

(3) ***Taste :*** Acidic (sour taste).

(4) ***Nature :*** *Hygroscopic and fumes in air.* Hence, the bottle containing nitric acid should always be stoppered.

(5) ***Density :*** 1·54 g/cm^3 (98%) but the commercial nitric acid (68%) has lower density 1.42 g/cm^3.

(6) ***Boiling point and melting point :*** It boils at 86°C and melts at –42°C.

An aqueous solution of nitric acid (68% concentration) forms a **constant boiling mixture** at 121°C.

(7) ***Solubility :*** Soluble in water in all proportions.

(8) ***Physiological action :*** Non-poisonous. It has a corrosive action on the skin and causes painful blisters. It stains the skin yellow as it reacts with protein of the skin and forms **xanthoproteic acid**.

(B) Chemical properties

(1) ***Stability :*** Pure nitric acid is unstable to heat or sunlight. It decomposes to give yellow solution due to the formation of nitrogen dioxide. Thus, 100% nitric acid is not generally used.

$$4HNO_3 \xrightarrow[\text{Sunlight}]{\Delta} 4NO_2 + 2H_2O + O_2$$

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In the presence of sunlight, nitric acid decomposes even at room temperature. *The nitric acid stored in a bottle turns yellow. This colour is due to dissolved NO_2 in HNO_3. To avoid the decomposition, nitric acid is normally stored in coloured bottles.*

(2) *Acidic properties :* Nitric acid is a very strong **monobasic acid.**

HNO_3 ionises almost completely in aqueous solution to produce hydrogen ions and nitrate ions.

$$HNO_3(aq) \rightleftharpoons H^+ + NO_3^-.$$

(i) ***It turns :***

blue litmus red,

methyl orange pink,

phenolphthalein remains colourless.

(ii) ***Reaction with alkalis :***

*It neutralises **alkalis** to form salt and water.*

All metallic oxides and hydroxides react with dilute nitric acid to form their respective soluble metallic nitrates and water only.

Base	+	Acid (dil)	→	Salt	+	Water
K_2O	+	$2HNO_3$	→	$2KNO_3$	+	H_2O
CuO	+	$2HNO_3$	→	$Cu(NO_3)_2$	+	H_2O
ZnO	+	$2HNO_3$	→	$Zn(NO_3)_2$	+	H_2O
$NaOH$	+	HNO_3	→	$NaNO_3$	+	H_2O
$Fe(OH)_3$	+	$3HNO_3$	→	$Fe(NO_3)_3$	+	$3H_2O$
$Mg(OH)_2$	+	$2HNO_3$	→	$Mg(NO_3)_2$	+	$2H_2O$

(iii) ***Reaction with carbonates and bicarbonates :***

It reacts with carbonates and bicarbonates to give salt, water and carbon dioxide.

Carbonate/ Bicarbonate	+	Acid	→	Salt	+	Water	+	Carbon dioxide
$CaCO_3$	+	$2HNO_3$	→	$Ca(NO_3)_2$	+	H_2O	+	$CO_2\uparrow$
$NaHCO_3$	+	HNO_3	→	$NaNO_3$	+	H_2O	+	$CO_2\uparrow$
Na_2CO_3	+	$2HNO_3$	→	$2NaNO_3$	+	H_2O	+	$CO_2\uparrow$
$CuCO_3$	+	$2HNO_3$	→	$Cu(NO_3)_2$	+	H_2O	+	$CO_2\uparrow$

(iv) ***Action with metallic sulphites and bisulphites :***

Metallic sulphites and bisulphites react with dilute nitric acid to form their soluble metallic nitrates, water and sulphur dioxide gas.

Metallic sulphite	+	Nitric acid (dil)	→	Metallic nitrate	+	Water	+	SO_2 (g)
K_2SO_3	+	$2HNO_3$	→	$2KNO_3$	+	H_2O	+	SO_2 (g)
$CuSO_3$	+	$2HNO_3$	→	$Cu(NO_3)_2$	+	H_2O	+	SO_2 (g)
$Ca(HSO_3)_2$	+	$2HNO_3$	→	$Ca(NO_3)_2$	+	$2H_2O$	+	$2SO_2$ (g)

(3) *Oxidising properties :* Nitric acid vigorously oxidises non-metals, metals, inorganic compounds and organic substances.

Nitric acid itself undergoes reduction to form different reduction products like NO, NO_2, N_2O, etc., These products depend on the acid concentration and the temperature of the reaction.

Oxidising properties are due to nascent oxygen which it gives on decomposition.

$$2HNO_3 \text{ (conc.)} \rightarrow 2NO_2 + H_2O + [O].$$

$$2HNO_3 \text{ (dil.)} \rightarrow 2NO + H_2O + 3[O].$$

Nitric acid is a powerful oxidising agent and the nascent oxygen formed on decomposition oxidises hydrogen to water.

(i) ***Action on non-metals***

Non-metal	*Acid [conc.]*	→	*Oxidised product*	*Water*	*Nitrogen dioxide*
a. Carbon					
C	+ $4HNO_3$	→	CO_2	+ $2H_2O$	+ $4NO_2$
b. Sulphur					
S	+ $6HNO_3$	→	H_2SO_4	+ $2H_2O$	+ $6NO_2$
c. Phosphorus					
P_4	+ $20HNO_3$	→	$4H_3PO_4$	+ $4H_2O$	+ $20NO_2$

(ii) ***Action on metals***

Nitric acid reacts with all metals except gold and platinum. The action of nitric acid on metals depends on the temperature and concentration of nitric acid.

(a) Cold and dilute nitric acid oxidises metals to their nitrates and liberates nitric oxide.

Metal	*Oxidised product*
1. Copper	$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$
2. Zinc	$3Zn + 8HNO_3 \rightarrow 3Zn(NO_3)_2 + 4H_2O + 2NO$
3. Iron	$3Fe + 8HNO_3 \rightarrow 3Fe(NO_3)_2 + 4H_2O + 2NO$

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(b) Concentrated nitric acid [or hot dilute nitric acid] – liberates nitrogen dioxide.

Metal	*Oxidised product*
1. Copper	
Cu + $4HNO_3$	→ $Cu(NO_3)_2$ + $2H_2O$ + $2NO_2$
2. Zinc	
Zn + $4HNO_3$	→ $Zn(NO_3)_2$ + $2H_2O$ + $2NO_2$
3. Iron	
Fe + $6HNO_3$	→ $Fe(NO_3)_3$ + $3H_2O$ + $3NO_2$

(c) *Metals like iron, aluminium, cobalt and nickel become passive (inert) when treated with pure concentrated nitric acid.*

It is due to the formation of extremely thin layer of insoluble metallic oxide (passivity) which stops the reaction.

***Passivity** can be removed, by rubbing the surface layer with sand paper, or by treating with strong reducing agents.*

***Note :** Dilute Nitric acid is generally considered a typical acid except for its reaction with metals since it does not liberate hydrogen. It is a powerful oxidising agent and the nascent oxygen formed oxidises the hydrogen to water.*

(d) Very dilute (about 1%) **acid** reacts with magnesium and manganese at room temperature to give their nitrates and hydrogen gas. Since the oxidising action of the acid is much reduced due to dilution.

$$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$$
(V. dilute)

$$Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2$$

These reactions prove that nitric acid contains hydrogen.

(4) ***Reaction as aqua regia :*** Conc. nitric acid (1 part by volume) when mixed with conc. hydrochloric acid (3 parts by volume) gives a mixture called **aqua regia** (meaning **royal water**).

$$HNO_3 + 3HCl \rightarrow NOCl + 2H_2O + 2[Cl]$$
(Nitrosyl chloride)

Nitric acid oxidises hydrochloric acid to chlorine.

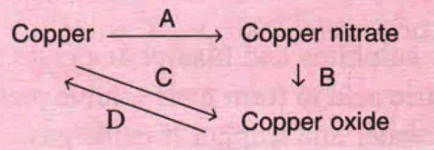
This mixture (aqua regia) reacts even with noble metals like gold and platinum to give their chlorides. Aqua regia contains nascent chlorine which attack these metals.

$$Pt + 4[Cl] \rightarrow PtCl_4 \text{ (soluble)}$$
Platinum — Platinum (IV) chloride

$$Au + 3[Cl] \rightarrow AuCl_3 \text{ (soluble)}$$
Gold — Gold (III) Chloride

Intext Questions

1. What is : (a) aqua fortis, (b) aqua regia (c) Fixation of Nitrogen ?
2. During thunderstorm, rain water contains nitric acid. Explain with reactions.
3. (a) Write a balanced chemical equation for the laboratory preparation of nitric acid.
   (b) In the preparation of nitric acid from KNO_3, concentrated hydrochloric acid is not used in place of concentrated sulphuric acid. Explain why ?
   (c) Conc. nitric acid prepared in laboratory is yellow in colour. Why ? How is this colour removed ?
   (d) Give reasons for the following :
   In the laboratory preparation of nitric acid, the mixture of concentrated sulphuric acid and sodium nitrate should not be heated very strongly above 200°C.
4. Nitric acid cannot be concentrated beyond 68% by the distillation of a dilute solution of HNO_3. State the reason.
5. What is passive iron ? How is passivity removed?
6. Name the products formed when :
   (a) carbon and conc. nitric acid is heated,
   (b) dilute HNO_3 is added to copper.
7. Give two chemical equations for each of the following :
   (a) Reactions of nitric acid with non-metals.
   (b) Nitric acid showing as acidic character.
   (c) Nitric acid acting as oxidising agent.
8. Write balanced equations and name the products formed when :
   (a) sodium hydrogen carbonate is added to nitric acid,
   (b) cupric oxide reacts with nitric acid,
   (c) zinc reacts with dilute nitric acid,
   (d) concentrated nitric acid is heated.
9. Write equation for the following conversions A, B, C and D.



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10. How will you prepare the following from nitric acid ?
    (a) Sodium nitrate (b) Copper nitrate
    (c) Lead nitrate (d) Magnesium nitrate
    (e) Ferric nitrate (f) Aqua regia
11. Correct the following, if required :
    (a) HNO_3 is a strong reducing agent.
    (b) $NaNO_3$ gives NO_2 and O_2 on heating.
    (c) Constant boiling nitric acid contains 80% nitric acid by weight.
    (d) Nitric acid remains colourless even when exposed to light.

10.6 USES OF NITRIC ACID

(i) *To etch designs on copper and brassware.*

Reason : Nitric acid acts as a *solvent* for a large number of metals except noble metals.

(ii) *To purify gold.*

Reason : Gold may contain Cu, Ag, Zn, Pb, *etc.*, as *impurities* which dissolve in nitric acid.

(iii) *It acts as a rocket fuel oxidant.*

(iv) *In preparation of fertilizers.*

Calcium nitrate $Ca(NO_3)_2$,
Ammonium nitrate NH_4NO_3,
Nitro chalk [$NH_4NO_3 + CaCO_3$] and
Basic calcium nitrate $CaO.Ca(NO_3)_2$.

(v) *In the preparation of aqua regia which dissolves noble metals.*

Industrial uses : In the manufacture of :

I. ***Explosives like T.N.T.***

II. ***Synthetic fibres like*** artificial silk, nylon, celluloid, plastics, photographic film, *etc.*

III. ***Important compounds like*** nitrates of potassium, ammonium, silver, *etc.*

IV. ***Dyes/Drugs/Perfumes.***

10.7 TESTS FOR NITRIC ACID AND NITRATES

1. *Conc. nitric acid gives brown fumes on heating.*

$$4HNO_3 \xrightarrow{\Delta} 2H_2O + 4NO_2 + O_2$$
(Brown fumes)

2. *Nitrates* (other than potassium, sodium and ammonium) produce ***reddish brown fumes of nitrogen dioxide*** **[NO_2].**

$$\text{Metallic nitrate} \xrightarrow{\Delta} \text{Metallic oxide} + NO_2 + O_2$$

3. On adding copper to HNO_3 or acidified nitrates

Dense reddish brown fumes of nitrogen dioxide **[NO_2]** ***are evolved.***

$$Cu + 4HNO_3 \xrightarrow{\Delta} Cu(NO_3)_2 + 2H_2O + 2NO_2$$

$$Cu + 4NaNO_3 + 4H_2SO_4 \xrightarrow{\Delta} 4NaHSO_4 + Cu(NO_3)_2 + 2H_2O + 2NO_2$$

4. Brown Ring Test

Procedure : To the aq. solution of a nitrate or nitric acid:

(i) Add freshly prepared saturated solution of iron [II] sulphate.

(ii) Now add conc. sulphuric acid carefully from the sides of the test tube, so that it should not fall dropwise in the test tube.

(iii) Cool the test tube in water.

(iv) *A brown ring appears at the junction of the two liquids.*

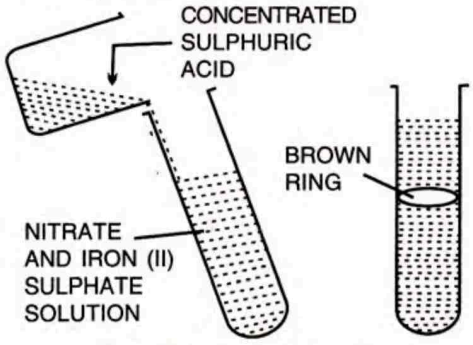

Fig. 10.2 Brown ring test

Reaction :

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$

$$FeSO_4 + NO \rightarrow FeSO_4{\cdot}NO$$
[Nitroso ferrous sulphate, a brown compound]

Note : (i) **A freshly prepared ferrous sulphate solution is used, because** on exposure to the atmosphere, it is oxidised to ferric sulphate which will not give the brown ring.

(ii) **The brown ring of nitroso ferrous sulphate is formed at the junction of the two liquids.** The conc. sulphuric acid being heavier settles down and the ferrous sulphate layer remains above it resulting in the formation of brown ring at the junction.

(iii) **The brown ring of nitroso ferrous sulphate decomposes on disturbing the test tube.** The heat evolved decomposes the unstable brown ring.

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10.8 EFFECTS OF HEAT ON NITRATES

(i) *Sodium and potassium nitrates or alkali metal nitrates.*

When heated, they melt into colourless liquids which decomposes on heating to give oxygen gas.

$$2NaNO_3 \xrightarrow{\Delta} \underset{\text{slightly deliquescent}}{2NaNO_2} + O_2\uparrow$$

$$2KNO_3 \xrightarrow{\Delta} \underset{\text{colourless crystalline}}{2KNO_2} + O_2\uparrow$$

When a glowing splinter is brought near the molten liquid, it bursts into flames because oxygen is formed.

(ii) *All other nitrates* except those of silver and mercury decompose into their oxides, nitrogen dioxide and oxygen.

1. Calcium nitrate

$$\underset{\text{Colourless, crystalline}}{2Ca(NO_3)_2} \xrightarrow{\Delta} \underset{\text{White}}{2CaO} + \underset{\text{reddish brown gas}}{4NO_2\,(g)} + O_2$$

2. Zinc nitrate (slightly decrepitates)

$$\underset{\text{Very deliquescent}}{2Zn(NO_3)_2} \xrightarrow{\Delta} \underset{\text{Yellow, when hot white, when cold}}{2ZnO} + 4NO_2 + O_2$$

3. Lead nitrate (decrepitates on heating)

$$\underset{\text{Colourless, crystalline}}{2Pb(NO_3)_2} \xrightarrow{\Delta} \underset{\text{Yellow solid fuses with glass}}{2PbO} + 4NO_2 + O_2$$

4. Copper nitrate

$$\underset{\text{Blue, crystalline}}{2Cu(NO_3)_2} \xrightarrow{\Delta} \underset{\text{Black}}{2CuO} + 4NO_2 + O_2$$

(iii) *Silver and mercury nitrates :* They are white corrosive salts, which decompose into their respective metals, nitrogen dioxide and oxygen.

1. Silver nitrate (Lunar caustic)

$$\underset{\text{White corrosive}}{2AgNO_3} \xrightarrow{\Delta} \underset{\text{Silver grey}}{2Ag} + 2NO_2 + O_2$$

2. Mercuric nitrate

$$\underset{\text{White, crystalline}}{Hg(NO_3)_2} \xrightarrow{\Delta} \underset{\text{Silver mirror}}{Hg} + 2NO_2 + O_2$$

(iv) *Ammonium nitrate :* Ammonium nitrate decomposes explosively leaving behind no residue.

$$\underset{\text{Colourless, crystalline}}{NH_4NO_3} \xrightarrow{\Delta} \underset{\text{[nitrous oxide] Laughing gas}}{N_2O[g]} + 2H_2O\ [vap.]$$

EXERCISE

1. Choose the correct answer :

(a) The nitrate salt which does not give a mixture of NO_2 and O_2 on heating is :

(i) $AgNO_3$, (ii) KNO_3,
(iii) $Cu(NO_3)_2$, (iv) $Zn(NO_3)_2$.

(b) The chemical used in the brown ring test is :

(i) $CuSO_4$, (ii) $FeSO_4$,
(iii) $Fe_2(SO_4)_3$, (iv) $ZnSO_4$.

(c) Lead nitrate decomposes on heating to give :

(i) NO, (ii) N_2O,
(iii) NO_2, (iv) N_2O_5.

2. Name :

(a) a nitrate of metal which on heating does not give nitrogen dioxide.
(b) a nitrate which on heating leaves no residue behind.
(c) a metal nitrate which on heating is changed into metal oxide.
(d) a metal nitrate which on heating is changed into metal.
(e) a solution which absorbs nitric oxide.
(f) the oxide of nitrogen which turns brown on exposure to air. How is it prepared ?

3. Mention three important uses of nitric acid. Give the property of nitric acid involved in the use.

4. (a) Explain with the help of a balanced equation, the brown ring test for nitric acid.

(b) Why is freshly prepared ferrous sulphate solution used for testing the nitrate radical in the brown ring test ?

5. From the following list of substances, choose one substance in each case which matches the description given below :

Ammonium nitrate, calcium hydrogen carbonate, copper carbonate, lead nitrate, potassium nitrate, sodium carbonate, sodium hydrogen carbonate, zinc carbonate.

(a) A nitrate which gives off only oxygen when heated.
(b) A nitrate which on heating decomposes into dinitrogen oxide [nitrous oxide] and steam.
(c) A nitrate which gives off oxygen and nitrogen dioxide when heated.

6. The action of heat on the blue crystalline solid X, gives a reddish brown gas Y, a gas which re-lights a glowing splint and leaves a black residue. When gas Z, which has

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a rotten egg smell, is passed through a solution of X, a black ppt. is formed.

(a) Identify X, Y and Z.

(b) Write equation for action of heat on X.

(c) Write equation between solution of X and gas Z.

7. X, Y and Z are three crystalline solids which are soluble in water and have a common anion.

To help you to identify X, Y and Z, you are provided with the following experimental observations. Copy and complete the corresponding inferences in (a) to (e).

(a) A reddish-brown gas is obtained when X, Y and Z are separately warmed with concentrated sulphuric acid and copper turning added to the mixture.

INFERENCE 1 : The common anion is the ..... ion.

(b) When X is heated, it melts and gives off only one gas which re-lights a glowing splint.

INFERENCE 2 : The cation in X is either ..... or .....

(c) The action of heat on Y produces a reddish-brown gas and a yellow residue which fuses with the glass of the test tube.

INFERENCE 3 : The metal ion present in Y is the ..... ion.

(d) When Z is heated, it leaves no residue. Warming Z with sodium hydroxide solution liberates a gas which turns moist red litmus paper blue.

INFERENCE 4 : Z contains the ..... cation.

(e) Write the equations for the following reactions:

(1) X and concentrated sulphuric acid (below 200° C). (One equation only for either of the cations given in INFERENCE 2).

(2) Action of heat on Y.

(3) Concentrated nitric acid is added to copper turnings kept in a beaker.

8. (a) Dilute nitric acid is generally considered a typical acid except for its reaction with metals. In what way is dilute nitric acid different from other acids when it reacts with metals?

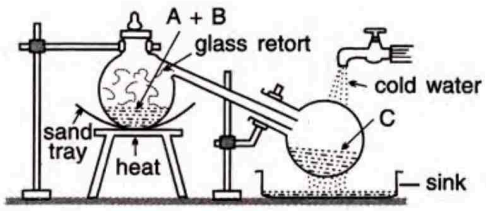
(b) Write the equation for the reaction of dilute nitric acid and conc. nitric acid with copper.

9. Explain why

(a) *Only all-glass apparatus should be used for the preparation of nitric acid by heating concentrated sulphuric acid and potassium nitrate.*

(b) Nitric acid is kept in a reagent bottle for a long time.

10. The figure given below illustrates the apparatus used in the laboratory preparation of nitric acid.



(a) Name A (a liquid), B (a solid) and C (a liquid). (Do not give the formulae).

(b) Write an equation to show how nitric acid undergoes decomposition.

(c) Write the equation for the reaction in which copper is oxidized by concentrated nitric acid.

11. (a) A dilute acid B does not normally give hydrogen when reacted with metals but does give a gas when reacts with copper. Identify B. Write equation with copper.

(b) Complete the table :

Name of Process	Inputs	Equation	Output
	Ammonia + Air		Nitric acid

(c) What is the property of nitric acid which allows it to react with copper.

2012

(a) Name – the gas produced when copper reacts with conc. HNO_3.

(b) State observation – zinc nitrate crystals are strongly heated.

(c) Correct the statement :

Magnesium reacts with nitric acid to liberate hydrogen gas.

(d) Iron is rendered passive with fuming HNO_3. Give reason.

(e) Give balanced equation for dilute nitric acid and copper carbonate.

2013

(a) Identify the gas evolved when.

(i) Sulphur is treated with conc. nitric acid.

(ii) A few crystals of KNO_3 are heated in a hard glass test tube.

(b) State two relevant observations for lead nitrate crystals are heated in a hard glass test tube.

(c) Give a balanced equation for : oxidation of carbon with conc. HNO_3.

2014

(a) Fill in the blank :

Cold dil. nitric acid reacts with copper to form ..... (hydrogen, nitrogen dioxide, nitric oxide).

(b) Give balanced equations for the following

(i) Laboratory preparation of nitric acid.

(ii) Action of heat on a mixture of copper and nitric acid.

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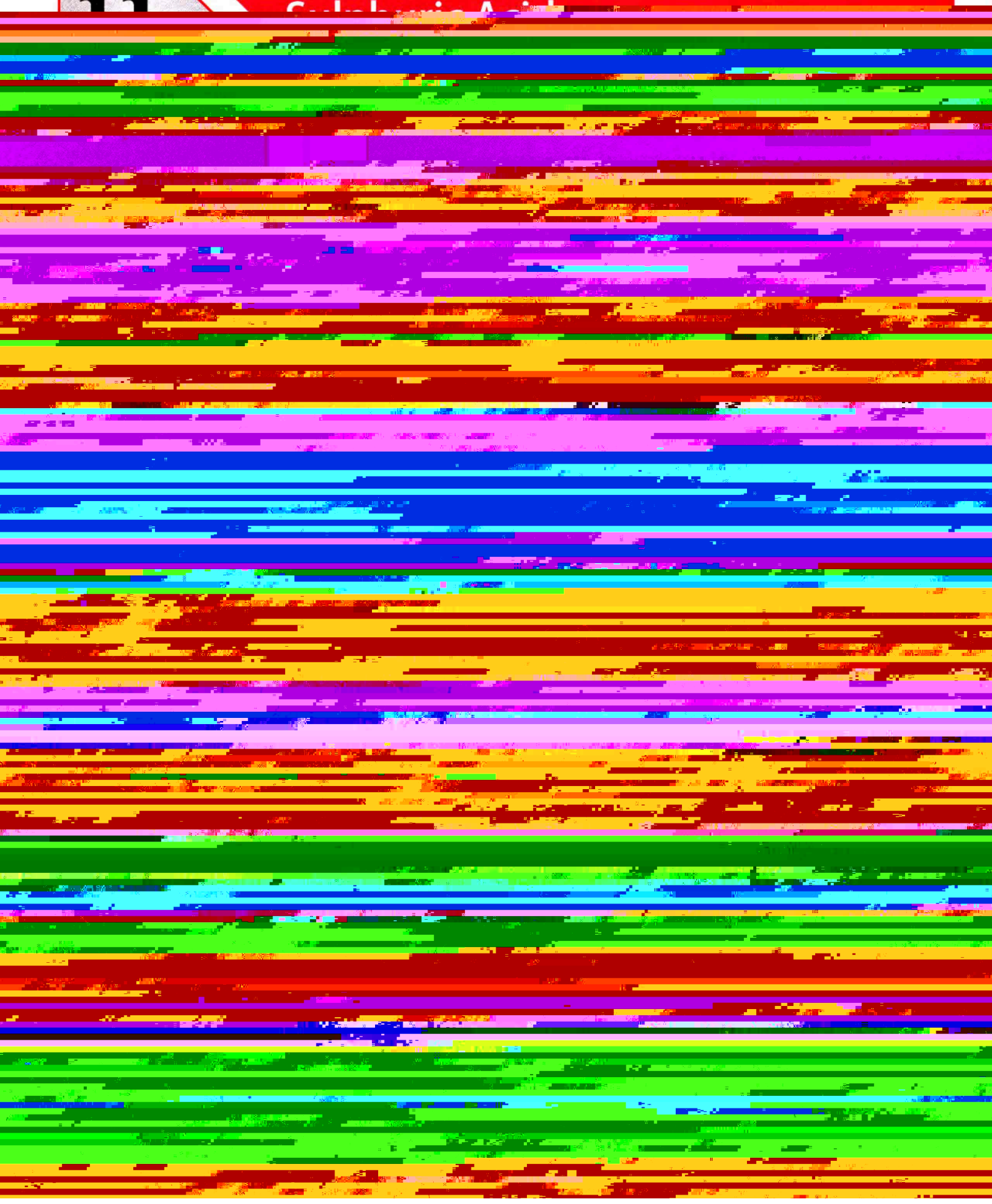

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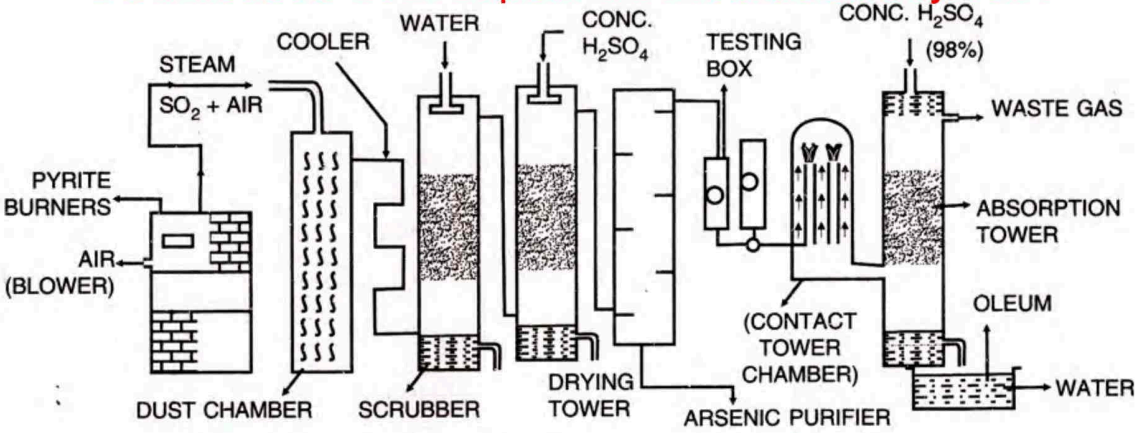

Fig. 11.1. Contact process for the manufacture of H_2SO_4 acid

Parts of the Plant	Function
Blower	For passage of purified air or oxygen.
Sulphur or iron pyrite burners	For production of sulphur dioxide by burning sulphur or iron pyrites.
Dusting chamber	For removal of dust particles by downward blow of steam.
Cooling pipes	For cooling the gases by passage through isolated pipes.
Scrubbing tower	For further removal of impurities and dust particles by downward spray of water.
Drying tower	For drying moist gases by downward spray of conc. H_2SO_4.
Arsenic purifier	For removing arsenic oxide by passage over ferric hydroxide.
Testing box	For testing the purity of the gases by passage of light through a darkened box.
Contact tower	For catalytic oxidation of SO_2 to SO_3 by passage of sulphur dioxide and oxygen through iron tower packed with vanadium pentoxide catalyst initially heated to 450°C.
Absorption tower	For absorption of sulphur trioxide vapours by descending stream of conc. H_2SO_4[98%].
Dilution tank	For dilution of oleum by flow of required amount of water.

The mixture is passed through a purifier called **electric precipitator,** consisting of a chamber with wires at high electric potential. The electric charge attracts solid particles which are removed. The mixture is then led to a **water scrubber** where it is completely freed from dust particles. It is then dried by a spray of concentrated sulphuric acid in another chamber. It then goes into the **arsenic purifier** where every trace of arsenic oxide is removed (Fig. 11.1).

(3) ***Catalytic oxidation of sulphur dioxide.*** The clean dried gaseous mixture of sulphur dioxide and air is passed through **contact tower** loosely packed with vanadium pentoxide or platinum on perforated shelves. The catalyst is placed in vertical iron pipes inside a cylindrical iron tower called the **converter**. Here, the preheated mixture of sulphur dioxide and air (oxygen) on passing through catalyst pipes forms sulphur trioxide.

$$2SO_2 + O_2 \underset{450°C}{\overset{V_2O_5}{\rightleftharpoons}} 2SO_3.$$

Since the reaction is highly exothermic, the catalyst is heated only in the beginning to about 450°C. This temperature is maintained by the heat evolved during the reaction.

(4) ***Absorption of sulphur trioxide in sulphuric acid.*** The gas is cooled in a heat exchanger and is then absorbed in **absorption tower** in concentrated sulphuric acid (98%).

$$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7 \text{ (oleum or pyrosulphuric acid)}$$

(5) ***Dilution of oleum to obtain sulphuric acid.*** It is diluted in **dilution tank** by adding calculated amount of water to obtain sulphuric acid of the desired strength.

$$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4.$$

Since sulphur trioxide does not dissolve in water satisfactorily and it gives a lot of heat and forms misty droplets of sulphuric acid, it is not directly absorbed by water.

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Favourable conditions for the conversion of SO_2 to SO_3 (contact process)

(i) ***Exothermic reactions are favoured at low temperature :*** The temperature should be as low as possible. The yield has been found to be maximum at about 410-450°C.

(ii) ***High pressure :*** High pressure favours the reaction because the product formed has less volume than reactant. But the acid-resistant towers which are able to withstand high pressure are difficult to build. Hence the pressure of 1 - 2 atmosphere is used.

(iii) ***Excess of oxygen :*** This increases the production of sulphur trioxide.

(iv) ***A suitable catalyst :*** Platinum is more efficient as a catalyst than vanadium pentoxide (V_2O_5), but platinum is more expensive and also it easily gets poisoned by impurities like arsenic (III) oxide.

Vanadium pentoxide is used as a catalyst.

SUMMARY OF CONTACT PROCESS

Formation of Sulphur dioxide :

$S + O_2 \rightarrow SO_2$; or $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

Conversion of purified SO_2 to SO_3 :

$$2SO_2 + O_2 \underset{400-450°C}{\overset{V_2O_5/Pt}{\rightleftharpoons}} 2SO_3$$

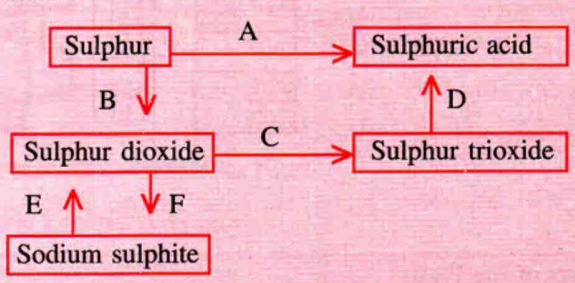
Conversion of sulphur trioxide into oleum :

$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$

Dilution of oleum :

$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$.

Intext Questions

1. Comment, sulphuric acid is referred to as :
   (a) King of chemicals, (b) Oil of vitriol.
2. Sulphuric acid is manufactured by Contact process.
   (a) Give two balanced equations to obtain SO_2 in this process,
   (b) Give the conditions for the oxidation of SO_2,
   (c) Why H_2SO_4 is not obtained by directly reacting SO_3 with water ?
   (d) Name the chemical used to dissolve SO_3 and also name the product formed. Give all the main reactions of this process.
   (e) Name a gas that can be oxidized to sulphur.
3. Why the impurity of arsenic oxide must be removed before passing the mixture of SO_2 and air through the catalytic chamber ?
4. 

   (a) Name the catalyst which helps in the conversion of sulphur dioxide to sulphur trioxide in step C.
   (b) In the Contact process for the manufacture of sulphuric acid, sulphur trioxide is not converted to sulphuric acid by reacting it with water. Instead a two-steps procedure is used. Write the equations for the two steps involved in D.
   (c) What type of substance will liberate sulphur dioxide from sodium sulphite in step E ?
   (d) Write the equation for the reaction by which sulphur dioxide is converted to sodium sulphite in step F.

11.5 PROPERTIES OF SULPHURIC ACID

11.5.1 Physical properties

1.	**Colour**	Colourless.
2.	**Odour**	Odourless.
3.	**Taste**	Slightly sour in taste.
4.	**Nature**	Dense, oily, hygroscopic liquid. It absorbs moisture, so H_2SO_4 should always be kept stoppered.
5.	**Density**	*Pure acid :* 1.85 g/cc.
6.	**Boiling point**	338°C.
7.	**Melting point**	Pure sulphuric acid freezes to colourless crystals at 10.4°C.
8.	**Solubility**	Soluble in water in all proportions.
9.	**Conductivity**	*Pure acid :* Almost a non-conductor of electricity. *Dilute acid :* Good conductor of electricity.
10.	**Constant boiling mixure**	It forms a *constant boiling mixture* at 338°C containing 98.5% of the acid.
11.	**Physiological nature**	Concentrated acid is highly corrosive in nature and chars the skin black.

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11.5.2 Chemical properties

Pure (100% H_2SO_4) hydrogen sulphate is not an acid as it does not ionise.

Sulphuric acid when dissolved in water form hydronium ion and so shows acidic properties.

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-.$$

$$HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}.$$

It ionises in two stages and thus it is a **dibasic acid**. The chemical properties of sulphuric acid depend on whether the acid is dilute or concentrated.

Dilution of H_2SO_4 : Water is never poured on acid to dilute it as large amount of heat is evolved which changes poured water to steam. The steam so formed causes spurting of acid which can cause burn injuries, so dilution is done by pouring acid on a given amount of water in a controlled manner by continuous stirring, else acid being heavier will settle down. The evolved heat is dissipated in the water itself and hence the spurting of the acid is minimized.

(a) Properties of dilute sulphuric acid

Acidic property :

(1) Dilute sulphuric acid reacts **with metals, which are above** hydrogen in the activity series to form metallic sulphate and hydrogen at ordinary temperature.

Active metal	+	Acid	→	Metal sulphate	+	Hydrogen
Mg	+	H_2SO_4	→	$MgSO_4$	+	$H_2\uparrow$
Zn	+	H_2SO_4	→	$ZnSO_4$	+	$H_2\uparrow$
Fe	+	H_2SO_4	→	$FeSO_4$	+	$H_2\uparrow$

Note : Noble metals like gold and platinum do not react with H_2SO_4 under any condition.

(2) It **neutralises** bases (oxides and hydroxides) to form salts and water.

Base/Alkali	+	Acid	→	Salt	+	Water
CuO	+	H_2SO_4	→	$CuSO_4$	+	H_2O
NaOH	+	H_2SO_4	→	$NaHSO_4$	+	H_2O
2NaOH	+	H_2SO_4	→	Na_2SO_4	+	$2H_2O$
FeO	+	H_2SO_4	→	$FeSO_4$	+	H_2O

(3) It liberates carbon dioxide from metallic **carbonates** and **bicarbonates**.

Carbonate/ bicarbonate	+	Acid	→	Salt	+	Water	+	Carbon dioxide
Na_2CO_3	+	H_2SO_4	→	Na_2SO_4	+	H_2O	+	$CO_2\uparrow$
$2NaHCO_3$	+	H_2SO_4	→	Na_2SO_4	+	$2H_2O$	+	$2CO_2\uparrow$
$2KHCO_3$	+	H_2SO_4	→	K_2SO_4	+	$2H_2O$	+	$2CO_2\uparrow$

(4) It evolves hydrogen sulphide from metal **sulphides**.

Metal sulphide	+	Acid	→	Salt	+	Hydrogen sulphide
Na_2S	+	H_2SO_4	→	Na_2SO_4	+	$H_2S\uparrow$
FeS	+	H_2SO_4	→	$FeSO_4$	+	$H_2S\uparrow$
ZnS	+	H_2SO_4	→	$ZnSO_4$	+	$H_2S\uparrow$

(5) It evolves sulphur dioxide from **sulphites** and hydrogen sulphites (bisulphites).

Sulphite / bisulphites	+	Acid	→	Salt	+	Water	+	Sulphur dioxide
Na_2SO_3	+	H_2SO_4	→	Na_2SO_4	+	H_2O	+	$SO_2\uparrow$
$2NaHSO_3$	+	H_2SO_4	→	Na_2SO_4	+	$2H_2O$	+	$2SO_2\uparrow$

Note : Sulphur dioxide is poisonous and, is therefore, responsible for polluting environment.

*It **adversely** affects the human health and vegetation.*

(b) Properties of concentrated sulphuric acid

(1) ***Non-volatile nature***

Concentrated sulphuric acid has a high boiling point (338°C) and so, it is considered to be a non-volatile acid. It is, therefore, used for preparing volatile acids like hydrochloric acid, nitric acid and acetic acid from their salts by double decomposition.

Salt of volatile acid	+	Acid (conc.)	→	Acid salt	+	Volatile acid
NaCl	+	H_2SO_4	→	$NaHSO_4$	+	HCl
KCl	+	H_2SO_4	→	$KHSO_4$	+	HCl
$NaNO_3$	+	H_2SO_4	→	$NaHSO_4$	+	HNO_3
KNO_3	+	H_2SO_4	→	$KHSO_4$	+	HNO_3
CH_3COONa Sodium acetate	+	H_2SO_4	→	$NaHSO_4$	+	CH_3COOH Acetic acid

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(2) *As an oxidising agent*

The oxidising property of concentrated sulphuric acid is due to the fact that on *thermal decomposition*, it yields nascent oxygen [O].

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$

Nascent oxygen oxidises non-metals, metals and inorganic compounds.

(a) Non-metals

1. *Carbon.*

$$C + 2H_2SO_4 \rightarrow CO_2 + 2H_2O + 2SO_2\uparrow$$

2. *Sulphur.*

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O.$$

3. *Phosphorus.*

$$2P + 5H_2SO_4 \rightarrow 2H_3PO_4 + 2H_2O + 5SO_2\uparrow$$

(b) Metals

1. *Copper.*

$$Cu + 2H_2SO_4 \rightarrow CuSO_4 + 2H_2O + SO_2\uparrow$$

2. *Zinc.*

$$Zn + 2H_2SO_4 \rightarrow ZnSO_4 + 2H_2O + SO_2\uparrow$$

(c) Inorganic compounds

1. *Hydrogen bromide.*

$$2HBr + H_2SO_4 \rightarrow Br_2 + 2H_2O + SO_2\uparrow$$

2. *Hydrogen sulphide.*

$$H_2S + H_2SO_4 \rightarrow S + 2H_2O + SO_2\uparrow$$

Note : Sulphur dioxide gas is oxidising as well as reducing agent. It also shows temporary bleaching property due to its reducing nature in presence of moisture.

$$SO_2 + 2H_2O \rightarrow H_2SO_4 + \underset{\text{Nascent hydrogen}}{2[H]}.$$

Coloured material + [H] Nascent hydrogen → Colourless product

This colourless material, if remains in air, regains its original colour.

Colourless material + Oxygen (air) → Coloured (original) material

It is not a strong bleaching agent as chlorine. Hence, it is **used to bleach only delicate materials as coloured clothes, coloured wet flower petals or fresh green grass.**

(3) *As a dehydrating agent*

H_2SO_4 has a great affinity for water. It readily removes elements of water from other compounds *i.e.*, it acts **as a dehydrating agent.**

(a) Organic acids and organic compounds are dehydrated by conc. H_2SO_4 as follows :

$$HCOOH \xrightarrow{\text{Conc. } H_2SO_4} CO + H_2O$$

$$(COOH)_2 \xrightarrow{\text{Conc. } H_2SO_4} CO + CO_2 + H_2O$$

$$C_2H_5OH \xrightarrow[170°C]{\text{Conc. } H_2SO_4} \underset{\text{Ethylene}}{C_2H_4\uparrow} + H_2O$$

(b) All **carbohydrates** such as glucose, sugar, and cellulose (paper, cotton, wood, *etc.*) react immediately to give a *black spongy mass of carbon* which rises up. Steam is given off and the whole mass gets heated due to an exothermic reaction. They are said to get *charred.*

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Conc. } H_2SO_4} 6C + 6H_2O$$

$$\underset{\text{Cane Sugar}}{C_{12}H_{22}O_{11}\,(s)} \xrightarrow{\text{Conc. } H_2SO_4} \underset{\text{Sugar Charcoal}}{12C(s)} + 11H_2O$$

Carbon obtained in this reaction is very pure and is called **sugar charcoal.**

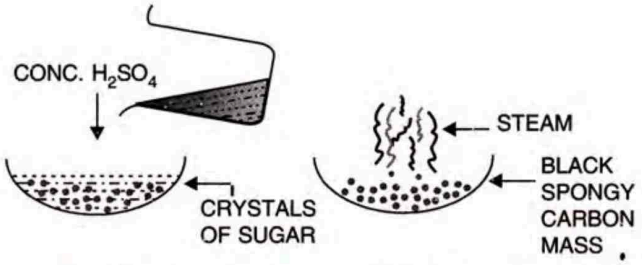

Fig. 11.2. Addition of conc. H_2SO_4 to sugar

Note : *Conc. H_2SO_4 reacts with skin to give* ***blisters****, because of removal of water, and chars the skin black.*

(c) **Conc. sulphuric acid removes water of crystallisation from salts.**

For example, blue copper (II) sulphate $CuSO_4.5H_2O$ taken in a test tube becomes white anhydrous copper sulphate, when a few drops of **conc.** sulphuric acid are added.

$$\underset{\text{Blue}}{CuSO_4\cdot 5H_2O} \xrightarrow{H_2SO_4} \underset{\text{Dirty white}}{CuSO_4(s)} + \underset{\text{(Removed by acid)}}{5H_2O}$$

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Difference between dilute and conc. sulphuric acid

Dilute	Concentrated
1. It ionises to great extent so behaves as strong acid.	1. It is a weak acid
2. Strong electrolyte.	2. Weak electrolyte.
3. It cannot act as an oxidising agent.	3. Conc. H_2SO_4 act as an oxidising agent.
4. Dilute sulphuric acid cannot be used as a drying or dehydrating agent.	4. It is used to dry gases like HCl and even dehydrate substances like $CuSO_4 \cdot 5H_2O$

(4) *Preparation of insoluble sulphates*

Sulphuric acid precipitates the insoluble sulphates of lead, barium and calcium from the solutions of their salts.

Soluble salt	+ Acid	→	Insoluble salt	+ Acid
$Pb(NO_3)_2$	+ H_2SO_4	→	$PbSO_4 \downarrow$	+ $2HNO_3$
$BaCl_2$	+ H_2SO_4	→	$BaSO_4 \downarrow$	+ $2HCl$

11.6 USES OF SULPHURIC ACID

1. It is used in the preparation of :

(a) Halogens, (b) Carbon monoxide, (c) Carbon dioxide, (d) Hydrogen, (e) Sulphur dioxide.

2. Metallurgy

(i) Extraction — Its reaction with metallic compounds gives sulphates which on electrolysis give the metal in the pure form.

(ii) Pickling metals — Removes metallic impurities like oxides and carbonates from the surface of metals before galvanizing.

3. In lead accumulators — It undergoes electrolysis in the aqueous state.

4. Oil refining — Sulphuric acid is used to remove harmful impurities in purification of oil products, e.g., petrol, kerosene and lubricants.

5. Industrial uses : in the manufacture of :

(i) Fertilizers like
- (a) **Ammonium sulphate** $[(NH_4)_2SO_4]$,
- (b) **Superphosphate of lime** $[Ca(H_2PO_4)_2 + CaSO_4]$,

(ii) Dyes, drugs
- (a) **Artificial fibres :** [I] Rayon, [II] Nylon.
- (b) **Dyes, drugs** [from coal tar derivatives].

(iii) Explosives like
- (a) **Tri-nitro toluene** [T.N.T.],
- (b) **Picric acid,**
- (c) **Tri-nitro glycerine** $[C_3H_5O_3(NO_2)_3]$.

(iv) Acids like
- (a) **Nitric acid** $[HNO_3]$,
- (b) **Hydrochloric acid** [HCl],
- (c) **Phosphoric acid** $[H_3PO_4]$,
- (d) **Acetic acid** $[CH_3COOH]$.

(v) Compounds like
- (a) **Sodium sulphate** [glass industry],
- (b) **Ferrous sulphate** [ink industry].

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11.7 TESTS FOR SULPHURIC ACID AND SULPHATES

(1) **Conc. sulphuric acid** on heating with copper evolves sulphur dioxide.

$$Cu + \underset{\text{(Conc.)}}{2H_2SO_4} \rightarrow CuSO_4 + 2H_2O + SO_2 \uparrow$$

Sulphur dioxide turns *acidified potassium dichromate solution green* (Reducing property of SO_2).

$$\underset{\text{(Orange)}}{K_2Cr_2O_7} + H_2SO_4 + 3SO_2 \rightarrow \underset{\text{(Green)}}{Cr_2(SO_4)_3} + K_2SO_4 + H_2O$$

(2) Conc. sulphuric acid on heating with NaCl evolves pungent fumes of HCl gas.

$$NaCl + \underset{\text{(Conc.)}}{H_2SO_4} \xrightarrow{\Delta} NaHSO_4 + HCl \uparrow$$

HCl gas gives dense white fumes of ammonium chloride, with a glass rod dipped in ammonium hydroxide.

$$HCl + NH_4OH \rightleftharpoons \underset{\text{(dense white fumes)}}{NH_4Cl} + H_2O$$

(3) When barium chloride solution is added to sulphuric acid or any soluble metal sulphate, **white precipitate** of barium sulphate is formed.

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HCl$$

$$BaCl_2 + MgSO_4 \rightarrow BaSO_4\downarrow + MgCl_2$$

The **white precipitate of barium sulphate is insoluble in any mineral acid like hydrochloric acid or nitric acid.**

Note : Precipitates of both barium sulphate and barium sulphite are white in colour but barium sulphite precipitate dissolves in mineral acids while barium sulphate precipitate does not.

EXERCISE

1. Why is water not added to concentrated H_2SO_4 in order to dilute it ?
2. Give two balanced reactions of each type to show the following properties of sulphuric acid :
   (a) Acidic nature, (b) Oxidising agent,
   (c) Dehydrating nature, (d) Non-volatile nature.
3. Give a chemical test to distinguish between :
   (a) dilute sulphuric acid and dilute hydrochloric acid,
   (b) dilute sulphuric acid and conc. sulphuric acid.
4. Name the products formed when hot and concentrated sulphuric acid reacts with the following :
   (a) Sulphur, (b) NaOH, (c) Sugar,
   (d) Carbon, (e) Copper.
5. Why is :
   (a) concentrated sulphuric acid kept in air tight bottles ?
   (b) H_2SO_4 not a drying agent for H_2S ?
   (c) sulphuric acid used in the preparation of HCl and HNO_3 ? Give equations in both cases.
6. What property of conc. H_2SO_4 is made use of in each of the following cases? Give an equation for the reaction in each case :
   (a) in the production of HCl gas when it reacts with a chloride,
   (b) in the preparation of CO from HCOOH,
   (c) as a source of hydrogen by diluting it and adding a strip of magnesium,
   (d) in the preparation of sulphur dioxide by warming a mixture of conc. sulphuric acid and copper-turnings,
   (e) hydrogen sulphide gas is passed through concentrated sulphuric acid.
7. What is the name given to the salts of :
   (a) sulphurous acid, (b) sulphuric acid ?
8. Give reasons for the following :
   (a) Sulphuric acid forms two types of salts with NaOH,
   (b) Red brown vapours are produced when concentrated sulphuric acid is added to hydrogen bromide,
   (c) A piece of wood becomes black when concentrated sulphuric acid is poured on it,
   (d) Brisk effervescence is seen when oil of vitriol is added to sodium carbonate.
9. Copy and complete the following table :

Column 1 Substance reacted with acid	Column 2 Dilute or concentrated acid	Column 3 Gas
		Hydrogen
		Carbon dioxide
		Only chlorine

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10. Give the odour of the gas evolved and name the gas produced when sodium sulphide is added to solution of HCl in water.

11. (a) Which property of sulphuric acid accounts for its use as a dehydrating agent ?

(b) Concentrated sulphuric acid is both an oxidizing agent and a non-volatile acid. Write one equation each to illustrate the above mentioned properties of sulphuric acid.

13. Some properties of Sulphuric acid are listed below. Choose the property A, B, C or D which is responsible for the reactions (i) to (v). Some properties may be repeated :

A. Acid B. Dehydrating agent

C. Non-volatile acid D. Oxidizing agent

(i) $C_{12}H_{22}O_{11} + nH_2SO_4 \rightarrow 12C + 11H_2O + nH_2SO_4$

(ii) $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

(iii) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

(iv) $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

(v) $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$

13. (a) Name the acid formed when sulphur dioxide dissolves in water.

(b) Name the gas released when sodium carbonate is added to a solution of sulphur dioxide.

2008

(a) Dilute sulphuric acid will produce a white precipitate when added to a solution of :

(i) Copper nitrate (ii) Zinc nitrate

(iii) Lead nitrate (iv) Sodium nitrate

(b) Identify the following substance : Liquid E can be dehydrated to produce ethene.

(c) Copy and complete the following table relating to an important industrial process and its final output.

Name of process	Inputs	Catalyst	Equation for catalyzed reaction output
Contact process	Sulphur dioxide + oxygen		

(d) Making use only of substances given : dil. sulphuric acid, sodium carbonate, zinc, sodium sulphite, lead, calcium carbonate :

Give equations for the reactions by which you could obtain :

(i) hydrogen (ii) sulphur dioxide

(iii) carbon dioxide (iv) zinc carbonate [2 steps]

(e) What property of conc. H_2SO_4 :

(i) is used in the action when sugar turns black in its presence.

(ii) allows it to be used in the preparation of HCl and HNO_3 acids.

(f) Write the equations for :

(i) dil. H_2SO_4 and barium chloride

(ii) dil. H_2SO_4 and sodium sulphuide

2009

(a) Name the gas evolved [formula is not acceptable]. The gas that can be oxidised to sulphur.

2010

(a) Give the equation for :

(i) Heat on sulphur with conc. sulphuric aicd

(ii) Reaction of – sugar with conc. sulphuric acid.

(b) Give a balanced equation for the conversion of zinc oxide to zinc sulphate.

(c) Select the correct answer from A, B, C.

A. Sodium hydroxide solution

B. A weak acid

C. Dilute sulphuric acid.

The solution which liberates sulphur dioxide gas, from sodium sulphite.

2011

(a) State your observation when – Sugar crystals are added to a hard glass test tube containing conc. sulphuric aicd.

(b) Choose the correct answer from the choices – The gas evolved when dil. sulphuric acid reacts with iron sulphide.

(i) Hydrogen sulphide (ii) Sulphur dioxide

(iii) Sulphur trioxide (iv) Vapour of sulphuric acid

(c) Give a balanced equation for :

(i) Dilute sulphuric acid is poured over sodium sulphite.

(ii) Manufacture of sulphuric acid by the – contact process.

(d) State the property of sulphuric acid shown by the reaction of conc. sulphuric acid when heated with

(i) potassium nitrate (ii) carbon

2012

(a) Name – The gas produced on reaction of dilute sulphuric acid with a metallic sulphide.

(b) Some properties of sulphuric acid are listed below. Choose the role played by sulphuric acid as A, B, C

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or D which is responsible for the reactions (i) to (v). Some role/s may be repeated.

(1) Dilute acid (2) Dehydrating agent

(3) Non-volatile acid (4) Oxidising agent

(i) $CuSO_4 \cdot 5H_2O \xrightarrow{conc.H_2SO_4} CuSO_4 + 5H_2O$

(ii) $S + 2H_2SO_4$ [conc.] $\rightarrow 3SO_2 + 2H_2O$

(iii) $NaNO_3 + H_2SO_4 \xrightarrow{<200°C} NaHSO_4 + HNO_3$

[conc.]

(iv) $MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$

(v) $Zn + 2H_2SO_4$ [conc.] $\rightarrow ZnSO_4 + SO_2 + 2H_2O$

(c) Give balanced equation for the reaction : Zinc sulphide and dilute sulphuric acid.

2013

(a) State one appropriate observation for : Conc. H_2SO_4 is added to a crystal of hydrated copper sulphate.

(b) In the given equation – $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$:

(c) Give a balanced equation for : Dehydration of concentrated sulphuric acid with sugar crystals.

(d) Identify the substance underlined : <u>A dilute mineral acid</u> which forms a white precipitate when treated with barium chloride solution.

2014

(a) Write balanced equations for the following : Action of concentrated sulphuric acid on carbon.

(b) Distinguish between the following pairs of compounds using the test given within brackets.

Dilute sulphuric acid and dilute hydrochloric acid [using barium chloride solution].

(c) State the conditions required for the following reactions to take place :

The conversion of sulphur dioxide to sulphur trioxide.

(d) Give one equation each to show the following properties of sulphuric acid :

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12 Organic Chemistry

SYLLABUS

(i) Introduction to Organic compounds.
Unique nature of Carbon atom – tetravalency, catenation, formation of single, double and triple bonds, straight chain, branched chain and cyclic compounds.

(ii) Structure and Isomerism.
Structure of compounds with single, double and triple bonds; Isomerism – structural (chain, position)

(iii) Homologous series — characteristics with examples
Alkane, alkene, alkyne series and their gradation in properties and the relationship with the molecular mass or molecular formula.

(iv) Simple nomenclature.
Simple nomenclature – of the hydrocarbons with simple functional groups – (double bond, triple bond, alcoholic, ether, aldehydic, keto, carboxylic group) longest chain rule and smallest number for functional groups rule – trivial and IUPAC names.

(v) Hydrocarbons : alkanes, alkenes, alkynes.
Alkanes – general formula; methane (green house gas) and ethane – methods of preparation from sodium ethanoate (sodium acetate), sodium propanoate (sodium propionate), from iodomethane (methyl iodide) and bromoethane (ethyl bromide). Oxidation of methane and ethane in presence of oxygen under suitable conditions reaction of methane and ethane with chlorine through substitution.
Alkenes – (unsaturated hydrocarbons with a double bond); ethene as an example. Methods of preparation of ethene by dehydro halogenation reaction and dehydration reactions.
Alkynes – (unsaturated hydrocarbons with a triple bond); ethyne as an example of alkyne; Methods of preparation from calcium carbide and 1, 2 dibromoethane ethylene dibromide). Only main properties, particularly addition products

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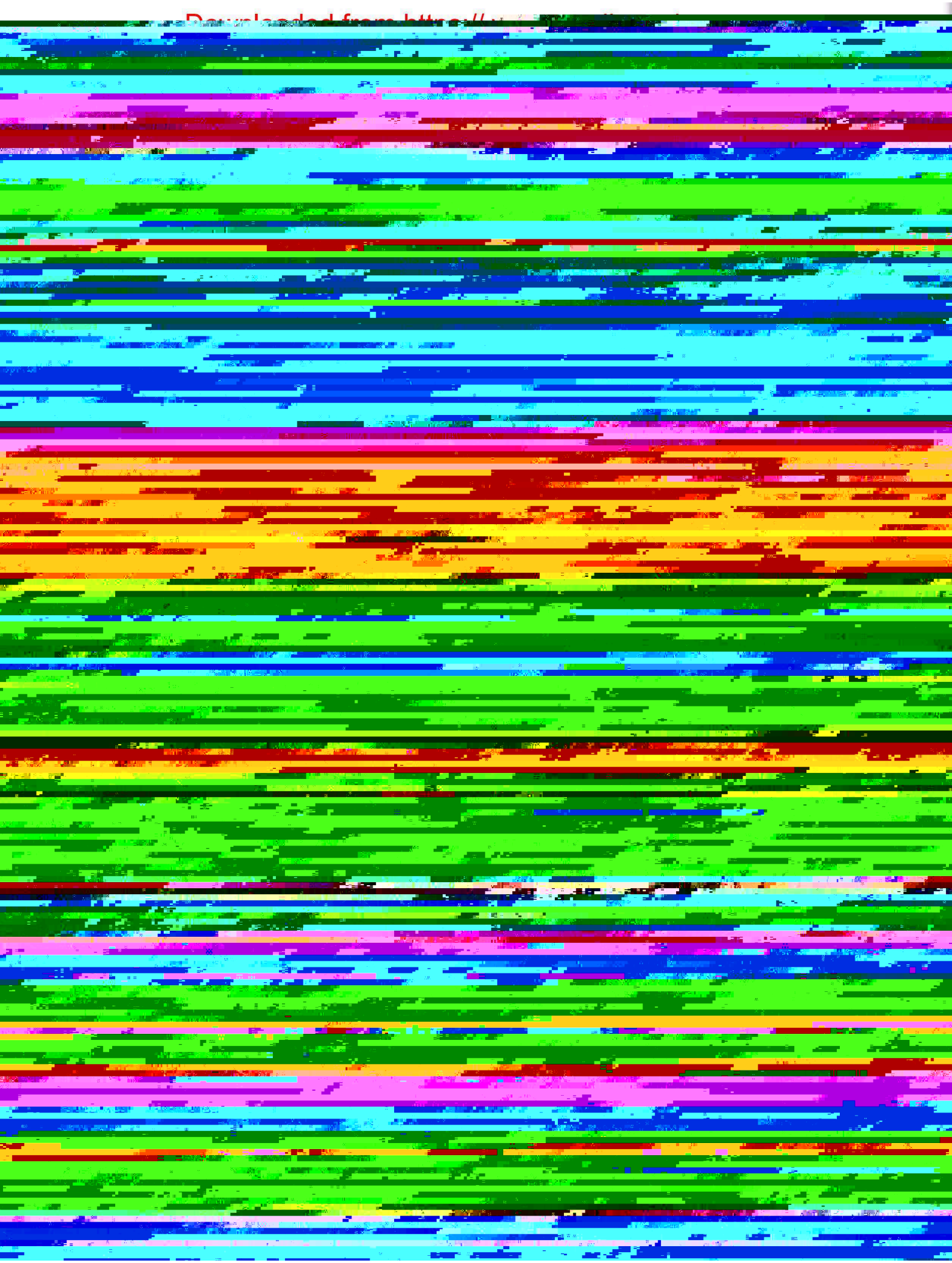

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Application of Organic Chemistry

Organic compounds are extremely useful to us in our daily life.

The soaps and shampoos we use while taking bath, the powders, perfumes, etc., we apply on the body, the clothes we wear, food we eat *i.e.*, carbohydrates, proteins, fats, vitamins, etc., fuels we use, natural gas, petroleum products, medicines, explosives, dyes, insecticides, etc., are all organic compounds. There is hardly any walk of life where we do not need organic compounds.

12.3 UNIQUE NATURE OF CARBON ATOMS

Carbon shows some unique properties like tetravalency and catenation, which are discussed below.

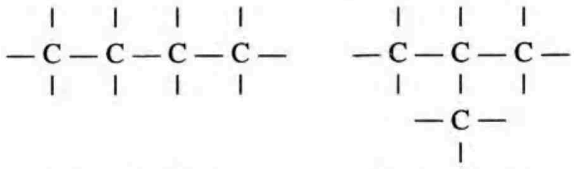
(i) Tetravalency of the carbon atom : Carbon has four valence electrons (*At. no. of C = 6; Elec. Config. = 2, 4*). Since it can neither lose nor gain electrons to attain octet, it forms covalent bonds by sharing its four electrons with other atoms. *This characteristic of the carbon atom, by virtue of which it forms four covalent bonds, is called the* ***tetravalency of carbon.***

(ii) Catenation : Carbon atom possesses an unique property to link together (self linking) to form very long chains. This property is referred to as catenation.

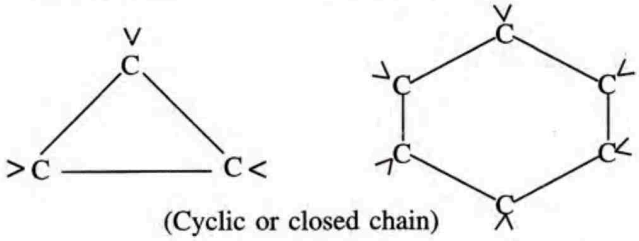
The property of ***self-linking*** *of atoms of an element through covalent bonds in order to form straight chains, branched chains and cyclic chains of different sizes is known as* **catenation**.

The property of catenation is shown by other elements also but carbon exhibits this property to maximum extent. This is due to greater strength of carbon-carbon bond and due to tetra-covalency of carbon. In this process of catenation carbon atoms form straight, branch and cyclic chains of atoms, and can involve single (–), double (=) or triple (≡) covalent bonds.

(a) Formation of straight, branched and cyclic chains of carbon atoms : The combination of carbon atoms with one another gives rise to *straight* or *branched* or *cyclic chains.*



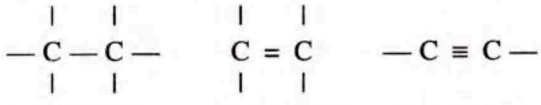
(Straight chain) (Branched chain)



(Cyclic or closed chain)

The carbon-carbon chain can be very long. Organic molecules containing as many as seventy carbon atoms joined together, one after the other, are known to exist.

(b) Formation of single, double and triple covalent bonds : The valency of carbon is four, *i.e.* it is tetravalent. In order to satisfy its valency, it forms single, double and triple covalent bonds by sharing one, two or three pairs of electrons respectively between two carbon atoms as well as with other atoms like oxygen, nitrogen etc.



(single bond) (double bond) (triple bond)

(double bond with oxygen atom) (triple bond with nitrogen atom)

This **unique nature** of carbon atom (catenation and tetravalency) gives rise to the formation of a large number of compounds. **More than 5 million organic compounds are known today and thousands of new compounds are added every year to the existing lot.**

All this demands a separate branch of chemistry, *i.e.*, organic chemistry.

Organic chemistry may be defined as the chemistry of hydrocarbons and their derivatives.

Features of covalent bonding involving carbon

- Carbon atoms can join to each other to form long chains. Atoms of other elements can then attach to the chain.

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- The carbon atoms in a chain can be linked by single, double or triple covalent bonds.
- Carbon atoms can also arranged themselves in rings.

12.4 TYPES OF ORGANIC COMPOUNDS

The organic compounds have been classified as hydrocarbons whereas their derivatives are classified as alcohol, aldehydes, carboxylic acids, ethers, halides, etc. These compounds may contain any number of other elements, including hydrogen, nitrogen, oxygen, the halogens as well as phosphorus, silicon and sulphur.

Table 12.1 : Organic Compounds

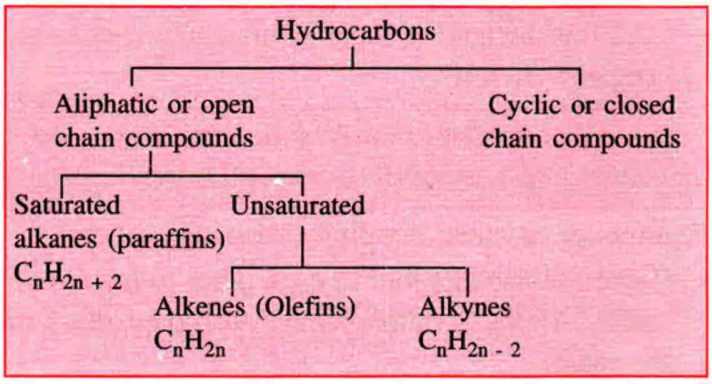
Types of compound	*Identified by the presence of*	*Example*
1. Hydrocarbon		
(a) alkane	—C—C— bond	Ethane $H_3C—CH_3$
(b) alkene	—C = C— bond	Ethene $H_2C = CH_2$
(c) alkyne	—C ≡ C— bond	Ethyne HC ≡ CH
2. Alcohol	—OH group	Methanol CH_3OH
3. Aldehyde	—CHO group	Ethanal CH_3CHO
4. Ketone	> C = 0	Propanone CH_3COCH_3
5. Carboxylic acid	—COOH group	Ethanoic acid CH_3COOH
6. Ether	C—O—C group	Dimethyl ether $H_3C—O—CH_3$
7. Halide	—X (F, Cl, Br, I)	Chloroethane C_2H_5Cl

12.5 HYDROCARBONS

Hydrocarbons are compounds that are made up only of carbon and hydrogen atoms.

Classification of Hydrocarbons

Hydrocarbons are sub-divided into two main groups, the aliphatic (open) and cyclic (closed) chain compounds.



The open chain compounds or aliphatic compounds are further sub-divided into ***saturated compounds*** and ***unsaturated compounds.***

Saturated Compounds

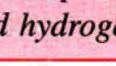
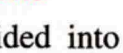
The simplest open chain hydrocarbon is **alkane.** It is represented by the formula C_nH_{2n+2}, where n represents natural number.

It is a saturated hydrocarbon, as all the four valencies of carbon are satisfied by a single covalent bond. (For further details refer section 12B).

Unsaturated Compounds

Open chain compounds where all the four valencies are not satisfied by single covalent bonds, double or triple bonds are required to satisfy the valencies. These compounds are known as **unsaturated compounds** *e.g.*, **Alkenes** are the hydrocarbons with double bond and **Alkynes** are the hydrocarbons with triple bond between two carbon atoms. (For further details refer sections 12C & 12D).

Comparison of Saturated and Unsaturated Hydrocarbons

Saturated organic compounds	*Unsaturated organic compounds*
1. All the four valencies of each carbon atom are *satisfied* by forming single covalent bonds with carbon and with hydrogen atoms.	1. The valencies of at least two carbon atoms are *not fully satisfied* by the hydrogen atoms.
2. Carbon atoms are joined only by a ***single covalent bond.*** 	2. Carbon atoms are joined by ***double covalent bonds*** or by ***triple covalent bonds.*** 
3. They are *less reactive* due to the **non-availability of electrons** in the single covalent bond, and therefore they undergo *substitution reaction**.	3. They are *more reactive* due to **the presence of electrons** in the double or the triple bond, and therefore undergo *addition reaction***.

* **Substitution reaction :** A reaction in which one atom of a molecule is replaced by another atom (or group of atoms) is called a substitution reaction. *e.g.* $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$.

** **Addition reaction :** A reaction involving *addition* of atom(s) or molecule(s) to the double or the triple bond of an unsaturated compound so as to yield a *saturated* product is known as an addition reaction, *e.g.* $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$.

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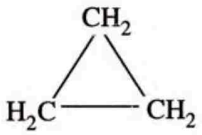
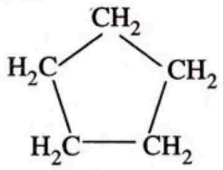
Cyclic or closed chain hydrocarbons contain three or more carbon atoms in their molecules. These have properties similar to open chain hydrocarbons. They are also called **carbocyclic** compounds.

Examples : (i) Cyclic compounds containing single bond are called cycloalkanes.

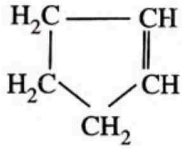


Cyclopropane (C_3H_6)

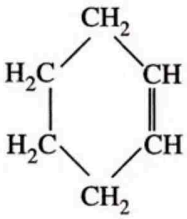


Cyclopentane (C_5H_{10})

(ii) Cyclic compounds containing double bond are called cycloalkenes.

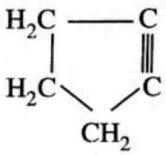


Cyclopentene (C_5H_8)

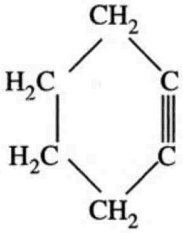


Cyclohexene (C_6H_{10})

(iii) Cyclic compounds containing triple bond are called cycloalkynes.



Cyclopentyne (C_5H_6)

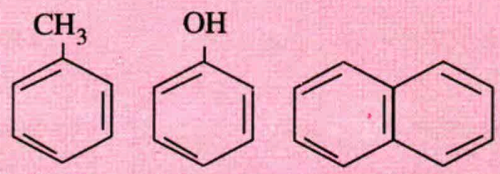


Cyclohexyne (C_6H_8)

Some hydrocarbons contain at least one carbon atoms having carbon-carbon single and carbon-carbon double bonds in alternate positions.

Most of these compounds have pleasant smell.

They are also known as aromatic compounds. Few other members of this class are :



Toluene **Phenol** **Naphthalene**

12.6 ALKYL GROUP

An alkyl group is obtained by removing one atom of hydrogen from an alkane molecule.

$$\underset{\text{Alkane}}{C_nH_{2n+2}} \xrightarrow{-H} \underset{\text{Alkyl}}{-C_nH_{2n+1}}$$

Table 12.2 : Alkyl radicals

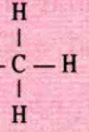
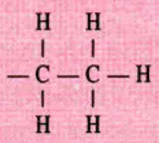
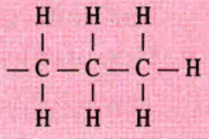
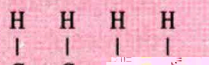
Name	Molecular formula	Abbreviated formula	Structural formula
Methyl	$-CH_3$	$-CH_3$	
Ethyl	$-C_2H_5$	$-CH_2-CH_3$	
Propyl	$-C_3H_7$	$-CH_2-CH_2-CH_3$	
Butyl	$-C_4H_9$	$-CH_2-CH_2-CH_2-CH_3$	

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Table 12.3 : Some common functional groups

R represents the alkyl group ($-C_nH_{2n+1}$)

Functional group	General formulae	Types of Organic compounds	Suffix	Examples with common names and IUPAC names	
Halide - X (F, Cl, Br, I)	R—X	Haloalkanes	ane	CH_3Cl Methyl chloride Chloromethane	C_2H_5Cl Ethyl chloride Chloroethane
Hydroxyl - OH	R—OH	Alcohols	ol	CH_3OH Methyl alcohol Methanol	C_2H_5OH Ethyl alcohol Ethanol
Aldehyde -CHO	$\begin{matrix}H\\R\end{matrix}>C=O$	Aldehydes	al	HCHO Formaldehyde Methanal	CH_3CHO Acetaldehyde Ethanal
Carboxyl - COOH	$R-C(=O)-O-H$	Carboxylic acids	oic acid	CH_3COOH Acetic acid Ethanoic acid	CH_3CH_2COOH Propionic acid Propanoic acid
Keto $-\overset{O}{\overset{\|}{C}}-$	$R-\overset{O}{\overset{\|}{C}}-R'$	Ketones	one	CH_3COCH_3 Acetone Propanone	$C_2H_5COC_2H_5$ Diethyl ketone Pentanone
Ethers $-\overset{\|}{\underset{\|}{C}}-O-\overset{\|}{\underset{\|}{C}}-$	R—O—R′	Ethers	oxy	$CH_3—O—CH_3$ Dimethyl ether Methoxy methane	$CH_3—O—C_2H_5$ Ethyl methyl ether Methoxy ethane

12.7 FUNCTIONAL GROUP

A functional group is an atom or a group of atoms that defines the structure (or the properties of a particular family) of organic compounds.

Alkanes do not contain any functional group. The functional group of alkenes is considered as double bond > C=C < while in alkynes it is triple bond —C≡C—. For other examples refer table 12.3.

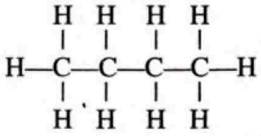
12.7.1 Characteristics of functional groups:

(i) The chemical properties of the compounds containing the same functional group are similar. Therefore, the compounds of the same functional groups are identified using the same type of tests.

(ii) The physical and chemical properties of the compounds of different functional groups are different.

For example : Butane (C_4H_{10}) can be represented in the following way :

Structural formula



A molecule having the bigger structural formula is often cumbersome. We represent it by **abbreviated formula**.

Abbreviated or Condensed formula of butane

$CH_3—CH_2—CH_2—CH_3$

A structure that shows only the linking of carbon atoms in a molecule is called the **carbon skeleton.**

Carbon skeleton of batane

C—C—C—C

12.9 ISOMERS

Compounds having the same molecular formula but different structural formula are known as ISOMERS and the phenomenon is known as ISOMERISM (ISO = same, meros = parts).

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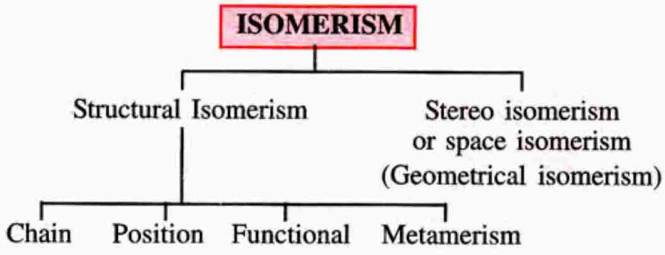
The following flow chart shows different types of isomerism :



There are two main causes of isomerism

1. Difference in the mode of linking of atoms :

For example : $C_4H_{10}O$ shows different types of linkages and thus it stands for three different types of *ether*.

This type of isomerism is structural isomerism.

$C_2H_5-O-C_2H_5$	$CH_3-O-C_3H_7$	$CH_3-O-CH(CH_3)_2$
Diethyl ether	**Methyl propyl ether**	**Methyl iso-propyl ether**

2. Difference in the arrangement of atoms or groups in space

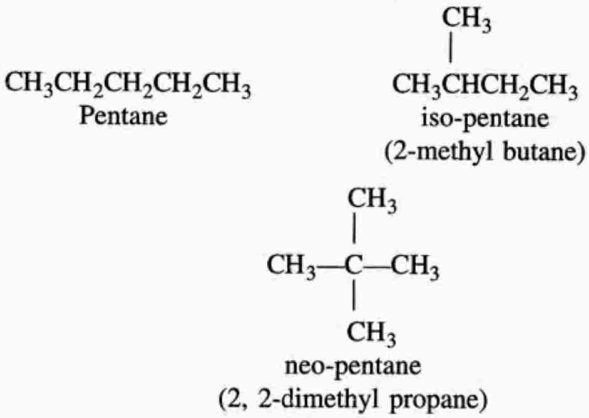
This type of isomerism is stereo isomerism.

For example : 1, 2-dichloro ethene

Different types of structural isomerism are :

(i) *Chain Isomerism :* When two or more compounds have a similar molecular formula but are different in the arrangement of carbon atoms in straight or branched chains the compounds are referred to as **chain isomers** and the phenomenon is termed as **chain isomerism**.

For example : Pentane C_5H_{12}



(ii) *Position isomerism :* When two or more compounds with the same molecular formula differ in the position of substituent atom or functional group on the carbon atom, they are called position isomers and this phenomenon is termed position isomerism.

For example :

(i) $CH_3—CH_2C \equiv C—H$ (but-1-yne) and $CH_3—C \equiv C—CH_3$ (but-2-yne)

(ii) $CH_3CH_2—\overset{\overset{O}{\|}}{C}—CH_2CH_3$ (pentan-3-one) and $CH_3—\overset{\overset{O}{\|}}{C}—CH_2CH_2CH_3$ (pentan-2-one)

(iii) *Functional isomerism : Two or more compounds with the same molecular formula but different functional groups are called functional isomers and this phenomenon is termed as functional isomerism

For example :

CH_3CH_2OH (ethanol) and $CH_3—O—CH_3$ (dimethyl ether)

(iv) *Metamerism : It arises due to unequal distribution of alkyl groups on either side of the functional groups in the molecules.

For example :

$CH_3OC_3H_7$ (methoxy propane, methyl propyl ether) and $C_2H_5OC_2H_5$ (ethoxyethane, Diethyl ether)

Note : The number of isomers increases with the number of carbon atoms; butane with 4 carbon atoms has 2 isomers, hexane has 5 isomers. An organic compound having formula $C_{21}H_{44}$ has about a million isomers and so on.

12.10 HOMOLOGOUS SERIES

A **homologous series** *is a group of organic compounds having a similar structure and similar chemical properties in which the successive compounds differ by a* **CH_2** *group.*

By adding CH_2 group (known as methylene group) in a given hydrocarbon, another hydrocarbon with a higher number of carbon atoms is obtained, *e.g.*,

$$\underset{\text{Methane}}{CH_4} + CH_2 \rightarrow \underset{\text{Ethane}}{C_2H_6}.$$

* *Not in syllabus*

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Similarly, by removing CH_2 group from a hydrocarbon, another hydrocarbon with a lesser number of carbon atoms is obtained, *e.g.*,

$$\underset{\text{Butane}}{C_4H_{10}} \rightarrow \underset{\text{Propane}}{C_3H_8} + CH_2$$

12.10.1 Characteristics of a homologous series

1. Each member of the series differs from the preceding one by the addition of CH_2 group and by molecular mass of 14 a.m.u.
2. All members of a *homologous series* share the *general formula.* (Same elements and same functional group).

   For example, the general formula for alkane is $C_n H_{2n+2}$, alkene is $C_n H_{2n}$.
3. The physical properties of the members change gradually i.e. show *gradation in properties* as the number of carbon atoms per molecule increases, *i.e.,* as molecular mass increases.

   For example, melting point, boiling point and the density of the successive members of the homologous series *increase with the increase in molecular mass.*

   Examples :

	Alkane	*Molar mass*	*Boiling point*	*Physical state*
(i)	Methane	16	–161.4°C	gas
(ii)	Ethane	30	–88.3°C	gas
(iii)	Propane	44	–44.5°C	gas
(iv)	Butane	58	0°C	gas
(v)	Pentane	72	36.2°C	liquid

4. Members of a homologous series have similar chemical properties. *For example :* Methane reacts with chlorine to form methyl chloride.

   $$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl.$$

   Similarly, ethane reacts to form ethyl chloride

   $$C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl.$$
5. All the members of a homologous series can be prepared by using the same general method of preparation.

   For example : Alcohols are prepared by alkyl halides.

   $$CH_3Br + KOH \xrightarrow[\text{(aqueous)}]{\text{boil}} CH_3OH + KBr.$$

   $$C_2H_5Br + KOH \xrightarrow[\text{(aqueous)}]{\text{boil}} C_2H_5OH + KBr.$$

Examples of Homologous Series.

1. ALKANE :

Gen. formula $[C_nH_{2n+2}]$	Molecular formula
Methane	CH_4
Ethane	C_2H_6
Propane	C_3H_8
Butane	C_4H_{10}
Pentane	C_5H_{12}
Hexane	C_6H_{14}
Heptane	C_7H_{16}
Octane	C_8H_{18}
Nonane	C_9H_{20}
Decane	$C_{10}H_{22}$

2. ALKENE :

Gen. formula $[C_nH_{2n}]$	Molecular formula
Ethene	C_2H_4
Propene	C_3H_6
Butene	C_4H_8
Pentene	C_5H_{10}
Hexene	C_6H_{12}
Heptene	C_7H_{14}
Octene	C_8H_{16}
Nonene	C_9H_{18}
Decene	$C_{10}H_{20}$

3. ALKYNE :

Gen. formula $[C_nH_{2n-2}]$	Molecular formula
Ethyne	C_2H_2
Propyne	C_3H_4
Butyne	C_4H_6
Pentyne	C_5H_8
Hexyne	C_6H_{10}
Heptyne	C_7H_{12}
Octyne	C_8H_{14}

4. ALCOHOLS :

Gen. formula $[C_nH_{2n+1}OH]$	Molecular formula
Methanol	CH_3OH
Ethanol	C_2H_5OH
Propanol	C_3H_7OH
Butanol	C_4H_9OH
Pentanol	$C_5H_{11}OH$
Hexanol	$C_6H_{13}OH$
Heptanol	$C_7H_{15}OH$
Octanol	$C_8H_{17}OH$

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12.10.2 Significance of the homologous series

(i) The nature of any member of that family of compounds can be ascertained, if the properties of the first few members of that series are known.

(ii) It helps to predict the properties of even those members of the series that are yet to be prepared.

(iii) Knowledge of homologous series is useful for the systematic study of organic chemistry, since it saves learning time.

Note : In an organic compound the alkyl group determines mainly the physical properties, whereas the functional group is responsible for the chemical properties or reactivity of the compounds.

Intext Questions

1. (a) What are organic compounds?
   (b) What is vital force theory? Why was it discarded?
2. (a) Name a few sources of organic compounds.
   (b) Give the various applications of organic chemistry.
3. Organic chemistry plays a key role in all walks of life. Discuss.
4. Carbon shows some unique properties, name them.
5. Explain the following :
   (a) tetravalency,
   (b) catenation.
6. Write any four properties of organic compounds that distinguish them from inorganic compounds.
7. Why are organic compounds studied as a separate branch of chemistry ?
8. What are hydrocarbons ? Compare saturated and unsaturated hydrocarbons?
9. Give reason for the existence of the large number of organic compounds.
10. Give at least one example in each case to show structure of isomers of :
    (a) single bond compound,
    (b) double bond compound,
    (c) triple bond compound.
11. Name a compound of each type and draw the figure.
    (a) Cyclic compound with single bond.
    (b) Cyclic compound with triple bond.
12. Give the name of one member of each of the following :
    (a) saturated hydrocarbons,
    (b) unsaturated hydrocarbons.
13. Define substitution and addition reaction. Give an example for each.
14. Define or explain chain isomerism and position isomerism with examples in each case.
15. (a) Define the term isomerism. State two main causes of isomerism.
    (b) Draw the chain isomers of hexane (C_6H_{12}).
    (c) Draw position isomers of butene (C_4H_8).
16. Define a functional group and give the structural formula of the following functional groups:
    (a) ketone, (b) alcohols, (c) aldehydes.
17. Write the name and formula of fourth member of the following homologous series :
    (a) Alkyne (b) Alcohol
18. Which part of an organic compound determines (i) physical properties (ii) chemical properties ?
19. Name the alkyl radical and the functional group of the following organic compounds :
    (a) CH_3OH, (b) C_2H_5OH,
    (c) C_3H_7CHO, (d) C_4H_9COOH,
    (e) CH_3COOH (f) HCHO.
20. (a) What is an alkyl group ?
    (b) Give the names of any three alkyl radicals. How are they formed ?
21. Give the names and the structural formula of the first three members of the homologous series of alkanes.
22. (a) What is a homologous series ?
    (b) What is the difference in the molecular formula of any two adjacent homologues :
    (i) in terms of molecular mass,
    (ii) in terms of number and kind of atoms per molecule?

12.11 NOMENCLATURE

Nomenclature is the system of assignment of names to organic compounds.

System of nomenclature are Trivial system and IUPAC (International Union of Pure and Applied Chemistry) system.

In the early days, the name of a compound depended on the source from which it was obtained. For example, CH_4 was named *marsh gas,* since it was obtained from marshy places. Acetic acid (Latin : *acetum* = vinegar) was obtained from

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vinegar, methyl alcohol (*methus* = spirit, *hule* = wood) was obtained from wood, and citric acid from the botanical source *citrus*. Such names are called **common names** or **trivial names.**

However, the nomenclature was systematized by the chemists of IUPAC in 1957.

According to this system, the name of an organic compound consists of three parts :

(i) root word, (ii) suffix, (iii) prefix.

(i) Root word : It depends upon the number of carbon atoms present in the longest carbon chain selected (Table 12.4).

Table 12.4

Number of carbon atoms		Root Word (Greek name)
One carbon atom	C_1	Meth
Two carbon atoms	C_2	Eth
Three carbon atoms	C_3	Prop
Four carbon atoms	C_4	But
Five carbon atoms	C_5	Pent
Six carbon atoms	C_6	Hex
Seven carbon atoms	C_7	Hept
Eight carbon atoms	C_8	Oct
Nine carbon atoms	C_9	Non
Ten carbon atoms	C_{10}	Dec

(ii) Suffix : The root word is followed by an appropriate suffix, which represents the nature of the bond in a carbon — carbon atom *i.e.*

C—C [single bond] ***–ane***

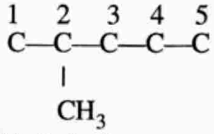
C=C [double bond] ***–ene***

C≡C [triple bond] ***–yne***

Table 12.5

Nature of bond	Suffix	General name	General formula
Single bond (C—C)	– ane	Alkane	C_nH_{2n+2}
Double bond (C=C)	– ene	Alkene	C_nH_{2n}
Triple bond (C≡C)	– yne	Alkyne	C_nH_{2n-2}
Group (R –)	– yl	Alkyl	C_nH_{2n+1}

(iii) Prefix : It denotes the substituent, alkyl or functional group and its position in the carbon chain.



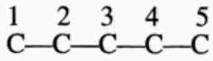
2-Methyl (because methyl is attached to second C)

Prefix also indicates additional substituent or functional group. *Di-, tri-, tetra-, are used for two, three and four groups of the same type respectively.*

In naming an organic compound, the following simple rules are followed :

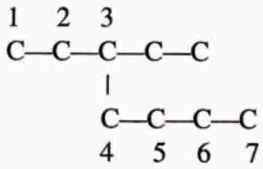
1. *Selection of carbon chains :* The longest continuous chain of 'C' atoms, known as ***parent chain,*** is selected. *The longest chain need not be straight. For example :*

(i)



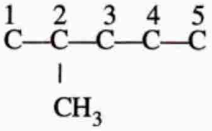
The longest chain is of 5 carbon atoms so root word is 'pent'.

(ii)



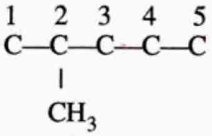
Here the longest chain is of 7 carbon atoms so root word is 'hept'. The remaining two carbon atoms (unnumbered) are for substituent.

2. The branch chains are considered to be *substituents,* and their positions are indicated by the number of carbon atoms to which they are attached. *For example :*

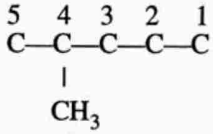


2-Methyl (because methyl is attached to second C)

3. The carbon atoms of the longest chain are numbered in such a way that the alkyl groups (substituents) get the smallest possible number. *For example :*



2-Methyl
Right



4-Methyl
Wrong

4. In case, any functional group is also present in the chain, then the carbon atoms are numbered in such a way that the functional group gets the smallest possible number.

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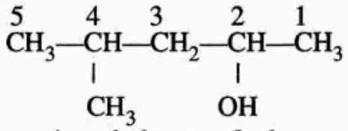
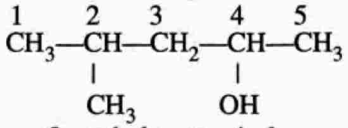
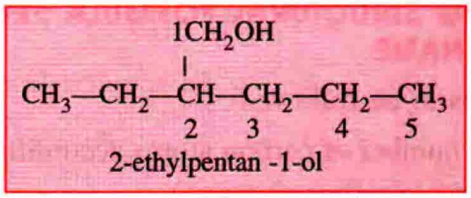
For example :



4-methylpentan-2-ol **Right**



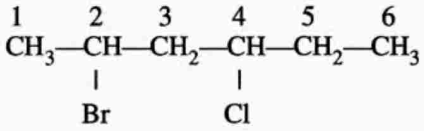
2-methylpentan-4-ol **Wrong**



2-ethylpentan -1-ol

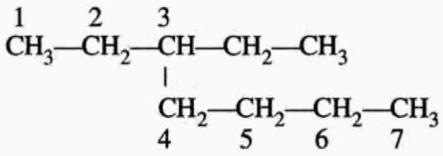
Note : In nomenclature, 'e' of alkane is replaced by suffix of functional group indicating the position.

5. In case, different types of substituents are attached to the chain, they are arranged and named alphabetically.



2-Bromo 4-chloro hexane

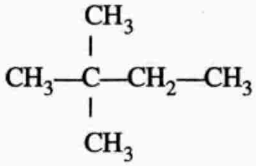
6. The position(s) of alkyl group(s) are indicated by writing the position and name of the alkyl group just before the name of parent hydrocarbon.



3-ethylheptane

7. Multiple alkyl groups are labelled with the Greek numerical prefixes such as 'di' for two, 'tri' for three, 'tetra' for four, 'penta' for five.

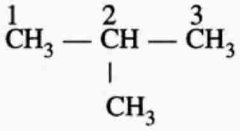
   If two alkyl groups are on the same carbon atom, the numeral is repeated.



2, 2-dimethyl butane

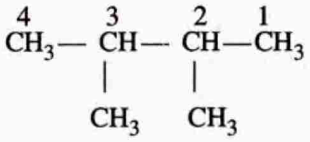
Examples of IUPAC names

Example 1 :



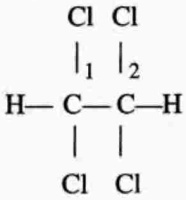
Word Root : Prop–
Suffix : –ane
Prefix : 2 methyl
IUPAC name : 2-methyl propane

Example 2 :



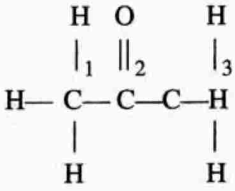
Word Root : But–
Suffix : –ane
Prefix : 2, 3-dimethyl
IUPAC name : 2, 3-dimethyl butane

Example 3 :



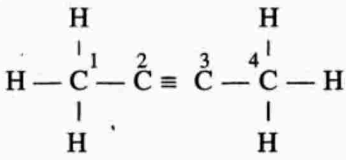
Word Root : Eth–
Suffix : –ane
Prefix : 1, 1, 2, 2-tetrachloro
IUPAC name : 1, 1, 2, 2-tetrachloro ethane

Example 4 :



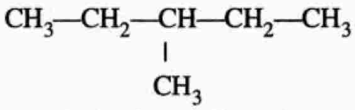
Word Root : Prop
Suffix : –ane
Prefix : 2-one
IUPAC name : Propan-2-one
Common name : Acetone

Example 5 :



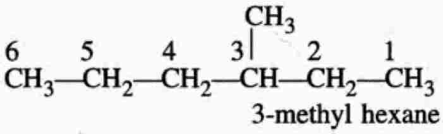
Word Root : But–
Suffix : –yne
Prefix : 2
IUPAC name : But-2-yne

Example 6 :



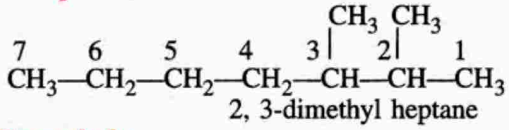
The longest chain is of 5 carbon atoms so root word is Pent. The nature of carbon chain is C – C [*single bond*] – Suffix -ane and the substituent is an (alkyl) methyl – CH_3 group. Thus name of compound is : ***3-methyl pentane***.

Example 7 :



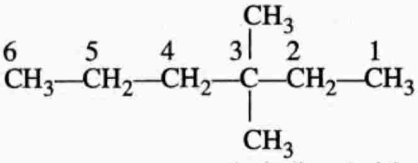
3-methyl hexane

Example 8 :



2, 3-dimethyl heptane

Example 9 :



3, 3-dimethyl hexane

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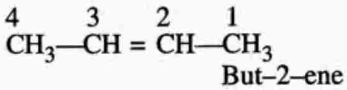
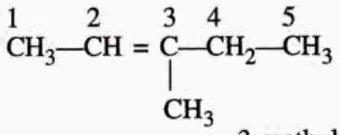
Example 10 :



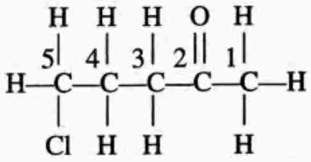
But–2–ene

Example 11 :



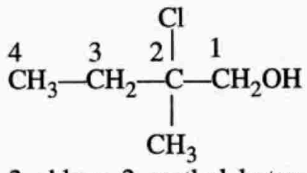
3-methylpent–2–ene

Example 12 :



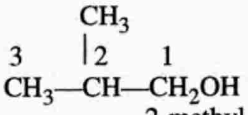
5–chloropentan–2–one

Example 13 :



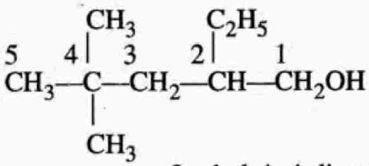
2–chloro–2–methyl butan–1–ol.

Example 14 :



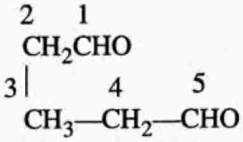
2-methyl propan–1–ol.

Example 15 :



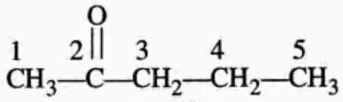
2-ethyl 4, 4-dimethyl pentan-1-ol.

Example 16 :



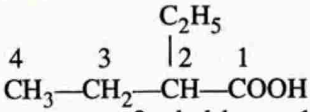
Pentan-1, 5-dial

Example 17 :



pentan-2-one.

Example 18 :



2-ethyl butan–1–oic acid

12.12 WRITING STRUCTURAL FORMULA FROM IUPAC NAME

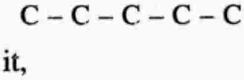
Following steps to be followed :

1. Write the number of carbon atoms according to the word root (carbon skeleton).
2. Number the carbon atoms from any end.
3. According to the suffix ane, ene or yne, the position of the bond is specified in the parent chain.
4. Attach the substituent or functional group at the mentioned carbon atom.
5. Satisfy the four valencies of carbon atom by attaching hydrogen atoms.

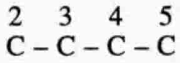
For example :

2–Bromo–4–methyl pent–2–ene

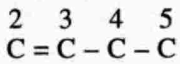
Step 1 : Write the carbon atoms of the parent chain,



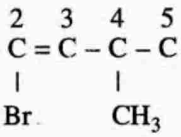
Step 2 : Number it,



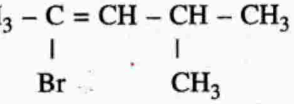
Step 3 : Locate the suffix,



Step 4 : Attach substituents,



Step 5 : Satisfy valencies by adding hydrogen atoms,



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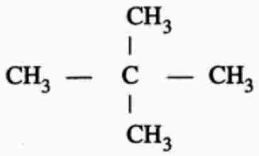
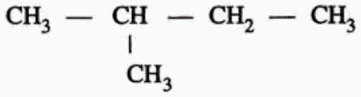
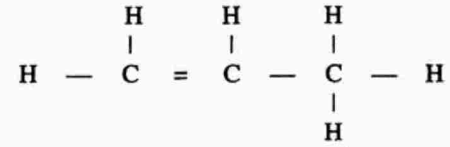
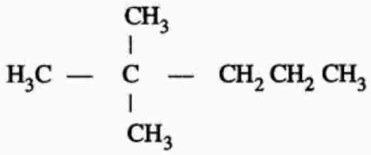
EXERCISE 12A

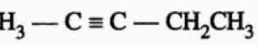
1. Write the IUPAC name of the following :

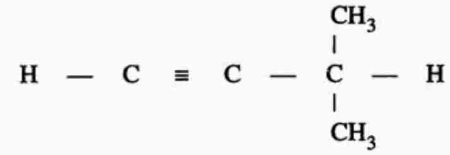
(a) 

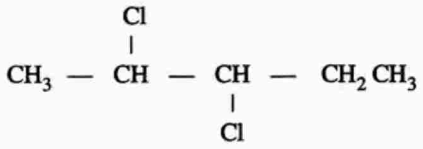
(b) 

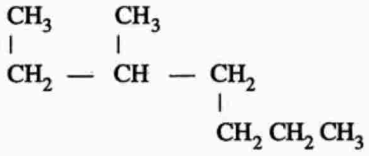
(c) 

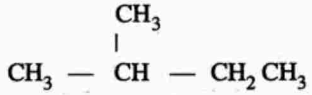
(d) 

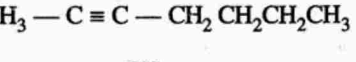
(e) 

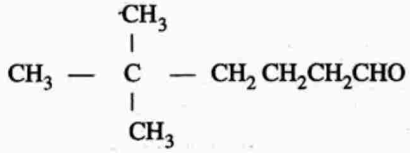
(f) 

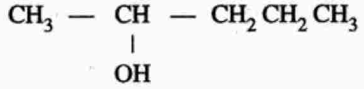
(g) 

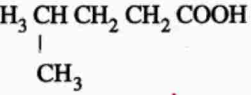
(h) 

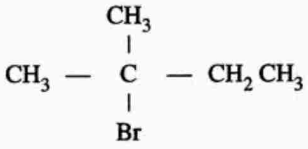
(i) 

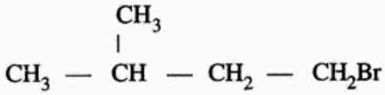
(j) 

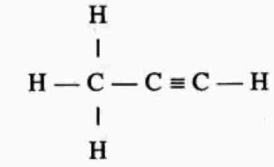
(k) 

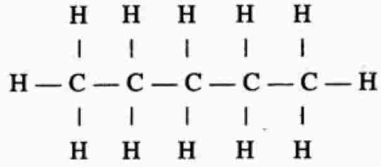
(l) 

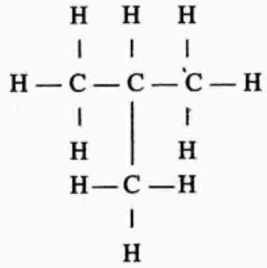
(m) 

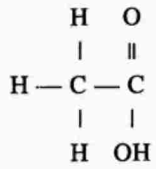
(n) 

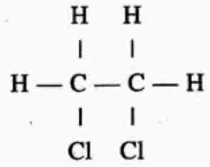
(o) 

(p) 

(q) 

(r) 

(s) 

(t) 

2. Write the structures of the following compounds:

(a) Prop-1-ene, (b) 2, 3-dimethylbutane,

(c) 2-methylpropane, (d) 3-hexene,

(e) Prop-1-yne, (f) 2-methylprop-1-ene,

(g) Alcohol with molecular formula $C_4H_{10}O$.

3. Choose the correct answer :

(a) C_5H_{11} is an

(i) alkane (ii) alkene

(iii) alkyne (iv) alkyl group

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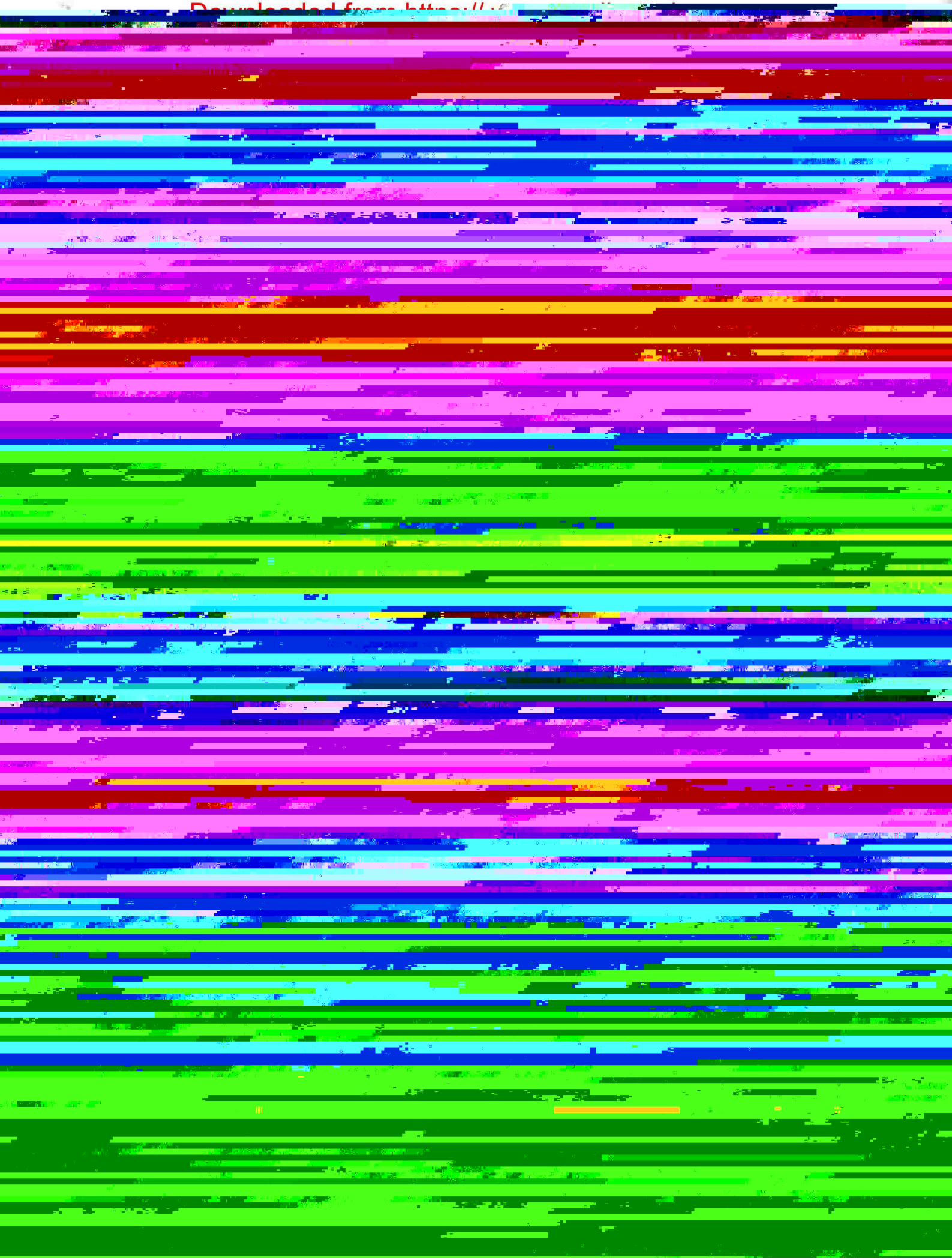

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12B HYDROCARBONS : ALKANES

12.13 ALKANES

Alkanes are hydrocarbons in which all the linkages between the carbon atoms are **single covalent bonds.**

These compounds are known as saturated hydrocarbons since all the four valencies of carbon are fully satisfied in the formula C_nH_{2n+2} by a **single bond (saturated by C – C and C – H single bonds)** (refer to Table 12.6).

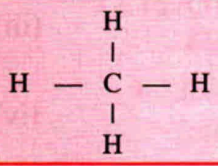
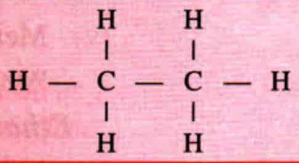
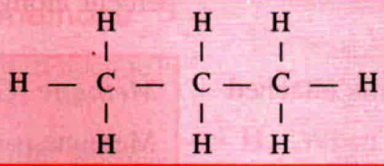
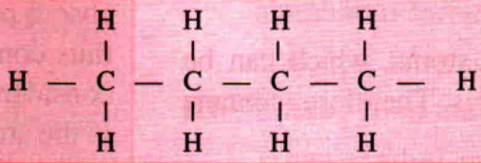
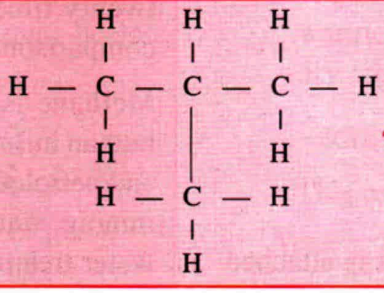
The simple alkanes share many properties in common. All enter into combustion reactions with oxygen to produce carbon dioxide and water vapour. In other words, many alkanes are flammable. This makes them good fuels. For example, methane is the principle component of natural gas, and butane is common lighter fluid.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

The combustion of methane

These hydrocarbons are relatively unreactive under ordinary conditions and so, they are also known as **paraffins** (*parum*–little, *affinis*–affinity).

Table 12.6 Homologous series of alkanes (General formula C_nH_{2n+2})

Name	Molecular formula	Structural formula	Abbreviated formula (Condensed formula)
Methane	CH_4		CH_4
Ethane	C_2H_6		$CH_3\,CH_3$
Propane	C_3H_8		$CH_3\,CH_2\,CH_3$
Normal butane	C_4H_{10}		$CH_3\,CH_2\,CH_2\,CH_3$
Isobutane (common name) or 2-methyl propane (IUPAC name)	$C_4\,H_{10}$		$CH_3\,CH\,CH_3$ $\quad\quad\quad

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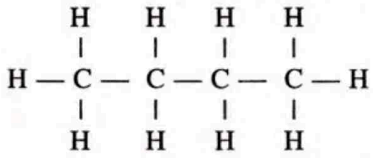
12.13.2 Isomerism in alkanes, *i.e.*, single bond hydrocarbons

Alkanes with *more than three carbon atoms form isomers.* The various isomers differ in the framework of the carbon chains. Therefore, they show chain isomerism.

Examples :

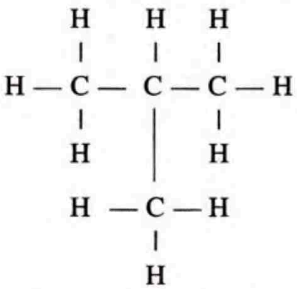
Isomers of butane (C_4H_{10})

Butane has four carbon atoms. These four carbon atoms can be arranged in two distinct ways. Therefore, there are two isomers of butane.



n–butane (C_4H_{10}), where '*n*' stands for *normal*

IUPAC Name : Butane



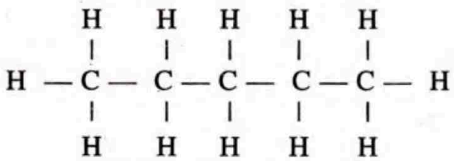
Iso-butane (C_4H_{10})

IUPAC Name : 2 Methyl propane

['iso', *i.e.,* at least one carbon atom is attached to three other carbon atoms].

Isomers of pentane (C_5H_{12})

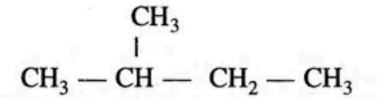
Pentane has five carbon atoms, which can be arranged in three different ways. Therefore, pentane can form three isomers.



IUPAC Name : Pentane

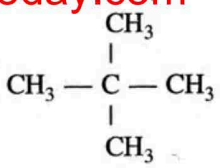
Common name : *n*-pentane

['*n*', means normal *i.e.* carbon atom is attached to maximum two carbon atoms].



IUPAC Name : 2–methyl butane

Common name : iso-pentane



IUPAC Name : 2, 2-dimethyl propane

Common name : neo-pentane

['neo', *i.e.,* at least one carbon atom is attached to 'four' other carbon atoms].

12.14 METHANE AND ETHANE

12.14.1 Occurrence of methane and ethane

Methane (Marsh gas)

(i) Marsh gas is formed at the bottom of marshes due to the fermentation of cellulose by a special type of bacteria.

(ii) It is also present in the air exhaled by animals whose food contains cellulose.

(iii) It is contained in intestinal gases and in the blood of animals and human beings.

(iv) Cavities in coal contain 90% methane. It is called **'fire-damp'**.

(v) Methane is produced also by dry distillation of wood, peat (coal).

***Ethane** : It occurs* to the extent of 10-20 percent alongwith methane.

Methane (Green house gas)

Methane is a primary constituent of natural gas. It absorb outgoing heat radiation from the earth, and thus contribute to the *green house effect* and so considered as a green house gas. Methane remains in the atmosphere for approximately 10 years. It is twenty times more effective in trapping heat in comparison to carbondioxide.

Methane is emitted from variety of natural and human influenced sources like landfills, natural gas and petroleum systems, agricultural activities, coal mining, stationary and mobile combustion, waste water treatment and certain industrial process.

Methane is an important energy source so efforts to prevent or utilize methane emissions can provide significant economical and environmental benefits.

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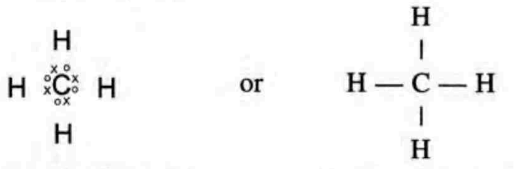
12.14.2 Structure of methane

The methane molecule is three-dimensional. The four outer electrons in the carbon atom move as far apart from each other as possible, because electrons repel each other (remember that particles with similar charge always repel each other). It is a fact that the electrons will have the greatest distance between themselves when the carbon-hydrogen bonds are in the shape of a **tetrahedron**. *A tetrahedron ('tetra' is a Greek word meaning "four") is pyramid-shaped.*



Electron dot formula Structural formula

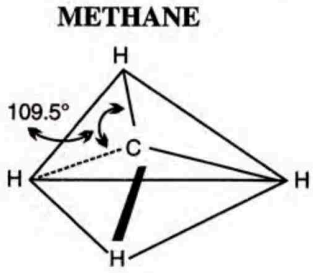


Fig. 12.1 The tetrahedral arrangement of four atoms of hydrogen around a carbon atom in a molecule of methane

12.14.3 Laboratory preparation of methane

Reactants : *Sodium ethanoate* (sodium acetate) and *soda lime.*

(Soda lime is in the form of a white porous mass or granules. It is a mixture of sodium hydroxide and calcium oxide).

Procedure : A mixture of sodium ethanoate (sodium acetate) and soda lime is taken in a hard glass test tube (as shown in Fig. 12.2), and heated with a bunsen flame.

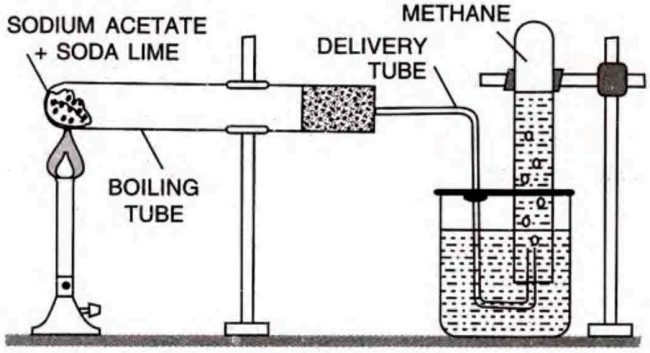


Fig. 12.2 Preparation of methane

Reaction :*

$$CH_3COONa + NaOH \xrightarrow[300°C]{CaO} Na_2CO_3 + CH_4\uparrow$$

Collection : The gas evolved is collected by downward displacement of water since it is slightly soluble in water and is lighter than air.

12.14.4 Laboratory preparation of ethane

Reactants : Sodium propionate and soda lime.

Procedure : A mixture of sodium propionate and soda lime is taken in a boiling tube and heated with a bunsen flame (apparatus similar to Fig. 12.2).

Reaction :*

$$C_2H_5COONa + NaOH \xrightarrow[300°C]{CaO} Na_2CO_3 + C_2H_6\uparrow$$

Collection : The gas evolved is collected by downward displacement of water.

12.15 OTHER METHODS OF PREPARATION OF METHANE AND ETHANE

I. ***Iodomethane*** (methyl iodide) or ***bromoethane*** (ethyl bromide) is reduced by nascent hydrogen at ordinary room temperature.

$$\underset{\text{Iodomethane}}{CH_3I} + 2[H] \rightarrow CH_4 + HI$$

$$\underset{\text{Bromoethane}}{C_2H_5Br} + 2[H] \rightarrow \underset{\text{Ethane}}{C_2H_6} + HBr$$

Nascent hydrogen is produced by the action of Zn powder and dil. HCl.

II. When water is added at room temperature to aluminium carbide, methane is prepared.

$$Al_4C_3 + 12H_2O \rightarrow 3CH_4\uparrow + 4Al(OH)_3\downarrow$$

III. ***Ethane from alkyl halides :***

When methyl iodide or methyl bromide is warmed with sodium metal in the presence of dry ether, ethane is produced.

$$2CH_3I + 2Na \xrightarrow[ether]{dry} H_3C - CH_3 + 2NaI$$

This reaction is referred to as ***Wurtz reaction***.

Note : *Methane cannot be prepared by this method. This method is not suitable for the preparation of alkanes with odd number of carbon atoms.*

* This reaction is known as **decorboxylation** reaction as carbon dioxide molecule is eleminated from a carboxylic acid. CO_2 is eleminated as carbonate.

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12.15.1 Physical properties

Methane

(i) It is a colourless and an odourless gas.

(ii) Its melting point is –183 °C and its boiling point is –162 °C.

(iii) It is negligibly soluble in water and soluble in organic solvents.

Ethane

(i) It is a colourless, odourless, tasteless and non-poisonous gas.

(ii) Its b.p. is –89 °C and its m.p. is –172 °C.

(iii) It is sparingly soluble in water but wholly soluble in organic solvents like alcohols, acetone and ether.

12.15.2 Chemical properties

Methane and ethane are saturated hydrocarbons. *They do not react with bases such as sodium hydroxide, or with oxidising agents such as potassium permanganate, or with reducing agents such as sodium metal.*

They undergo substitution reactions as well as combustion reaction.

1. Substitution reaction

(i) *Reaction with halogens*

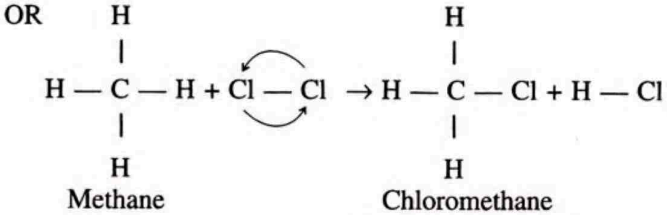
Alkanes react with **chlorine, bromine** or **iodine** in the presence of sunlight or ultraviolet light to give halogen substituted products that are known as *alkyl halides*.

Alkyl halides contain one or more halogen atoms.

Methane reacts with chlorine in diffused sunlight, or when heated to 600 K, to give chloromethane.

$$CH_4 + Cl_2 \xrightarrow[\text{or } 600K]{\text{diffused sunlight}} CH_3Cl + HCl$$

OR



In this reaction, CH_3Cl is formed by the substitution of hydrogen atom of a methane molecule, by a chlorine atom. This type of reaction is known as **substitution reaction** and the product formed is known as **substitution product**.

The product formed (chloromethane) further reacts with chlorine, substituting one more hydrogen atom and producing dichloromethane. This further produces trichloromethane and finally, tetrachloromethane.

$$\underset{\text{Chloromethane}}{CH_3Cl} + Cl_2 \rightarrow \underset{\text{Dichloromethane}}{CH_2Cl_2} + HCl$$

$$\underset{\text{Dichloromethane}}{CH_2Cl_2} + Cl_2 \rightarrow \underset{\substack{\text{Trichloromethane}\\(\text{Chloroform})}}{CHCl_3} + HCl$$

$$\underset{\text{Trichloromethane}}{CHCl_3} + Cl_2 \rightarrow \underset{\substack{\text{Tetrachloromethane}\\(\text{Carbon tetrachloride})}}{CCl_4} + HCl$$

The substitution products can be separated by fractional distillation because of the appreciable differences in their respective boiling points.

Ethane also reacts with chlorine in the same way, producing successively chloroethane, dichloroethane, trichloroethane, tetrachloroethane, pentachloroethane and hexachloroethane.

Alkane	+	Halogen →	Substituted product	+	Acid
C_2H_6 (ethane)	+	Cl_2 →	C_2H_5Cl (monochloroethane)	+	HCl
C_2H_5Cl (chloroethane)	+	Cl_2 →	$C_2H_4Cl_2$ (dichloroethane)	+	HCl
$C_2H_4Cl_2$ (dichloroethane)	+	Cl_2 →	$C_2H_3Cl_3$ (trichloroethane)	+	HCl
$C_2H_3Cl_3$ (trichloroethane)	+	Cl_2 →	$C_2H_2Cl_4$ (tetrachloroethane)	+	HCl
$C_2H_2Cl_4$ (tetrachloroethane)	+	Cl_2 →	C_2HCl_5 (pentachloroethane)	+	HCl
C_2HCl_5 (pentachloroethane)	+	Cl_2 →	C_2Cl_6 (hexachloroethane)	+	HCl

Note : *All alkanes react with chlorine, bromine and iodine in a similar manner, producing the corresponding substituted products.*

(ii) *Reaction with oxygen*

Methane and *ethane* burn in air with a *bluish non-sooty flame* to form carbon dioxide and

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water vapour. A large amount of heat is also given out, so alkanes are good gaseous fuels.

Methane :	CH_4	+	$2O_2$	→	CO_2 +	$2H_2O$
Ethane :	$2C_2H_6$	+	$7O_2$	→	$4CO_2$ +	$6H_2O$

During combustion, due to the low carbon content in methane and ethane, they get fully oxidised to carbon dioxide, and *burn with a non-sooty flame.*

Insufficient supply of air

When alkanes burn in an *insufficient supply of air* (oxygen), they form carbon monoxide and water.

$$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$$

$$2C_2H_6 + 5O_2 \rightarrow 4CO + 6H_2O$$

$CH_4 + O_2 \rightarrow \underset{\text{(soot)}}{C} + 2H_2O$ **(with still less oxygen).**

Soot is used in the manufacture of printing inks and tyres.

2. Decomposition of alkanes (Cracking or pyrolysis)

The decomposition of a compound by heat in the absence of air is called ***Pyrolysis***. When pyrolysis occurs in alkanes, the process is termed **cracking**.

The alkanes, on heating under high temperature or in the presence of a catalyst in absence of air are broken down into lower alkanes, alkenes and hydrogen.

$$\underset{\text{Methane}}{2CH_4} \xrightarrow{1500°C} \underset{\text{Ethyne}}{HC \equiv CH} + 3H_2$$

$$\underset{\text{Ethane}}{H_3C - CH_3} \xrightarrow[Al_2O_3]{500°C} \underset{\text{Ethene}}{H_2C = CH_2} + H_2$$

$$\underset{\text{pentane}}{C_5H_{12}} \xrightarrow[\text{Alumina or silica}]{400–600°C} \underset{\text{propane}}{C_3H_8} + \underset{\text{ethene}}{C_2H_4}$$

$$\underset{\text{heptane}}{C_7H_{16}} \xrightarrow[\text{Alumina or silica}]{400–600°C} \underset{\text{butane}}{C_4H_{10}} + \underset{\text{propene}}{C_3H_6}$$

3. Catalytic oxidation of alkanes

On controlled oxidation or catalytic oxidation, alkanes give alcohols or aldehydes or carboxylic acids, *depending upon the reaction conditions.*

(i) When a mixture of methane and oxygen in the ratio 9:1 by volume is compressed to about 120 atm pressure and passed over copper tubes at 475 K, **methyl alcohol** is formed.

$$2CH_4 + O_2 \xrightarrow[120\,atm.]{475\,K\,Cu\ tube} 2CH_3OH$$

(ii) When a mixture of methane and oxygen is passed through heated *molybdenum oxide* (MoO), the mixture is oxidised to **formaldehyde**.

$$CH_4 + O_2 \xrightarrow{MoO} HCHO + H_2O$$

(iii) When a manganese based catalyst is used at 100°C, methane can be oxidised to formic acid.

$$2CH_4 + 3O_2 \xrightarrow{Mn\ compound} 2HCOOH + 2H_2O$$

Similarly, ethane oxidises to either ethyl alcohol or acetaldehyde or acetic acid.

$$2C_2H_6 + O_2 \xrightarrow[Cu\ tubes\ 475K]{120\,atm} \underset{\text{Ethyl alcohol}}{2C_2H_5OH}$$

$$C_2H_6 + O_2 \xrightarrow{MoO} \underset{\text{Acetaldehyde}}{CH_3CHO} + H_2O$$

$$C_2H_5OH + O_2 \xrightarrow[300°C]{Pt} \underset{\text{Acetic acid}}{CH_3COOH} + H_2O$$

4. Slow combustion

Methane or ethane can be treated with oxidising agents like acidified $K_2Cr_2O_7$ or acidified $KMnO_4$ at high pressure and at comparatively low temperature.

$$\text{Alkane} \xrightarrow[K_2Cr_2O_7]{[O]} \text{Alcohol} \xrightarrow[K_2Cr_2O_7]{[O]} \text{Aldehyde} \xrightarrow[K_2Cr_2O_7]{[O]} \underset{\text{acid}}{\text{Carboxylic}} \rightarrow \underset{\text{water vapour}}{\text{Carbon dioxide and}}$$

or

$$\underset{\text{Methane}}{CH_4} \xrightarrow{[O]} \underset{\text{alcohol}}{CH_3OH} \xrightarrow{[O]} \underset{\text{dehyde}}{\text{Formal-}}HCHO \xrightarrow{[O]} \underset{\text{Formic acid}}{HCOOH} \xrightarrow{[O]} CO_2 + H_2O$$

([O] represents contribution of the oxidising agent)

Other alkanes also form similar products when reacts with oxidising agent.

12.15.3 Uses

Methane

(i) Methane is a source of carbon monoxide and hydrogen.

(ii) It is used in the preparation of useful compounds like ethyne (acetylene), methanal

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(formal-dehyde), methanol, chloro-methane and tetrachloro-methane (carbon tetrachloride).

Earlier chloroform was used for anaesthesia. Since its higher dose can kill a person therefore, now-a-days HALOTHANE ($CF_3CHBrCl$) is used for anaesthetic purposes.

(iii) It is employed as a domestic fuel.

Ethane

(i) Ethane is used in the preparation of ethene, ethanol, ethanal (acetaldehyde) and ethanoic acid (acetic acid).

(ii) It forms ethyl chloride, which is used to make tetraethyllead. 1, 1, 1-trichloroethane is one solvent that is used a lot, in dry cleaning.

(iii) Ethane is also a good fuel.

EXERCISE 12B

1. State the sources of Alkanes.
2. Methane is a green house gas. Comment.
3. Give the general formula of alkanes.
4. Draw the structures of isomers of :
   (a) butane, (b) pentane.
   Write the IUPAC and common names of these isomers.
5. Write the :
   (a) molecular formula,
   (b) electron dot formula and
   (c) structural formula of methane and ethane.
6. How is :
   (a) methane and
   (b) ethane prepared in the laboratory ?
7. How are methane and ethane prepared from methyl iodide and ethyl bromide ?
8. What is a substitution reaction ?
   Give the reaction of chlorine with ethane and name the product formed.
9. Name the compounds formed when methane burns in :
   (a) sufficient air,
   (b) insufficient air.
   Give a balanced equation.
10. Write the names and the formula of the products formed when :
    (a) methane, (b) ethane,
    reacts with : (i) chlorine (ii) bromine
    Write the chemical equations.
11. Name the compound prepared from :
    (a) sodium propionate, (b) methyl iodide and
    (c) ethyl bromide.
    Write a balanced equation for the same.
12. What is pyrolysis or cracking ? Explain with example.
13. Convert :
    (a) Methane into chloroform,
    (b) Sodium acetate into methane,
    (c) Methyl iodide into ethane,
    (d) Aluminium carbide into methane.
14. Give three uses of :
    (a) methane, (b) ethane.
15. Under what conditions does ethane get converted to :
    (a) ethyl alcohol, (b) acetaldehyde,
    (c) acetic acid.
16. Give the inter-relationship of methane, methyl alcohol, formaldehyde and formic acid with conditions.

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12C HYDROCARBONS : ALKENES

12.16 ALKENES

Alkenes are also called ***olefins*** (oil-forming) because the lower members of alkenes form oily products when they are treated with chlorine or bromine. Alkenes form a homologous series having the **general formula C_nH_{2n}** (refer to Table 12.7).

Table 12.7 : Homologous series of alkenes, general formula C_nH_{2n}

Molecular formula	*Condensed formula*	*Structural formula*	*Trivial name (Common name)*	*IUPAC name*
C_2H_4	$CH_2 = CH_2$	H H | | H—C = C—H	Ethylene	Ethene
C_3H_6	$CH_3 - CH = CH_2$	H H H | | | H—C—C = C—H | H	Propylene	Propene
C_4H_8	$CH_3–CH_2–CH = CH_2$	H H H H | | | | H—C—C—C = C—H | | H H	Butylene	Butene

Alkenes are unsaturated aliphatic hydrocarbons that contain one double bond.

Occurrence : Since alkenes are reactive, they seldom occur free in nature. Lower alkenes occur in minute quantities in coal gas. They are produced on a large scale by the cracking of petroleum.

Isomers in alkenes

Alkenes with 4 or more than 4 carbon atoms can form isomers.

For example : Butene has three isomers :

(i) $CH_3CH_2CH = CH_2$
But-1-ene

(ii) $CH_3CH = CHCH_3$
But-2-ene

(iii) $CH_2 = C(CH_3) - CH_3$
2–methyl propene

Types of isomerism shown by Alkenes

(i) Chain Isomerism :
Example
$CH_3CH_2CH = CH_2$ — 1-butene
$CH_3 — C(CH_3) = CH_2$ — 2-methyl propene

(ii) Position isomerism :
Example
$CH_3CH_2CH = CH_2$ — 1-butene
$CH_3CH = CHCH_3$ — 2-butene

12.16.1 Ethene (Ethylene) C_2H_4

It is the first member of the alkene series. It is present in natural gas. In small amounts, it occurs as a plant hormone and is responsible for the **ripening of fruits**.

12.16.2 Structure of ethene

Each carbon atom of ethene is attached to two hydrogen atoms by single covalent bonds and to another carbon atom by a double covalent bond. There are four C–H single covalent bonds and one C=C double covalent bond. It is a **planar** (flat) **molecule**. All bond angles (H–C–H and H–C=C) *are* of 120°, as shown in Fig. 13.3.

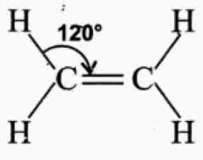

Fig. 12.3 Structure of ethylene

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12.16.3 Preparation of ethene (ethylene)

(i) Dehydration of ethyl alcohol

Reactants : Ethanol and conc. sulphuric acid.

Procedure : Arrange the apparatus as shown in Fig. 12.4. Take one part of ethyl alcohol in a flask. Add two parts of concentrated sulphuric acid and heat gradually to about 170°C. Ethylene is evolved. The gas is passed through NaOH solution which removes CO_2 and SO_2 (impurities formed by oxidation of the acid). Ethylene is collected by downward displacement of water (*add some aluminium sulphate to the water in order to avoid frothing*).

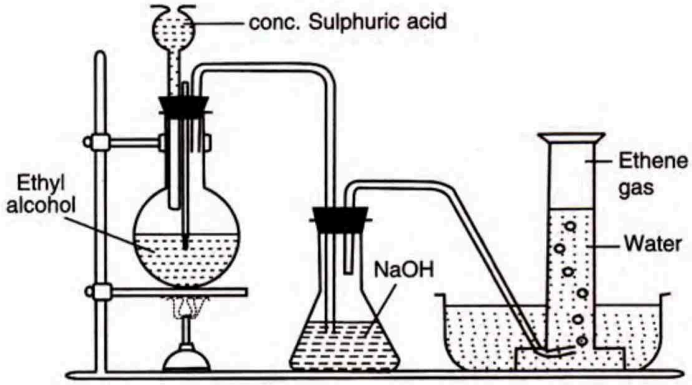

Fig. 12.4 Preparation of ethene from ethyl alcohol

Reaction :

$$\underset{\text{(Ethanol)}}{CH_3CH_2OH} + H_2SO_4 \rightarrow \underset{\text{(Ethyl hydrogen sulphate)}}{CH_3CH_2HSO_4} + H_2O$$

$$CH_3CH_2HSO_4 \xrightarrow[170°C]{\text{excess } H_2SO_4} \underset{\text{Ethene}}{CH_2 = CH_2} + H_2SO_4$$

Or

$$C_2H_5OH \xrightarrow[170°C]{\text{Conc. } H_2SO_4} C_2H_4 + H_2O$$

Collection : The gas is collected by downward displacement of water because :

(iii) Dehydrohalogenation :

$$C_2H_5Cl + KOH \rightarrow C_2H_4 + KCl + H_2O$$
(alc. hot and conc.)

$$C_2H_5Br + KOH \rightarrow C_2H_4 + KBr + H_2O$$
(alcoholic)
(hot and conc.)

$$C_2H_5I + KOH \rightarrow C_2H_4 + KI + H_2O$$
(alcoholic)
(hot and conc.)

12.16.4 Properties of Alkenes

A. Physical properties

(i) Ethene is a colourless and inflammable gas with a peculiar odour (faint sweetish smell).

(ii) Its boiling point is –102°C and melting point is –169°C.

(iii) It is sparingly soluble in water but highly soluble in organic solvents like alcohol, ether and chloroform.

(iv) It produces an anaesthetic effect upon inhalation.

B. Chemical properties

1. Addition Reactions

(i) Addition of hydrogen (hydrogenation)

When ethene and hydrogen are passed over finely divided catalysts such as **platinum or palladium** at ordinary temperature or **nickel at 200°C**, the two atoms of hydrogen molecule are added to the unsaturated molecule, which thus becomes a saturated one.

$$\underset{\text{Ethene}}{C_2H_4} + H_2 \xrightarrow[Ni]{200°C} \underset{\text{Ethane}}{C_2H_6}.$$

(ii) Addition of halogens (halogenation)

Reaction with fluorine is explosive, hence the compound formed is not stable.

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$$\underset{\text{Ethene}}{CH_2 = CH_2} + Br_2 \rightarrow \underset{\substack{| \\ Br}}{CH_2} - \underset{\substack{| \\ Br}}{CH_2}$$

1, 2-dibromo ethane
(ethylene bromide)

Bromine solution in CCl_4 has an orange colour. When added dropwise to ethene, the orange colour of bromine disappears due to the formation of the colourless ethylene bromide.

$$C_2H_4 + I_2 \rightarrow C_2H_4I_2$$

1, 2-di-iodoethane
(ethylene iodide)

Reaction with iodine is very slow and ethylene iodide formed is unstable. It eliminates the iodine readily and regenerates the ethene.

(iii) Addition of water (hydration)

A water molecule gets added to alkenes in the presence of acids (e.g., sulphuric acid) to form alcohols.

$$CH_2 = CH_2 + H_2O \xrightarrow{H^+} C_2H_5OH.$$

(iv) Addition of HCl

When ethene gas is treated with HCl, chloroethane is formed.

$$\underset{\text{Ethene}}{CH_2 = CH_2} + HCl\ (aq) \rightarrow \underset{\substack{\text{Ethyl chloride} \\ \text{(chloroethane)}}}{CH_3 - CH_2Cl}.$$

If chloroethane is treated with aqueous potassium hydroxide, ethanol is formed.

$$CH_3CH_2Cl + KOH \rightarrow CH_3CH_2OH + KCl.$$

2. Reaction with ozone (ozonolysis) : When a stream of ozone is passed through a solution of ethene in ether, ethene ozonide is formed.

$$\underset{\text{Ethene}}{H_2C = CH_2} + \underset{\text{Ozone}}{O_3} \rightarrow H_2C \underset{\text{Ethene ozonide}}{\overset{O}{\underset{O}{|}}} CH_2.$$

3. Polymerization : When two or more molecules of the same compound associate to form a bigger molecule, the reaction is called polymerisation.

Ethene polymerises to produce polyethene

$$\underset{\text{(ethene)}}{n\ H_2C = CH_2} \xrightarrow[\text{high pressure catalyst}]{\text{high temperature}} \underset{\text{(Polyethene)}}{[H_2C - CH_2]_n}.$$

4. Oxidation : Ethene is oxidised with alkaline $KMnO_4$ at room temperature.

$$\underset{\text{Ethene}}{H_2C = CH_2} + H - O - H + [O] \rightarrow \underset{\substack{| \quad | \\ OH\ OH \\ \text{1, 2-Ethanediol}}}{H_2C - CH_2}.$$

Cold alkaline $KMnO_4$ solution [oxidising agent] (Ethylene glycol)

The purple colour of $KMnO_4$ decolourises.

5. Combustion of ethene : Ethene burns in air with a *sooty flame* producing a large amount of heat. However, if the supply of air is in excess, it burns with a *pale blue flame* to produce carbon dioxide, water and a large amount of heat.

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O + \text{Heat}.$$

12.16.5 Uses of ethene

(i) Ethene is used in making **polythene.**

(ii) It is converted into ethanol which is used as a starting material for other products, mainly cosmetics and toiletry preparations.

(iii) It is used for **ripening of fruits.**

(iv) It is used in making **epoxyethane** (used in the manufacture of detergents).

(v) It is used for producing **oxy-ethylene flame,** which is used for **cutting and welding of metals.**

(vi) It is used in the manufacture of synthetic chemicals.

EXERCISE 12C

1. Write : (a) molecular formula, (b) electron dot formula and (c) structural formula of ethene (ethylene).
2. The molecules of alkene family are represented by a general formula C_nH_{2n}. Answer the following:
   (a) What do n and $2n$ signify ?
   (b) What is the name of alkene when $n = 4$?
   (c) What is the molecular formula of alkene when $n = 4$?
   (d) What is the molecular formula of the alkene if there are ten H atoms in it ?
   (e) What is the structural formula of the third member of the alkene family ?

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(f) Write the molecular formula of lower and higher homologous of an alkene which contains four carbon atoms.

3. Draw the structures of butene and write IUPAC names.

4. Give a balanced equation for the lab. preparation of ethylene. How is the gas collected ?

5. How is ethene prepared by :
   (a) dehydrohalogenation reaction ?
   (b) dehydration reaction ?
   Give equations and name the products formed.

6. (a) Ethylene when reacts with halogens (chlorine and bromine) form saturated products. Name them and write balanced equations.
   (b) Give the conditions and the main product formed by hydrogenation of ethylene.

7. How is ethanol converted into ethene using
   (a) solid dehydrating agent
   (b) hot conc. H_2SO_4 ? Give only balanced equations.

8. Write the following properties of ethene :
   (a) Physical state, (b) Odour,
   (c) Density as compared to air, (d) Solubility.

9. How would you convert :
   (a) ethene into 1, 2-dibromoethane ?
   (b) ethene into ethyl bromide ?

10. Give balanced equations when :
   (a) ethene is burnt in excess of oxygen.
   (b) ethene reacts with chlorine.
   (c) ethene combines with hydrogen chloride.
   (d) a mixture of ethene and hydrogen is passed over nickel at 200°C.

11. Give the formula and names of A, B, C and D in the following equations :
   (a) $CH_4 \xrightarrow[-HCl]{Cl_2} A \xrightarrow[-HCl]{Cl_2} B \xrightarrow[-HCl]{Cl_2} C \xrightarrow[-HCl]{Cl_2} D$
   (b) $C_2H_2 \xrightarrow{H_2} A \xrightarrow{H_2} B \xrightarrow[-HBr]{Br_2} C \xrightarrow[-HBr]{Br_2} D$
   (c) $C_2H_4 + Cl_2 \rightarrow A$
   (d) $C_2H_4 + B \xrightarrow[Ni]{200°C} C_2H_6$

12. Write the name and formula of the product formed in each case below :
   (a) $C_2H_4 + Cl_2 \longrightarrow$ .....
   (b) $C_2H_5Br + KOH\ (alc.) \xrightarrow{\Delta}$ .....
   (c) $H_2C = CH_2 \xrightarrow{alk.KMnO_4}$ .....
   (d) $H_2C = CH_2 + HBr \longrightarrow$ .....
   (e) $H_2C = CH_2 + O_3 \longrightarrow$ .....

13. What do you observe when ethylene is passed through alkaline $KMnO_4$ solution ?

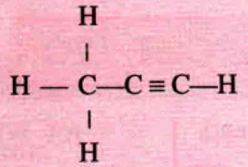
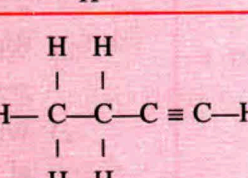
14. Name three compounds formed by ethylene and give the use of these compounds.

12D HYDROCARBONS : ALKYNES

12.17 ALKYNES

The aliphatic hydrocarbons that contain a triple bond (–C ≡ C–), *i.e.*, **acetylenic bond,** between two carbon atoms are known as alkynes. They are unsaturated compounds, due to the triple bond between two carbon atoms. They form a homologous series, with the general formula $\mathbf{C_nH_{2n-2}}$ (Refer to Table 12.8).

Table 12.8 : Homologous series of alkynes (General formula : C_nH_{2n-2})

Molecular formula	Condensed formula	Structural formula	Trivial name	IUPAC name of the compound
C_2H_2	CH ≡ CH	H — C ≡ C–H	Acetylene	Ethyne
C_3H_4	$CH_3 - C \equiv CH$		Allylene	Propyne
C_4H_6	$CH_3 - CH_2\ C \equiv CH$		Crotonylene	Butyne

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Sources of alkynes : Natural gas and petroleum.

Isomers in alkynes

Alkynes with four or more than four carbon atoms can form isomers. Alkynes show position isomerism as well as chain isomerism. *For example,*

Butyne

It shows position isomerism.

(i) $CH_3 - CH_2 - C \equiv CH$
IUPAC Name : But–1–yne

(ii) $CH_3 - C \equiv C - CH_3$
IUPAC Name : But–2–yne

12.17.1 Ethyne

(Common name : acetylene)

Molecular formula : C_2H_2.

Ethyne (Acetylene) is the first member of the alkyne series. Ethyne being an unsaturated hydrocarbon, is not found in free state.

Sources : Traces of ethyne are present in coal gas and in gases obtained by the decomposition of certain complex organic compounds. It is obtained by cracking of alkanes from various fractions of petroleum.

12.17.2 Structure of ethyne (acetylene)

Each carbon atom in acetylene is attached to one hydrogen atom by single covalent bond and to another carbon atom by a triple covalent bond.

Electron dot formula

$H - C \equiv C - H$
Structural formula

The shape of ethyne molecule is **linear.**

12.17.3 Laboratory preparation of ethyne (acetylene)

Reactants : Water and calcium carbide.

Procedure : Arrange the apparatus as shown in Fig. 12.5. Take a few pieces of calcium carbide in a conical flask. Add a few drops of water through the thistle funnel. Calcium carbide reacts with water to produce colourless gas with ether like odour (acetylene gas). The reaction is exothermic.

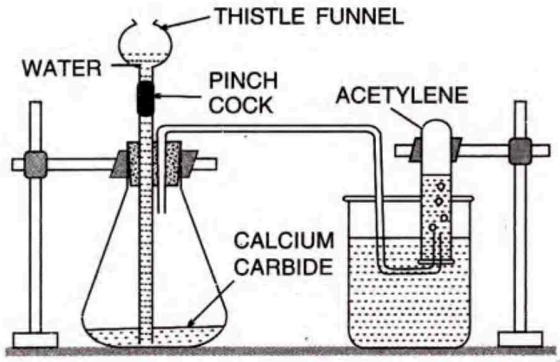

Fig. 12.5 Lab. preparation of (acetylene) ethyne.

Reaction :

$$\underset{\text{Calcium carbide}}{CaC_2} + 2H_2O \rightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2} + \underset{\text{acetylene}}{C_2H_2}\uparrow$$

Collection : The gas is collected by downward displacement of water, since it is insoluble in water.

Purification : Impurities like phosphine, hydrogen sulphide, ammonia and arsenic are formed along with acetylene due to the contamination of calcium sulphide and calcium phosphide in calcium carbide. On passing through water, all impurities except *phosphine* are absorbed. Phosphine is *absorbed by passing through acidified copper sulphate solution or acidified potassium dichromate.*

Note : Flask should not contain air as acetylene forms explosive mixture with air. The air of the flask is displaced with oil gas.

Preparation from 1, 2-dibromoethane (ethylene dibromide).

When 1, 2-dibromoethane (ethylene dibromide) is boiled with alcoholic potassium hydroxide, ethyne is formed.

$$\underset{\text{1, 2-dibromoethane}}{\begin{matrix}CH_2Br\\|\\CH_2Br\end{matrix}} + 2KOH \xrightarrow[\text{boiling}]{200^\circ C} \underset{\text{(Ethyne)}}{\begin{matrix}CH\\\|\\CH\end{matrix}} + 2KBr + 2H_2O$$

12.17.4 Properties of acetylene (ethyne)

A. *Physical properties*

(i) Acetylene is a colourless gas with an ether-like odour when it is pure *(the one prepared from calcium carbide has garlic odour due to the presence of traces of phosphine and hydrogen sulphide).*

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(ii) It is negligibly soluble in water but highly soluble in organic solvents like acetone and alcohol.

(iii) It is lighter than air (V.D. = 13).

(iv) It liquefies at –84 °C.

(v) Its boiling point is –75 °C.

B. *Chemical properties*

Ethyne is a highly reactive compound due to the presence of triple bond between its two carbon atoms.

1. Oxidation of ethyne (combustion)

Ethyne contains a greater proportion of carbon content than does alkanes (ethane) and alkenes (ethene). Since all the carbon particles do not get oxidised completely, it burns with a **sooty flame**.

Ethyne burns in excess air with a *brilliant white flame* to produce carbon dioxide, water vapours and a large amount of heat.

$$2CH \equiv CH + 5O_2 \rightarrow 4CO_2 + 2H_2O + \text{Heat}$$

2. Addition reactions

Alkynes are unsaturated compounds, so they are associated with addition reactions, since triple bonds break up easily. The breaking up of a triple bond gives a product that is still unsaturated (double bond). Further breaking up of the double bond gives a saturated product.

(i) *Addition of hydrogen (catalytic hydrogenation)*

In the presence of nickel, platinum or palladium, ethyne first takes up two atoms of hydrogen to change to ethene, which further takes up two atoms of hydrogen to give ethane.

First stage of reaction

$$\underset{\text{Ethyne}}{C_2H_2} + H_2 \xrightarrow{Ni} \underset{\text{Ethene}}{C_2H_4}$$

Second stage

$$\underset{\text{Ethene}}{C_2H_4} + H_2 \xrightarrow{Ni} \underset{\text{Ethane}}{C_2H_6}$$

Therefore,

$$\underset{\text{Ethyne}}{HC \equiv CH} \xrightarrow[Ni]{+H_2} \underset{\text{Ethene}}{CH_2 = CH_2} \xrightarrow[Ni]{+H_2} \underset{\text{Ethane}}{CH_3 - CH_3}$$

(ii) *Addition of halogen*

Reaction with chlorine : Ethyne in an inert solvent of carbon tetra chloride [CCl_4] adds chlorine to change into an ethene derivative with carbon-carbon double bond, and then to an ethane derivative with carbon-carbon single bond.

$$\underset{\substack{\text{Acetylene}\\\text{ethyne}}}{\begin{matrix}CH\\|||\\CH\end{matrix}} \xrightarrow{Cl_2} \underset{\substack{\text{Acetylene dichloride}\\\text{1, 2-dichloro ethene}}}{\begin{matrix}CHCl\\||\\CHCl\end{matrix}} \xrightarrow{Cl_2} \underset{\substack{\text{Acetylene tetrachloride}\\\text{1, 1, 2, 2 – tetrachloro ethane}}}{\begin{matrix}CHCl_2\\|\\CHCl_2\end{matrix}}$$

Acetylene reacts vigorously with chlorine gas in the presence of sunlight to give out flames.

$$C_2H_2 + Cl_2 \rightarrow 2C + 2HCl.$$

Reaction with bromine : When bromine, in carbon tetrachloride, is added to ethyne, *its brown colour disappears due to the formation of addition products.* The reaction is similar to the addition of chlorine.

$$\underset{\substack{\text{Ethyne}\\\text{(Acetylene)}}}{C_2H_2} \xrightarrow{+Br_2} \underset{\substack{\text{Acetylene}\\\text{dibromide}}}{C_2H_2Br_2} \xrightarrow{+Br_2} \underset{\substack{\text{Acetylene}\\\text{tetrabromide}}}{C_2H_2Br_4.}$$

Reaction with iodine : Iodine reacts slowly in the presence of alcohol to form di-iodo derivative.

$$CH \equiv CH + I_2 \rightarrow 1CH = CHI.$$

(1, 2,–di-iodoethene)

(iii) *Reaction with* HCl

Ethyne reacts with hydrochloric acid to first form chloroethene and finally 1, 1-dichloro ethane.

$$\underset{\text{Acetylene}}{\begin{matrix}CH\\|||\\CH\end{matrix}} \xrightarrow{+HCl} \underset{\text{Chloro ethene}}{\begin{matrix}CH_2\\||\\CHCl\end{matrix}} \xrightarrow{+HCl} \underset{\text{1,1-dichloro ethane}}{\begin{matrix}CH_3\\|\\CHCl_2\end{matrix}}$$

(iv) *Reaction with ozone*

Ethyne reacts with ozone at room temperature to produce acetylene ozonide.

$$\underset{\text{Ethyne}}{HC \equiv CH} + O_3 \rightarrow \underset{\text{(Acetylene ozonide)}}{H-\overset{O}{\overset{/\ \backslash}{C}}-\overset{O}{\overset{/\ \backslash}{C}}-H \quad (O{-}O)}$$

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12.17.5 Uses of ethyne (acetylene)

It is used

(i) for oxy-acetylene welding at very high temperatures. These temperatures are obtained when ethyne burns in oxygen.

(ii) as an illuminant in oxy-acetylene lamp.

(iii) for artificial ripening and preservation of fruits.

(iv) also for the manufacture of synthetic products like polymers, artificial rubber and oxalic acid, *etc.*

(v) for the manufacture of important organic compounds like acetaldehyde, acetic acid, plastic and rubber, *etc.*

12.17.6 Chemical Tests to distinguish between Alkanes, Alkenes and Alkynes

No.	Test	Alkanes (Methane and ethane) Saturated compound	Alkenes (Ethylene) Unsaturated compound	Alkynes (Acetylene) unsaturated compound
1	On adding a few drops of bromine solution in carbon tetrachloride to the hydrocarbon.	No change is observed.	The reddish brown colour of bromine solution gets decolorised.	The reddish brown colour gets decolorised.
2	On adding a few drops of alkaline potassium permanganate (purple colour) to the hydrocarbon.	No change is observed.	The purple colour fades.	The purple colour fades. (Baeyer's test) $CH \equiv CH + 4[O]$ Ethyne $\xrightarrow[\text{potassium permangana te}]{\text{alkaline}}$ $\begin{matrix}COOH\\ \vert \\ COOH\end{matrix}$ oxalic acid
3	On adding a few drops of ammonical cuprous chloride to the hydrocarbon.	No change is observed.	No change is observed.	Red precipitate of copper acetylide is formed.
4.	On adding ammonical Silver nitrate	No observation	No observation	White precipitate of silver acetylide is formed.

EXERCISE 12D

1. What are the sources for alkynes ? Give the general formula of alkynes.
2. Give an example of isomers shown by triple bond hydrocarbon (alkynes) and write its IUPAC name.
3. How is acetylene prepared in the laboratory ?
   (a) Draw a diagram. (b) Give an equation.
   (c) How is pure dry gas collected ?
4. Give the method of preparation of ethyne by : 1, 2-dibromoethene.
5. Name the hydrocarbon which :
   (a) is a tetrahedral molecule,
   (b) is a planar molecule,
   (c) is a linear molecule,
   (d) forms a red precipitate with ammoniacal solution of copper (I) chloride,
   (e) is known as paraffin,
   (f) is known as olefin.
   (g) a compound which will give acetylene (Ethene) gas when treated with water.
6. Classify the following compounds as alkanes, alkenes and alkynes. C_3H_4, C_3H_8, C_5H_8, C_3H_6
7. Give a chemical test to distinguish between
   (a) saturated and unsaturated compounds.
   (b) ethane and ethene,
   (c) ethene (ethylene) and ethyne (acetylene).
8. Name the products formed and write an equation when ethyne is added to the following in an inert solvent :
   (a) chlorine, (b) bromine,
   (c) iodine, (d) hydrogen.
   (e) excess of hydrochloric acid.

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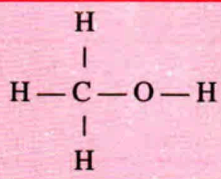
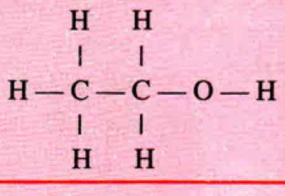
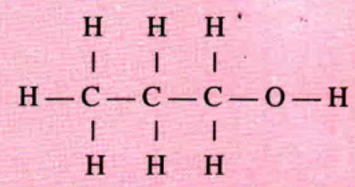
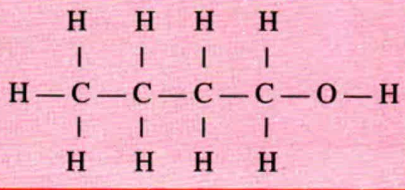
12E ALCOHOLS

12.18 ALCOHOLS

Alcohols are the hydroxyl derivatives of alkanes. They are formed by replacing one or more hydrogen atoms of the alkane with OH group.

Homologous series of alcohols and structural formula are given in Table 12.9.

TABLE 12.9 : HOMOLOGOUS SERIES OF ALCOHOLS (General formula : $C_nH_{2n+1}OH$)

Common Name	Molecular formula	Abbreviated formula	Structural formula	IUPAC Name
Methyl alcohol	CH_3OH	CH_3–OH		Methanol
Ethyl alcohol	C_2H_5OH	$CH_3 - CH_2 - OH$		Ethanol
Propyl alcohol	C_3H_7OH	$CH_3 - CH_2 - CH_2 - OH$		Propanol
Butyl alcohol	C_4H_9OH	$CH_3 - CH_2 - CH_2 - CH_2 - OH$		Butanol

Alcohols can be ***monohydric,*** *i.e.,* with one OH group attached to the carbon atom.

Example :

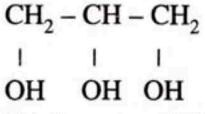
Methyl alcohol (CH_3OH) and **ethyl alcohol** (C_2H_5OH).

They can be ***dihydric,*** *i.e.,* with two OH groups attached to the carbon atom.

Example : **Glycol** $C_2H_4(OH)_2$.

They can be ***trihydric,*** *i.e.,* with three OH groups attached to the carbon atom.

Example : **Glycerol**



IUPAC name : 1, 2, 3–propane triol.

12.19 ETHANOL

12.19.1 Occurrence

Since alcohols are not found naturally in the earth's atmosphere, so they are obtained by synthesis.

For example, methanol (wood spirit) is obtained from destructive distillation of wood, while ethanol is obtained from fermentation* of sugar.

$$C_6H_{12}O_6\ (aq.) \xrightarrow[37°C]{\text{enzymes}} 2C_2H_5OH + 2CO_2$$

Cracking of petroleum is a source of ethane, which is used for preparing ethanol.

* Fermentation is an anaerobic process. It takes place under conditions where there is no air or oxygen available.

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Ethyl alcohol is an essential constituent of all wines and is called SPIRIT OF WINE. It is obtained from starchy grain and is also known as GRAIN ALCOHOL.

12.19.2 Laboratory preparation

Alcohol can be prepared by the ***hydrolysis of alkyl halides*** *(haloalkanes) on reaction with a hot dilute alkali.* [R stands for alkyl group]

Alkyl halide	+	Alkali	→	Alcohol	+	Alkali halide
RX	+	KOH (aqueous)	$\xrightarrow{\text{boil}}$	ROH	+	KX
CH_3Br	+	KOH (aqueous)	$\xrightarrow{\text{boil}}$	CH_3OH	+	KBr
C_2H_5Cl	+	KOH (aqueous)	$\xrightarrow{\text{boil}}$	C_2H_5OH	+	KCl
C_2H_5Br	+	NaOH (aqueous)	$\xrightarrow{\text{boil}}$	C_2H_5OH	+	NaBr

12.19.3 Industrial method (large scale method)

Hydration of ethene : When concentrated sulphuric acid is added to ethene (obtained from cracking of petroleum) at a temperature of 80°C and pressure of 30 atm, ethyl hydrogen sulphate is produced. Ethyl hydrogen sulphate, on hydrolysis with boiling water, gives ethanol.

$$\underset{\text{Ethene}}{C_2H_4} + H_2SO_4 \xrightarrow[30\,\text{atm}]{80°C} C_2H_5HSO_4$$

$$\underset{\text{Ethyl hydrogen sulphate}}{C_2H_5HSO_4} + H_2O \xrightarrow[\text{(Boiling)}]{} \underset{\text{Ethanol}}{C_2H_5OH} + H_2SO_4$$

Alternatively, ethanol is produced when ethene is heated with water at 300°C and 60 atmosphere pressure in presence of phosphoric acid catalyst.

$$C_2H_4 + H_2O \xrightarrow[300°C,\ 60\ \text{atm.}]{H_3PO_4} C_2H_5OH$$

12.19.4 Properties of alcohols

A. *Physical properties :*

(i) **Nature :** They are inflammable volatile liquids.

(ii) **Boiling point :** Their boiling point increases with an increase in molecular weight.

Examples : CH_3OH = 64·5°C

CH_3CH_2OH = 78·3°C

(iii) **Solubility :** They are soluble in water as well as in organic solvents.

(iv) **Density :** Ethanol is lighter than water its specific gravity is 0·79 at 293 K.

(v) They are **colourless** and have a faint odour and a burning taste.

(vi) They are toxic.

For example : methyl alcohol, *if consumed, causes blindness and even death.*

Ethyl alcohol affects *that part of the brain which controls our muscular movements and then gives temporary relief from tiredness. But it damages the liver and kidney too.*

B. *Chemical properties :*

Alcohols are a reactive class of compounds.

1. Combustion (burning)

Alcohol burns readily in air to produce carbon dioxide and water vapours. A lot of heat is produced during the combustion of alcohols.

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

Ethanol burns with a pale blue flame.

2. Oxidation with acidified potassium dichromate (oxidising agents)

Alcohols react with oxidising agents like acidified *potassium dichromate* and acidified *potassium* manganate.

These oxidising agent supply nascent oxygen [O] for oxidation and thus alcohols first get converted into aldehydes and later change into their respective acids.

Alcohol	*Oxidising agent*	*Aldehyde*	*Oxidising agent*	*Carboxylic acid*
CH_3OH Methyl alcohol	$\xrightarrow[K_2Cr_2O_7]{[O]}$	$HCHO + H_2O$ Formaldehyde (methanal)	$\xrightarrow[K_2Cr_2O_7]{[O]}$	HCOOH Formic acid (methanoic acid)
C_2H_5OH Ethyl alcohol	$\xrightarrow[K_2Cr_2O_7]{[O]}$	$CH_3CHO + H_2O$ Acetaldehyde (ethanal)	$\xrightarrow[K_2Cr_2O_7]{[O]}$	CH_3COOH Acetic acid (ethanoic acid)

If wine or beer is left open in the air, ethanol is oxidised to ethanoic acid. This is why wine tastes like vinegar after it's been left open for a couple of days.

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3. Action with sodium

When sodium reacts with methyl alcohol and ethyl alcohol at room temperature, hydrogen is evolved with the formation of sodium methoxide and sodium ethoxide respectively.

$$2CH_3OH + 2Na \rightarrow \underset{\text{Sodium methoxide}}{2CH_3ONa} + H_2$$

$$2C_2H_5OH + 2Na \rightarrow \underset{\text{Sodium ethoxide}}{2C_2H_5ONa} + H_2$$

Effervescence of Hydrogen serves as a test for alcohols.

4. Reactions with acetic acid

When alcohols react with acetic acid in the presence of conc. H_2SO_4 at high temperature, the respective esters are produced and the process is known as *esterification*.

$$CH_3OH + CH_3COOH \xrightarrow{\text{conc. } H_2SO_4} \underset{\text{(Methyl acetate)}}{CH_3COOCH_3} + H_2O$$

$$C_2H_5OH + CH_3COOH \xrightarrow{\text{conc. } H_2SO_4} \underset{\text{(Ethyl acetate)}}{CH_3COOC_2H_5} + H_2O$$

Esters can be detected by their fruity smell.

5. Dehydration with conc. sulphuric acid

When concentrated sulphuric acid is added to ethyl alcohol and the mixture is heated upto 170°C, it causes dehydration of ethyl alcohol to give ethene.

$$\underset{\text{Ethyl alcohol}}{C_2H_5OH} \xrightarrow[170°C]{\text{conc. } H_2SO_4} \underset{\text{Ethene}}{CH_2 = CH_2} + H_2O$$

Ethanol can also be dehydrated by aluminium oxide at 350°C.

$$C_2H_5OH \xrightarrow[350°C]{Al_2O_3} C_2H_4 + H_2O$$

At 140°C conc. H_2SO_4 dehydrates alcohol to give the respective ethers.

$$\underset{\text{Methyl alcohol}}{2CH_3OH} \xrightarrow[140\,°C]{\text{conc. } H_2SO_4} \underset{\text{Dimethyl ether}}{CH_3 - O - CH_3} + H_2O$$

$$\underset{\text{Ethyl alcohol}}{2C_2H_5OH} \xrightarrow[140\,°C]{\text{conc. } H_2SO_4} \underset{\text{Diethyl ether}}{C_2H_5 - O - C_2H_5} + H_2O$$

6. Reaction with phosphorus halide

Alcohols react with phosphorus trichloride at room temperature to give chloroalkane and phosphorus acid.

$$3CH_3OH + PCl_3 \rightarrow \underset{\text{Chloromethane}}{3CH_3Cl} + H_3PO_3$$

$$3C_2H_5OH + PCl_3 \rightarrow \underset{\text{Chloroethane}}{3C_2H_5Cl} + \underset{\text{Phosphorus acid}}{H_3PO_3}$$

12.19.5 Uses of ethyl alcohol

(i) It is a good solvent for gums and resins.

(ii) It is used in thermometers and as preservative for biological specimens, due to its low freezing point.

(iii) It is used in the manufacture of chemicals such as chloroform, iodoform, ether, acetic acid and synthetic products like dyes, esters (perfumes), antiseptics, preservatives, etc.

(iv) Ethyl alcohol is used in alcoholic drinks (beverages) like whisky, wine and beer. Beers contain about 4% ethanol, wines about 11% and spirits (rum, brandy and whisky) 40% or more.

(v) As a fuel in form of power alcohol and as an antifreeze for automobile radiators.

Different commerical forms of ethanol

The alcohol obtained by distillation is 95% pure. It forms a constant boiling mixture of 5% water and 95% ethanol. Absolute alcohol may be obtained by distilling moist alcohol with benzene (an organic solvent). The mixture of water and benzene distills off and anhydrous alcohol is left behind.

1. Denatured alcohol or methylated spirit

Ethyl alcohol is a part of drinking beverages and is also a widely used solvent in industries. In order to make it undrinkable, certain poisonous substances like **pyridine, methyl alcohol** and **copper sulphate** are added to it.

Methylated spirit or denatured alcohol is ethyl alcohol with 5% methyl alcohol, a coloured dye and some pyridine. It is used for industrial applications.

2. Spurious alcohol

This is illicit liquor made by improper distillation. It contains large proportions of methanol in a mixture of alcohols. It is fatal for human consumption.

Spurious alcohol, though impure, can be used as a solvent for paints and varnishes.

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EXERCISE 12E

1. (a) What are alcohols ? State their sources.
   (b) Give general formulae of monohydric alcohol.
2. Give the :
   (a) dot diagram,
   (b) abbreviated formula,
   (c) structure of second member of the alcohol group.
   (d) structure of alcohol with 4 carbon atoms.
3. State the method of preparation of ethanol :
   (a) by hydrolysis of ethene,
   (b) by hydrolysis of ethyl bromide.
4. Halo alkanes reacts with alkalies to produce alcohol. Give the equation for the preparation of second member of homologous series of alcohol. State under what condition the reaction occur.
5. (a) How do the boiling point and melting point change in the homologous series of alcohols ?
   (b) Name the product formed when ethanol reacts with acetic acid. Give an equation.
   (c) What is the name given to this type of reaction ?
6. Complete and balance the following equations. State the conditions wherever necessary.
   (a) $\begin{matrix} CH \\ \| \\ CH \end{matrix} + H_2 \rightarrow _____ + H_2 \rightarrow _____$
   (b) $C_2H_4 + Cl_2 \rightarrow _____$
   (c) $C_2H_4 + HCl \rightarrow _____$
   (d) $CaC_2 + H_2O \rightarrow _____$
   (e) $C_2H_2 + Br_2 \rightarrow _____$
   (f) $C_2H_5OH \xrightarrow[K_2Cr_2O_7]{[O]} _____$
7. What is the effect of ethanol on human body ?
8. How are the following obtained :
   (a) absolute alcohol, (b) spurious alcohol,
   (c) methylated spirit ?
9. Name the products formed and give appropriate chemical equations for the following :
   (a) Sodium reacting with ethyl alcohol.
   (b) Ethanol oxidised by acidified potassium dichromate.
10. Give the trivial (common) names and the IUPAC names of the following :
    (a) C_3H_6, (b) C_2H_4,
    (c) C_2H_2, (d) CH_3OH,
    (e) C_2H_5OH.
11. Ethanol can be oxidised to ethanoic acid. Write the equation and name the oxidising agent.
12. Name an organic compound which is :
    (a) used for illuminating country houses,
    (b) used for making a household plastic material,
    (c) called 'wood spirit',
    (d) poisonous and contain OH group,
    (e) consumed as a drink,
    (f) made from water gas,
    (g) solvent for ethanol.

12F. CARBOXYLIC ACIDS

12.20 CARBOXYLIC ACIDS

An organic compound containing the carboxyl group (–COOH) is known as carboxylic acid. These compounds possess acidic properties.

General formula $C_nH_{2n+1}COOH$ (or RCOOH)

Functional group $-\overset{\overset{O}{\|}}{C}-OH$

An organic compound may contain one or more of these groups. Those which contain one —COOH group are called **monocarboxylic acid**.

For example : HCOOH (Formic acid), CH_3COOH (Acetic acid)

Those which contain two COOH groups are called **dicarboxylic acid**.

For example : $\begin{matrix} COOH \\ | \\ COOH \end{matrix}$ OR $(COOH)_2$

Oxalic acid

IUPAC Name : Ethane-di-oic acid

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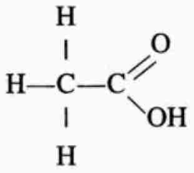
Table 12.10 : Homologous series of monocarboxylic acid

Formula	Common name	IUPAC name
$H-\overset{\overset{O}{\|}}{C}-OH$	Formic acid	Methanoic acid
$CH_3-\overset{\overset{O}{\|}}{C}-OH$	Acetic acid	Ethanoic acid
$CH_3CH_2-\overset{\overset{O}{\|}}{C}-OH$	Propionic acid	Propanoic acid
$CH_3CH_2CH_2-\overset{\overset{O}{\|}}{C}-OH$	Butyric acid	Butanoic acid

IUPAC name : IUPAC name is given by replacing '*e*' of the corresponding alkane by 'oic acid', because they are derived from alkanes by replacing one hydrogen atom with the carboxyl (–COOH) group. Thus they are also known as **Alkanoic acid**.

12.21 ACETIC ACID CH_3COOH

Structural formula :



The name acetic acid comes from latin word **'ACETUM'**. Acetic acid (IUPAC name ethanoic acid) occurs in free state in many fruits and as esters in a number of essential oils.

Dilute (4-5 percent) solution of acetic acid is also called vineger. The presence of a colouring matter gives vinegar a greyish colour while the presence of some other organic compound imparts it the usual taste and flavour. Vinegar is used for flavouring and preserving foods.

12.21.1 Lab preparation

Acetic acid can be prepared in the lab by oxidation of ethyl alcohol or acetaldehyde with acidified **potassium dichromate** or acidified sodium dichromate

$$\underset{\text{(ethanol)}}{CH_3CH_2OH} \xrightarrow{[O]} \underset{\text{(ethanal)}}{CH_3CHO} \xrightarrow{[O]} \underset{\text{ethanoic acid}}{CH_3COOH.}$$

12.21.2 Manufacture

(i) From acetylene

Acetylene is first converted into acetaldehyde by passing through 40% H_2SO_4 at 60°C in the presence of 1% $HgSO_4$ (catalyst).

$$C_2H_2 + H_2O \xrightarrow[HgSO_4]{H_2SO_4\ (dil.)} CH_3CHO.$$

The acetaldehyde is oxidised to acetic acid by passing a mixture of acetaldehyde vapours and air over **manganous acetate** at 70°C.

$$2CH_3CHO + O_2 \xrightarrow[\text{Catalyst}]{\Delta} 2CH_3COOH.$$

(ii) From ethanol

Acetic acid can also be prepared by passing alcohol vapours over platinum black as catalyst at 300°C in the presence of oxygen.

$$C_2H_5OH + O_2 \xrightarrow[\text{Platinum}]{300°C} CH_3COOH + H_2O$$

12.21.3 Physical properties

1. **Physical state :** Pure acetic acid is a colourless liquid.
2. **Odour :** Characteristic pungent smell.
3. **Boiling point :** It boils at 118°C.
4. **Melting point :** The anhydrous acid on cooling forms crystalline mass resembling ice; melting point 17°C and for this reason, it is called *glacial acetic acid.*
5. **Nature :** It is *hygroscopic liquid*, sp. gr. at 0°C is 1·08.
6. **Solubility :** It is miscible with water, alcohol and ether in all proportions.

12.21.4 Chemical properties

1. It is a weak acid, hence

(i) turns blue litmus red.

(ii) reacts with active metals (*i.e.*, Zn and Mg) evolving hydrogen.

$$2CH_3COOH + Zn \rightarrow \underset{\text{Zinc acetate}}{(CH_3COO)_2Zn} + H_2\uparrow$$

$$\underset{\text{Accetic acid}}{2CH_3COOH} + Mg \rightarrow \underset{\text{Magnesium acetate}}{(CH_3COO)_2Mg} + H_2\uparrow$$

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(iii) forms salt and water with bases

Acid	+	Base	→	Salt	+	Water
CH_3COOH	+	NaOH	→	CH_3COONa	+	H_2O
Acetic acid				Sod. acetate		

(iv) liberates carbon dioxide with carbonates or hydrogen carbonates.

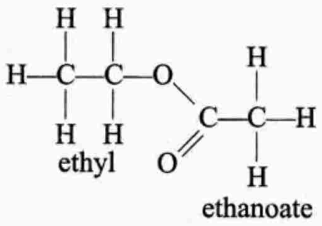
$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2\uparrow$$
$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2\uparrow$$

2. Acetic acid forms ester (pleasant fruity smelling compound) on reacting with alcohol in the presence of conc. H_2SO_4.

$$\underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Ethanol}}{HOC_2H_5} \xrightarrow{H_2SO_4} \underset{\text{Ethyl acetate}}{CH_3COOC_2H_5} + H_2O$$

The reaction is known as reaction of esterification.

Structure of Ethyl acetate (ethyl ethanoate)



3. Acetic Acid reacts with Phosphorus penta chloride and get converted to **acetylchloride**.

$$CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$$

4. It forms **acetic anhydride** on heating with phosphorus pentoxide as, it loses water molecule.

$$2CH_3COOH \xrightarrow[P_2O_5]{\Delta} (CH_3CO)_2O + H_2O$$

5. *Reduction :* A strong reducing agent such as lithum aluminium hydride ($LiAlH_4$) will reduce acetic acid to ethanol.

$$CH_3COOH + 4[H] \rightarrow C_2H_5OH + H_2O$$

12.21.5 Test

1. On adding acetic acid to carbonates and bicarbonates, carbon dioxide is evolved.
2. When warmed with ethyl alcohol and conc. sulphuric acid, a pleasant fruity smell of ethyl acetate is produced.
3. On adding neutral iron (III) chloride, wine red colour is produced.

12.21.6 Acetic acid is used

1. as a solvent for resins, cellulose, *etc.*
2. as a laboratory reagent.
3. as vinegar.
4. in medicines.
5. in the production of dyes, perfumes, esters, etc.
6. For coagulating rubber from latex.
7. For the manufacture of cellulose acetate which is used for making packaging materials, varnishes, rayons, *etc.*

EXERCISE 12F

1. What are carboxylic acids ? Give their general formula.
2. Write the common name, IUPAC name and formula of one monocarboxylic acid and one dicarboxylic acid.
3. Write the names of :
   (a) first three members of carboxylic acid series.
   (b) three compounds which can be oxidised directly, or in stages to produce acetic acid.
4. Vinegar is greyish in colour with a particular taste. Explain.
5. Give the structural formulae and IUPAC name of acetic acid. What is glacial acetic acid ?
6. Complete :
   (a) Vinegar is prepared by the bacterial oxidation of ..... .
   (b) The organic acid present in vinegar is ..... .
   (c) The next higher homologue of ethanoic acid is ..... .
7. How is acetic acid prepared from
   (a) ethanol (b) acetylene ?
8. What do you notice when acetic acid reacts with
   (a) litmus, (b) metals,
   (c) alkalies, (d) alcohol ?
9. Acetic acid is a typical acid. Write one equation in each case for its reaction with
   (a) a metal, (b) a base/alkali,
   (c) a carbonate, (d) a bicarbonate.

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10. Give *two* tests to show that CH_3COOH is acidic in nature.

11. What do you observe when acetic acid is added to :
    (a) sodium bicarbonate,
    (b) ethyl alcohol in the presence of sulphuric acid,
    (c) neutral $FeCl_3$ solution ?

12. Name :
    (a) compound formed when acetic acid and ethanol react together.
    (b) reducing agent used to convert acetic acid into ethanol.
    (c) substance used to change acetic acid to acetic anhydride.

MISCELLANEOUS

1. (a) Write an equation for the laboratory preparation of (i) an unsaturated hydrocarbon from calcium carbide. (ii) an alcohol from ethyl bromide.
   (b) What would you see, when ethyne is bubbled through a solution of bromine in carbon tetrachloride ?
   (c) Name the addition product formed between ethene and water.

2. Ethanol can be converted into ethene which can be changed into ethane. Choose the correct word or phrase from the brackets to complete the following sentences.
   (a) The conversion of ethanol into ethene is an example of _____ (dehydration, dehydrogenation).
   (b) Converting ethanol into ethene requires the use of _____ (conc. HCl, conc. HNO_3, conc. H_2SO_4).
   (c) The conversion of ethene into ethane is an example of _____ (hydration, hydrogenation).
   (d) The catalyst used in the conversion of ethene into ethane is commonly _____ (iron, nickel, cobalt).

3. Give reasons :
   (a) Ethyne is more reactive than ethene.
   (b) Ethene is more reactive than ethane.
   (c) Hydrocarbons are excellent fuels.

4. (a) Write balanced equations
      (i) when butane is burnt in oxygen
      (ii) preparation of ethylene from ethyl alcohol.
   (b) (i) Convert ethane to acetic acid
      (ii) Convert acetylene to ethane
      (iii) Convert acetic acid to ethyl alcohol
      (iv) Convert acetic acid to ethyl acetate.

5. Write the equations for the following lab. preparations:
   (a) Ethane from sodium propionate,
   (b) Ethene from Iodoethane,
   (c) Ethyne from calcium carbide,
   (d) Methanol from Iodomethane.

6. (a) Write the equation for the preparation of carbon tetrachloride from methane.
   (b) Draw the structural formula of ethyne.
   (c) How is the structure of alkynes different from that of alkenes ?

7. Fill in the blanks with the correct words from the brackets :
   Alkenes are the (a) ..... (analogous/homologous) series of (b) ..... (saturated/unsaturated) hydrocarbons. They differ from alkanes due to the presence of (c) ..... (double/single) bonds. Alkenes mainly undergo (d) ..... (addition/substitution) reactions.

8. (a) Draw the structural formulae of the two isomers of Butane. Give the correct IUPAC name of each isomer.
   (b) State one use of acetylene.

9. Copy and complete the following table which relates to three homologous series of hydrocarbons:

General Formula	C_nH_{2n}	C_nH_{2n-2}	C_nH_{2n+2}
IUPAC name of the homologous series			
Characteristic bond type			Single bonds
IUPAC name of the first member of the series			
Type of reaction with chlorine.		Addition	

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2008

(a) **Name** the organic compound prepared by each of the following reactions :

(i) $C_2H_5COONa + NaOH \rightarrow$

(ii) $CH_3I + 2H \rightarrow$

(iii) $C_2H_5Br + KOH$ (alcoholic solution) $\rightarrow$

(iv) $CO + 2H_2$ (Zinc oxide catalyst) $\rightarrow$

(v) $CaC_2 + 2H_2O \rightarrow$

(b) Write the equations for the following reactions :

(i) Calcium carbide and water.

(ii) Ethene and water (steam).

(iii) Bromoethane and an aqueous solution of sodium hydroxide.

(c) Distinguish between the saturated hydrocarbon ethane and the unsaturated hydrocarbon ethene by drawing their structural formulae.

(d) Addition reactions and substitution reactions are types of organic reactions. Which type of reaction is shown by :

(i) ethane (ii) ethene ?

(e) (i) Write the equation for the complete combustion of ethane.

(ii) Using appropriate catalysts, ethane can be oxidized to an alcohol, an aldehyde and an acid. Name the alcohol, aldehyde and acid formed when ethane is oxidized.

(f) (i) Why is pure acetic acid known as glacial acetic acid ?

(ii) What type of compound is formed by the reaction between acetic acid and an alcohol ?

2009

(a) Which of the following statements is wrong about alkanes ?

(i) They are all saturated hydrocarbon.

(ii) They can undergo addition as well as substitution reaction.

(iii) They are almost non polar in nature.

(iv) On complete combustion give out carbon dioxide and water.

(b) The organic compound obtained as the end product of the fermentation of sugar solution is :

(i) Methanol (ii) Ethanol

(iii) Ethane (iv) Methanoic acid

(c) Find the odd one out and explain :

C_3H_8, C_5H_{10}, C_2H_6, CH_4

(d) Give chemical equation for :

(i) The laboratory preparation of methane from sodium acetate.

(ii) The industrial preparation fo methanol from water gas.

(iii) The reaction of one mole of ethene with one mole of chlorine gas.

(iv) The preparation of ethyne from 1, 2-dibromoethane.

(e) State how the following conversions can be carried out :

(i) Ethyl chloride to Ethyl alcohol

(ii) Ethyl chloride to Ethene

(iii) Ethene to Ethyl alcohol

(iv) Ethyl alcohol to Ethene

(f) (i) Define isomerism

(ii) Give the IUPAC name of the isomer C_4H_{10} which has a branched chain.

2010

(a) An organic compound undergoes addition reactions and gives a red colour precipitate with ammoniacal cuprous chloride. Therefore, the organic compound could be :

(i) Ethane (ii) Ethene

(iii) Ethyne (iv) Ethanol

(b) An organic weak acid is :

(i) Formic acid (ii) Sulphuric acid

(iii) Nitric acid (iv) Hydrochloric acid

(c) The organic compound mixed with ethanol to make it spurious is :

(i) Methanol (ii) Methanoic acid

(iii) Methanal (iv) Ethanoic acid

(d) Draw the structural formula for each of the following:

(i) Ethanoic acid (ii) But-2-yne

(e) Compound A is bubbled through bromine dissolved in carbon tetrachloride and the product is $CH_2Br - CH_2Br$.

(i) Draw the structural formula of A.

(ii) What type of reaction has A undergone ?

(iii) What is your observation ?

(iv) Name (not formula) the compound formed when steam reacts with A in the presence of phosphoric acid.

(v) What is the procedure for converting the product of (e) (iv) back to A ?

2011

(a) The functional group present in acetic acid is :

(i) Ketonic $>C = O$ (ii) Hydroxyl – OH

(iii) Aldehydic – CHO (iv) Carboxyl – COOH

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(b) The unsaturated hydrocarbons undergo :

(i) a substitution reaction

(ii) an oxidation reaction

(iii) an addition reaction

(iv) none of the above

(c) The number of C-H bonds in ethane molecule are :

(i) Four (ii) Six

(iii) Eight (iv) Ten

(d) Choose the correct word/phrase from within the brackets to complete the following sentences :

(i) The catalyst used for conversion of ethene to ethane is commonly ..... (nickel / iron / cobalt)

(ii) When acetaldehyde is oxidized with acidified potassium dichromate, it forms ..... (ester / ethanol / acetic acid)

(iii) Ethanoic acid reacts with ethanol in presence of concentrated H_2SO_4, so as to form a compound and water. The chemical reaction which takes place is called ..... (dehydration / hydrogenation / esterification)

(iv) Write the equation for the reaction taking place between 1, 2–dibromoethane and alcoholic potassium hydroxide.

(v) The product formed when ethene gas reacts with water in the presence of sulphuric acid ..... (ethanol /ethanal / ethanoic acid)

(e) Write balanced chemical equations for the following :

(i) Monochloro ethane is hydrolysed with aqueous KOH.

(ii) A mixture of sodalime and sodium acetate is heated.

(iii) Ethanol under high pressure and low temperature is treated with acidified potassium dichromate.

(iv) Water is added to calcium carbide.

(v) Ethanol reacts with sodium at room temperature.

2012

(a) Give the structural formula for the following :

(i) Methanoic acid (ii) Ethanal

(iii) Ethyne (iv) Acetone

(iv) 2-methyl propane.

(b) From the following organic compounds given below, choose one compound in each case which relates to the description [i] to [iv] :

[*Ethyne, ethanol, acetic acid, ethene, methane*].

(i) An unsaturated hydrocarbon used for welding purposes.

(ii) An organic compound whose functional group is carboxyl.

(iii) A hydrocarbon which on catalytic hydrogenation gives a saturated hydrocarbon.

(iv) An organic compound used as a thermometric liquid.

(c) (i) Why is pure acetic acid known as glacial acetic acid ?

(ii) Give a chemical equation for the reaction between ethyl alcohol and acetic acid.

2013

(a) (i) Give a chemical test to distinguish ethene gas and ethane gas.

(ii) Identify the statement that is incorrect about alkanes :

(A) They are hydrocarbons.

(B) There is a single covalent bond between carbon and hydrogen.

(C) They can undergo both substitution as well as addition reactions.

(D) On complete combustion they produce carbon dioxide and water.

(b) Give the structural formulae for the following :

(i) An isomer of n-butane.

(ii) 2-propanol.

2014

(a) The I.U.P.A.C. name of acetylene is :

(i) propane (ii) propyne

(iii) ethene (iv) ethyne

(b) Name hydrocarbons containing a $-\overset{\overset{O}{\|}}{C}-$ functional group.

(c) Give preparation of ethane from sodium propionate.

(d) Distinguish ethane and ethene (using alkaline potassium permanganate solution).

(e) Give the structural formula of the following :

(i) ethanol (ii) 1-propanal

(iii) ethanoic acid (iv) 1, 2, dichloroethane.

(f) Give preparation of ethanol from monochloroethane and aq. sodium hydroxide.

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13 Practical Work

SYLLABUS

Internal Assessment of Practical Work

Candidates will be asked to observe the effect of reagents and/or of heat on substances supplied to them. The exercises will be simple and may include the recognition and identification of certain gases and ions listed below. The examiners will not, however, be restricted in their choice to substances containing the listed ions.

Gases : Hydrogen, Oxygen, Carbon dioxide, Chlorine, Hydrogen chloride, Sulphur dioxide, Hydrogen sulphide, Ammonia, Water vapour, Nitrogen dioxide.

Ions : Calcium, Copper, Iron, Lead, Zinc and Ammonium, Carbonate, Chloride, Nitrate, Sulphide, Sulphite and Sulphate.

Knowledge of a formal scheme of analysis is not required. Semi-micro techniques are acceptable but candidates using such techniques may need to adapt the instructions given to suit the size of the apparatus being used.

Candidates are expected to have completed the following minimum practical work.

1. Make a solution of the unknown substance : add sodium hydroxide solution or ammonium hydroxide solution, make observations and give your deduction. Warming the mixture may be needed. Choose from substances containing Ca^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}, Pb^{2+}, Zn^{2+}, NH_4^+.
2. Supply a solution of a dilute acid and alkali. Determine which is acidic and which is basic, giving two tests for each.
3. Add concentrated hydrochloric acid to each of the given substances, warm, make observations, identify any product and make deductions :

   (a) copper oxide (b) manganese dioxide.
4. Use of *p*H in soil analysis, water analysis, medical field – simple identification with universal indicator.

13.1 RECOGNITION AND IDENTIFICATION OF GASES

Preparation of Gases	Recognition and Identification of Gas
1. Hydrogen *Add dilute HCl or dilute H_2SO_4 to the reactive metals* (metals above hydrogen in the activity series) *like magnesium, zinc, iron, etc.* Active metal + dil Acid → Salt + Hydrogen Mg + H_2SO_4 → $MgSO_4$ + H_2 Zn + 2HCl → $ZnCl_2$ + H_2	(a) The evolved gas is colourless, odourless and neutral to litmus. (b) Pure hydrogen burns with a **pale blue flame** when a burning splint is brought near it. (c) Hydrogen-air burns with a ***pop sound*** when a burning taper is brought near it.
2. Oxygen *Heat higher metallic oxides or metal nitrates* Heavy metal oxide $\xrightarrow{\Delta}$ Metal oxide/ Metal + oxygen $2Pb_3O_4 \xrightarrow{\Delta} 6PbO + O_2$ $2HgO \xrightarrow{\Delta} 2Hg + O_2$ Metal nitrates $\xrightarrow{\Delta}$ Metal oxide + Nitrogen dioxide + Oxygen $2Cu(NO_3)_2 \xrightarrow{\Delta} 2CuO + 4NO_2 + O_2$ $2Zn(NO_3)_2 \xrightarrow{\Delta} 2ZnO + 4NO_2 + O_2$	(a) The gas is colourless, odourless and neutral to litmus. (b) It ***rekindles a glowing splinter.*** (c) The gas is absorbed in colourless alkaline solution of pyrogallol and turns it dark brown.

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Preparation of Gases	Recognition and Identification of Gas
3. Carbon dioxide *Heat metallic carbonate* **(except sodium carbonate and potassium carbonate)** *or add dilute acid to any carbonate or hydrogen carbonate.* Metallic carbonate $\xrightarrow{\Delta}$ Metal oxide + carbon dioxide $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$ $CuCO_3 \xrightarrow{\Delta} CuO + CO_2$ Metal carbonate/ *hydrogen* carbonate + Acid (dil) → Salt + Water + Carbon dioxide $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	(a) The gas is colourless and odourless. (b) It turns ***moist blue litmus faint red***. (c) When the gas is passed through the lime water, it turns milky due to the formation of white precipitate of calcium carbonate. $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$ The milkiness disappears on passing excess of CO_2 $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$ ppt (Soluble) (d) The gas has no effect on filter paper dipped in acidified $K_2Cr_2O_7$ or $KMnO_4$.
4. Chlorine *Add conc. HCl to oxidising agents like Pb_3O_4, PbO_2, MnO_2, etc.* Oxidising agent + Conc. HCl → Metal chloride + Water + Chlorine $Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + 4H_2O + Cl_2$ $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$	(a) The gas is *greenish yellow* having a ***sharp pungent choking odour***. (b) It turns a ***moist blue litmus paper red*** and finally bleaches *i.e.*, ***decolourises it***. $Cl_2 + H_2O \rightarrow HCl + HClO$ $HClO \rightarrow HCl + [O]$ Colouring matter + [O] → Colourless or bleached product (c) It turns ***moist starch iodide paper (KI + Starch solution) blue black.*** $Cl_2 + 2KI \rightarrow 2KCl + I_2$ Starch + I_2 → Blue black colour (d) Pass the gas through ***silver nitrate solution***, a ***white ppt.*** is formed.
5. Hydrogen chloride *Add conc. H_2SO_4 to metal chlorides like NaCl, KCl, etc.* Metal chloride + Conc. sulphuric acid → Metal sulphate + Hydrogen chloride $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$ $KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$	(a) The gas is colourless with a pungent choking odour. (b) The gas turns moist *blue litmus paper red.* (c) If a glass rod dipped in ammonia solution is brought near the gas ***dense white fumes of ammonium chloride are formed.*** $NH_3(aq) + HCl \rightarrow NH_4Cl$ (d) Gas when passed through silver nitrate solution white precipitate is formed. $AgNO_3(aq) + HCl \rightarrow AgCl\downarrow + HNO_3$ The ppt dissolves in excess of NH_4OH. $AgCl + 2NH_4OH \rightarrow Ag(NH_3)_2Cl + 2H_2O$ soluble compound
6. Sulphur dioxide *Add dil. HCl or dil. H_2SO_4 to metallic sulphites.* Metal sulphite\ hydrogen sulphite + dil Acid → Salt + Water + Sulphur dioxide $Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$ $Na_2SO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + SO_2$ $2NaHSO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2SO_2$	(a) The gas is colourless with a smell of burning sulphur *i.e.*, suffocating odour. (b) It turns moist blue litmus red and finally bleaches it. (c) The gas turns lime water turbid (milky). (Similar to carbon dioxide). $Ca(OH)_2 + SO_2 \rightarrow CaSO_3\downarrow + H_2O$ (d) It ***decolourises pink potassium permanganate solution.*** $2KMnO_4 + 2H_2O + 5SO_2 \rightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$ (pink) (colourless) (e) It changes ***orange/yellow solution of acidified potassium dichromate green.*** $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow Cr_2(SO_4)_3 + K_2SO_4 + H_2O$ (orange) (green) (f) It has no effect on lead acetate paper.

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Preparation of Gases	Recognition and Identification of Gas
7. Hydrogen sulphide *Add dil HCl or dil H_2SO_4 to metallic sulphides like ZnS, FeS.* Metal sulphide + dil Acid → Salt + Hydrogen sulphide $FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$ $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$	(a) The gas is colourless having a *foul smell of rotten eggs.* (b) It turns ***moist blue litmus red.*** (c) It turns ***lead acetate paper black.*** $(CH_3COO)_2Pb + H_2S \rightarrow PbS\downarrow + 2CH_3COOH$ (black) (d) It also turns ***lead nitrate solution black.*** $Pb(NO_3)_2 + H_2S \rightarrow PbS\downarrow + 2HNO_3$
8. Ammonia ***Add alkali to ammonium salt like ammonium chloride, ammonium sulphate.*** Ammonium salt + Alkali → Salt + Water + Ammonia $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$ $NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$	(a) The gas is colourless with a sharp ***pungent characteristic smell.*** (b) It turns ***moist red litmus blue.*** (c) If a glass rod dipped in conc. HCl is brought near the gas, ***dense white fumes*** of ammonium chloride are formed. (d) The gas turns colourless ***Nesslers reagent*** *i.e.* (K_2HgI_4) potassium mercuric iodide brown.
9. Water vapour *Heat a crystalline substance like hydrated sodium carbonate or hydrated copper sulphate.* Hydrated salt $\xrightarrow{\Delta}$ Anhydrous salt + Water vapour $Na_2CO_3 \cdot 10H_2O \xrightarrow{\Delta} Na_2CO_3 + 10H_2O$ $CuSO_4 \cdot 5H_2O$ (blue) $\xrightarrow{\Delta}$ $CuSO_4$ (white) $+ 5H_2O$	(a) Colourless, odourless forms a clear liquid on the cooler parts of the test-tube. (b) The liquid is neutral to litmus. (c) This liquid ***turns anhydrous copper sulphate from white to blue.*** $CuSO_4$ (white) $+ 5H_2O \rightarrow CuSO_4 \cdot 5H_2O$ (blue) (d) It turns blue cobalt chloride paper pink. $CoCl_2$ (blue) $+ 6H_2O \rightarrow CoCl_2 \cdot 6H_2O$ (pink)
10. Nitrogen dioxide *Heat heavy metal nitrate like copper nitrate, lead nitrate.* Metal nitrate $\xrightarrow{\Delta}$ Metal oxide + Oxygen + Nitrogen dioxide $2Cu(NO_3)_2 \xrightarrow{\Delta} 2CuO + O_2 + 4NO_2$ $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + O_2 + 4NO_2$ **Note :** Sodium nitrate and potassium nitrate do not produce nitrogen dioxide on heating	(a) The gas is *brown* in colour having an irritating (pungent) odour and non-combustible. (b) It turns moist blue litmus paper red. (c) It turns starch Iodide paper from colourless to blue black. $2KI + 2NO_2 \rightarrow 2KNO_2 + I_2$ Turns potassium iodide paper brown. (d) It turns green ***acidified ferrous sulphate solution brown.***

13.2 RECOGNITION OF SUBSTANCES BY COLOUR, ODOUR, PHYSICAL STATE AND ACTION OF HEAT

A. Colour :	Suspected ion
1. Blue or bluish green colour.	Cu^{2+}
2. Light green colour.	Fe^{2+}
3. Yellow or yellowish brown.	Fe^{3+}
4. White colour (or colourless).	Pb^{2+}, Zn^{2+}, Ca^{2+} Na^+, K^+, or NH_4^+

B. Odour :	
1. Smell of ammonia gas.	NH_4^+
2. Smell of hydrogen sulphide gas.	S^{2-}
3. Smell of sulphur dioxide gas.	SO_3^{2-}

C. State :	
1. Amorphous salt.	CO_3^{2-}
2. Hygroscopic or deliquescent nature.	Cl^- or NO_3^-

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D. Action of Heat :

1. **Ammonium salt** when heated with alkali (except ammonium hydroxide) produces Ammonia gas. Ammonium nitrate (explosive) and Ammonium chloride leaves no residue behind on heating. Ammonium dichromate on heating leaves greenish grey mass.

   $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Cr_2O_3 + N_2 + 4H_2O.$
   (orange) greenish grey
2. PbO_2, Pb_3O_4, HgO, KNO_3, $NaNO_3$ on heating produces oxygen gas.
3. Carbonate and bicarbonate on heating evolves carbon dioxide (except K_2CO_3 and Na_2CO_3).
4. Hydrated salts on heating produces water vapour.
5. Sulphites and some sulphates when heated produce sulphur dioxide gas.
6. Lead compounds decompose to give **lead monoxide PbO (litharge)**. PbO is brown when hot, yellow when cold and sticks to the glass test tube.
7. Zinc compounds decompose on heating to give zinc oxide. $ZnCO_3 \rightarrow ZnO + CO_2$

   Zinc oxide is yellow when hot, white when cold. $ZnO \underset{\text{Cool}}{\overset{\Delta}{\rightleftharpoons}} ZnO$
   white yellow
8. Copper compounds decompose to give black copper oxide. $CuCO_3 \xrightarrow{\Delta} CuO + CO_2$
   (green) (black)

Note : **Dry test** involves – colour, density, physical state, dry heating and flame test.
Wet test involves adding reagents to identify the substance.

Table of solubility of salts and bases in water (Roman numerals indicate the valency of the radical or metal)

Cation→ Anion ↓	K (I)	Na (I)	Ba (II)	Ca (II)	Mg (II)	Al (III)	Zn (II)	Fe (II)	Fe (III)	Mn (II)	Pb (II)	Cu (II)	Ag (I)	Hg (II)
NO_3(I)	S	S	S	S	S	S	S	S	S	S	S	S	S	S
CH_3COO(I)	S	S	S	S	S	Sp	S	S	S	S	S	S	S	S
Cl(I)	S	S	S	S	S	S	S	S	S	S	N	S	N	N
SO_4(II)	S	S	N	Sp	S	S	S	S	S	S	N	S	Sp	S
OH(I)	S	S	S	Sp	Sp	N	N	N	N	N	N	N	—	—
S(II)	S	S	S	Sp	S	—	N	N	—	N	N	N	N	N
SO_3(II)	S	S	N	N	N	—	N	—	N	N	N	N	N	N
PO_4(III)	S	S	N	N	N	N	N	N	N	N	N	N	N	N
CO_3(II)	S	S	N	N	N	—	N	N	N	N	N	N	N	N

Note : S : Soluble; N : Insoluble; — : Does not exist; Sp : Sparingly soluble

13.3 IDENTIFICATION OF IONS

Identification of ions is usually done in solution state.

Salt is dissolved in water to make solution. If the salts do not dissolve in water then Nitric acid is added to form nitrate. All nitrates are soluble in water.

Cations are tested by the action of alkalies as they give characteristic coloured metallic hydroxide precipitates (see chapter 4). When adding the alkali, add it slowly at first (one drop at a time). If it is added too quickly, it is easy to miss a precipitate that redissolves in excess.

Anions are tested by the reaction of acids or by specific reagents.

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IDENTIFICATION OF CATIONS BY ACTION OF ALKALIES (SEE CHAPTER 4 ALSO)

Sodium hydroxide solution			Ammonia solution		
Metal	***Colour of precipitate***	***With excess sodium hydroxide solution***	***Metal***	***Colour of precipitate***	***With excess ammonium hydroxide solution***
Calcium	White curdy	Insoluble	Calcium	No ppt.	No change
Lead	White chalky	Soluble	Lead	White chalky	Insoluble
Zinc	White gelatinous	Soluble	Zinc	White gelatinous	Soluble
Copper	Pale blue	Insoluble	Copper	Pale blue	Soluble - deep blue solution
Iron (II)	Pale green turning brown	Insoluble	Iron (II)	Pale green turning brown	Insoluble
Iron (III)	Reddish brown	Insoluble	Iron (III)	Rust (brown)	Insoluble

Test for NH_4^+ salt

(i) When caustic alkali (NaOH or KOH) is added to any ammonium salt, ammonia gas is evolved.

(ii) If Nessler's reagent is added to any ammonium salt solution, it turns brown.

IDENTIFICATION OF ANIONS

(A) By adding dilute sulphuric acid

	Experiment	Observation	Inference
1. (a)	Take a small quantity of salt in a test tube and add dilute H_2SO_4. Warm, if no action in cold.	Brisk effervescence and a *colourless* and *odourless* gas evolved. It *does not support* combustion *i.e.*, burning splinter get extinguished. It turns *moist blue litmus faint red.*	The gas evolved is carbon dioxide. The salt contains **Carbonate** ion (CO_3^{2-}).
(b)	Pass the gas through lime water.	Lime water turns milky.	
(c)	To the above precipitate, pass the gas in excess or add dilute nitric acid in excess.	The white precipitate dissolves to form colourless solution.	
2. (a)	Take a small quantity of salt in a test-tube and add dil. H_2SO_4 acid.	**Rotten egg smelling** gas evolved. It turns moist blue litmus paper red.	The gas evolved is hydrogen Sulphide.
(b)	Bring a moist lead acetate paper near the gas.	Moist lead acetate paper **turns black.**	The salt contains **Sulphide** ion (S^{2-}).
3. (a)	Take a small quantity of salt in a test-tube and add dilute H_2SO_4 acid (warm if necessary).	Gas evolved has **suffocating odour of burning sulphur.**	The gas evolved is Sulphur dioxide.
(b)	Bring a filter paper moistened with acidified $K_2Cr_2O_7$ (Potassium dichromate) near the gas.	**Golden yellow or orange** colour paper turns **green.**	The salt contains **Sulphite** ion (SO_3^{2-}).

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(B) By adding concentrated sulphuric acid

	Experiment	Observation	Inference
1. (a)	Take a small amount of salt in a test-tube and add conc. H_2SO_4. Warm it gently.	Colourless gas (HCl) evolves with *pungent* odour.	***Chloride** ion (Cl^-)* may be present.
(b)	Bring a glass rod carrying a drop of ammonia solution near the gas evolved in (a).	*Dense white fumes* are produced.	***Chloride** ion (Cl^-)* confirmed.
(c)	Add a pinch of manganese dioxide to the salt followed by conc. H_2SO_4 and heat.	Greenish yellow gas evolves with a pungent odour, and turns moist starch iodide paper blue black.	***Chloride** ion (Cl^-)* confirmed.
(d)	Add silver nitrate ($AgNO_3$) solution	White precipitate of AgCl is formed which dissolves in excess of NH_4OH.	***Chloride** ion (Cl^-)* confirmed.
2. (a)	Take a small amount of salt in a test-tube and add conc. H_2SO_4 and warm gently.	*Reddish brown fumes* evolve. The fumes becomes thick on adding copper turnings.	The gas evolved is nitrogen dioxide and salt contains. ***Nitrate** (NO_3^-)*
(b)	To the salt solution, add freshly prepared ferrous sulphate solution, then cautiously pour a few drops of conc. H_2SO_4 along the side of the test tube. **This test is known as *brown ring-test.***	A *brown ring appears* at the junction of the two liquid layers. The brown ring disappears on shaking.	***Nitrate** ion (NO_3^-)* confirmed.

(C) Test for sulphate ion

	Experiment	Observation	Inference
1.	To the salt solution, add little nitric acid and then add barium chloride solution.	White precipitate is obtained, which is insoluble in mineral acid.	***Sulphate** ion (SO_4^{2-})* is confirmed.
2.	To the solution, add acetic acid and lead acetate solution.	White precipitate is obtained, which is soluble in excess of ammonium acetate solution.	***Sulphate** ion (SO_4^{2-})* confirmed.

General Identification of Anions

	Add to the salt or salt solution	Observation	Anion Present
1.	dil HCl or dil H_2SO_4 and heat	(i) Carbon dioxide gas is evolved. (ii) Sulphur dioxide gas is evolved. (iii) Hydrogen Sulphide gas is evolved.	CO_3^{2-} SO_3^{2-} S^{2-}
2.	Conc. H_2SO_4 and heat	(i) Hydrogen chloride gas is evolved. (ii) Nitrogen dioxide gas is evolved.	Cl^- NO_3^-
3.	Silver Nitrate solution	Curdy white ppt., is insoluble in dil. HNO_3 but dissolves in NH_4OH	Cl^-
4.	Lead acetate solution	White ppt insoluble in dil. HCl, dil. HNO_3	SO_4^{2-}
5.	Barium nitrate solution or Barium chloride solution	(i) White precipitate soluble in dil. HCl or dil. HNO_3. (ii) White precipitate insoluble in dil. HCl or dil. HNO_3.	CO_3^{2-} or SO_3^{2-} SO_4^{2-}

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13.4 DISTINCTION BETWEEN COLOURLESS SOLUTIONS OF DILUTE ACIDS AND ALKALIS.

	TEST	ACID SOLUTION	ALKALIS
1. (a)	Moist litmus paper	*Blue turns red*	*Red turns blue*
(b)	Methyl orange	Orange turns *pink*	Orange turns *yellow*
(c)	Phenolphthalein	Colourless remains *colourless*	Colourless turns *pink*
2.	**Chemical tests**		
(a)	On adding sodium carbonate	Carbon dioxide is evolved [test with lime water]	Carbon dioxide is not evolved
(b)	On adding ammonium carbonate	No ammonia gas evolved	Ammonia gas evolved [test with red litmus and bring glass rod dipped in HCl in contact with the gas]

13.5 DISTINGUISH BETWEEN BLACK COPPER OXIDE AND BLACK MANGANESE DIOXIDE

	Experiment	Manganese dioxide	Copper oxide
(a)	Add conc. HCl to black powder and heat.	Greenish yellow gas *chlorine* is evolved.	Chlorine gas is not evolved.
(b)	Filter the above solution.	*Filtrate* is brownish in colour.	*Filtrate* is bluish in colour.
(c)	On adding ammonium hydroxide to the above *filtrate*.	No precipitate is formed.	The pale blue precipitate is formed, which is soluble in excess of ammonium hydroxide, giving an azure blue colour (deep blue) to the solution.

13.6 FLAME TEST

Procedure : Concentrated HCl is added to the salt to form a paste and to convert the salt into its chloride, since chlorides of metals generally volatize and ionize when heated giving a characteristic colour in the non luminous flame of a bunsen burner.

A thin platinum wire is first cleaned thoroughly by **dipping** it into the **concentrated hydrochloric acid**. It is then heated in the non-luminous flame of the burner. *When the wire imparts no colour to the flame, it is ready for use.* Now, a little paste of the substance is taken on the wire. It is then introduced into the non-luminous part of the flame and colour imparted to the flame is observed.

Table 13.1 : Colour imparted to the flame by salts of different metals

	Colour imparted to the flame (Colour observed with naked eye)	Colour seen through the blue glass.	Name of metal
(a)	Golden yellow (persistent)	Yellow colour vanishes.	***Sodium ion*** (Na^+)
(b)	Violet (Lilac)	Violet or pink.	***Potassium ion*** (K^+)
(c)	Brick red	Light green.	***Calcium ion*** (Ca^{2+})
(d)	Peacock bluish green	Bluish green.	***Copper ion*** (Cu^{2+})

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FLAME TESTS — COLOURS

SODIUM SALT
Golden Yellow Flame

POTASSIUM SALT
Lilac Flame

CALCIUM SALT
Brick Red Flame

COPPER SALT
Peackock Bluish Green Flame

13.7 HYDROGEN ION SCALE OR THE *PH* SCALE (Refer Chapter 3)

The acidic and basic strengths of solutions are compared on the hydrogen ion scale or the pH scale, and gives a value called pH value.

A solution of *p*H 7 is neutral (water), less than 7 is acidic and more than 7 is basic.

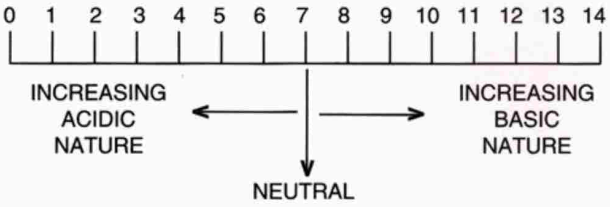

Fig. 13.1 The pH scale.

13.8 INDICATORS

Indicator	Colour in neutral solution	Colour in acidic solution	Colour in alkaline solution
Litmus	Purple	Red	Blue
Methyl orange	Orange	Pink	Yellow
Phenolphthalein	Colourless	Colourless	Pink
Alkaline phenolphthalein	Pink	Colourless	Pink

Universal indicator

It gives different colours with solutions of different *p*H values. Thus, *one universal indicator produces a green colour in neutral solution, pH = 7. It changes in basic solution from blue to indigo and to violet when pH increases from 7 to 14.* The colour change in acidic solution is from yellow to pink and then to red as *p*H decreases from 7 to 1 (Fig. 13.2).

Experiment : *How to find the pH of different substances ?*

Make solution of the substance to be tested and put a drop of it on *p*H paper. *For example* take hydrochloric acid, distilled water, sodium hydroxide solution, and ammonium hydroxide on different strips of *p*H paper. You will observe different shades on different strips. By comparing the colours given by the manufacturer, you can find the *p*H of different solutions.

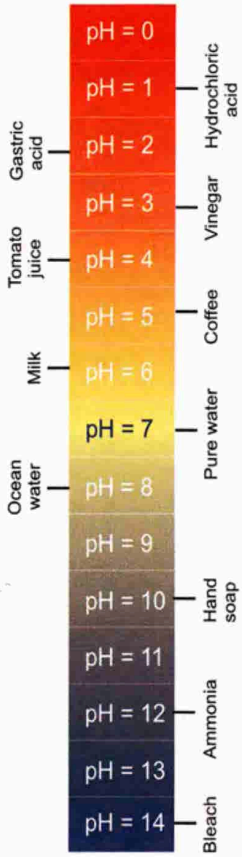

Fig. 13.2. Colour changes of universal indicator at different pH values

EXERCISE

1. (a) Give only one suitable chemical test to identify the following gases.
   (i) Ammonia (ii) Sulphur dioxide
   (iii) Hydrogen chloride (iv) Chlorine
   (v) Carbon dioxide (vi) Oxygen
   (vii) Hydrogen

   (b) Select a basic gas mentioned in Q.1 (a). How is the basic nature suspected ?

   (c) Select acidic gases from the gases mentioned in Q.1(a). How is the acidic nature suspected?

   (d) The two gases A and B are bleaching agents. A is greenish yellow and bleaches due to its oxidising

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property while B a colourless gas bleaches due to reduction. Identify A and B.

(e) Which gas turns blue cobalt chloride paper light pink ?

Give one similarity in test between (i) Cl_2 and HCl (ii) SO_2 and CO_2.

2. Name the gases which

(a) extinguishes burning wooden splinter.

Ans. : NH_3, HCl, SO_2, H_2S, CO_2, NO_2, Cl_2.

(b) turns moist red litmus blue.

(c) do no effect moist litmus.

(d) affects the acidified $K_2Cr_2O_7$ paper and also turns lime water dirty milky.

3. Name :

(a) Two carbonates which do not produce carbon dioxide on heating.

(b) A colourless gas which bleaches

(c) Gases which have sour taste

(d) Greenish yellow gas which also bleaches

(e) Gas with rotten egg smell.

4. From the following list of substances *choose* those which meet the description given in part (a) below. Ammonium chloride, ammonium nitrate, chlorine, dilute hydrochloric acid, iron, lead nitrate, manganese (IV) oxide, silver nitrate, sodium nitrate, sodium nitrite and sulphur.

Two compounds whose aqueous solutions give white precipitates with dilute hydrochloric acid.

5. Name the *anion* present in each of the following compounds :

(a) Compound A when warmed with concentrated sulphuric acid gives a gas which fumes in moist air and which gives dense white fumes with ammonia.

(b) When barium chloride solution is added to a solution of compound B, a white precipitate insoluble in dilute hydrochloric acid is formed.

(c) The action of heat on the insoluble compound C produces a gas which turns lime water turbid.

(d) Compound D when warmed with dilute sulphuric acid gives a gas which turns acidified dichromate solution green.

6. A given white crystalline salt was tested as follows :

(a) It dissolved in water and the resulting solution of the salt turned blue litmus red.

(b) Addition of barium chloride solution into this solution gave a white precipitate.

(c) A flame test on the salt gave a persistent golden-yellow colourisation.

What conclusions can be drawn for each observation ?

7. (a) Sodium hydroxide solution is added to solution A. A white precipitate is formed which is insoluble in excess sodium hydroxide solution. Name the metal ion present in solution A.

(b) When ammonium hydroxide is added to solution B, a pale blue precipitate is formed. This pale blue precipitate dissolves in excess ammonium hydroxide giving an inky blue solution. Name the cation present in solution B.

(c) When an ammonium salt is warmed with sodium hydroxide solution, ammonia gas is evolved. State *three* ways in which you could identify this gas.

8. Complete the following table and write your observations.

	Hydrogen sulphide	Ammonia	Sulphur dioxide	Hydrogen chloride
Shake the gas with red litmus solution				
Shake the gas with blue litmus solution				
Apply a burning splint to the gas				

9. Use the information given in (a) to (h) to identify the substances P to W selecting your answers from the given list.

List :

Calcium	Oxygen	Copper (II) oxide
Carbon	Calcium hydroxide	Copper (II) nitrate
Lead (II) oxide	Hydrogen chloride	Chlorine
Lead (II) nitrate	Calcium oxide	Ammonium chloride

(a) P is a white solid. When heated produces white fumes (sublime).

(b) P and R on warming produce an alkaline gas.

(c) On adding water to T, heat is evolved and R is formed.

(d) Q burns brightly in air to form T.

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(e) When S is heated, it gives off brown fumes and leaves a black residue of U.

(f) A solution of S is formed by warming U with dilute nitric acid.

(g) V is a gaseous non metallic element that reacts with hydrogen to form W.

(h) A solution of W will neutralise the solution of R.

10. Copy and complete the following table which refers to the action of heat on some carbonates :

Carbonate	Colour of residue on cooling
Zinc carbonate	
Lead carbonate	
Copper carbonate	

11. Distinguish by a chemical test,

(a) Sodium carbonate and sodium sulphite

(b) Sodium chloride and sodium sulphide

(c) Sodium hydroxide solution and ammonium hydroxide solution.

(d) Ammonium sulphate and sodium sulphate.

(e) Sulphuric acid from nitric acid and hydrochloric acid.

12. Sodium hydroxide solution is added first in a small quantity, then in excess to the *aqueous salt solutions* of copper (II) sulphate, zinc nitrate, lead nitrate, calcium chloride and iron (III) sulphate. Copy the following table and write the colour of the precipitate in (i) to (v) and the nature of the precipitate (soluble or insoluble) in (vi) to (x).

Aqueous salt solution	Colour of precipitate when NaOH is added in a small quantity	Nature of precipitate (soluble or insoluble) when NaOH is added in excess
Copper (II) sulphate	(i)	(vi)
Zinc nitrate	(ii)	(vii)
Lead nitrate	(iii)	(viii)
Calcium chloride	(iv)	(ix)
Iron (III) sulphate	(v)	(x)

13. State your observations when :

(a) lead nitrate solution and sodium chloride solution are mixed.

(b) zinc chloride solution, zinc nitrate solution and zinc sulphate solutions are added individually to

(i) barium chloride solution,

(ii) lead nitrate solution.

(c) Decomposition of bicarbonates by dil. H_2SO_4

$2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2$

$2KHCO_3 + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O + 2CO_2$

14. The questions (i) to (v) refer to the following salt solutions listed A to F.

A. Copper nitrate B. Iron (II) sulphate

C. Iron (III) chloride D. Lead nitrate

E. Magnesium sulphate F. Zinc chloride

(i) Which two solutions will give a white precipitate when treated with dilute hydrochloric acid followed by barium chloride solution?

(ii) Which two solutions will give a white precipitate when treated with dilute nitric acid followed by silver nitrate solution?

(iii) Which solution will give a white precipitate, when either dilute hydrochloric acid or dilute sulphuric acid is added to it?

(iv) Which solution becomes a deep/inky blue colour when excess of ammonium hydroxide is added to it?

(v) Which solution gives a white precipitate with excess ammonium hydroxide solution?

15. Mention the colour changes observed when the following indicators are added :

Solution	Acids	Alkalies
(a) Alkaline phenolphthalein solution,		
(b) Methyl orange solution,		
(c) Neutral litmus solution.		

16. Salts A, B, C, D and E undergo reactions (i) to (v) respectively. Identify the anion present in these salts on the basis of these reactions.

(a) When silver nitrate solution is added to a solution of A, a white precipitate, insoluble in dilute nitric acid, is formed.

(b) Addition of dilute hydrochloric acid to B produces a gas which turns lead acetate paper black.

(c) When a freshly prepared solution of ferrous sulphate is added to a solution of C and concentrated sulphuric acid is gently poured from the side of the test-tube, a brown ring is formed.

(d) When dilute sulphuric acid is added to D, a gas is produced which turns acidified potassium dichromate solution from orange to green.

(e) Addition of dilute hydrochloric acid to E produces an effervescence. The gas produced turns limewater milky but does not affect acidified potassium dichromate solution.

2008

(a) The salt which in solution gives a pale green precipitate with sodium hydroxide solution and a white precipitate with barium chloride solution is :

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(i) Iron (III) sulphate (ii) Iron (II) sulphate
(iii) Iron (II) chloride (iv) Iron (III) chloride

(b) Identify the following substances :
(i) An alkaline gas A which gives dense white fumes with hydrogen chloride.
(ii) A dilute acid B which does not normally give hydrogen when reacted with metals but does give a gas when it reacts with copper.
(iii) Gas C has an offensive smell like rotten eggs.
(iv) Gas D is a colourless gas which can be used as a bleaching agent.
(v) Liquid E can be dehydrated to produce ethene.

2009

(a) Carbon dioxide and sulphur dioxide gas can be distinguished by using.
(i) moist blue litmus paper
(ii) lime water
(iii) acidified potassium dichromate paper
(iv) none of the above.

(b) *Identify the substance 'R' based on the information given below :*
The pale green solid 'R' turns reddish brown on heating. Its aqueous solution gives a white precipitate with barium chloride solution. The precipitate is insoluble in mineral acids.

(c) Give one chemical test to distinguish between the following pairs of compounds.
(i) Zinc sulphate soln. and zinc chloride soln.
(ii) Iron (II) chloride soln. and iron (III) chlorde soln.
(iii) Calcium nitrate soln. and calcium chloride soln.

2010

(a) Select the correct answer from A, B, C, D and E.
A. Nitroso iron (II) sulphate B. Iron (III) chloride
C. Chromium sulphate D. Lead (II) chloride
E. Sodium chloride.
The compound which is responsible for the green colour formed when SO_2 is bubbled through acidified potassium dichromate solution.

(b) State your observation :
(i) a piece of moist blue litmus paper.
(ii) paper soaked in potassium permanganate solution – is introduced in each case into a jar of sulphur dioxide.

(c) Write the equation for the reaction of magnesium sulphate solution with barium chloride solution.

2011

(a) Choose from the list of substances – Acetylene gas, aqua fortis, coke, brass, barium chloride, bronze, platinum.
An aqueous salt solution used for testing sulphate radical.

2012

(a) Name – the gas which turns acidified potassium dichromate solution green.

(b) Identify the anion present in the following compounds :
(i) Compound X on heating with copper turnings and conc. sulphuric acid liberates a reddish brown gas.
(ii) When a solution of compound Y is treated with silver nitrate solution a white precipitate is obtained which is soluble in excess of ammonium hydroxide solution.
(iii) Compound Z which on reacting with dilute sulphuric acid liberates a gas which turns lime water milky, but the gas has no effect on acidified potassium dichromate solution.
(iv) Compound L on reacting with barium chloride solution gives a white precipitate insoluble in dilute hydrochloric acid or dilute nitric acid.

(c) State one chemical test between each of the following pairs :
(i) Sodium carbonate and sodium sulphite.
(ii) Ferrous nitrate and lead nitrate.
(iii) Manganese dioxide and copper (II) oxide.

(d) State your observation : A zinc granule is added to copper sulphate solution.

(e) Give balanced equation for the reaction : Silver nitrate solution and sodium chloride solution.

2013

(a) Give a chemical test to distinguish between :
(i) Sodium chloride soln. and sodium nitrate soln.
(ii) Hydrogen chloride gas and hydrogen sulphide gas.
(iii) Calcium nitrate soln. and zinc nitrate soln.
(iv) Carbon dioxide gas and sulphur dioxide gas.

(b) Which one of the following will not produce an acid when made to react with water.
(i) Carbon monoxide (ii) Carbon dioxide
(iii) Nitrogen dioxide (iv) Sulphur trioxide

2014

(a) Distinguish between : Sodium nitrate and sodium sulphite [using dilute sulphuric acid].

(b) State your observation : When moist starch iodide paper is introduced into chlorine gas.

(c) The flame test with a salt P gave a brick red flame. What is the cation in P.

(d) A gas Q turns moist lead acetate paper silvery black. Identify the gas Q.

(e) pH of liquid R is 10. What kind of substance is R ?

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